



Software User Guide for the Advanced Electrolyte Model (AEM) Developed by Idaho National Laboratory (INL)

Prepared and Distributed by:

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Advanced Electrolyte Model (AEM)

Revision Record

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1.0	May 31, 2023	First release.
1.1	June 07, 2023	Second release with notes for Trial Version.
1.2	February 05, 2024	Third release with new salts and solvents with AEM version 2.24.1.
1.3	July 12, 2024	Fourth release with AEM ACCC Module and AEM Python API.
1.4	Aug 23, 2024	Fifth release with updates and ACCC functionality for Python API.
1.5	January 21, 2025	Updated release package with AEM version 2.24.3, AEM Compare, updated reports, automatic .csv files, and other minor updates.
1.6	July 21, 2025	Updated release package with AEM version 2.25.1, lower salt concentration ranges, dual ACCC salts, improved matrix optimization runs, custom recipe input mode, and other minor updates.
1.7	April 2, 2026	Updated release package with AEM version 2.26.1, updates to report 3, report 6, new report 21, and new compounds.

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1 Introduction to the Advanced Electrolyte Model (AEM)

Software License Agreement Background:

BATTELLE ENERGY ALLIANCE, LLC (BEA), having its principal place of business at 2525 N. Fremont Avenue, Idaho Falls, ID 83415, is the management and operating contractor of the Idaho National Laboratory (INL) under the United States Department of Energy (DOE) Contract No. DE-AC07-05ID14517. BEA has granted RIDGETOP GROUP INC. exclusive distribution rights to make and distribute copies of the commercial product version of the Advanced Electrolyte Model (AEM) comprising of a jointly developed Graphical User Interface (GUI), AEM OBJECT CODE, and engineering services to support customer training and understanding of AEM data output.

Prior to software use and operation please review all terms and conditions covered in the standard AEM Software License Agreement (SLA).

About Idaho National Laboratory:

As one of 17 national labs in the U.S. Department of Energy complex, Idaho National Laboratory is home to more than 5,700 researchers and support staff focused on innovations in nuclear research, renewable energy systems, and security solutions that are changing the world. From discoveries in advanced nuclear energy to carbon-free energy options and to protecting our nation's most critical infrastructure assets, our talented team at INL is constantly pushing the limits to redefine what's possible. To find out more information about INL visit our website at www.inl.gov.

About Ridgetop Group:

Ridgetop Group Inc., founded in 2000, is an established engineering and technology company that provides condition-based maintenance (CBM), prognostic health management (PHM), and reliability engineering solutions to government and commercial organizations around the world. Ridgetop and INL have formed a strong technology and business collaboration partnership that complements Ridgetop's background and reputation of providing customers with innovative products and solutions for aerospace, defense, transportation, energy, and industrial applications. Ridgetop is headquartered in Tucson, Arizona and has a dedicated staff of highly qualified researchers, engineers, and data scientists who support the business development team to design, develop, and deploy the most innovative solutions. Ridgetop is also an AS9100D and ISO9001:2015 certified organization by the Management Certification of North America (MCNA). More information on Ridgetop's complete list of products and services can be reviewed on the company website at www.ridgetopgroup.com.

1.1 Technology Introduction and Overview

Welcome, and congratulations on your purchase of the Advanced Electrolyte Model (AEM)! This innovative technology is a scientific modeling tool capable of exploring activity at the molecular level to simulate, diagnose, and predict electrolyte behaviors.

Alessandro Volta introduced the voltaic pile in 1800, ushering in the age of portable power and kicking off a relentless pursuit of a better battery that continues unabated today. A far cry from crude stacks of copper and zinc plates separated by brine-soaked paper, today's batteries are marvels of modern engineering that utilize exotic materials in novel configurations. Containing a complex mixture of chemical elements in its four primary working components (the cathode, anode, separator, and electrolyte solution), effective batteries are limited to key combinations within the periodic table that represent the best levels of performance, value, safety, and environmental impact. While the possible combinations are finite, small tweaks to their relative proportions can produce fundamentally different properties. Developing new and improved battery configurations using traditional methods is a time-consuming and expensive process.

Looking to overcome this, researchers at Idaho National Laboratory have developed sophisticated modeling software that quickly and accurately assesses both macro-scale effects and molecular-level interactions of electrolytic solutions, analyzing and reporting on more than 35 key parameters. The Advanced Electrolyte Model (AEM) can handle multi-component solutions and can be run on a laptop computer, delivering results orders of magnitude faster than competing technologies. The resulting modeled predictions have been experimentally verified to be within a 5 to 10-percent deviation of lab data, often less.

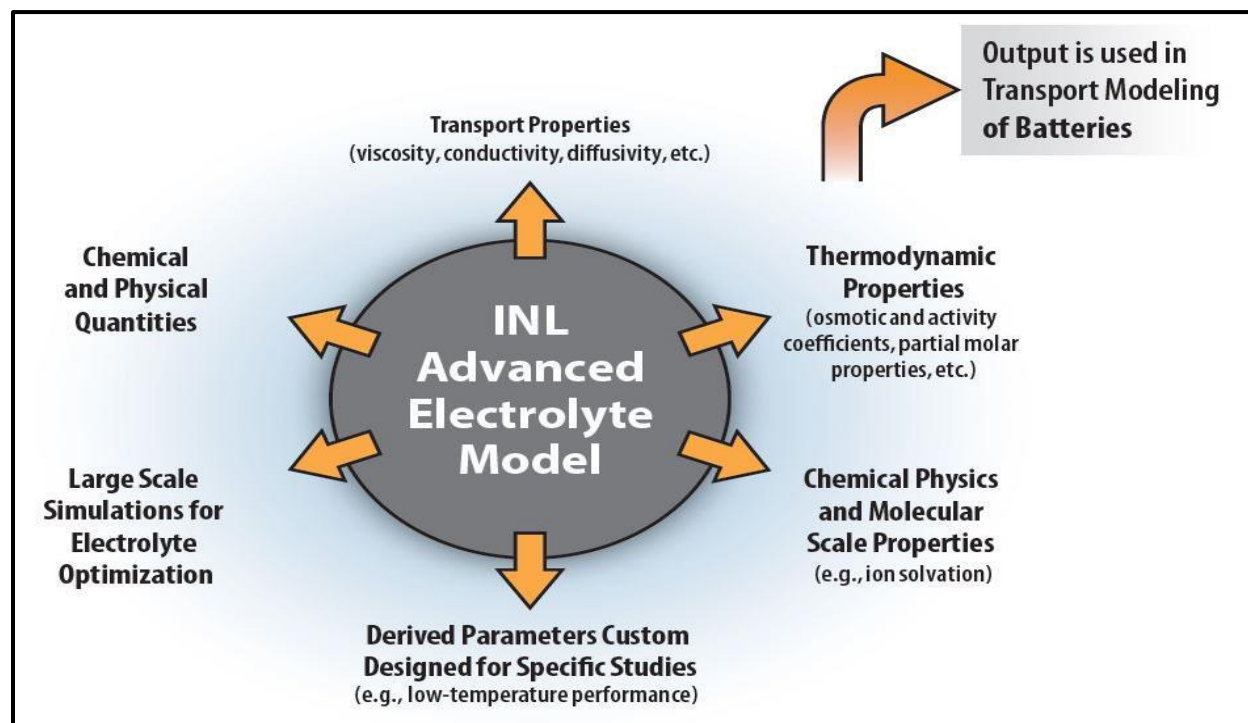


Figure 1. AEM Framework

Advanced Electrolyte Model (AEM)

The result is a faster and less expensive workflow, one that is highly accurate, responsive to a user's specific needs, and applicable to real-world scenarios. The technology won an **R&D 100 Award** in **2014**, has been successfully utilized by Dow Chemical Company, XALT Energy, and others, and was recently licensed by Dalhousie University for its work on high-tech battery development.

AEM is a proven capability designed to explore molecular-to-macroscale level aspects of electrolyte behavior and can be used to drastically reduce the time required to characterize and optimize electrolytes. Although it is applied most frequently to Li-ion and Na-ion battery systems, it is general in its theory and can be used toward numerous other targets.

At the heart of every battery is the electrolyte (salts dissolved in a solution), the material responsible for transporting electrically charged ions across the two electrodes and allowing the flow of electricity in the process. Modern battery electrolytes are composed of numerous solvents and salts, in ratios specific to their intended usage: A battery designed for robust cold weather output will be designed differently than one designed for, say, rapid recharging, even if the chemicals involved are identical. The AEM can analyze systems with multiple solvents and dual salts to find optimum values of chosen parameters such as conductivity, diffusivity, and ion desolvation energy, removing the guesswork about an electrolyte's qualifications for a specific application. And since batteries can experience wide changes of internal and external conditions, the AEM provides evaluations of electrolyte properties over wide ranges of temperature, salt concentration, and solvent composition. Much quicker than quantum chemical methods of electrolyte analysis, density functional theory (DFT), and molecular dynamics (MD), the AEM provides robust predictions without having to depend on step-by-step simulations along a timeline. This results in a model that reduces required lab work and allows for rapid exploration of new components, configurations, and usage conditions.

For any given usage scenario, a battery is judged on its safety, longevity, energy density (the amount of energy it can store compared to its weight), how reliably and steadily it discharges, how quickly it can be recharged, how many times it can be recharged, its environmental impact, and, of course, its cost to the consumer. Virtually all these metrics have a connection to the chosen electrolyte formulation. Small tweaks identified by the AEM to existing battery chemistry can alter any of these variables, resulting in an improved battery for a given application.

More than just a tool to find a superior battery, AEM can be used to improve any electrolytic application, from petroleum and gasoline refining to large-scale water processing and desalination projects. It can even be used in medical research to better understand the role of electrolyte composition in the human body's metabolic functions. This disruptive capability is unique, powerful, relevant to present and future electrolyte development, and without peer. It redefines electrolyte modeling for highly complex contemporary systems, wherein significant steps have been taken to capture the reality of electrolyte behavior in the electrochemical cell environment. This capability can have a very positive impact on accelerating domestic battery development to support aggressive vehicle and energy goals in the 21st century.

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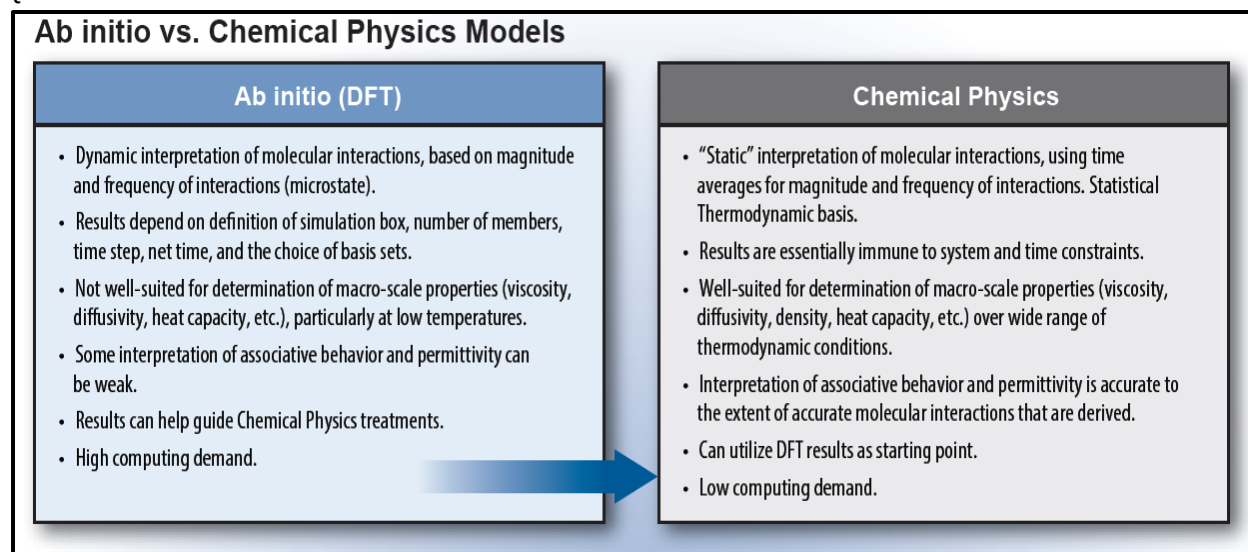


Figure 2. Chemical Physics approaches, such as the Non-Primitive Associative form of the Mean Spherical Approximation (NPAMSA), offer significant advantages in streamlining the computational process, while yielding a wide array of accurate property predictions in a fraction of the time required by ab initio models (density functional theory – DFT).

The AEM generates numerous separate reports, covering issues and quantities such as follows:

1. Transport Properties (viscosity, conductivity, diffusivity, transference numbers, ionic hopping)
2. Thermodynamic Properties (activity coefficients, osmotic coefficient, osmotic pressure, etc.)
3. Ion Association Speciation (ion pairs, triple ions, solid solvates)
4. Solvent-to-ion Binding Energies and solvation numbers for both cations and anions
5. Ligand-wise and net Lithium Desolvation energy and kinetics
6. Relative molal enthalpy
7. Solvent and Solution Permittivity, and related Dielectric Depression
8. Transport Analysis of Double-Layer Regions and related electrolyte properties therein
9. Attenuation of properties under Faradaic transport conditions
10. Field Effects on solvent dipole orientation, considering both Ionic and surface charge fields
11. Arrhenius Analysis (activation energies) of ten foremost electrolyte properties
12. Large-Scale Optimization of solvent composition, given a multi-solvent system (there are about ten distinct optimization parameters that are considered)
13. Summary of non-convergent cases
14. Many Other quantities

[Section 3.4.2](#) and [Section 7.1](#) provide more discussion of the output quantities generated by AEM.

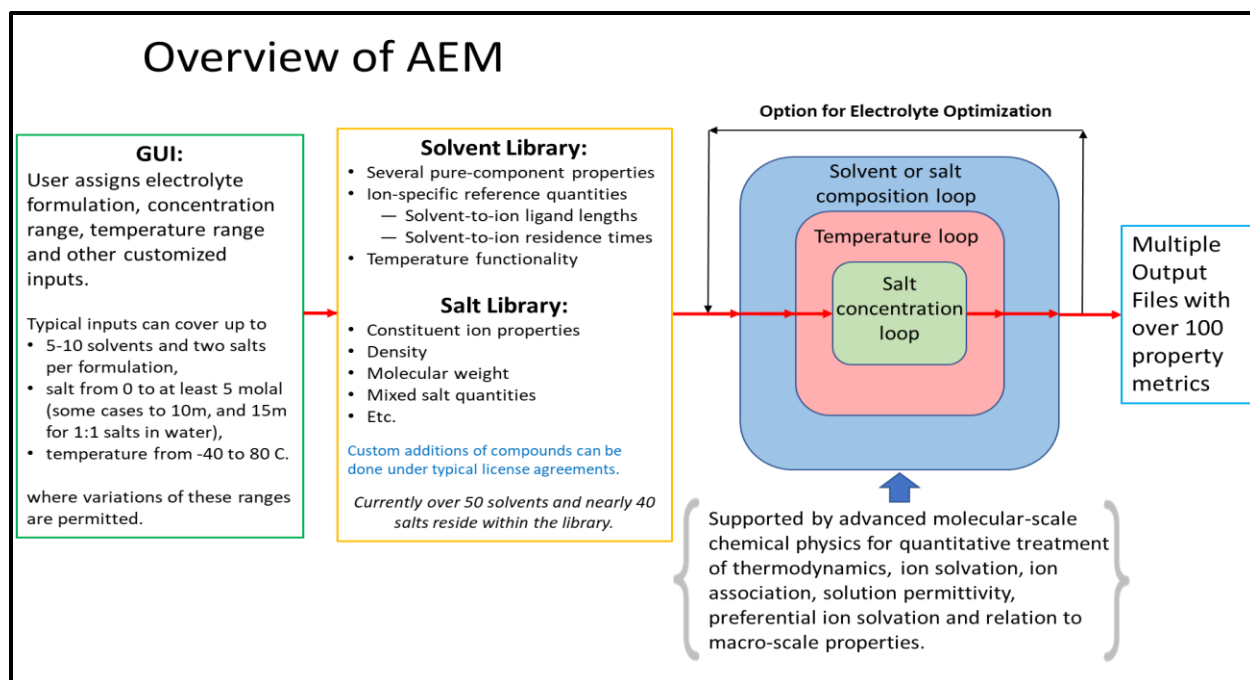


Figure 3. AEM Overview

1.2 Selected Key Features

- The chemical physics or statistical-mechanical basis of the AEM is found in the non-primitive non-restricted associative form of the Mean Spherical Approximation (NPNRAMSA), a powerful computational approach to accurate interpretation of chemical physics behind real electrolyte systems. The NPNRAMSA enables robust predictions over solvent composition, salt concentration, temperature, and permittivity domains.
- Added to this is an ion-solvation equation of state (IS-EOS) that renders accurate values of various quantities tied to ion solvation in such systems, such as effective solvated ion sizes, solvent-ion binding energies, solvation numbers, etc. The effects of ion solvation are central to the accurate prediction of all properties of electrolytes and cation (e.g., lithium) solvation/desolvation parameters.
- To address relevant contemporary electrolytes, the AEM accurately predicts the properties of multiple-solvent systems (up to 5 solvents or more) and handles two-salt systems.
- Key interfacial issues are targeted, including double-layer behavior, field effects on the solvent in the presence of the SEI, and lithium desolvation.
- The double-layer regions can and should exhibit local regions of non-electroneutrality, and hence, the AEM was developed to predict properties for such conditions.
- Contributing theories and mathematics are seamlessly integrated and self-consistent.
- AEM Component Database covers many battery-relevant solvents and salts and is growing.
- Predictions for many systems have average percent deviations with lab data that fall within 5-10%, with many conditions below 5% deviation. The model has been validated with scores of electrolyte systems over hundreds of unique conditions.

- The AEM has been used to help a growing list of private sector companies and DOE partners with electrolyte characterization and optimization, with more collaborations under development.

1.3 Technology Summary

WHAT IS AEM?

The Advanced Electrolyte Model (AEM) is a molecular-based computer simulation program created by Dr. Kevin Gering, a Distinguished Staff Scientist at Idaho National Laboratory, and is exclusively distributed by Ridgetop Group. It offers comprehensive and accurate theoretical relationships for the properties of complex electrolyte formulations and their influence on battery performance. Battery researchers use AEM to optimize electrolyte chemistries quickly and reduce energy storage costs worldwide. By lowering the cost of batteries, global dependence on fossil fuels can be diminished, and electric vehicle adoption and grid-scale battery system deployment can be accelerated. AEM can determine over 100 property metrics with each run, making it a virtual laboratory for genome-level electrolyte properties. It offers millions of combinations of over 60 solvents (and growing) and over 30 salts, allowing users to match their needs with a plausible set of candidates while balancing consequences for different choices of electrolytes. Over 20 reports are generated with each AEM run, covering properties that span transport, thermodynamic, physical, thermophysical, kinetic, molecular, and macroscale metrics.

Benefits

- Accelerates battery development by quickly screening a wide array of materials for key properties, leading to time-to-market advantages.
- Provides a significant cost advantage compared to expensive empirical data collection techniques, reducing laboratory expenses for labor, analysis, and material requisition and disposal.
- Saves users significant time and resources, potentially translating to hundreds of thousands or even millions of dollars.
- Explores and reports with certainty and clarity on molecular-to-macroscale level aspects of electrolyte behavior, removing the guesswork in electrolyte qualification for specific applications.

Applications:

- Battery manufacturers
- Automotive manufacturers
- Battery research institutions
- Lithium-ion and sodium-ion battery developers
- Energy system companies (petroleum and gas refining)
- Water processing and desalination industries
- Medical research institutions studying electrolytic functions in human cells
- Crystallization process industries
- Ion exchange system manufacturers

2 AEM Installation

2.1 System Requirements

The following requirements must be met to ensure proper system operation for executing both the AEM Graphical User Interface (GUI) and the legacy AEM Command Line Interface (CLI) software programs:

1. **Version:** AEM Version greater than or equal to **2.1.9 (released August 2024)**
2. **Compatibility:**
 - a. PC with **Windows 7 – Windows 11** operating system (**either 32 or 64-bit**). AEM is also adaptable to **MacOS** systems.
 - b. Laptop or Desktop architectures can be used. There is no need for a supercomputer or a special CPU/GPU design.
3. **Storage Requirements:**
 - a. Operating/executable files: **less than 500 MB**.
 - b. Report files generated during each run will vary in size depending on user inputs. In practical terms the total report file storage requirement is generally between **5 to 500 MB**.

Important Note: There is no specialized test equipment, hardware, or additional third-party software that needs to be purchased to run the AEM GUI. INL will provide initial guidance on software usage and oversee periodic updates to AEM during the license term. Ridgetop Group Inc. will facilitate AEM licensing, software distribution, and notification of updates. Updates can include additional compounds within the AEM library, new property sets, and overall improvements made to the GUI layout, output formatting and computing architecture.

2.2 Software Installation and License Activation

The AEM GUI can be installed by downloading the **AEM – Graphical User Interface** folder from an online filesharing system such as email, Dropbox, Box, or Ridgetop Group's website. Once downloaded, the following procedure can be followed to install and activate the software.

1. Download and extract the **AEM – Graphical User Interface** folder from the shared download source. Figure 4 provides a visual of this folder in Users default Downloads directory.

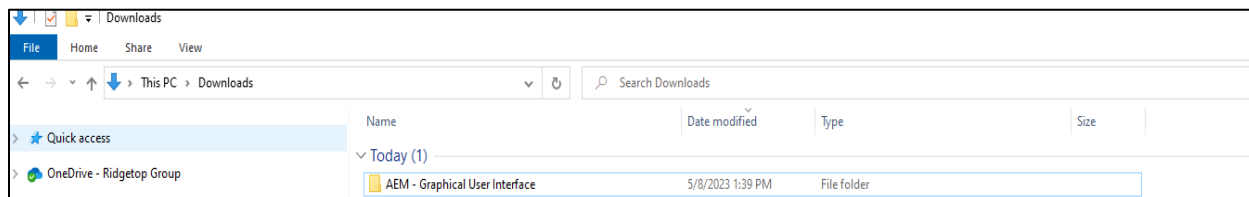


Figure 4. AEM installation folder saved in This PC > Downloads.

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2. Use the Windows File Explorer to open the **AEM – Graphical User Interface** folder and verify that the AEM User Guide and AEM Setup executable is inside the downloaded folder.

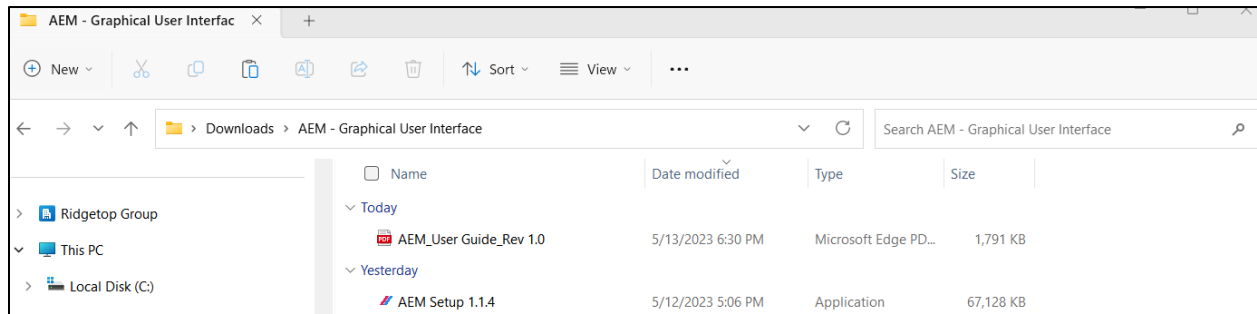


Figure 5. The folder is located at This PC\Downloads\[AEM Download Path].

3. Right-click on the **AEM Setup x.x.x.exe** or **AEM Trial Setup x.x.x.exe** and select **Run as Administrator**.

Important Note: x.x.x represents the AEM GUI version, and your version may be different if it was compiled during a later software release.

4. Verify that the installation wizard initializes as shown in Figure 6.

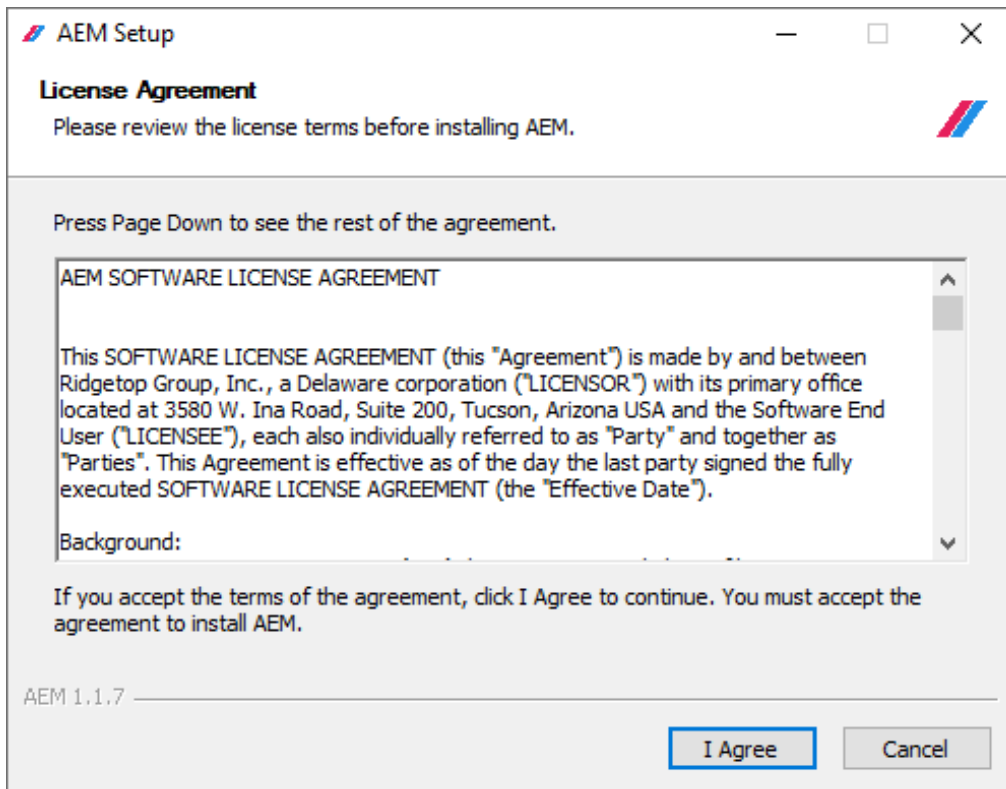


Figure 6. AEM installation wizard initializing.

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5. Accept the AEM Software License Agreement terms by clicking **I Agree** and wait for the installation wizard to complete as shown in Figure 7:

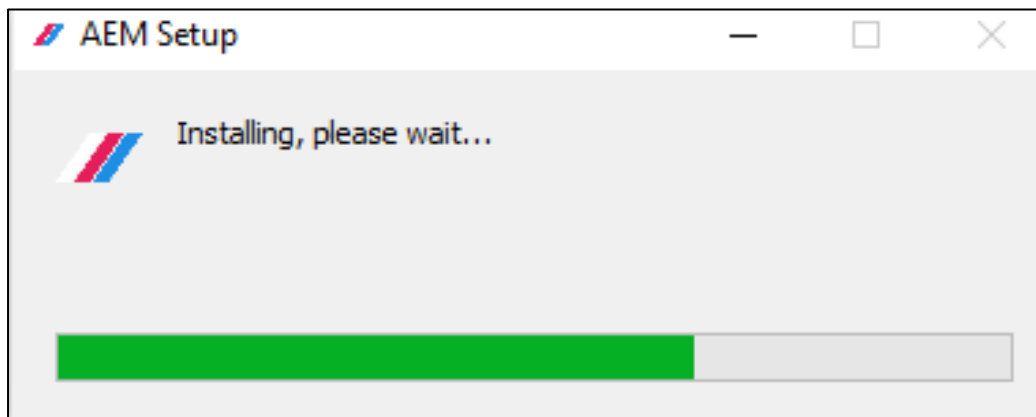


Figure 7. View of the installation wizard for the AEM installation directory. Click Next>> to continue.

6. Verify that AEM GUI starts as shown in Figure 8.

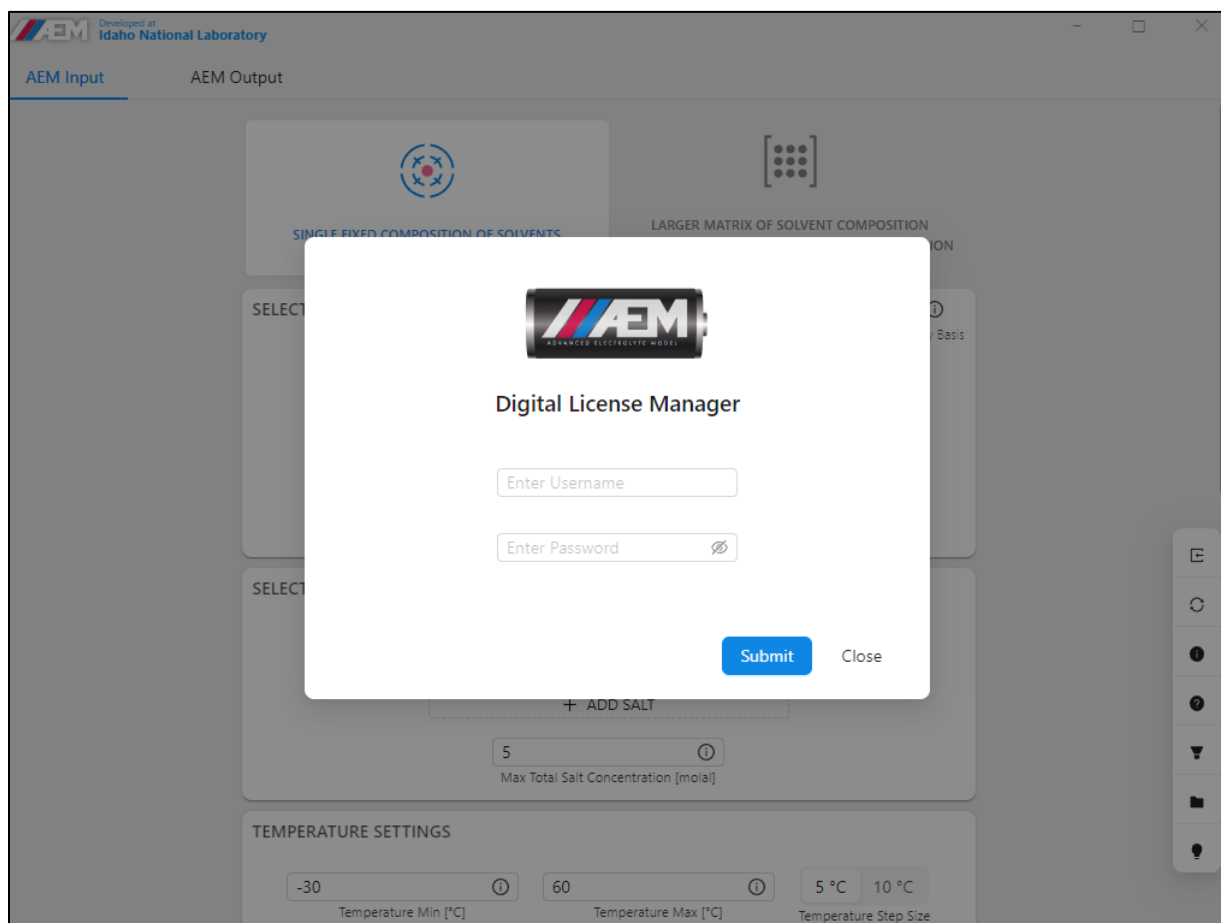


Figure 8. View of the AEM GUI when it starts after a fresh installation.

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7. Enter your license username and password to activate your AEM software and verify that the pop-up message in Figure 9 appears.

Important Note: Each set of license credentials is linked to a unique computer ID for the term of the software license agreement (SLA) or the software trial period. Contact Ridgetop or INL directly if the license needs to be transferred to a new computer, or create a support ticket on the Ridgetop Group website: <https://www.ridgetopgroup.com/contact-us/support/>

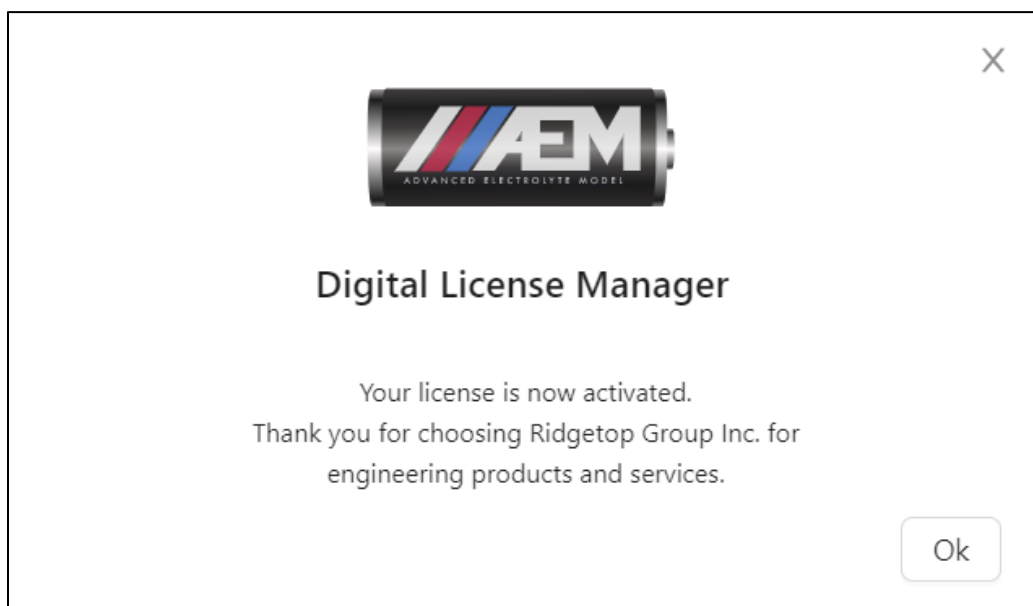


Figure 9. AEM GUI pop-up indicating successful software license activation.

8. After successfully activating your AEM software license click **Ok** to continue and the AEM GUI shall prompt the user to specify their AEM User Directory as shown in Figure 10. The default program directory is outlined below:

C:\Users\user-name\Documents\AEM\		Executable subdirectory
1.	ACCC\	AEM ACCC Composition Files
2.	CLI\	AEM Command Line Interface Directory
3.	DATA\	AEM Data Output Files
3.	EXAMPLES\	AEM Example Files

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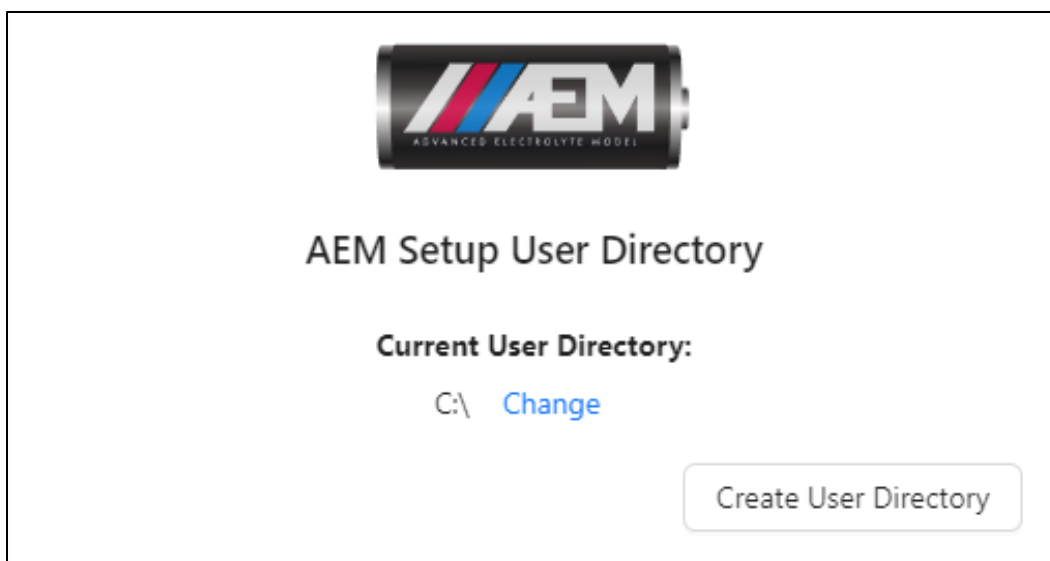


Figure 10. Selecting Default Program Directory during AEM Installation Process.

9. Click the **About Icon** as shown in Figure 11 to verify that your AEM GUI installation is **greater than or equal to Version AEM V1.2.5** and that your AEM CLI Version is **greater than or equal to v2.23.1**.

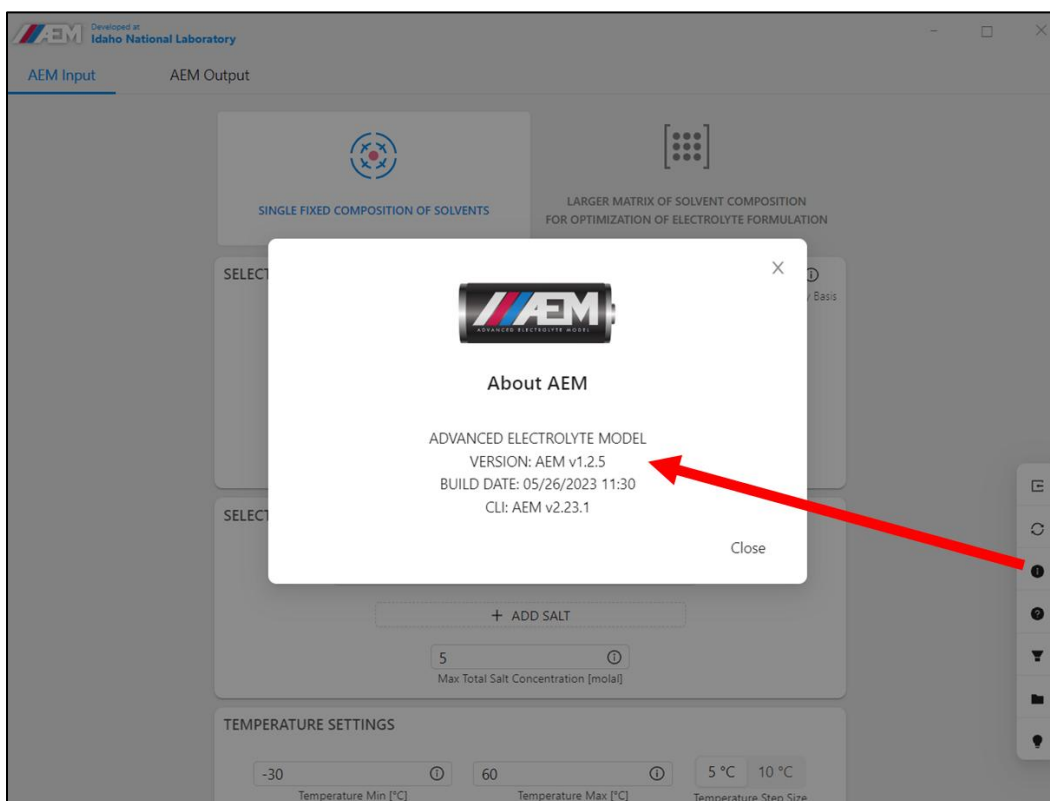


Figure 11. Verifying AEM Software Version.

3 AEM Graphical User Interface (GUI)

This User Guide provides details regarding the AEM Graphical User Interface (GUI) software application, which is an extended product offering from the traditional AEM Command-Line Interface (CLI) program developed at Idaho National Laboratory (INL).

3.1 AEM GUI Overview

The AEM GUI is a Microsoft Windows-based Graphical User Interface (GUI) program that allows users to easily interact with the AEM CLI input conditions. The AEM GUI also allows users to save previous simulation runs and offers a built-in text previewing and plotting capability to analyze AEM data output reports and results.

There are two main tabs in the AEM GUI as listed below:

1. AEM Input
2. AEM Output
3. AEM Compare

Each tab offers a variety of checkboxes and parameters that a user can specify to run a simulation. The following sub-sections provide an overview of each tab.

3.2 AEM Input Tab

The AEM GUI has the following global simulation conditions as shown in Figure 12 on the first GUI tab.

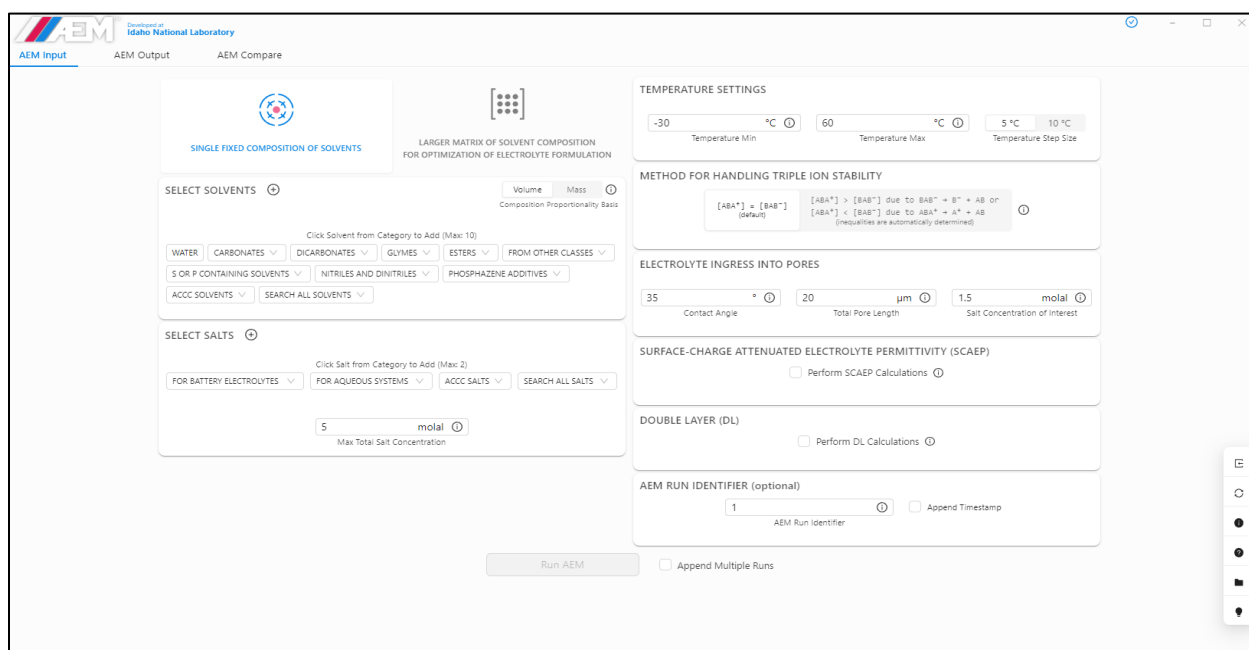


Figure 12. Tab 1 – AEM Input.

3.2.1 SOLVENT COMPOSITION

There are 2 available options for solvent composition to run an AEM simulation as listed below.

1. **SINGLE FIXED COMPOSITION OF SOLVENTS:** This will run a finite number of solvents to formulate the composition of the electrolyte.
2. **LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION:** This will run a matrix of solvents optimizing the process of formulating the electrolyte. This takes a longer time to run due to a larger time to process inputs.

The user can toggle between these options using the mode toggle switches shown in Figure 13 below. The option that is selected is displayed in a colored format with a box in the background.

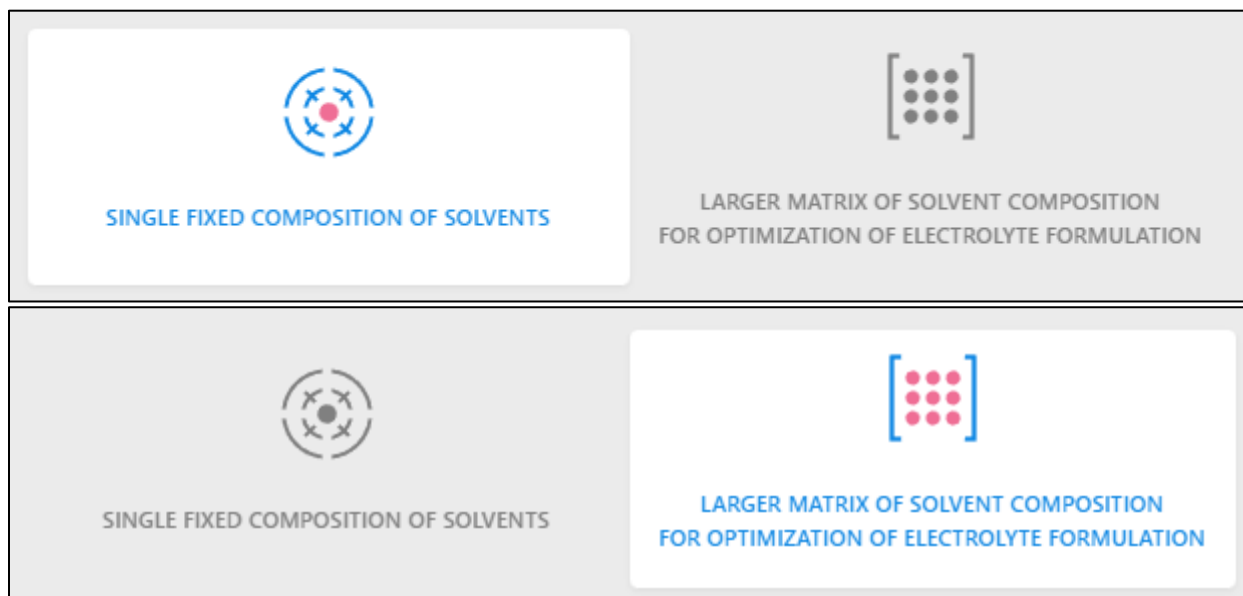


Figure 13. Solvent Composition Mode Toggle Switches in the AEM Input tab.

3.2.2 Custom Recipe Input Mode

The Custom Recipe Input Mode was developed to support AEM users who work primarily in laboratory or formulation-centric environments, where electrolyte compositions are traditionally defined by weight percent (wt%) for solvents and additives, and molality (mol/kg of solvent) for salts. This feature bridges the gap between experimental workflows and simulation requirements by allowing users to input familiar recipe formats—such as primary solvents, co-solvents, additives, and dual-salt systems—without the need to manually convert into volume fractions or molar ratios.

Historically, AEM required users to define electrolyte components using either normalized mole or volume fractions for solvents and total salt concentration expressed in molality. While accurate, this approach posed a barrier to users who preferred to enter recipes based on standard lab conventions. The Custom Recipe Input Mode now solves this by handling the conversion internally using compound-specific physical data, including molecular weights, densities, and reference

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temperatures. This ensures that user inputs remain physically meaningful while maintaining AEM's core modeling accuracy.

General Process:

1. User selects *Custom Recipe Mode* from the Input Mode dropdown.
2. User selects primary solvents and co-solvents/additives, assigning each a mass percentage. The total solvent/additive mass must sum to 100%.
3. User specifies the primary and secondary salts in units of molality (mol/kg of solvent). One or both salts may be used depending on the formulation.
4. User previews the conversion table, which displays:
 - Total molality and ionic strength
 - Solvent/salt mole fractions and mass ratios
 - Reference compound data (molecular weight, density, reference temperature)The user may also enter a Run Identifier to name and archive the formulation.
5. User proceeds with the remaining AEM input settings, including:
 - Maximum allowable salt concentration (e.g., 5 molal)
 - Temperature range for simulation (e.g., -30 °C to 60 °C)
 - Optional enabling of SCAEP, Debye Length, or Pore Transport Models

Note: Volume-based inputs are intentionally excluded due to non-ideal mixing behaviors observed in electrolyte systems. All internal calculations are based on mass- and molar-scale quantities, ensuring consistent and physically accurate simulation results.

RECIPE BREAKDOWN

Composition Summary

Solvents/Salts Properties

1000 grams ⓘ

Primary Solvent Mass

Category	Component	Input Solvent Mass %	Input Salt Molality	Mass (g)	Moles	Total Mass %	Total Moles %	AEM Solvent Mass %	AEM Salt Molar %
Solvent	EC	35%	---	437.50	4.97	31.33%	36.22%	43.75%	---
Solvent	EMC	30%	---	375.00	3.60	26.86%	26.26%	37.50%	---
Solvent	DEC	15%	---	187.50	1.59	13.43%	11.57%	18.75%	---
Co-Solvent	DME	15%	---	150.00	1.66	10.74%	12.14%	15.00%	---
Co-Solvent	THF	5%	---	50.00	0.69	3.58%	5.06%	5.00%	---
Salt	LiPF ₆	---	0.8m	121.52	0.80	8.70%	5.83%	---	66.67%
Salt	LiFSI	---	0.4m	74.83	0.40	5.36%	2.92%	---	33.33%
Total		100%	1.200000	1396.35	13.72	100.00%	100.00%	120.00%	100.00%

Figure 14. Recipe Breakdown when using Custom Recipe Input Mode.

Benefits of this Approach:

The Custom Recipe Input Mode offers several key advantages that strengthen both the usability and fidelity of AEM simulations. First, it integrates experimental realism with simulation rigor by

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allowing users to define electrolyte formulations in the same way they would in a laboratory—using mass percent for solvents and molality for salts—thereby reducing the disconnect between experimental development and digital twin validation. Importantly, this feature preserves AEM's validated input structures; all conversions take place in a dedicated pre-processing layer, so there are no changes to the underlying modeling engine or input schema. This approach also improves transparency and minimizes entry errors by generating automated conversion tables that clearly display underlying assumptions and compound-specific parameters, giving users the opportunity to verify all inputs before simulation. Finally, it enables batch standardization for real-world electrolyte screening by allowing recipes to be saved, shared, and reused across different formulations and operating conditions, supporting systematic exploration of multi-salt and multi-solvent systems.

3.2.3 SELECT SOLVENTS

This is the section where the user defines the number and composition of solvents that are combined to formulate the desired electrolyte. The input to this section is highly dependent on what option the user decides to choose for the solvent composition as described in the previous section. Note that when the GUI is in the **Expanded Format**, the solvent selection list will be organized based on categories. When the GUI is in the **Compact Format**, the solvent selection list will be supported with a search capability to quickly type in the desired solvent name and/or abbreviation. Additional details on the **Expanded Format** vs. **Compact Format** are covered in [Section 3.6](#) where a visual aid was prepared in Figure 79.

FOR SINGLE FIXED COMPOSITION OF SOLVENTS:

The maximum number of solvents allowed as inputs to the program in this option's case is **ten (10)**. The user can select the desired solvent using the drop-down list buttons classified according to the solvents' chemistries. The user can also add their own solvents using the "+" shown with **red** box in Figure 15 below. The user also must supply an additional input by choosing the solvents' composition proportionality basis. This is achieved by using another toggle mode switch through which the user toggles between either **Volume** or **Mass**. A visual of this section with the drop-down list buttons (**green** box) and the mode toggle switch (**blue** box) is shown in Figure 15 below.

SELECT SOLVENTS +

Volume Mass ⓘ
Composition Proportionality Basis

Click Solvent from Category to Add (Max: 10)

WATER CARBONATES ▾ DICARBONATES ▾ GLYMES ▾ ESTERS ▾ FROM OTHER CLASSES ▾

S OR P CONTAINING SOLVENTS ▾ NITRILES AND DINITRILES ▾ PHOSPHAZENE ADDITIVES ▾

ACCC SOLVENTS ▾ SEARCH ALL SOLVENTS ▾

Figure 15. SELECT SOLVENTS section for the SINGLE FIXED COMPOSITION OF SOLVENTS option of the AEM Input tab.

SELECT SOLVENTS (+)

Volume Mass ⓘ
Composition Proportionality Basis

Click Solvent from Category to Add (Max: 10)

WATER CARBONATES ▾ DICARBONATES ▾ GLYMES ▾ ESTERS ▾ FROM OTHER CLASSES ▾

S OR P CONTAINING SOLVENTS ▾ NITRILES AND DINITRILES ▾ PHOSPHAZENE ADDITIVES ▾

ACCC SOLVENTS ▾ SEARCH ALL SOLVENTS ▾

SELECT SALTS (+)

Click Salt from Category to Add (Max: 2)

FOR BATTERY ELECTROLYTES ▾ FOR AQUEOUS SYSTEMS ▾ ACCC SALTS ▾ SEARCH ALL SALTS ▾

Propylene Carbonate (PC)
Ethylene Carbonate (EC)
Ethylmethyl Carbonate (EMC)
Diethyl Carbonate (DEC)
Dimethyl Carbonate (DMC)
Fluoroethylene Carbonate (FEC)
Vinylene Carbonate (VC)
Vinyl Ethylene Carbonate (VEC)

5 molal ⓘ
Max Total Salt Concentration

Figure 16. Drop-down List Button to select solvents for the SINGLE FIXED COMPOSITION OF SOLVENTS option of the AEM Input tab.

Once the solvents have been defined and the composition proportionality has been set, there is one more input that the user must supply to the GUI. For each chosen solvent, the user must enter the percentage of the solvent by either **% Mass** or **% Volume** based on what was chosen for the composition proportionality basis. This is shown in Figure 17 in which both the **Mass** and **Volume** cases are considered.

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The figure displays two identical screenshots of the 'SELECT SOLVENTS' interface in the Advanced Electrolyte Model (AEM) software. Each screenshot features a header with 'SELECT SOLVENTS' and a plus icon, and a sub-header with 'Volume' and 'Mass' tabs, with 'Volume' selected in the top panel and 'Mass' selected in the bottom panel. Below the tabs, a note indicates 'Composition Proportionality Basis'. The main section is titled 'Click Solvent from Category to Add (Max: 10)' and contains several categories of solvents: WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, FROM OTHER CLASSES, S OR P CONTAINING SOLVENTS, NITRILES AND DINITRILES, PHOSPHAZENE ADDITIVES, and ACCC SOLVENTS. A 'SEARCH ALL SOLVENTS' button is also present. At the bottom of each panel, two solvents are listed in green boxes: 'CARBONATES / Ethylene Carbonate (EC)' and 'CARBONATES / Ethylmethyl Carbonate (EMC)'. To the right of each solvent name is a text input field for the percentage (e.g., '% Volume' or '% Mass') and a red minus icon to remove the solvent.

Figure 17. Percentage of Solvent input depending on Volume/Mass for each solvent for the SINGLE FIXED COMPOSITION OF SOLVENTS option of the AEM Input tab.

FOR A LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION:

The maximum number of solvents allowed as inputs to the program in this option's case is **five (5)**. The user can select the desired solvent using the drop-down list buttons classified according to the solvents' chemistries. A visual of this section with the drop-down list buttons (**green** box) is shown in Figure 18 below.

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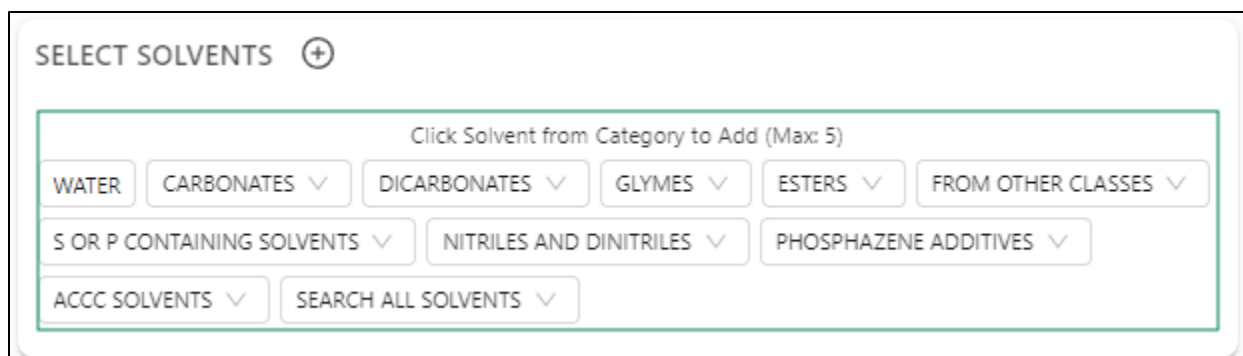


Figure 18. SELECT SOLVENTS section for the LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION option of the AEM Input tab.

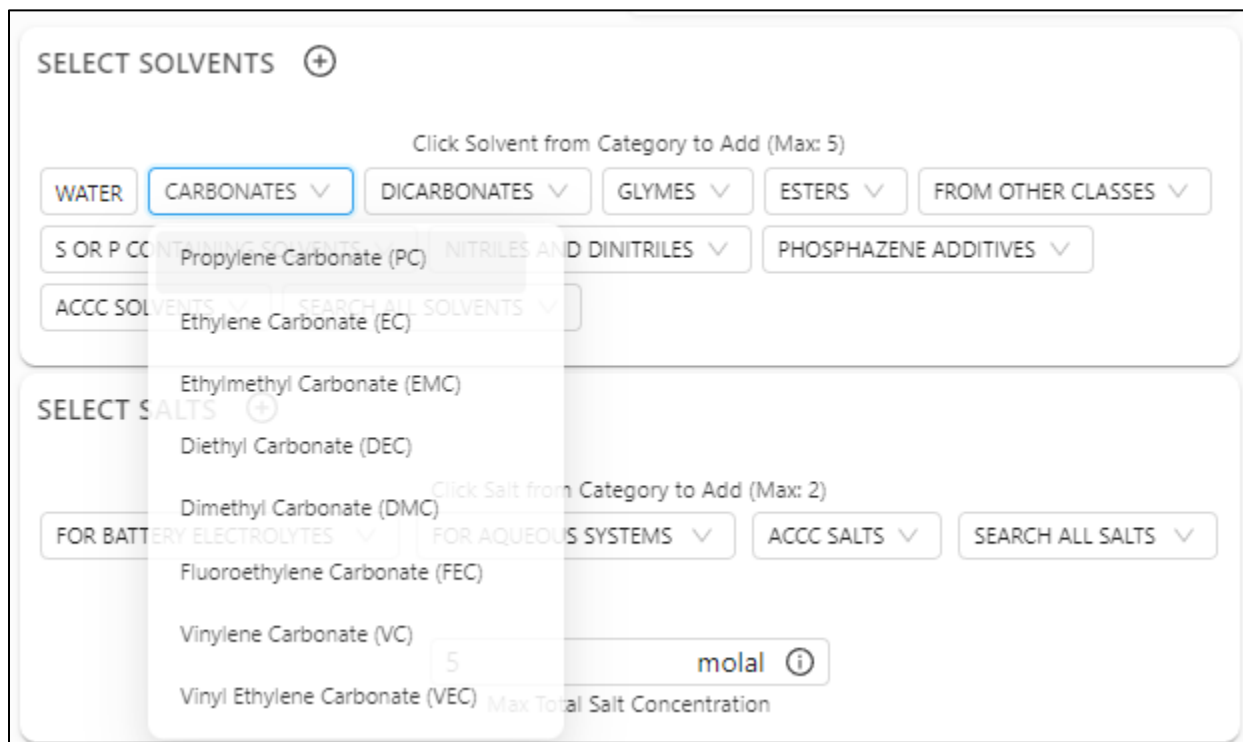


Figure 19. Drop-down List Button to select solvents for the LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION option of the AEM Input tab.

Once the solvents have been defined, there is no more input that the user must supply to the GUI, unless the user selects two (2) or more solvents. In that case, the user is given a choice to set the mass fraction of one (1) of the chosen solvents constant and define it between zero (0) and (1). For ≤ 2 solvents, the visual of this section is shown in Figure 20. For ≤ 2 solvents, the visual of this section is shown in Figure 21, where, for the solvent whose mass fraction is set constant, the user must define its value in the text box provided.

Advanced Electrolyte Model (AEM)

The screenshot shows the 'SELECT SOLVENTS' interface with a '+' icon. Below the title is the instruction 'Click Solvent from Category to Add (Max: 5)'. There are several category buttons: WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, FROM OTHER CLASSES, S OR P CONTAINING SOLVENTS, NITRILES AND DINITRILES, PHOSPHAZENE ADDITIVES, ACCC SOLVENTS, and SEARCH ALL SOLVENTS. Below these, two solvents are listed in green-bordered boxes: 'ESTERS / Methyl Butyrate (MB)' and 'OTHER / 1NM2-Siloxane (1NM2)'. Each has a red minus icon to its right.

Figure 20. SELECT SOLVENTS section for the LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION option of the AEM Input tab in the case of less than or equal to two (2) total solvents.

This screenshot shows the same 'SELECT SOLVENTS' interface but with three solvents listed: 'ESTERS / Methyl Butyrate (MB)', 'OTHER / 1NM2-Siloxane (1NM2)', and 'PHOSPHAZENE ADD. / INL Additive PA5'. Each solvent entry has a checkbox and a text field to its right. For the first solvent, the checkbox is checked and the text field says 'Mass Fraction'. For the other two, the checkboxes are unchecked and the text field says 'Set Constant Mass Fraction'. Each entry also has a red minus icon to its right.

Figure 21. SELECT SOLVENTS section for the LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION option of the AEM Input tab in the case of greater than two (2) total solvents.

AEM currently houses over seventy (70) solvents of varying chemistries as shown in the following list. Ridgetop Group is currently working with INL to expand this list of available solvents.

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A.3: Choose solvent(s) from the following library list:

1: water

CARBONATES:

2: propylene carbonate (PC)
3: ethylene carbonate (EC)
6: ethylmethyl carbonate (EMC)
7: diethyl carbonate (DEC)
8: dimethyl carbonate (DMC)
15: fluoroethylene carbonate (FEC)
39: vinylene carbonate (VC)
45: vinyl ethylene carbonate (VEC)
71: methyl trifluoroethyl carbonate (FEMC)

DICARBONATES:

49: Dimethyl 2,5, Dioxahexanedioate (DMOHC)
50: Diethyl 2,5, Dioxahexanedioate (DEOHC)

GLYMES

5: 1,2-dimethoxyethane (DME)
59: 1,2-dimethoxypropane (DMP)
20: DEGDME (diglyme)
21: 3-TEGDME (triglyme)
22: 4-TEGDME (tetraglyme)

S or P CONTAINING SOLVENTS:

16: ethylmethoxyethyl sulfone (EMES)
28: ethylmethyl sulfone (EMS)
34: sulfolane (TMS)
35: trimethyl phosphate (TMP)
55: triethyl phosphate (TEP)
46: (proprietary, unavailable)
60: dimethyl sulfite (DMS)

FROM OTHER CLASSES:

4: gamma-butyrolactone (GBL)
10: 1,3 dioxolane
14: ethylene glycol (EG)
17: Dowanol-PMA (DOW-PMA)
18: 2-methoxyethyl acetate (MEA)
19: 1NM2-siloxane (1NM2)
47: Tris(2,2,2-trifluoroethyl)-orthoformate (TFEO)
51: 7-oxabicyclo[2.2.1]heptane-2-carbonitrile (OCN)
52: tetrahydrofuran (THF)
53: (proprietary, unavailable)
54: (proprietary, unavailable)
56: Bis(2,2,2-trifluoroethyl) ether (BTFE)
57: 1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether (TTE)
63: Fluorobenzene (FBZ)
64: 7,8-Dihydroxyphenazine-2-sulfonic acid (DHPS)
65: 1,8-dihydroxy-2,7-dicarboxymethyl-9,10-anthraquinone (DCDHAQ)
66: 2,2,2-Trifluoro-N,N-dimethylacetamide (FDMA)
67: water in trace amounts (<1%)
72: tetraethyl orthosilicate (TEOS)

ESTERS:

9: ethyl acetate (EA)
11: methyl butyrate (MB)
12: ethyl propionate (EP)
13: n-propyl acetate (nPA)
29: methyl acetate (MA)
30: methyl propionate (MP)
31: methyl formate (MF)
32: ethyl formate (EF)
37: methyl monofluoroacetate (M1FA)
38: methyl difluoroacetate (M2FA)
58: ethyl difluoroacetate (E2FA)
40: n-propyl propionate (nPP)
42: n-butyl acetate (nBA)
43: n-butyl propionate (nBP)
44: n-butyl butyrate (nBB)

NITRILES AND DINITRILES:

26: glutaronitrile (GLN)
27: adiponitrile (ADN)
33: propionitrile (PN)
36: acetonitrile (AN)
41: butyronitrile (BN)
48: succinonitrile (SN)
61: isovaleronitrile (IVN)
62: isobutyronitrile (IBN)

PHOSPHAZENE ADDITIVES:

23: INL Additive FM2
24: INL Additive SM4
25: INL Additive PA5

Figure 22. Solvent list in AEM Version 2.26.1 with recent additions including 64-DHPS, 65-DCDHAQ, 66-FDMA, 67-water in trace amounts, 71-FEMC, and 72-TEOS. Note that only select solvents are available in the AEM Trial Version.

3.2.4 SELECT SALTS

This is the section where the user defines the number and composition of salts that are combined along with the chosen solvents to formulate the desired electrolyte. The input to this section is highly dependent on the number of salts chosen by the user – 1 or 2.

As previously mentioned with the solvent selection process, the **Expanded Format** and **Compact Format** will change how salts are selected for AEM simulation. When the GUI is in the **Expanded Format**, the salt selection list will be organized based on categories. When the GUI is in the **Compact Format**, the salt selection list will be supported with a search capability to quickly type in the desired salt name or abbreviation. Additional details on the **Expanded Format** vs. **Compact Format** are covered in [Section 3.6](#) where a visual aid was prepared in Figure 79.

Important Note: The maximum number of salts the user is allowed to choose is **two (2)**.

The user can select the desired salts using the drop-down list buttons classified according to the salts' chemistries – **FOR BATTERY ELECTROLYTES** and **FOR AQUEOUS SYSTEMS**. The user can also create their own salt definition by clicking on the "+" sign (**red** box). The user also must supply an additional text-box input by defining the maximum total salt concentration (**For battery electrolytes, this value falls between 6m to 8m, where m is molal units. This value can reach upwards of 15m when dealing with aqueous systems**). A visual of this section with the drop-down list buttons (**green** box) and the text box (**blue** box) is shown in Figure 23.

Figure 23. SELECT SALTS section of the AEM Input tab.

Advanced Electrolyte Model (AEM)

SELECT SALTS (+)

Click Salt from Category to Add (Max: 2)

FOR BATTERY ELECTROLYTES ▾ FOR AQUEOUS SYSTEMS ▾ ACCC SALTS ▾ SEARCH ALL SALTS ▾

LIPF₆

LiBF₄ 5 molal ⓘ

LiBOB

LiTFSI

Li₂(B₁₂F₁₂)

LiDFOB

LiDFBOP

LiFSI

Li-Triflate

LiAsF₆

LiPO₂F₂ (additive only)

NaClO₄

NaPF₆

Max Total Salt Concentration

Run AEM

Figure 24. Drop-down List Button to select salts for the SELECT SALTS section of the AEM Input tab.

FOR ONE (1) SALT:

In the case that the user selects only one (1) salt, the only other input that must be supplied to the GUI is the maximum total salt concentration. This is shown in Figure 25, in which this case is considered with NaCl – common salt.

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The screenshot shows the 'SELECT SALTS' section of the AEM Input tab. At the top, there is a 'SELECT SALTS' label with a plus icon. Below it, a instruction 'Click Salt from Category to Add (Max: 2)' is followed by four category buttons: 'FOR BATTERY ELECTROLYTES', 'FOR AQUEOUS SYSTEMS', 'ACCC SALTS', and 'SEARCH ALL SALTS'. A green-bordered box contains the text 'FOR BATTERY ELECTROLYTES / LiPF₆' with a minus icon to its right. Below this box is a text input field with the value '5' and the unit 'molal', followed by an information icon. At the bottom, it says 'Max Total Salt Concentration'.

Figure 25. SELECT SALTS section of the AEM Input tab for 1 salt.

FOR TWO (2) SALTS:

In the case that the user selects two (2) salts, there is one other input that must be supplied to the GUI other than the maximum total salt concentration. The GUI provides a toggle mode switch through which the user toggles between either **Single** or **Range** composition proportion mode for the salts. This is shown in Figure 26 in which this case is considered with LiBF₄ and NaCl – common salt with the toggle mode switch for the composition proportion mode (blue box).

The screenshot shows the 'SELECT SALTS' section of the AEM Input tab for two salts. At the top, there is a 'SELECT SALTS' label with a plus icon. Below it, a instruction 'Click Salt from Category to Add (Max: 2)' is followed by four category buttons: 'FOR BATTERY ELECTROLYTES', 'FOR AQUEOUS SYSTEMS', 'FOR AQUEOUS FLOW BATTERIES', 'ACCC SALTS', and 'SEARCH ALL SALTS'. A blue-bordered box at the top right contains a toggle switch with 'Single' and 'Range' options, and an information icon. Below the box, it says 'Composition Proportion Mode'. A green-bordered box contains two salt entries: 'LiPF₆ FOR BATTERY ELECTROLYTES' and 'LiBF₄ FOR BATTERY ELECTROLYTES', each with a minus icon to its right. Below this box are two buttons: 'Default' and 'Dilute', followed by the text 'Molal Concentration Mode'. At the bottom, there is a text input field with the value '5' and the unit 'molal', followed by an information icon. At the very bottom, it says 'Max Total Salt Concentration'.

Figure 26. SELECT SALTS section of the AEM Input tab for 2 salts.

Single mode will simulate **only one (1) mixed salt molar proportion**, while the **Range** mode will simulate **several proportions, covering mole fractions (0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0)**. If the user selects the **Single** mode, the user must specify the percentage molar proportions or relative amount for each salt in the text boxes provided by the GUI. This is shown

Advanced Electrolyte Model (AEM)

in Figure 27 in which this case is considered with LiPF_6 and NaCl – common salt with the text boxes for the molar percentages of the salts (blue box).

The figure displays two screenshots of the 'SELECT SALTS' interface in the AEM software, comparing 'Single' and 'Range' composition proportion modes. Both screenshots show the same set of controls: a 'Composition Proportion Mode' selector (Single/Range), category filters (FOR BATTERY ELECTROLYTES, FOR AQUEOUS SYSTEMS, FOR AQUEOUS FLOW BATTERIES), search buttons (ACCC SALTS, SEARCH ALL SALTS), a list of selected salts (LiPF₆ and LiBF₄), a 'Molar %' input field (highlighted with a blue box in the top screenshot), a 'Default/Dilute' toggle for 'Molal Concentration Mode', and a 'Max Total Salt Concentration' input (set to 5 molal).

Figure 27. Single vs Range Composition Proportion Mode for the SELECT SALTS section of the AEM Input tab for 2 salts.

As illustrated in Figure 27, a new input parameter has been introduced to define the Molal Concentration Mode, giving users greater control over how salt concentration steps are applied in AEM simulations. This feature allows users to choose between two distinct options tailored for different experimental regimes:

- **Option 1** enables the use of AEM’s default molal step sizes, which span a standard range of salt concentrations commonly encountered in conventional electrolyte systems—ideal for general-purpose simulations.
- **Option 2** activates a fine-resolution mode optimized for dilute solution conditions, applying smaller molal increments ranging from 0.001 m to 0.1 m total salt concentration. This is particularly useful for investigating subtle property changes at low salt concentrations or validating electrolyte behavior under highly diluted regimes.

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This enhancement expands AEM's flexibility across both standard and low-concentration use cases, allowing researchers to better align simulation granularity with experimental needs. AEM currently houses over forty (40) salts of varying chemistries as shown in the following list. Ridgetop Group is currently working with INL to expand this list of available salts.

A.5: Choose salt(s) from the following partial sample list:

For Battery Electrolytes:

- 6: LiClO₄
- 7: LiPF₆
- 8: LiBF₄
- 9: LiBOB
- 10: LiTFSI
- 11: Li₂(B₁₂F₁₂)
- 12: LiDFOB
- 38: LiDFOBOP
- 13: LiFSI
- 14: Li-Triflate
- 15: LiAsF₆
- 16: LiP₀₂F₂ (additive only)
- 17: NaClO₄
- 18: NaPF₆
- 19: NaBF₄
- 20: NaTFSI
- 37: NaFSI
- 21: PA5-PF₆ (RTIL)
- 22: PYR₁₃-FSI (RTIL)
- 23: EMI-FSI or EMIM-FSI (RTIL)
- 24: CsPF₆
- 25: Mg(TFSI)₂
- 43: TEABF₄
- 44: DMPBF₄

For Aqueous Systems:

- 17: NaClO₄
- 26: NaCl
- 27: NaNO₃
- 28: Na₂SO₄
- 29: LiNO₃
- 30: Li₂SO₄
- 31: MgSO₄
- 32: Al(NO₃)₃
- 33: LiBr
- 34: NaOH
- 35: KOH
- 36: LiOH
- 37: NaFSI

Intended for Aqueous flow battery electrolytes:

- 39: NaOH/KOH (1:1)
- 40: NaK(OOCH/HC₃)
- 41: NaKCO₃
- 42: Na₃PO₄/K₃PO₄ (1:1)

Note 1: aqueous systems outside salts 25-31 have been partially validated.

Note 2: if an RTIL is chosen in a salt mixture, input the RTIL FIRST.

Figure 28. Salt list in AEM Version 2.26.1 with recent additions including 38-44 Note that only select Salts are available in the AEM Trial Version.

3.2.5 TEMPERATURE SETTINGS

This is the section where the user defines the temperature range along with the step size for that range in which the electrolyte will be used. The input from this section is essential for the calculation of properties over temperature to derive important thermodynamic quantities. The user can specify the **Temperature Min (°C)** and **Temperature Max (°C)** using text boxes provided by the GUI. The user must also supply an additional toggle mode switch input through which the user toggles between either **5°C** or **10°C** for the **Temperature Step Size**. A **5°C** step size **produces a finer grid and provides slightly better fidelity in Arrhenius calculations**. A **10°C** step size is recommended for the **LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION** option for **SOLVENT COMPOSITION** as it **reduces matrix by nearly half and provides coarser fidelity in Arrhenius calculations**. A visual of this section with the text boxes for the temperature inputs and the toggle mode switch for the temperature step size is shown in Figure 29.

Important Note: The temperature range allowed by AEM will cover all conditions for electrolyte systems under conventional use conditions. A typical range for non-aqueous electrolytes is **-30°C to 60°C** with lower and higher temperatures possible. It is recommended not to exceed 100 °C.

The figure shows a screenshot of the 'TEMPERATURE SETTINGS' section in the AEM Input tab. It contains three input fields: 'Temperature Min [°C]' with a value of -30, 'Temperature Max [°C]' with a value of 60, and 'Temperature Step Size' with a toggle switch set to 5 °C. Each input field has an information icon (i) to its right.

Figure 29. TEMPERATURE SETTINGS section of the AEM Input tab.

3.2.6 METHOD FOR HANDLING TRIPLE ION STABILITY

This is the section where the user defines the preferred approach for triple ion (TI) stability, whether the default is used ($[ABA^+] = [BAB^-]$) or whether there is a subsequent conversion of either $[ABA^+]$ or $[BAB^-]$ into a single ion and an ion pair. The user must supply the preferred option using a toggle mode switch through which the user toggles between either **$[ABA^+] = [BAB^-]$ (default)** or **$[ABA^+] > [BAB^-]$ due to $ABA^+ \rightarrow A^+ + AB$, or $[ABA^+] < [BAB^-]$ due to $ABA^+ \rightarrow A^+ + AB$ (inequalities are automatically determined)**. If the **$[ABA^+] = [BAB^-]$ (default)** option is selected, the concentrations of cationic triple ions $[ABA^+]$ and anionic triple ions $[BAB^-]$ are assumed to be equal so that their ratio $[ABA^+]/[BAB^-]$ equals 1. Whereas, if the **$[ABA^+] > [BAB^-]$ due to $ABA^+ \rightarrow A^+ + AB$, or $[ABA^+] < [BAB^-]$ due to $ABA^+ \rightarrow A^+ + AB$ (inequalities are automatically determined)** option is selected, the ratio of $[ABA^+]/[BAB^-]$ is recalculated to determine if there is cationic TI dominance (ratio > 1) or anion TI dominance (ratio < 1). These distinctions within the TI population have an impact on some properties such as the cation transference number. A visual of this section with the toggle mode switch to select the triple ion stability method is shown in Figure 30.

METHOD FOR HANDLING TRIPLE ION STABILITY

$[ABA^+] = [BAB^-]$
(default)

$[ABA^+] > [BAB^-]$ due to $BAB^- \rightarrow B^- + AB$ or
 $[ABA^+] < [BAB^-]$ due to $ABA^+ \rightarrow A^+ + AB$
 (inequalities are automatically determined)

Figure 30. TEMPERATURE SETTINGS section of the AEM Input tab.

3.2.7 ELECTROLYTE INGRESS INTO PORES

This is the section where the user defines three inputs correlating to the permeation of the electrolyte in porous regions. The user can specify the **Contact Angle (°)**, **Total Length of Pore (μm)**, and **Salt Concentration of Interest (m)** using text boxes provided by the GUI. A visual of this section with the text boxes for the inputs described above is shown in Figure 29.

Important Note: The ranges allowed for the above inputs are as follows.

- **Contact Angle (°):** 0 – 90°
- **Total Length of Pore (μm):** 0.1 – 50μm
- **Salt Concentration of Interest (m):** Less than equal to ($\leq$) the Maximum Salt Concentration, usually 0.1 – 8m

ELECTROLYTE INGRESS INTO PORES

2 Contact Angle [°]

5 Total Pore Length [μm]

4 Salt Concentration of Interest

Figure 31. ELECTROLYTE INGRESS INTO PORES section of the AEM Input tab.

3.2.8 SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP)

This is the section where the user defines certain inputs for the calculation of surface-charge effects on permittivity for electrolyte neighboring an electrode surface. Local electrolyte permittivity will be reduced by the presence of an electrode surface charge as would be present during actual battery operation. The GUI provides a **Perform SCAEP Calculations** check box that allows the user to turn the SCAEP calculations on and off. A visual of this section with the check box not checked (SCAEP calculations turned off) (default) is shown in Figure 32.

SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP)

☐ Perform SCAEP Calculations

Figure 32. SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP) section of the AEM Input tab with a check box to perform SCAEP Calculations not checked (default).

If the check box is checked and the SCAEP calculations are turned on, the GUI prompts the user to specify some new inputs. Using a toggle mode switch, the user must specify the **Type of Pulse Condition** between either **Discharge** or **Charge**. Subsequently, the user can specify the **Cell Voltage of Interest (V)**, **Bulk Salt Concentration (molal)**, **Thickness of Cathode/Anode SEI (Å)**, **Relative Permittivity of Cathode/Anode SEI (%)**, and **Average Porosity of Cathode/Anode SEI (ϵ)** using text boxes provided by the GUI. A visual of this section with the toggle mode switch and text boxes for the inputs described above is shown in Figure 33.

Important Note: The ranges allowed for the above inputs are as follows.

- **Cell Voltage of Interest (V):** 3 – 4.5V [Specific to Li-ion Cells]
- **Bulk Salt Concentration (molal):** 0.2 – 3.8m
- **Thickness of Cathode/Anode SEI (Å):** 10 – 100Å, typically between 200 – 400Å
- **Relative Permittivity of Cathode/Anode SEI (%):** 2 – 100%, typically 10 – 20%
- **Average Porosity of Cathode/Anode SEI (ϵ):** 0.05 – 0.9, typically 0.2 – 0.4

The figure displays two identical GUI panels for the 'SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP)' section. Each panel features a checked 'Perform SCAEP Calculations' checkbox and an information icon. Below this is a toggle switch for 'Type of Pulse Condition' with 'Discharge' selected. The input fields are arranged in two rows of three. The top row contains 'Cell Voltage of Interest [V]' (3.5), 'Bulk Salt Concentration [m]' (3.5), and 'Ave. Porosity of Cathode SEI [ϵ ?]' (0.2). The bottom row contains 'Thickness of Cathode SEI [Å]' (200), 'Rel. Permittivity of Cathode SEI [%]' (10), and 'Ave. Porosity of Cathode SEI [ϵ ?]' (0.2). The bottom panel's bottom row labels are 'Thickness of Anode SEI [Å]', 'Rel. Permittivity of Anode SEI [%]', and 'Ave. Porosity of Anode SEI [ϵ ?]'.

Figure 33. SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP) section of the AEM Input tab with a check box to perform SCAEP Calculations checked.

3.2.9 DOUBLE LAYER (DL)

This is the section where the user specifies inputs to facilitate double-layer (DL) calculations. These calculations are done over a fine grid of spatial (1D, 5nm steps) and time dimensions, allowing the determination of electrolyte gradients and related properties in the anode and cathode double layers. These calculations are done over relatively short periods to reveal the emergent dynamic of the electrolyte gradients. The GUI provides a **Perform DL Calculations** check box that allows the user to turn the DL calculations on and off. A visual of this section with the check box not checked (DL calculations turned off) (default) is shown in Figure 34.

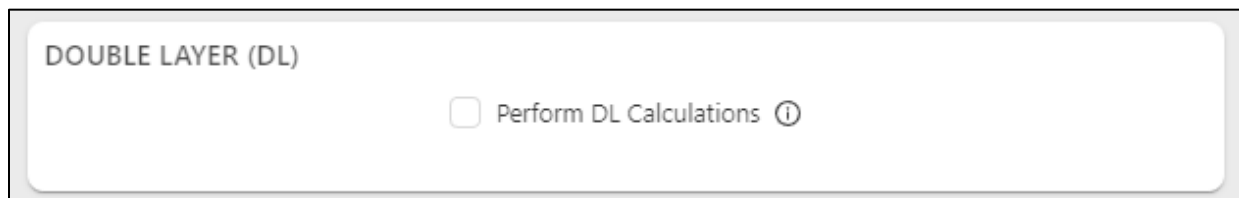


Figure 34. DOUBLE LAYER (DL) section of the AEM Input tab with check box to perform DL Calculations not checked (default).

If the check box is checked and the DL calculations are turned on, the GUI prompts the user to specify some new inputs. The user can specify the **Pre-Pulse Salt Concentration (molar)**, **Current Density (A/cm²)**, and **Temperature of Interest for DL (°C)** using text boxes provided by the GUI. A visual of this section with the text boxes for the inputs described above is shown in Figure 35.

Important Note: The ranges allowed for the above inputs are as follows.

- **Pre-Pulse Salt Concentration (molar):** 0.5 – 1.5 molar
- **Current Density (A/cm²):** 0.00001 – 0.1 A/cm²
- **Temperature of Interest for DL (°C):** Between $T_{\min}$ and $T_{\max}$ defined in Section 3.2.5. Should be at a decade of °C, for example, -10°C, 0°C, 10°C, and so on

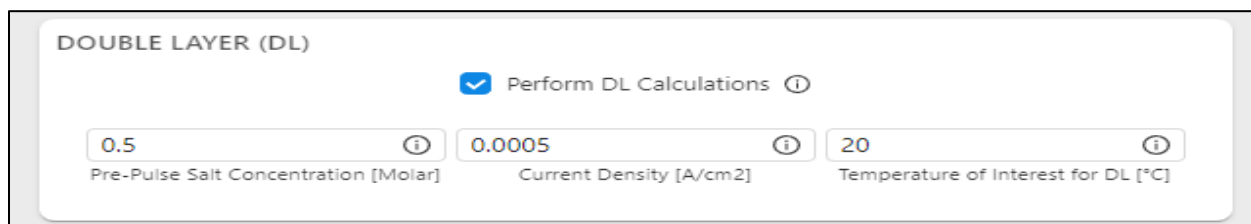


Figure 35. DOUBLE LAYER (DL) section of the AEM Input tab with a check box to perform DL Calculations checked.

3.2.10 AEM RUN IDENTIFIER

This is the section where the user can specify optional inputs to add a custom name to the AEM run and record the time when the AEM simulation was run. All the reports generated by AEM (as well as an import file) will be stored in **C:\Users\user-name\Documents\AEM\DATA** after a

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run. When the **Append Timestamp** check box is selected, the folder will be saved with your **AEM Run Identifier** followed by a timestamp in the form of **AEM Run Identifier-YYYYMMDD_hhmmss**, where **YYYYMMDD** is the format for **Date (i.e., YYYY-MM-DD)** and **hhmmss** is the format for **Time (i.e., hh:mm:ss)**. If the check box is not selected, then the folder will simply be named whatever is in the **AEM Run Identifier** text field. If the folder already exists for that run, the directory will append a number in parentheses depending on how many runs with the same name exist. A visual of this section is shown in Figure 36.

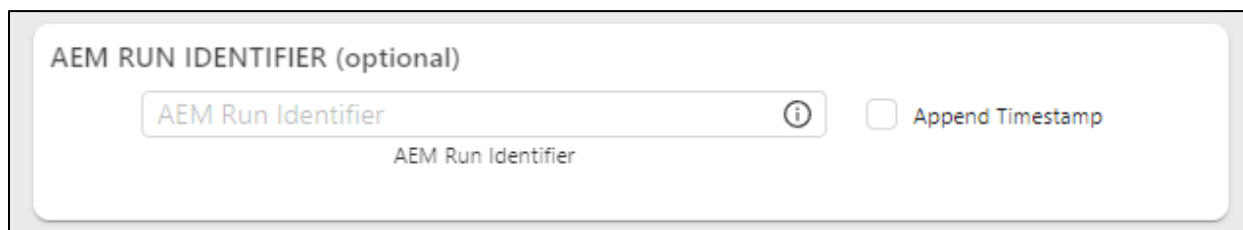


Figure 36. AEM RUN IDENTIFIER section of the AEM Input tab.

3.2.11 RUN AEM

This is the last step after the input conditions for a particular AEM simulation have been entered or imported using the **Import** button described in Section 3.6. The GUI provides a **Run AEM** as shown in Figure 37, which upon clicking initiates the start of the AEM simulation. Shown in Figure 38 is what happens after the button is clicked and Figure 39 is the prompt the GUI throws when the AEM simulation is complete.

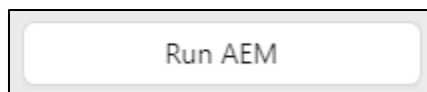


Figure 37. RUN AEM Button of the AEM Input tab.



Figure 38. Prompt showing AEM running after the RUN AEM Button of the AEM Input tab is clicked.

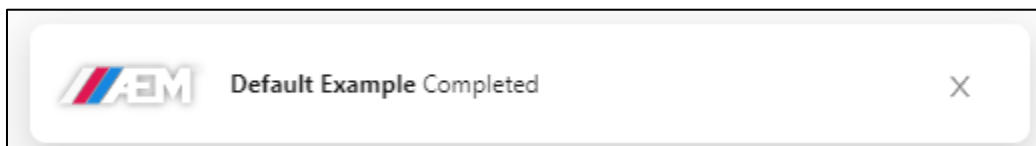


Figure 39. Prompt showing AEM Run Complete.

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The user can also choose to append multiple runs (up to 10) by ticking the check box in **blue** box as shown in Figure 40. This allows the user to add more simulations.



Figure 40. Append Multiple Runs

Once, the check box is marked, users can modify the simulation inputs and add it to the run as shown in Figure 41.

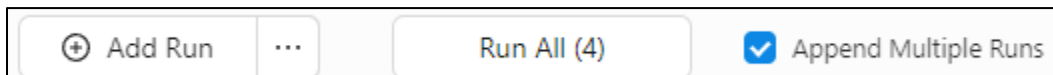


Figure 41. Append multiple runs checked

The individual runs can also be viewed or cleared by clicking on the 3 dots beside the add Run button. This is shown in Figure 42.

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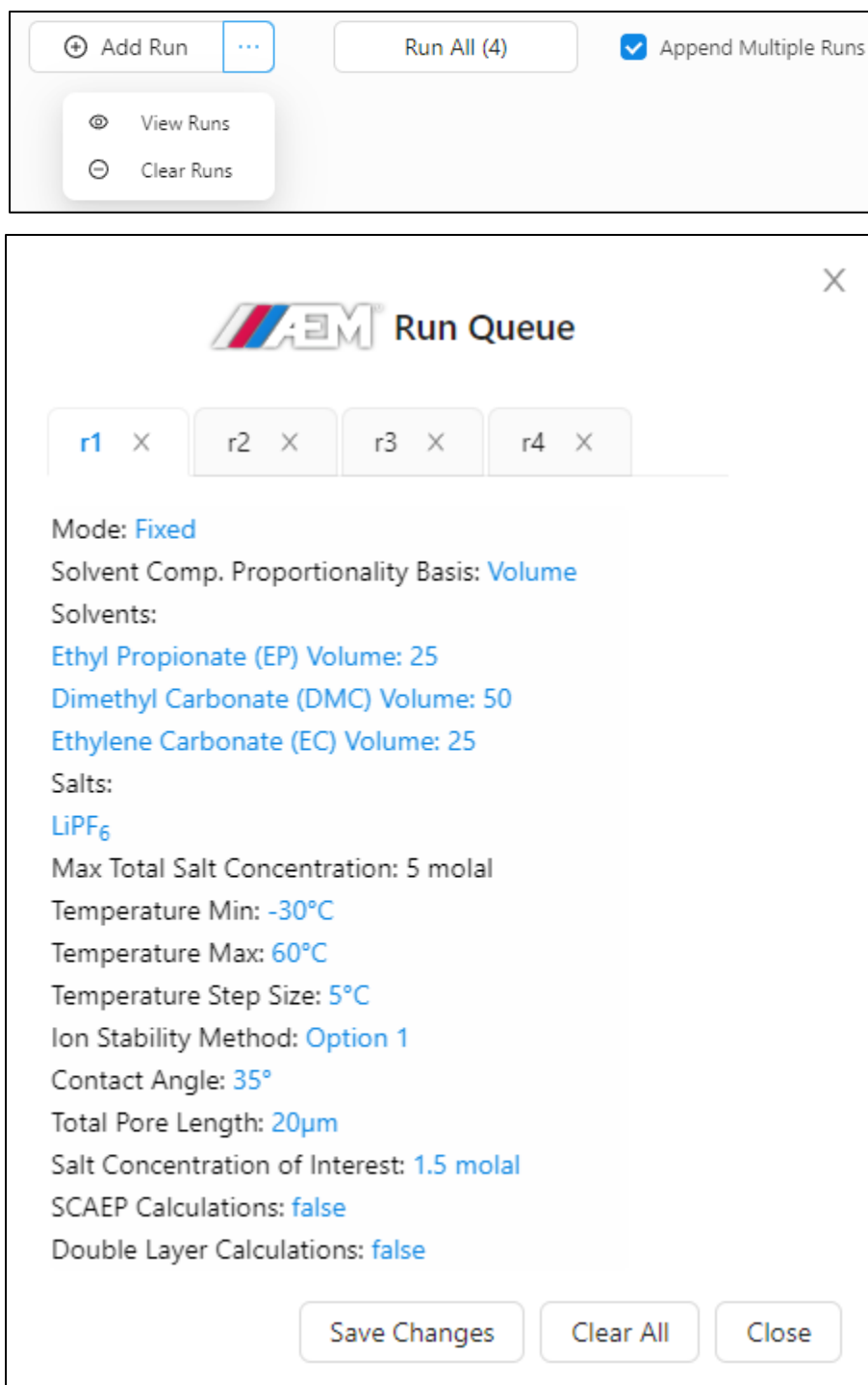


Figure 42. AEM Run Queue

In the case that the AEM simulation fails, the GUI will prompt an error message as shown in Figure 43 and record a **ERROR_OCCURED.txt** text file in the directory for the current simulation in **C:\Users\user-name\Documents\AEM\DATA**. A visual of where this file is located and how it looks in a text editor is shown in Figure 44 and Figure 45, respectively.

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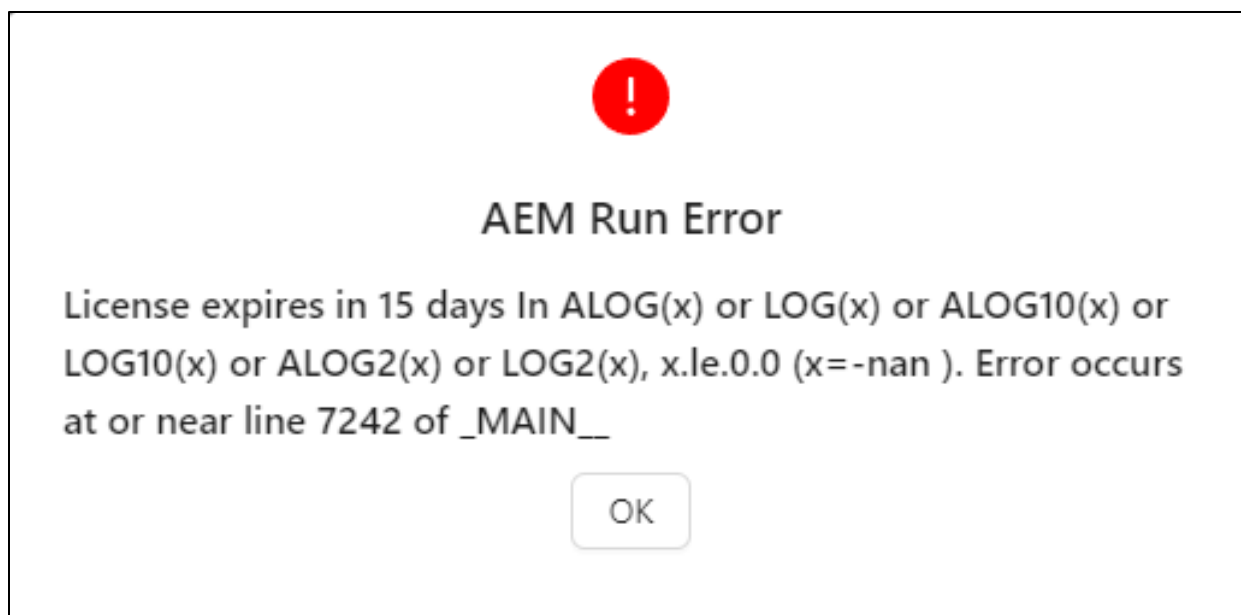


Figure 43. AEM Run Error.

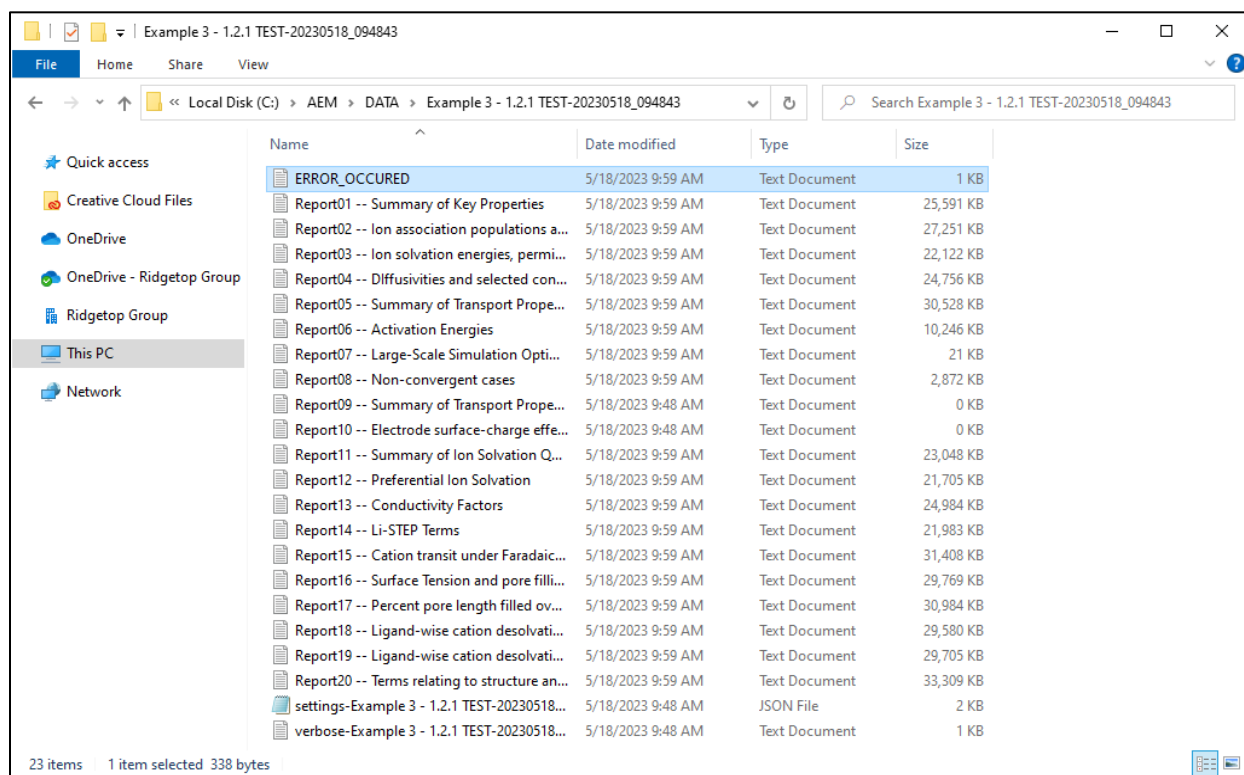
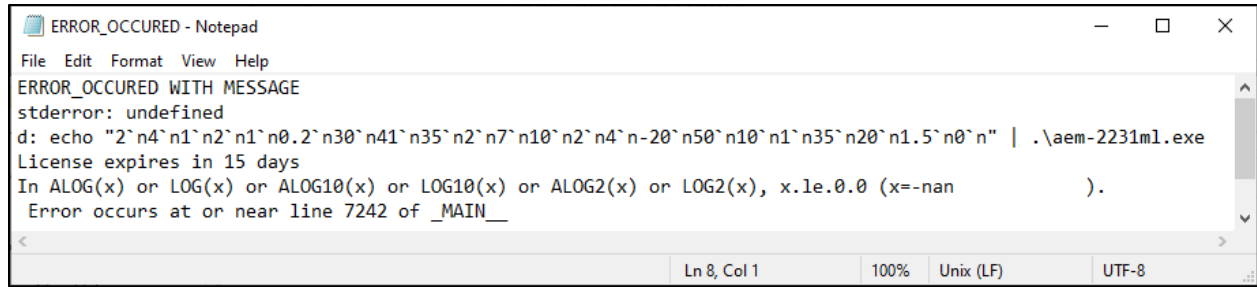


Figure 44. ERROR_OCCURED.txt located at C:\Users\user-name\Documents\AEM\DATA\[Simulation Folder Name].

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```
ERROR_OCCURED - Notepad
File Edit Format View Help
ERROR_OCCURED WITH MESSAGE
stderror: undefined
d: echo "2`n4`n1`n2`n1`n0.2`n30`n41`n35`n2`n7`n10`n2`n4`n-20`n50`n10`n1`n35`n20`n1.5`n0`n" | .\aem-2231m1.exe
License expires in 15 days
In ALOG(x) or LOG(x) or ALOG10(x) or LOG10(x) or ALOG2(x) or LOG2(x), x.1e.0.0 (x=-nan      ).
Error occurs at or near line 7242 of _MAIN__
Ln 8, Col 1  100%  Unix (LF)  UTF-8
```

Figure 45. ERROR_OCCURED.txt as viewed in a Text Editor.

3.3 AEM Arbitrarily Chosen Chemical Compounds (ACCC) Module

The Advanced Electrolyte Model (AEM) has long been recognized as a groundbreaking molecular simulation tool by the global battery R&D community, fundamentally transforming the way researchers approach electrolyte chemistry optimization. This robust platform functions as a virtual laboratory, allowing scientists to explore and fine-tune genome-level electrolyte properties with exceptional precision. AEM's comprehensive database includes over 50 solvents and 30 salts, which has consistently demonstrated high accuracy in predicting electrolyte behaviors. By accelerating the adoption of electric vehicles, enabling grid-scale battery systems, and advancing clean energy solutions, AEM plays a vital role in driving the clean energy revolution, guiding global efforts toward a more sustainable future.

Building on this foundation, the new Arbitrarily Chosen Chemical Compounds (ACCC) Module enhances AEM's capabilities by enabling customers to personalize their material libraries with unique salts and solvents. This optional add-on mirrors the functionality seen in the New Chemistry Import Feature for Ridgetop's CellSage Battery Health Modeling, Simulation, and Analysis (MS&A) software platform, where users can upload specific input parameters and baseline testing data to create new cell chemistry models. Both AEM and CellSage are equipped with default libraries that are updated regularly under the term of the customer's Software License Agreement (SLA), but the ACCC Module offers the flexibility to expand these libraries according to their specific research needs and interests. The AEM ACCC Module enables an extensive range of new compounds, significantly enriching the research possibilities within the realms of pure solvents, multi-solvent mixtures, and now—dual-salt systems. With the latest update, users can define and simulate formulations containing two custom ACCC salts, expanding the realism and applicability of AEM simulations to better match cutting-edge experimental systems.

The reliability of the AEM ACCC Module is underscored by rigorous upfront validation, which confirms that AEM simulations incorporating ACCC definition files achieve targeted results with at least 98% accuracy in conductivity compared to the default simulations using in-code compound parameters. This level of precision is maintained across various applications, including those involving aqueous systems and other complex solvent mixtures. AEM's commitment to accuracy and functionality makes it an indispensable tool in the field of battery research and development, which traditionally suffers from a lack of precise and efficient modeling tools. As the landscape of battery technology evolves rapidly, the need for such advanced tools becomes increasingly critical to understand the complex interactions within batteries and to optimize their electrolyte properties effectively.

As illustrated in Figure 46, the AEM ACCC Module provides a significant enhancement to the capabilities of AEM users. This innovative module allows users to circumvent the standard library of solvents and salts, enabling them to define custom classes of salts and/or solvents. These can then be utilized in electrolyte optimization simulations conducted over various temperature ranges and/or salt concentration gradients. This feature is particularly advantageous for researchers focused on developing new proprietary salts and solvents aimed at creating innovative electrolytes.

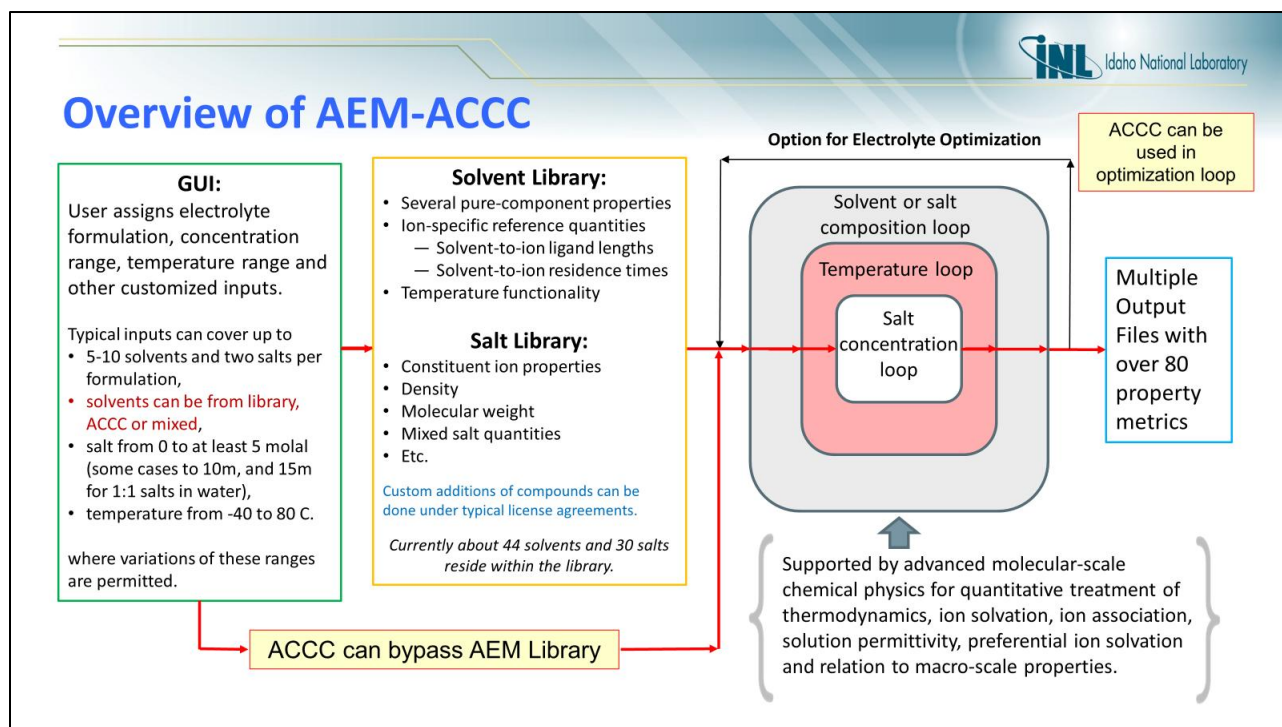


Figure 46. AEM ACCC Overview Diagram.

The simulation framework employed by the ACCC Module is grounded in advanced molecular-scale chemical physics. It offers a quantitative approach to the thermodynamics, ion solvation, ion associations, solution permittivity, and preferential ion solvation, which are crucial for understanding the relationship between micro-scale chemical interactions and macro-scale properties. This comprehensive framework ensures that simulations are both accurate and relevant to real-world applications.

Furthermore, the AEM ACCC Module includes several detailed data output reports, as discussed in [Section 3.4.2](#) of this User Guide. These reports provide users with deep insights into the outcomes of their simulations, facilitating a thorough analysis of new electrolyte formulations. This level of detail and customization makes the AEM ACCC Module an essential tool for advancing the frontier of electrolyte research.

For customers interested in upgrading their software license to include the AEM ACCC Module, we encourage you to reach out directly to Ridgetop Group or your regional sales representative. Upgrading your software license will unlock these enhanced capabilities, allowing you to leverage the full potential of AEM in your research and development efforts, thereby contributing to the advancement of next-generation energy storage solutions.

3.3.1 Resources and Links

To assist in creating ACCC definition files, the following resources are invaluable for sourcing accurate data on solvents and salts:

General Resources for Solvents and Salts:

1. **PubChem:**
 - A free chemistry database maintained by the National Center for Biotechnology Information (NCBI).
 - [PubChem](#)
2. **ChemSpider:**
 - A free chemical structure database providing fast access to over 100 million structures, properties, and associated information.
 - [ChemSpider](#)
3. **Journal of Chemical & Engineering Data:**
 - Publishes data on the properties of chemicals and materials.
 - [Journal of Chemical & Engineering Data](#)
4. **Chemical Reviews:**
 - Offers comprehensive reviews on various chemical topics, including solvent and salt properties.
 - [Chemical Reviews](#)
5. **Journal of Physical and Chemical Reference Data:**
 - Provides critically evaluated physical and chemical property data.
 - [Journal of Physical and Chemical Reference Data](#)

Other Resources For Solvents:

6. **Reaxys:**
 - A comprehensive chemistry database with detailed information on chemical compounds, reactions, and properties.
 - [Reaxys](#)
7. **CRC Handbook of Chemistry and Physics:**
 - A comprehensive reference for chemical and physical data, available through many library subscriptions or online.
 - CRC Press
8. **NIST Chemistry WebBook:**
 - Provides access to data compiled and distributed by the National Institute of Standards and Technology (NIST).
 - [NIST Chemistry WebBook](#)

Other Resources For Salts:

6. **Materials Project:**
 - Provides open-access to a vast database of material properties.
 - [Materials Project](#)
7. **Handbook of Inorganic Chemicals:**
 - Offers detailed information on inorganic compounds, including salts. Available through many library subscriptions.
8. **NIST Standard Reference Data:**
 - Offers a wide range of reference data for salts and other materials.
 - [NIST Standard Reference Data](#)

3.3.2 AEM ACCC Solvent Class

The AEM ACCC Solvent Class, as detailed in Figure 47 and Table 1, is an integral component of the AEM ACCC Module, designed to provide extensive customization in electrolyte simulations. Each solvent within this class is defined by 33 distinct parameters, allowing for precise control over its behavior in simulation environments. These parameters include physical properties such as molecular weight, density, viscosity, and dielectric properties, alongside advanced chemical properties like solvation numbers, residence times, and configurational entropy factors. This granular control enables researchers to simulate a wide range of solvent behaviors accurately.

Solvents in the ACCC Solvent Class are defined using unique ACCC solvent definition files, which associate one solvent with one salt, reflecting their specific ion solvation interactions. These solvent definitions are stored in text files (*.txt), which can be created and managed directly within the AEM graphical user interface (GUI). Once configured, these files can be easily integrated into the AEM GUI or command line interface (CLI) for simulation runs, with the system requiring only one ACCC definition file per simulation. This process is facilitated by a simple suffix system for solvent identification, where solvents such as EP and BN would be labeled as ep-bn or EP-BN in the file name, ensuring a match between user input and filename.

Furthermore, the flexibility of the ACCC Module allows for both ACCC and non-ACCC solvents to be used within the same simulation. This capability enables the use of up to ten ACCC solvents per simulation, significantly broadening the scope for electrolyte formulation. Solvents used in the simulation are flagged within the formulation tables of the output files, providing clear and actionable data for further analysis. This robust framework ensures that AEM users have the tools necessary to push the boundaries of electrolyte chemistry research, paving the way for the development of more efficient and effective energy storage solutions.

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BN 1

1

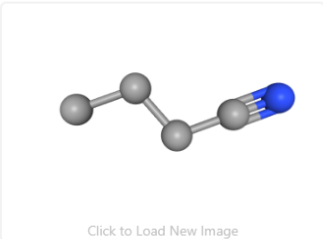
1. Solvent Class Name

2. Number of Solvents

IMPORT SOLVENT VALUES ▾

butyronitrile (BN) 1

3. Solvent 1 Identifier



Click to Load New Image

https://pubchem.ncbi.nlm.nih.gov/compound/8008

ACCC Solvent Class Reference Link

69.11 g/mole 1

4. Molecular Weight

25. °C 1

5. Reference Temperature For Density

0.787 kg/L 1

6. Density at Reference Temperature

20. °C 1

7. Reference Temperature for Relative Permittivity

24.56 1

8. Relative Permittivity at Reference Temperature

-3.85e-03 1

9. Parameter Related to the Temperature Dependence of Relative Permittivity

25. °C 1

10. Reference Temperature for Viscosity

0.553 cP 1

11. Viscosity at Reference Temperature

500. 1

12. Parameter Related to the Temperature Dependence of Viscosity

2. 1

13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations

0.58 ns 1

14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt

0.3 ns 1

15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt

1.383 1

16. Refractive Index (Na D line)

4.07 D 1

17. Dipole Moment, debye

0.71 1

18. π^* Solvatochromic parameter

0.364 1

19. Reichardt's Normalized Polarity Index $[E_T^{N}]$

3. 1

20. Reference Solvation Number for Cation at Infinite Dilution (estimate)

1.5 1

21. Reference Solvation Number for Anion at Infinite Dilution (estimate)

16.6 1

22. Donicity or Gutmann Donor Number

2 1

23. Index for Solvent Solvation Configuration Around Cation

4 1

24. Index for Solvent Solvation Configuration Around Anion

1.4 1

25. Parameter for Solvent Solvation Configuration Around Cation

1.15 1

26. Parameter for Solvent Solvation Configuration Around Anion

12 1

27. Number of Atoms in Solvent Molecule

1 1

28. Classification for Configurational Entropy Calculations as Related to Surface Tension

1. 1

29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion

4. 1

30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension

1. 1

31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension

1. 1

32. Default Factor for Configurational Entropy, as Related to Surface Tension

1 1

33. Index for Solvent Viscosity Mixing Rule

-100. °C 1

34. Pure Solvent Melting Point

0. kJ/mole 1

35. Pure Solvent Melting Point Activation Energy

Save

✕ Close

Figure 47. AEM ACCC Solvent Class GUI View.

Table 1. ACCC Solvent Class Input Parameters.

ID	Parameter Description	Comments
1	Solvent Class Name	User-defined identifier, Limited to 24 characters
2	Number of Solvents	Limit of 10 depending on the experiment
3	Solvent N Identifier	User-defined identifier

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4	Molecular weight, g/mole	Determined experimentally or from online sources (literature, datasets)
5	Reference T for density, degrees C	User defined value
6	Density at reference T, kg/L	Determined experimentally or from online sources (literature, datasets)
7	Reference T for relative permittivity (dielectric constant), degrees C	User defined value
8	Relative permittivity at reference T	Determined experimentally or from online sources (literature, datasets)
9	Parameter related to the temperature dependence of relative permittivity	Slope of d(relative permittivity)/dT on a log basis
10	Reference T for viscosity, degrees C	Determined experimentally or online sources (literature, datasets)
11	Viscosity at reference T, cP	Determined experimentally or online sources (literature, datasets)
12	Parameter related to the temperature dependence of viscosity	This parameter generally falls within values of 200-900. Larger values represent a steeper slope of d(viscosity)/dT.
13	Reference maximum average solvation sites within a single solvent molecule that promote interaction with cations	Estimate of maximum average solvation sites that comprise dipole interactions with a cation, such as lithium, at infinite dilution salt. Values would generally fall within 1 to 6, with common values of 1-3.
14	Reference value for average residence time of solvent around given cation at 25 °C, infinite dilution salt, with units of ns	Key molecular parameters unique to interactions found for each solvent-ion combination. Typically range is 0.5 to 5 ns
15	Reference value for average residence time of solvent around given anion at 25 °C, infinite dilution salt, with units of ns	Key molecular parameters unique to interactions found for each solvent-ion combination. Typical range is 0.2 to 1 ns.
16	Refractive index (Na D line)	Obtained from experiment/simulations; see ACCC examples for common values
17	Dipole moment, Debye	Estimated numerically from similar compounds; see ACCC examples for common values
18	π^* solvatochromic parameter	Estimated numerically from similar compounds; see ACCC examples for common values
19	Reichardt's normalized polarity index (E_T^N)	Estimated numerically from similar compounds; see ACCC examples for common values

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20	Reference solvation number for cation at infinite dilution (estimate)	Obtained from experiment/simulations; common range is 2 to 6
21	Reference solvation number for anion at infinite dilution (estimate)	Obtained from experiment/simulations; common range is 0.5 to 2
22	Donicity or Gutmann donor number	Estimated numerically from similar compounds; common range is 10 to 24
23	Index for solvent solvation configuration around cation	Refer to Figure 48.
24	Index for solvent solvation configuration around anion	Refer to Figure 48.
25	Parameter for solvent solvation configuration around cation	Solvent shape variable for solvent in proximity to cation that determines relative scale between major and minor axes. Default = 1, with typical values ranging from 0.5 to 4.
26	Parameter for solvent solvation configuration around anion	Solvent shape variable for solvent in proximity to anion that determines relative scale between major and minor axes. Default = 1, with typical values ranging from 0.5 to 4.
27	Number of atoms in solvent molecule	Obtained from solvent molecule description
28	Classification for configurational entropy calculations, as related to surface tension	'1' (default) when pivot-point carbons are used '2' when total free carbons are used
29	Parameter for oxygen (or S, P) species in configurational entropy, as related to surface tension	Number of (O, S, P) in molecule
30	Parameter for carbon species in configurational entropy, as related to surface tension	Number of C in molecule
31	Parameter for molecular pivot points in solvent for configurational entropy, as related to surface tension	Number of "pivot points" in molecular structure that enable increased configurational entropy.
32	Default factor for configurational entropy, as related to surface tension	Default value of '1'.
33	Index for solvent viscosity mixing rule	Most systems at '1', PC-based at '2', glyme-based at '3'. This assigns which set of viscosity mixing parameters are used for a pure solvent mixture.
34	Pure solvent melting point T, in °C	This parameter specifies the melting point of the pure solvent in degrees Celsius. It is particularly relevant for solvents with a melting point at or above 0°C. Accurate specification of this value is important in systems near their solid-liquid phase

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		transition, where melting behavior can significantly affect physical properties such as viscosity and ionic conductivity. Default = -100 if user chooses to skip these calculations.
35	Pure solvent melting point activation energy in kJ/mole	This parameter represents the activation energy associated with the melting behavior of the pure solvent, typically ranging between 10–25 kJ/mol. It is used to estimate the amount of free (liquid) solvent that becomes available as salt is added—particularly at or below the melting point. This calculation is primarily intended for single-component solvent systems with melting points at or above 0°C. • Note: This calculation is crucial for accurately modeling electrolyte behavior in systems near solid–liquid phase transitions, where the presence or absence of free solvent strongly impacts electrolyte transport properties. • Reference: See <i>Report11</i>, far-right column, for output details. Default = 0 if user chooses to skip these calculations.

Additional Notes for ACCC Solvent Class:

1. For Parameters with a Green ID, these values are typically known or easily derived.
2. For Parameters with a Red ID, these values are typically unknown but, in many cases, can be estimated from similar compounds. This requires some level of iteration to satisfy model accuracy for an array of properties.
3. For Parameters 16-22, these values will be unique to the chosen salt, and require some level of optimization to get accurate ion solvation characterization and numerous other properties.
4. For Parameters 23-26, these are key parameters for solvated ion molecular configurations based on average solvent-ion interactions. Unique to each solvent-ion combination at 25 °C, infinite dilution.
5. For Parameters 28-33, these parameters are part of a proprietary algorithm developed by Idaho National Laboratory.
6. Figure 48 refers to the options tied to parameters 23 and 24.

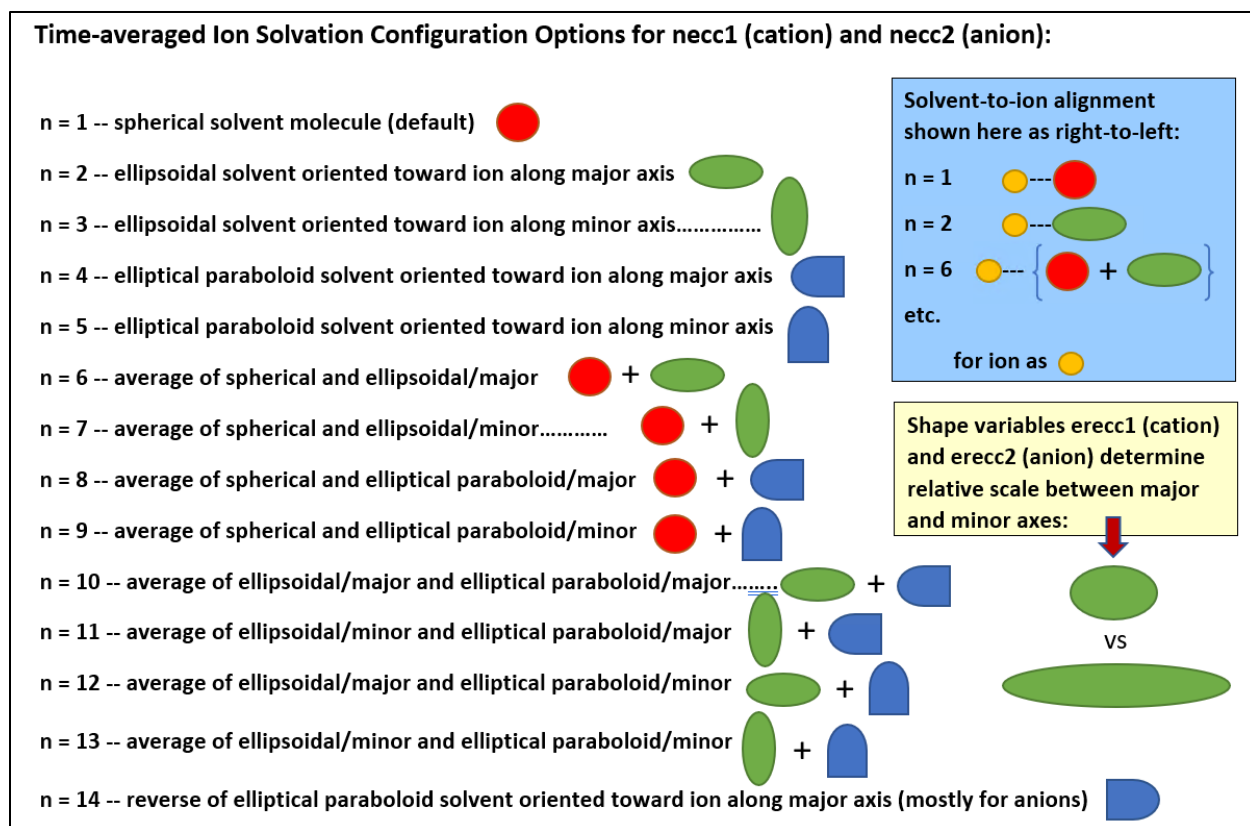


Figure 48. Note for time-average ion solvation.

3.3.3 Example of AEM ACCC Solvent Class

The Advanced Electrolyte Model (AEM) has introduced an innovative approach for simulating solvent interactions within electrolytes through its Arbitrarily Chosen Chemical Compounds (ACCC) Module. This section of the User Guide highlights the Solvent Examples used to test and validate this module. A comprehensive set of 16 Solvent Definition files were developed, showcasing the module's robustness and versatility. Figure 49 offers a visual reference for users to better understand the configuration and application of the ACCC Solven Class within AEM.

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Create New ACCC Solvent Class

BN

1

1. Solvent Class Name

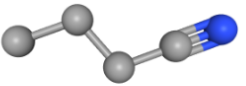
2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES

butyronitrile (BN)

3. Solvent 1 Identifier



Click to Load New Image

<https://pubchem.ncbi.nlm.nih.gov/compound/8008>

ACCC Solvent Class Reference Link

69.11	g/mole	25.	°C	0.787	kg/L	20.	°C
4. Molecular Weight		5. Reference Temperature For Density		6. Density at Reference Temperature		7. Reference Temperature for Relative Permittivity	
24.56		-3.85e-03		25.	°C	0.553	cP
8. Relative Permittivity at Reference Temperature		9. Parameter Related to the Temperature Dependence of Relative Permittivity		10. Reference Temperature for Viscosity		11. Viscosity at Reference Temperature	
500.		2.		0.58	ns	0.3	ns
12. Parameter Related to the Temperature Dependence of Viscosity		13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations		14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt		15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt	
1.383		4.07	D	0.71		0.364	
16. Refractive Index (Na D line)		17. Dipole Moment, debye		18. π^* Solvatochromic parameter		19. Reichardt's Normalized Polarity Index $[E_T^N]$	
3.		1.5		16.6		2	
20. Reference Solvation Number for Cation at Infinite Dilution (estimate)		21. Reference Solvation Number for Anion at Infinite Dilution (estimate)		22. Donicity or Gutmann Donor Number		23. Index for Solvent Solvation Configuration Around Cation	
4		1.4		1.15		12	
24. Index for Solvent Solvation Configuration Around Anion		25. Parameter for Solvent Solvation Configuration Around Cation		26. Parameter for Solvent Solvation Configuration Around Anion		27. Number of Atoms in Solvent Molecule	
1		1.		4.		1.	
28. Classification for Configurational Entropy Calculations as Related to Surface Tension		29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion		30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension		31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension	
1.		1		-100.	°C	0.	kJ/mole
32. Default Factor for Configurational Entropy, as Related to Surface Tension		33. Index for Solvent Viscosity Mixing Rule		34. Pure Solvent Melting Point		35. Pure Solvent Melting Point Activation Energy	

Figure 49. ACCC Solvent Definition for Butyronitrile (BN).

Figure 177 through Figure 191 in [Section 7.2](#) of the Appendix provide detailed screenshots for other ACCC solvent examples. The selection of solvents for the ACCC Module was influenced by existing solvent types already integrated into AEM, ensuring a seamless validation and demonstration process. The focus was on organic carbonates, esters, and nitriles, with specific examples including ethylene carbonate (EC), propylene carbonate (PC), dimethyl carbonate (DMC), methyl acetate (MA), ethyl acetate (EA), and acetonitrile (AN), among others. Additionally, optional solvents such as dimethoxyethane (DME), diglyme, and water were also considered.

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These choices reflect the diversity and flexibility of the AEM in handling various solvent types, enhancing the accuracy and applicability of the model in real-world scenarios.

In terms of technical details, each ACCC solvent definition within the module contains thirty-three (33) distinct parameters, meticulously defined to capture pure solvent properties, temperature dependence, and the unique ion solvation behaviors of different solvent-salt combinations. It is important to note that ACCC solvent definition files have parameters that are dependent on benchmarking data with different solvents and salts, and only one (1) salt can be paired with a given ACCC defined solvent or group of ACCC solvents used in the same simulation. This is exemplified by the ACCC solvent definition file for "EC_PC EMC_MA_BN, which is specifically highlighted as a multi-solvent electrolyte formulation:

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Create New ACCC Solvent Class

EC_PC EMC_MA_BN

5

1. Solvent Class Name

2. Number of Solvents

Sol1

Sol2

Sol3

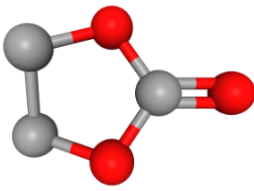
Sol4

Sol5

IMPORT SOLVENT VALUES

ethylene carbonate

3. Solvent 1 Identifier



Click to Load New Image

<https://pubchem.ncbi.nlm.nih.gov/compound/7303>

ACCC Solvent Class Reference Link

88.06 g/mole

39. °C

1.3214 kg/L

40. °C

4. Molecular Weight

5. Reference Temperature For Density

6. Density at Reference Temperature

7. Reference Temperature for Relative Permittivity

90.36

-5.175e-03

40. °C

1.93 cP

8. Relative Permittivity at Reference Temperature

9. Parameter Related to the Temperature Dependence of Relative Permittivity

10. Reference Temperature for Viscosity

11. Viscosity at Reference Temperature

800.

1.5

0.357 ns

0.286 ns

12. Parameter Related to the Temperature Dependence of Viscosity

13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations

14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt

15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt

1.42

4.51 D

0.84

0.552

16. Refractive Index (Na D line)

17. Dipole Moment, debye

18. π^* Solvatochromic parameter

19. Reichardt's Normalized Polarity Index $[E_T^N]$

4.

1.

16.4

10

20. Reference Solvation Number for Cation at Infinite Dilution (estimate)

21. Reference Solvation Number for Anion at Infinite Dilution (estimate)

22. Donicity or Gutmann Donor Number

23. Index for Solvent Solvation Configuration Around Cation

2

1.28

1.22

10

24. Index for Solvent Solvation Configuration Around Anion

25. Parameter for Solvent Solvation Configuration Around Cation

26. Parameter for Solvent Solvation Configuration Around Anion

27. Number of Atoms in Solvent Molecule

1

3.

3.

0.

28. Classification for Configurational Entropy Calculations as Related to Surface Tension

29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion

30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension

31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension

1.

1

-100. °C

0. kJ/mole

32. Default Factor for Configurational Entropy, as Related to Surface Tension

33. Index for Solvent Viscosity Mixing Rule

34. Pure Solvent Melting Point

35. Pure Solvent Melting Point Activation Energy

Figure 50. Example ACCC Solvent Definition File for multiple solvents.

The ACCC solvent definition files are stored in text files (*.txt) which are automatically recognized and processed by the AEM software program when properly assigned by the user. Although the ACCC solvent definition files can be edited in a text editor program, it is recommended that each solvent definition is edited using the AEM GUI. In the output files, ACCC solvents are distinctly flagged within the output Report files, providing ready identification of their role in electrolyte formulations.

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The integration of ACCC and non-ACCC solvents within the same simulation is a testament to the model's flexibility, allowing for a wide range of electrolyte formulations. This capability supports the use of anywhere from 0 to 10 ACCC solvents in a single simulation, significantly expanding the experimental scope. Solvents involved in the simulations are flagged within the formulation tables of the output files, ensuring clear and accessible data for analysis. This level of detail and accuracy, with simulation results matching the default simulations to an accuracy of $\geq 99\%$ based on conductivity, confirms the efficacy of the AEM-ACCC functionality in advancing electrolyte research and application.

3.3.4 AEM ACCC Salt Class

The AEM ACCC Salt Class, detailed in Figure 51 and Table 2, enriches the AEM compound library by providing a specialized framework for the precise characterization of salts used in electrolyte simulations. This class encompasses 24 distinct parameters for each salt, enabling a deep dive into the ion solvation behaviors that are critical for understanding and optimizing electrolyte formulations.

In the ACCC Salt Class, each salt is defined in conjunction with one or more solvents, capturing the unique solvation interactions specific to each salt-solvent combination. This is facilitated through the use of ACCC salt definition files, which are text files (*.txt) created and managed within the AEM graphical user interface (GUI). Once these files are set up, they are seamlessly integrated into the AEM GUI or command-line interface (CLI) for simulation purposes. Additionally, a corresponding ACCC solvent file may be included in the simulation as needed.

Salt assignment within the AEM system is straightforward, requiring users to input a simple suffix when prompted by the software. For instance, a salt like LiPF₆ used in an electrolyte combination with EC, DMC, and MP would have a suffix like LiPF₆_w_EC_DMC_MP, and all relevant ACCC data for this setup would be stored in a file named ACCC Inputs (Salt Class)--LiPF₆_w_EC_DMC_MP.txt. It is crucial that the suffix provided matches exactly between the user input and the filename to ensure proper file handling and simulation accuracy.

The ACCC framework now supports the use of up to two (2) ACCC salts in a given simulation, significantly expanding formulation flexibility. Users can define either a single ACCC salt or a combination of two ACCC salts, or pair one ACCC salt with a non-ACCC (library) salt. This dual-salt support allows researchers to explore synergistic effects in mixed-salt systems, which are increasingly common in next-generation electrolyte design. It is important to note that each ACCC salt has solvent dependent parameters that should be correlated with benchmarking data when adding new ACCC salts to the library. This capability is particularly advantageous as an ACCC salt can be used with default library solvents or ACCC solvents, enhancing the adaptability of the AEM to meet diverse research needs.

In the output files, ACCC salts are distinctly flagged within the formulation tables, providing clear visibility and ease of analysis for researchers evaluating the impacts of different salt configurations on the electrolyte's performance. This comprehensive approach ensures that the AEM ACCC Salt Class is a robust tool for advancing the field of electrolyte chemistry.

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Create New ACCC Salt Class
✕

ⓘ
1. Salt Class Name

ⓘ
2. Salt Identifier

Regular Salt

Ionic Liquid

3. Salt Type

Å ⓘ
4. Diameter of Cation from Crystalline Basis

Å ⓘ
5. Diameter of Anion from Crystalline Basis [Å]

Å ⓘ
6. Equivalent Diameter of Anion Accounting for Electrostatic Charge Distribution of Anion

ⓘ
7. Number of Charge Regions for Anion

ⓘ
8. Unit Charge of Cation

ⓘ
9. Unit Charge of Anion

g/mole ⓘ
10. Molecular Weight of Salt

g/mole ⓘ
11. Molecular Weight of Cation

g/cm³ ⓘ
12. Density of Pure Crystalline Salt

ⓘ
13. Relative Permittivity of Pure Crystalline Salt

Å ⓘ
14. Factor to Express Equivalent Diameter of Cation in RTIL Accounting for Electrostatic Charge Distribution of Cation

ⓘ
15. Number of Charge Regions for a Given Cation in RTIL

ⓘ
16. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

ns ⓘ
17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C

ns ⓘ
18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C

ⓘ
19. Reference Solvation Number for Cation at Infinite Dilution (estimate)

ⓘ
20. Reference Solvation Number for Anion at Infinite Dilution (estimate)

ⓘ
21. Index for Solvent Solvation Configuration Around Cation

ⓘ
22. Index for Solvent Solvation Configuration Around Anion

ⓘ
23. Parameter for Solvent Solvation Configuration Around Cation

ⓘ
24. Parameter for Solvent Solvation Configuration Around Anion

Save

Close

Figure 51. AEM ACCC Salt Class GUI View.

Table 2. ACCC Salt Class Input Parameters.

ID	Parameter Description	Comments
1	Salt Class Name	User-defined identifier, limited to 25 characters
2	Salt Identifier	User-defined salt name, limited to 12 characters
3	Salt Type	User-defined: '1' – regular salts '2' – ionic liquids
4	Diameter of cation from crystalline basis, Angstrom	Obtained from Pauling radii
5	Diameter of anion from crystalline basis, Angstrom	Obtained from Pauling radii
6	Equivalent diameter of anion accounting for electrostatic charge distribution of anion, Angstrom (this term $\leq \text{sig2p}$)	If anion has a localized charge, then estimate the effective diameter of the solvent defined at that charge locus, considering the closest approach to charge.
7	Number of charge regions for a given anion (default value = 1.; value	default value = 1.; value of 2 or higher used for non-uniform acentric charge.

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	of 2. or higher used for non-uniform acentric charge)	Use default value for simple anions where the effective electrostatic anion diameter is the same as the cryst. value . A value of 2 may be warranted for more complex anions.
8	Unit charge of cation	Positive integers 1, 2, 3 etc.
9	Unit charge of anion	Negative integers -1, -2, -3 etc.
10	Molecular weight of salt	Obtained from experiment/online sources
11	Molecular weight of cation	Obtained from experiment/online sources
12	Density of pure crystalline salt, subtracting void space contribution from particle packing fraction, g/cm ³ .	Obtained from experiment/online sources
13	Relative permittivity of pure crystalline salt	Obtained from experiment/online sources. Typical values range from 10-20.
14	Factor to express equivalent diameter of cation in RTIL accounting for electrostatic charge distribution of cation, Angstrom (this term between 0 to 1)	Accounts for charge dislocation and corresponding reduced electrostatic diameter of the cationic member of the RTIL. Theoretical values range from 0. to 1., with a practical range of 0.5 to 1.
15	Number of charge regions for a given cation in RTIL (default value = 1.; value of 2. or higher used for non-uniform acentric charge)	Analogous to Item 7 herein. (Default value = 1.; value of 2. or higher used for non-uniform acentric charge)
16	Number of Solvents	Specifies number of solvents that the ACCC salt is compatible with based on solvent related parameters 17-24.
17	Reference value for average residence time of solvent around given cation at 25 °C, infinite dilution, ns	Key molecular parameters unique to interactions found for each solvent-ion combination. Typical range is 0.2 to 1 ns.
18	Reference value for average residence time of solvent around given anion at 25 °C, infinite dilution, ns	Initial value provided by user (say, 4 to 8 nm); value is updated by code.
19	Reference solvation number for cation at infinite dilution (estimate)	Obtained from experiment/simulations; common range is 2 to 6
20	Reference solvation number for anion at infinite dilution (estimate)	Obtained from experiment/simulations; common range is 0.5 to 2
21	Index for solvent solvation configuration around cation	Refer to Figure 48.
22	Index for solvent solvation configuration around anion	Refer to Figure 48.

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23	Parameter for solvent solvation configuration around cation	Solvent shape variable for solvent in proximity to cation that determines relative scale between major and minor axes. Default = 1, with typical values ranging from 0.5 to 4.
24	Parameter for solvent solvation configuration around anion	Solvent shape variable for solvent in proximity to cation that determines relative scale between major and minor axes. Default = 1, with typical values ranging from 0.5 to 4.

Additional Notes for ACCC Salt Class:

1. For Parameters 17-24, these terms were discussed in the previous section for the ACCC Solvent Class.
2. Note that if an ACCC solvent and an ACCC salt are both used in a simulation, these terms must be identical within the two input definition files / strings covering the ACCC solvent and salt.
3. When using two ACCC salts, each must be defined independently with a complete ACCC Salt Definition File. The AEM GUI will prompt users to assign both files during simulation setup, and the model automatically handles combined interactions and outputs.
4. When simulating two ACCC salts, it is recommended that users cross-check solvent compatibility and ensure that shared solvents have consistent solvation parameters across both ACCC Salt Definition Files. This helps preserve thermodynamic balance and reduces the likelihood of unexpected property anomalies during simulation.
5. For users working in Custom Recipe Mode, the AEM GUI will guide the entry of two ACCC salts and integrate them automatically into the conversion table and simulation input deck. See Section 3.2.2 for details.

3.3.5 Example of AEM ACCC Salt Class

The Advanced Electrolyte Model (AEM) has further extended its functionality with the Arbitrarily Chosen Chemical Compounds (ACCC) Module, which includes a detailed implementation of salt definitions to enhance simulation capabilities. To validate this aspect of the module, six Salt Definition files were meticulously developed, encompassing a variety of commonly used lithium salts and ionic liquids. Figure 52 provides a visual demonstration for one example salt definition file, aiding users in comprehending the specific configurations and applications within the model. Figure 193 through Figure 197 in [Section 7.3](#) of the Appendix shows additional examples for each of the default ACCC salt definition files that were tested and validated. The salts tested include LiPF₆, LiFSI, LiTFSI, LiBOB, LiAsF₆, and LiBF₄, alongside room temperature ionic liquids (RTILs) such as PYR13-FSI and EMI-FSI.

Each salt in the ACCC Module is parameterized with 23 unique values that reflect its ion solvation behavior, which is crucial for accurately simulating the interaction between salts and solvents. The individual ACCC salt definitions are strategically structured to define interactions between a specific salt and one or more solvents, capturing the unique characteristics of each combination. This detailed approach ensures that the simulation results are highly accurate, achieving an

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impressive $\geq 98\%$ accuracy compared to default simulations that use built-in parameters, particularly noting the efficiency in simulating non-aqueous systems.

The flexibility of the AEM in handling various salt and solvent combinations is showcased through the capability to read and integrate ACCC salt definition files, which are stored as text files (*.txt) and are automatically recognized by the AEM once proper assignments are made. This process involves assigning a specific suffix when prompted by the software, which directs the model to the appropriate file containing all necessary ACCC data. For instance, if an electrolyte simulation involves LiPF₆ with solvents like ethylene carbonate (EC), dimethyl carbonate (DMC), and methyl propionate (MP), the user would input the suffix 'LiPF₆_w_EC_DMC_MP', correlating exactly with the filename 'ACCC Inputs (Salt Class)--LiPF₆_w_EC_DMC_MP.txt'.

Moreover, the AEM allows for the integration of both ACCC and non-ACCC salts in the same simulation, offering a broader range of electrolyte formulations to researchers. This inclusion now accommodates up to two ACCC salts per simulation, enhancing the model's adaptability to advanced electrolyte systems that involve salt co-formulations. For example, a user can now simulate an electrolyte containing both LiPF₆ and LiFSi — each defined as separate ACCC salts — with EC and DMC solvents, enabling side-by-side evaluation of transport and SEI-modifying effects from both anions. All salts, whether ACCC or not, are clearly flagged in the formulation tables within the output files, providing an easy reference for analyzing the results. This comprehensive system ensures that users can efficiently manipulate and utilize the AEM for advanced simulations, pushing the boundaries of electrolyte research.

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Edit ACCC Salt Class

LiBF4_w_EC EMC

1. Salt Class Name

LiBF4

2. Salt Identifier

Regular Salt

Ionic Liquid

3. Salt Type

1.36 Å

4. Diameter of Cation from Crystalline Basis

3.795 Å

5. Diameter of Anion from Crystalline Basis [Å]

3.795 Å

6. Equivalent Diameter of Anion Accounting for Electrostatic Charge Distribution of Anion

1.

7. Number of Charge Regions for Anion

1.

8. Unit Charge of Cation

-1.

9. Unit Charge of Anion

93.74 g/mole

10. Molecular Weight of Salt

6.941 g/mole

11. Molecular Weight of Cation

2.15 g/cm³

12. Density of Pure Crystalline Salt

16.

13. Relative Permittivity of Pure Crystalline Salt

0

14. Factor to Express Equivalent Diameter of Cation in RTIL Accounting for Electrostatic Charge Distribution of Cation

0

15. Number of Charge Regions for a Given Cation in RTIL

2

16. Number of Solvents

Sol1

Sol2

IMPORT SOLVENT VALUES

0.357 ns

17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C

0.7 ns

18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C

4.

19. Reference Solvation Number for Cation at Infinite Dilution (estimate)

2.

20. Reference Solvation Number for Anion at Infinite Dilution (estimate)

10

21. Index for Solvent Solvation Configuration Around Cation

4

22. Index for Solvent Solvation Configuration Around Anion

1.28

23. Parameter for Solvent Solvation Configuration Around Cation

2.2

24. Parameter for Solvent Solvation Configuration Around Anion

Delete

Save

Close

Figure 52. ACCC Salt Definition for LiBF4_w_EC EMC.

3.3.6 Example of ACCC Validation

This section demonstrates two example validation cases for the ACCC module, considering both cases of solvent and salts that are defined as an ACCC class. Additional validation cases for other solvents and salts can be reviewed in [Section 7.4](#) of the Appendix. As shown in Figure 53 and Figure 54, the validation process involved conducting two simulations: the first using the default library with non-ACCC solvent or salt compounds (represented by the black curves in the figures), and the second using the same compounds defined as ACCC solvents or salts (represented by the red-dashed curves in the figures). The results, focusing on electrolyte conductivity, indicate that the accuracy remains high between ACCC and non-ACCC cases, showing no significant differences in outcomes between the two methods. The agreement is generally within 99% accuracy for library versus ACCC bases, demonstrating that ACCC effectively bypasses the library. This consistency across various solvent and salt types indicates that the system's performance is robust and repeatable with the ACCC architecture.

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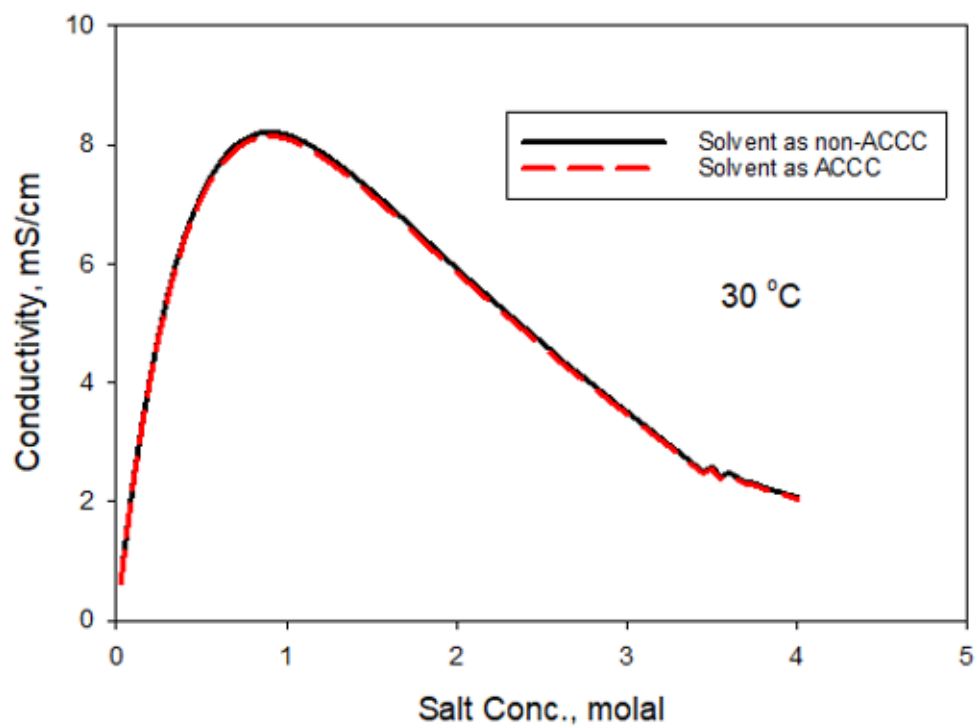


Figure 53. EC+LiPF₆ at 30⁰

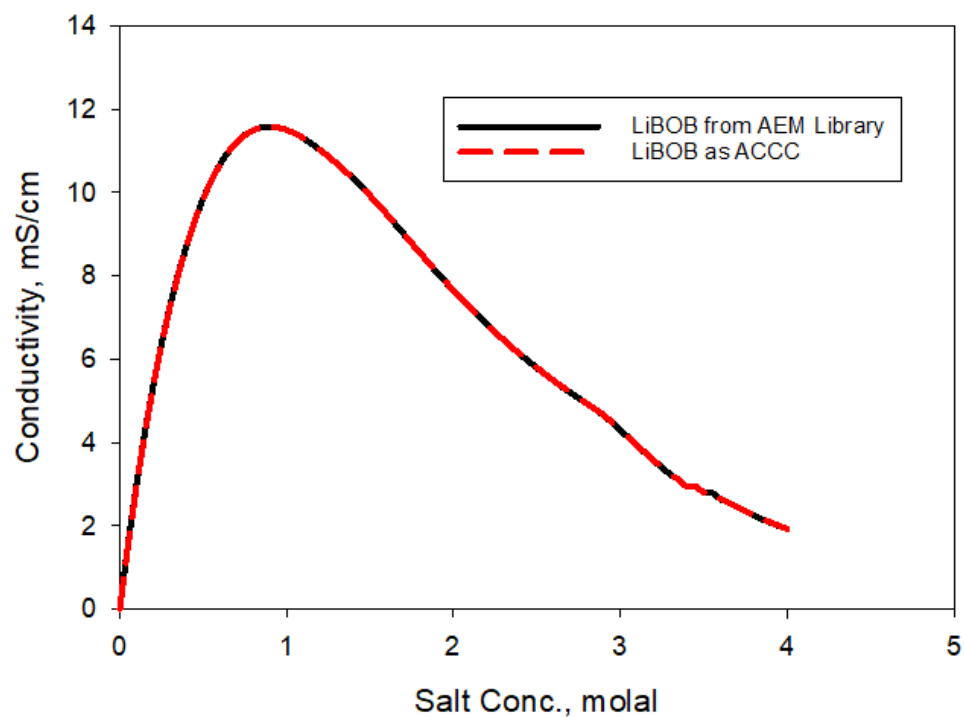


Figure 54. PC-EA (3:7 by mass) + LiBOB

3.4 AEM Output Tab

The second in the AEM GUI is the **AEM Output** tab, where the user can preview input conditions for a particular run, toggle between multiple different runs, and view data output records and plots for each individual run. The **AEM Output tab** is separated into the following components.

- Sidebar that contains three select drop-down sections – **AEM RUNS**, **AEM REPORTS**, **Data Display** and a text previewing window in the **AEM RUNS** section.
- **REPORT TEXT** section with a text viewer scroll box.
- **REPORT DATA VISUALIZATION (2D)** section with three select drop down sections – **Filter List**, **X-Axis Parameter**, **Y-Axis Parameter**, a list scroll box to display options based on what filter is chosen in the **Filter List** section, and a graphing chart to display output data plots based on what is chosen in the **X-Axis Parameter** and **Y-Axis Parameter** sections.
- **REPORT DATA VISUALIZATION (3D)** section with three select drop down sections – **Filter List**, **X-Axis Parameter**, **Y-Axis Parameter**, **Z-Axis Parameter**, a list scroll box to display options based on what filter is chosen in the **Filter List** section, and a graphing chart to display output data plots based on what is chosen in the **X-Axis Parameter**, **Y-Axis Parameter** and **Z-Axis Parameter** sections. Note: Y-axis is fixed to Temperature.

A visual of the AEM Output Tab with the **REPORT TEXT** section (Figure 55) and **REPORT DATA VISUALIZATION** section (Figure 56) is shown below.

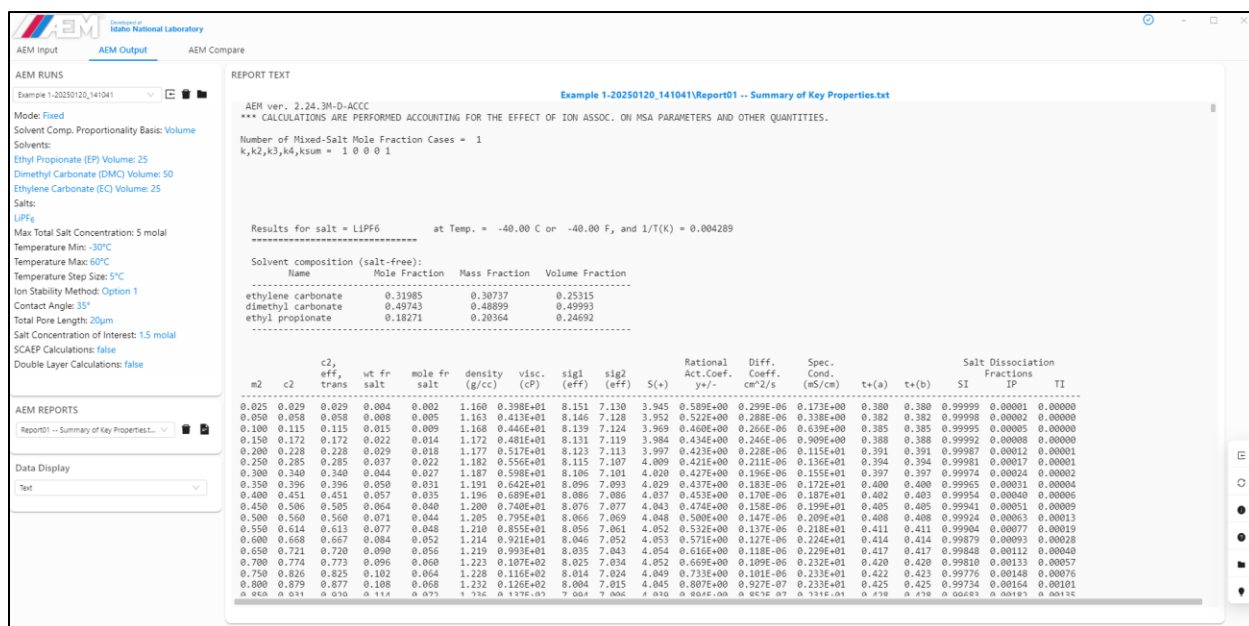


Figure 55. AEM Output Tab with REPORT TEXT section.

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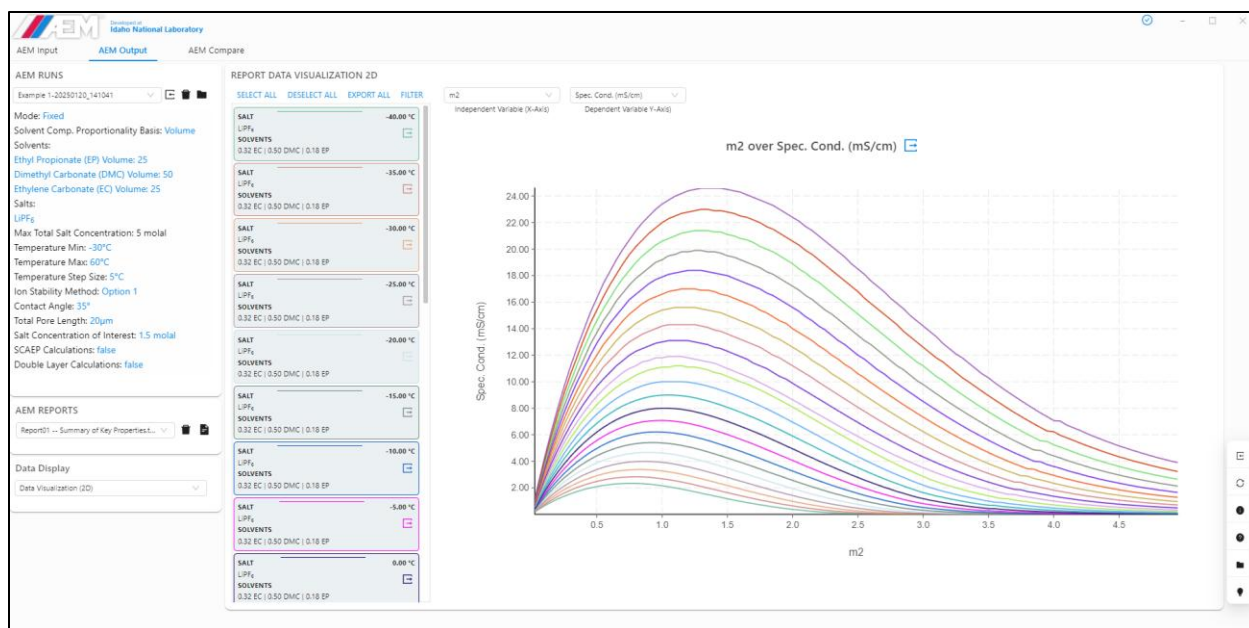


Figure 56. AEM Output Tab with REPORT DATA VISUALIZATION (2D) section.

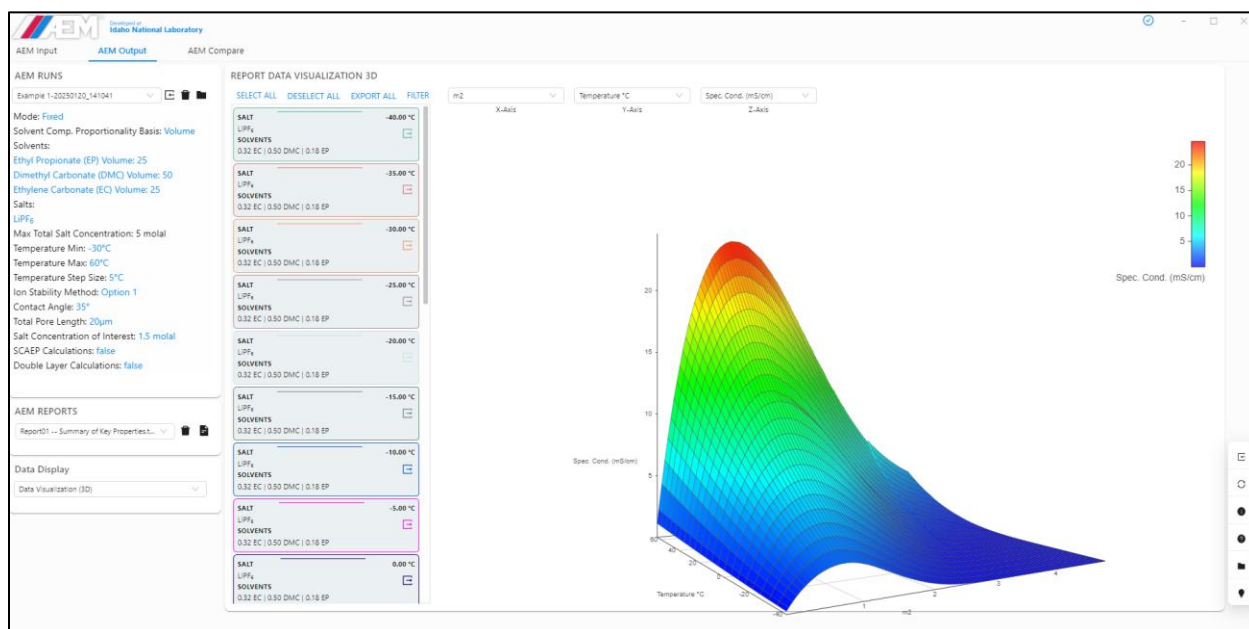


Figure 57. AEM Output Tab with REPORT DATA VISUALIZATION (3D) section

3.4.1 AEM RUNS

This is a dropdown select option where the user can access a list of available simulation runs that have been logged in the data output directory - **C:\Users\user-name\Documents\AEM\DATA**. The GUI also provides a text previewing window to preview a full set of simulation conditions from a .json file that is stored in the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory. The user can use the **Delete** button (🗑️) to delete the instance for a particular run and the **Reveal**

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in **File Explorer** button (■) to open the .json file in File Explorer on the user's local machine. A visual of this section is shown in Figure 58 below.

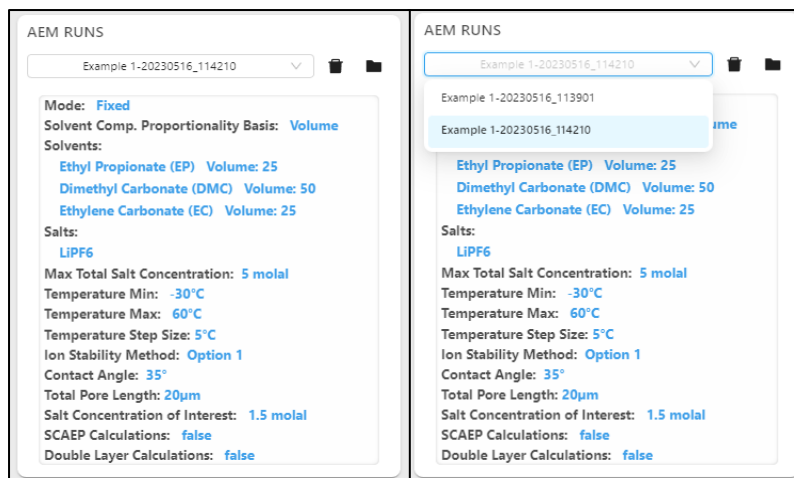


Figure 58. AEM RUNS section of the AEM Output tab.

3.4.2 AEM REPORTS

When all the simulations settings are supplied and the user clicks on the **Run AEM** button on the **AEM Input** tab, AEM starts the simulation run. It then updates the output report files with the simulation results. The user can choose to run the simulation with another set of conditions. The results of a subsequent simulation run will be appended to the output files. However, if the software is restarted, then those files will be overwritten. These files should be copied to another folder for future use. The next simulation run will overwrite any results in these files. The following output report files are created as a result. These are described in Section 7.1 later in detail.

- ◆ **Report1** -- Summary of Key Properties
- ◆ **Report2** -- Ion association populations and other thermodynamic terms
- ◆ **Report3** -- Ion solvation energies, permittivity and cation desolvation
- ◆ **Report4** -- Diffusivities and selected conductivity terms
- ◆ **Report5** -- Summary of Transport Properties and Walden analysis
- ◆ **Report6** -- Activation Energies
- ◆ **Report7** -- Large-Scale Simulation Optimization
- ◆ **Report8** -- Non-convergent cases
- ◆ **Report9** -- Double-Layer Regions transport analysis
- ◆ **Report10** -- Electrode surface-charge effects
- ◆ **Report11** -- Summary of Ion Solvation Quantities
- ◆ **Report12** -- Preferential Ion Solvation
- ◆ **Report13** -- Conductivity Factors
- ◆ **Report14** -- Li-STEP Terms
- ◆ **Report15** -- Cation transit under Faradaic conditions
- ◆ **Report16** -- Surface Tension and pore filling time over salt conc.
- ◆ **Report17** -- Percent pore length filled over time

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- ◆ **Report18** -- Ligand-wise cation desolvation energy and time
- ◆ **Report19** -- Ligand-wise cation desolvation energy and time (accounting for CS)
- ◆ **Report20** -- Terms relating to structure and Communal Ion Solvation (CS)

Other than the output report files, the AEM GUI generates two more output files described below.

- **settings-AEM Run Identifier-YYYYMMDD_hhmmss.json:** This **setting .json** file saves the user's current run settings, which allows user to import them in the future if they want to rerun the same or make small changes to the imported run. This file also is the backbone of the GUI to populate the settings pane in the output page, allowing users a quick glance of the settings for the currently selected run folder.
- **verbose-AEM Run Identifier-YYYYMMDD_hhmmss.txt:** This **verbose .txt** file is just a more readable version of the values being sent to the AEM command line executables.

In this section, the GUI provides a drop down select option for the user to choose what output report to view. The report chosen is displayed in the **REPORT TEXT** section described in Section 3.4.4, given that the **Text** option is selected in the **Data Display** section described in Section 3.4.3. The user can use the **Delete** button (🗑️) to delete the instance for a particular output report file and the **Open in System Editor** button (📄) to open the output report file in the default text editor on the user's local machine. A visual of this section is shown in Figure 59 below.

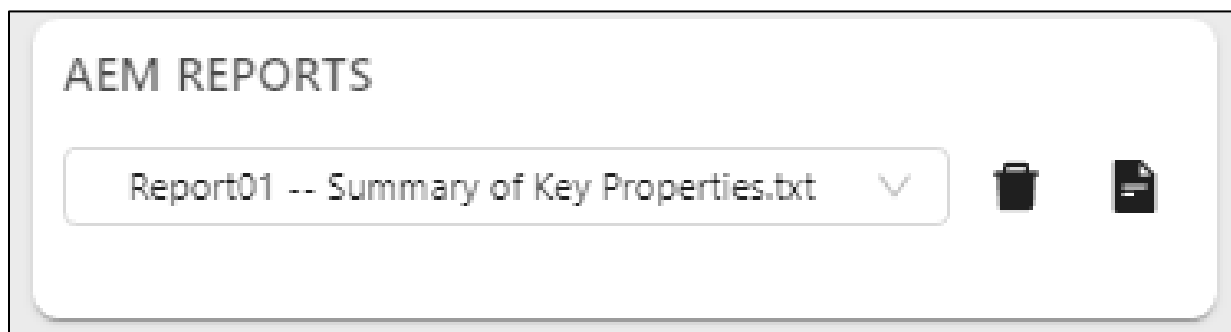


Figure 59. AEM REPORTS section of the AEM Output tab.

3.4.3 Data Display

This is a dropdown select option where the user can select whether the output files must be viewed in text format or be plotted and viewed as a graphical representation. The user has two options – **Text** that opens the **REPORT TEXT** section, **Data Visualization (2D)** and **Data Visualization (3D)** that opens the **REPORT DATA VISUALIZATION** section. A visual of this section is shown in Figure 60 below.

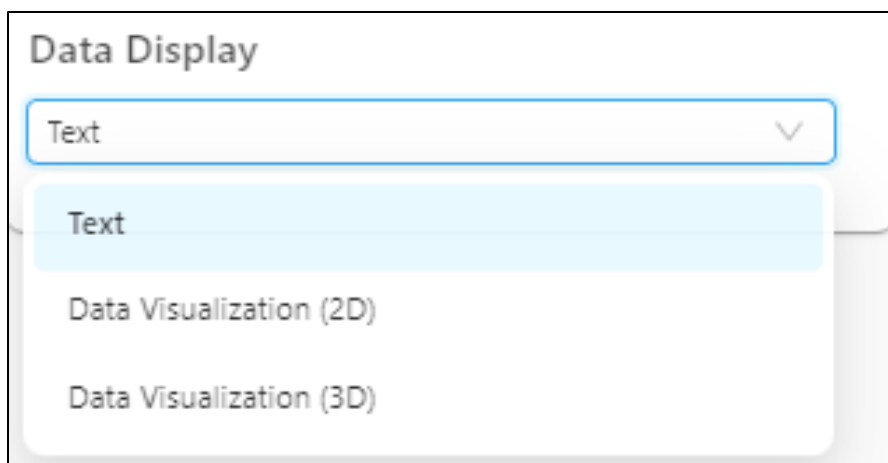


Figure 60. Data Display section of the AEM Output tab.

3.4.4 REPORT TEXT

This is the section where the user can view any report in **Report01 – Report20**, along with the **settings.json** and **verbose.txt** files using the GUI's inbuilt text viewer scroll box given that the user selects the **Text** option in the **Display Data** section. A visual of the **AEM Output** Tab with the **REPORT TEXT** section for viewing the **Report01 – Report20 Output Report** files (Figure 61), the **settings.json** file (Figure 62), and the **verbose.txt** file (Figure 63) is shown below.

REPORT TEXT

Example 1-20230516_113901/Report01 - Summary of Key Properties.txt

AEM ver. 2.23.1M
 *** CALCULATIONS ARE PERFORMED ACCOUNTING FOR THE EFFECT OF ION ASSOC. ON HSA PARAMETERS AND OTHER QUANTITIES.

Number of Mixed-Salt Mole Fraction Cases = 1
 k,k2,k3,k4,ksum = 1 0 0 0 1

Results for salt = LiPF6 at Temp. = -40.00 C or -40.00 F, and 1/T(K) = 0.004289

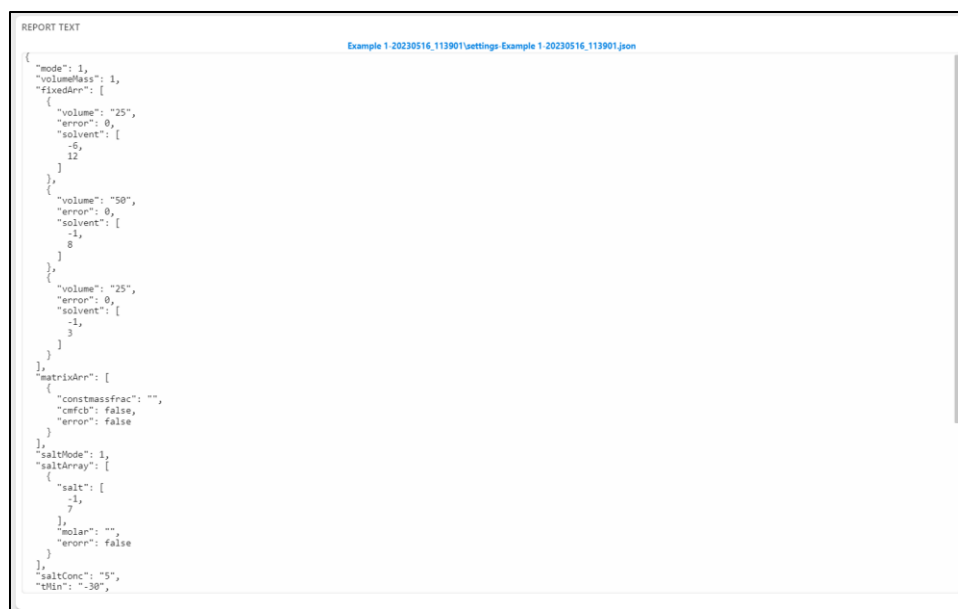
Solvent composition (salt-free):

Name	Mole Fraction	Mass Fraction	Volume Fraction
ethylene carbonate	0.31985	0.30737	0.25315
dimethyl carbonate	0.49743	0.48899	0.49993
ethyl propionate	0.18271	0.20364	0.24692

m2	c2	wt fr salt	mole fr salt	density (g/cc)	visc. (cP)	sig1 (eff)	sig2 (eff)	S(+)	Rational Act.Coeff. y+/-	Diff. Coeff. cm ² /s	Spec. Cond. (mS/cm)	t+(a)	t+(b)	dissoc (SI)	dissoc (TI)
0.025	0.025	0.004	0.002	1.160	3.98	8.151	7.131	3.945	0.589E+00	0.299E-06	0.173E+00	0.380	0.380	0.99999	0.00000
0.050	0.058	0.008	0.005	1.163	4.13	8.146	7.129	3.951	0.522E+00	0.288E-06	0.338E+00	0.382	0.382	0.99998	0.00000
0.100	0.115	0.015	0.009	1.168	4.46	8.139	7.126	3.969	0.468E+00	0.266E-06	0.640E+00	0.385	0.385	0.99995	0.00000
0.150	0.172	0.022	0.014	1.172	4.81	8.131	7.122	3.984	0.434E+00	0.246E-06	0.909E+00	0.388	0.388	0.99992	0.00000
0.200	0.228	0.025	0.018	1.177	5.18	8.123	7.117	3.997	0.423E+00	0.228E-06	0.115E+01	0.391	0.391	0.99987	0.00001
0.250	0.285	0.037	0.022	1.182	5.57	8.114	7.112	4.009	0.422E+00	0.211E-06	0.137E+01	0.395	0.395	0.99981	0.00001
0.300	0.340	0.044	0.027	1.187	5.98	8.105	7.106	4.020	0.427E+00	0.196E-06	0.156E+01	0.398	0.398	0.99973	0.00002
0.350	0.396	0.050	0.031	1.191	6.42	8.096	7.099	4.030	0.438E+00	0.182E-06	0.172E+01	0.401	0.401	0.99964	0.00004
0.400	0.451	0.057	0.035	1.196	6.90	8.086	7.092	4.038	0.454E+00	0.170E-06	0.187E+01	0.403	0.403	0.99953	0.00006
0.450	0.506	0.064	0.040	1.200	7.41	8.076	7.085	4.044	0.475E+00	0.158E-06	0.199E+01	0.406	0.406	0.99939	0.00009
0.500	0.560	0.071	0.044	1.205	7.96	8.066	7.077	4.049	0.502E+00	0.147E-06	0.210E+01	0.409	0.409	0.99922	0.00014
0.550	0.614	0.077	0.048	1.210	8.56	8.055	7.069	4.053	0.534E+00	0.136E-06	0.218E+01	0.412	0.412	0.99900	0.00020
0.600	0.668	0.084	0.052	1.214	9.22	8.045	7.060	4.054	0.573E+00	0.127E-06	0.225E+01	0.415	0.415	0.99874	0.00029
0.650	0.721	0.090	0.056	1.219	9.94	8.034	7.051	4.055	0.618E+00	0.117E-06	0.229E+01	0.418	0.418	0.99842	0.00042
0.700	0.774	0.096	0.060	1.223	10.74	8.024	7.043	4.054	0.672E+00	0.109E-06	0.232E+01	0.421	0.421	0.99801	0.00060
0.750	0.826	0.102	0.064	1.228	11.63	8.013	7.034	4.051	0.736E+00	0.100E-06	0.234E+01	0.424	0.424	0.99764	0.00080
0.800	0.879	0.108	0.068	1.232	12.63	8.003	7.024	4.046	0.811E+00	0.925E-07	0.233E+01	0.427	0.426	0.99720	0.00107
0.850	0.931	0.114	0.072	1.236	13.74	7.992	7.015	4.040	0.899E+00	0.850E-07	0.231E+01	0.429	0.429	0.99666	0.00143
0.900	0.982	0.120	0.076	1.241	14.98	7.981	7.005	4.032	0.100E+01	0.780E-07	0.227E+01	0.432	0.432	0.99597	0.00191
0.950	1.034	0.126	0.080	1.245	16.39	7.971	6.996	4.012	0.112E+01	0.714E-07	0.222E+01	0.435	0.435	0.99511	0.00254
1.000	1.085	0.132	0.084	1.249	17.99	7.960	6.986	4.000	0.126E+01	0.651E-07	0.216E+01	0.438	0.438	0.99402	0.00338
1.050	1.135	0.138	0.088	1.254	19.79	7.950	6.976	3.987	0.143E+01	0.592E-07	0.209E+01	0.440	0.440	0.99267	0.00445

Figure 61. REPORT TEXT section of the AEM Output tab – Viewing the Report01 – Report20 Output Report files.

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```
REPORT TEXT
Example 1-20230516_113901\settings-Example 1-20230516_113901.json
{
  "mode": 1,
  "volumeMass": 1,
  "fixedArr": [
    {
      "volume": "25",
      "error": 0,
      "solvent": [
        -6,
        12
      ]
    },
    {
      "volume": "50",
      "error": 0,
      "solvent": [
        -4,
        8
      ]
    },
    {
      "volume": "25",
      "error": 0,
      "solvent": [
        -1,
        3
      ]
    }
  ],
  "matrixArr": [
    {
      "constmassfrac": "",
      "coeff": false,
      "error": false
    }
  ],
  "saltMode": 1,
  "saltArray": [
    {
      "salt": [
        -1,
        7
      ],
      "molar": "",
      "error": false
    }
  ],
  "saltConc": "5",
  "tMin": "-30",
}
```

Figure 62. REPORT TEXT section of the AEM Output tab – Viewing the settings .json file.



```
REPORT TEXT
Example 1-20230516_113901\verbose-Example 1-20230516_113901.txt
SINGLE FIXED COMPOSITION OF SOLVENTS: 1
OPTION (1)Volume or (2) MASS: 1
NUMBER OF SOLVENTS: 3
Input Number for Solvent 1 : 12
Input Number for Solvent 2 : 8
Input Number for Solvent 3 : 3
Input Percent for Solvent 1 : 25
Input Percent for Solvent 2 : 50
Input Percent for Solvent 3 : 25
NUMBER OF salts: 1
Input Number for Solvent 1 : 7
Salt Conc: 5
TMIN: -30
TMAX: 60
TSTEP: 5
ION STABILITY: 1
CONTACT ANGLE: 35
PORE LENGTH: 20
SALT CONCENTRATION OF INTEREST: 1.5
USE SCAEP: 0
USE DOUBLE LAYER: 0
```

Figure 63. REPORT TEXT section of the AEM Output tab – Viewing the verbose .txt file.

3.4.5 REPORT DATA VISUALIZATION (2D)

This is the section where the user can plot the output parameters from **Report01 – Report20** using the GUI's built-in graphing capability. This functionality is enabled when the user selects the Data Visualization option. The output parameters for each output report are defined in detail in Section 7.1.1. The GUI provides the user with the following three drop down sections.

- **Filter List** - This is a drop down select option to filter the parameters being plotted based on Temperature, Salt Composition, and Solvent Composition.
- **X-Axis Parameter & Y-Axis Parameter** - These are drop down select options to plot various output parameters on the X-Axis and Y-Axis.

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A visual of the three drop down select options (Figure 64) and the entire **AEM Output** Tab with the **REPORT DATA VISUALIZATION (2D)** section (Figure 65) is shown in the following Figures. Also note that individual results from certain data output reports can be exported to a .csv file using the export button under the Temperature ID. This functionality allows for easy export and use into other software programs for further analysis and visualization.



Figure 64. REPORT DATA VISUALIZATION section of the AEM Output tab – 3 select drop down options.

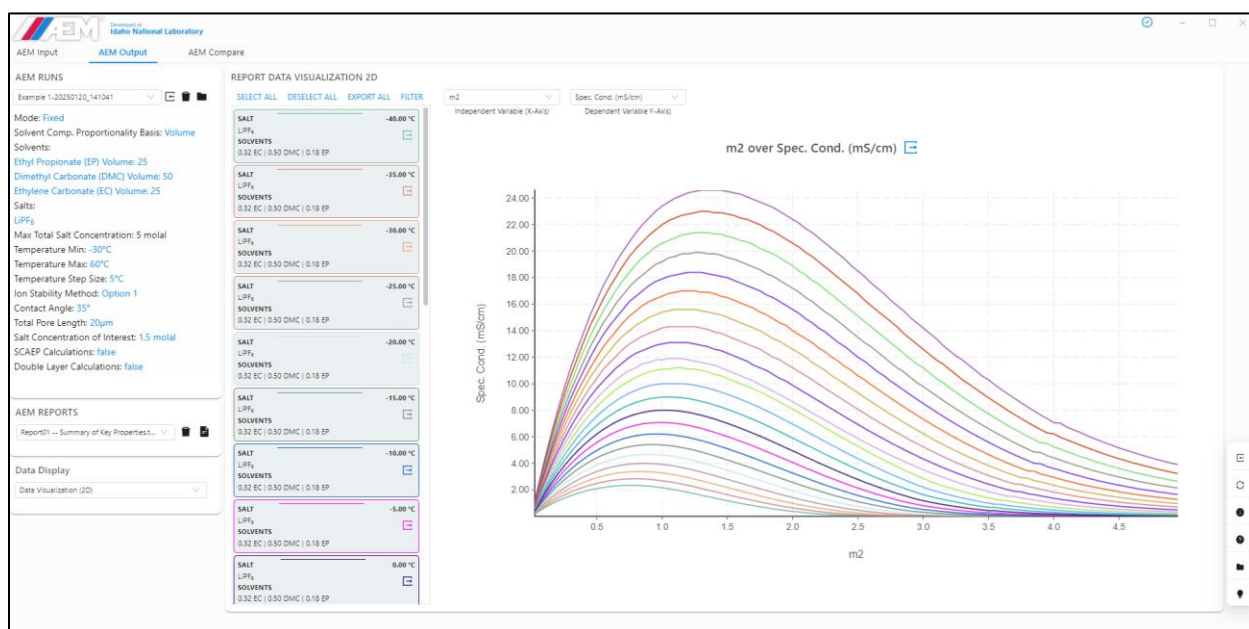


Figure 65. REPORT DATA VISUALIZATION section of the AEM Output tab.

Important Note: Not all generated data output in Report01 – Report20 have plotting and export to .csv capabilities. In such cases, only the text data output is displayed. The AEM GUI also has a built-in limit to only parse and display data output reports that are less than 500 MB.

3.4.6 REPORT DATA VISUALIZATION (3D)

This is the section where the user can plot the output parameters from reports using the GUI's built-in graphing capability. This functionality is enabled when the user selects the Data Visualization option. The output parameters for each output report are defined in detail in Section 7.1.1. The GUI provides the user with the following three drop down sections.

- **Filter List** - This is a drop down select option to filter the parameters being plotted based on Temperature, Salt Composition, and Solvent Composition.

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- **X-Axis Parameter, Y-Axis Parameter and Z-Axis Parameter** - These are drop down select options to plot various output parameters on the X-Axis, Y-Axis and Z-axis. Note: Y-axis is fixed to Temperature.

A visual of the **REPORT DATA VISUALIZATION (3D)** section (60) is shown in the following Figure 66. Also note that individual results from certain data output reports can be exported to a .csv file using the export button under the Temperature ID. This functionality allows for easy export and use into other software programs for further analysis and visualization.

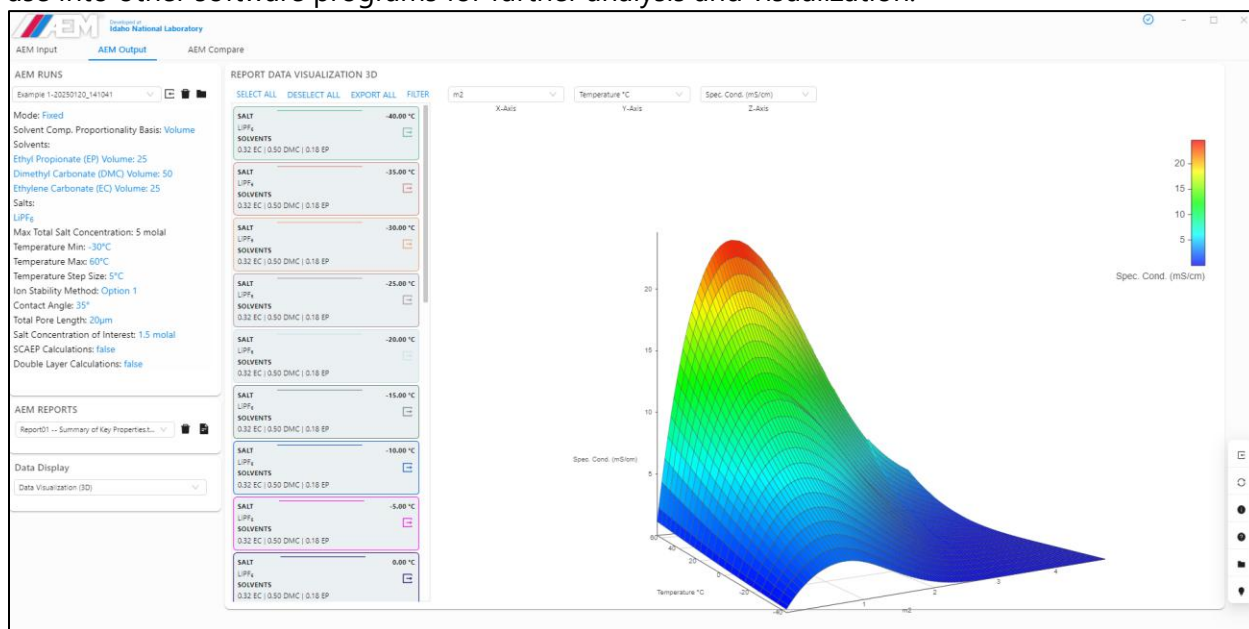


Figure 66. REPORT DATA VISUALIZATION (3D) section of the AEM Output tab.

3.4.7 REPORT DATA EXTRACTION

This is the section where the user can extract the output parameters from **Report01 – Report20** using the GUI's built-in graphing capability. This functionality is enabled when the user selects the Data Visualization option and then has a choice to either export data for a single parameter or all parameters for a particular report into a single CSV file. This feature would help users to extract meaningful data for external analysis. The two options are further described below with appropriate visualizations:

- **Single Parameter Export:** Users can export any parameter values for a particular report for either all temperatures or a range of temperatures to a csv file. The blue icon beside the figure title as shown in Figure 67, pops up a window that enables the user to save the file at a user specified location. The user can either choose the default name or can specify their own and hit on the "Save" button. A snapshot for the layout of the csv extracted for this feature is shown in Figure 68. The first column represents the X axis and represents the parameter as chosen in the GUI before exporting the file. The other columns represent the values of the parameter of interest for all range of temperature selected during export. This capability helps users for external analysis.

Advanced Electrolyte Model (AEM)

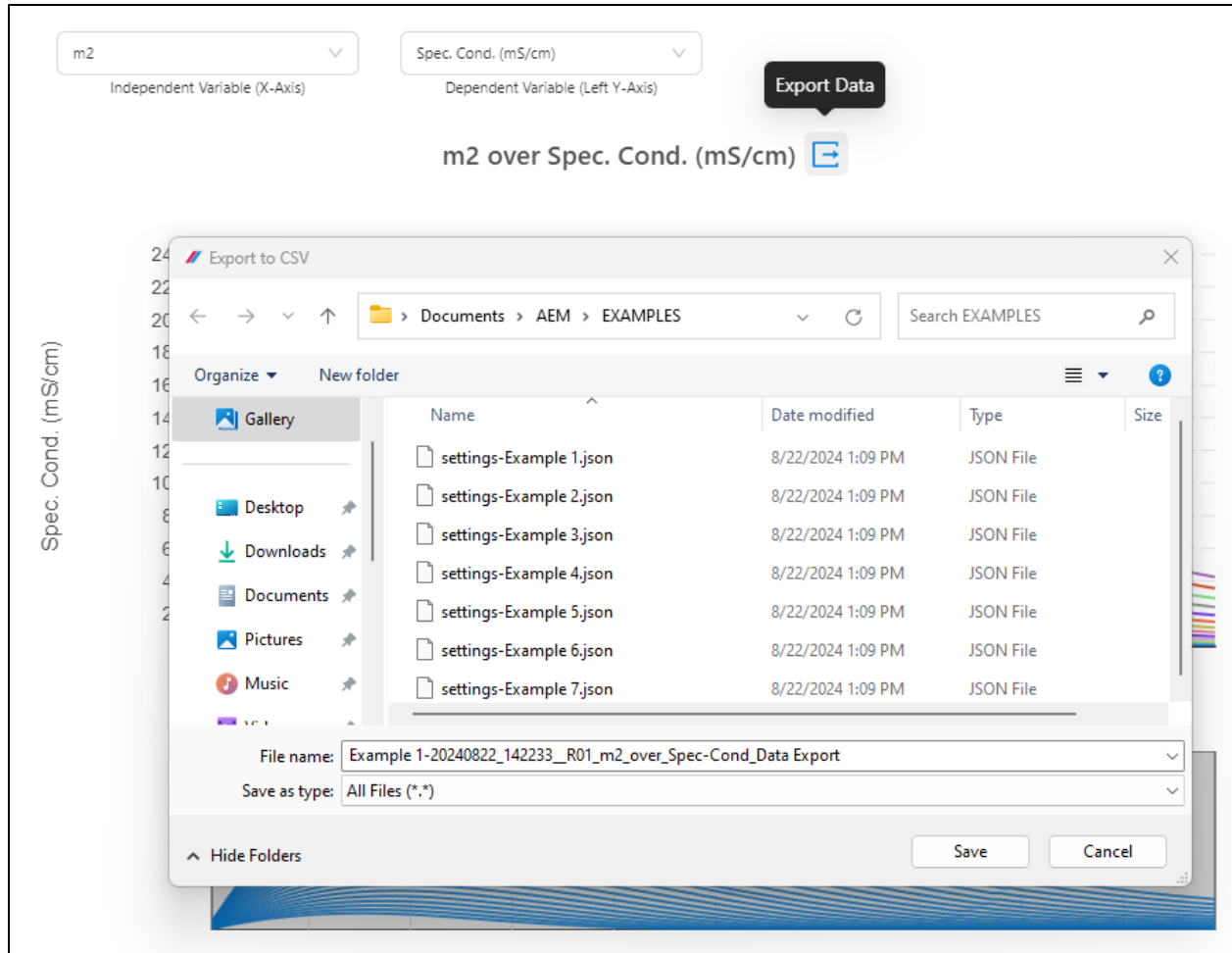


Figure 67. Single Paramter Export Feature

The screenshot displays the COMSOL 6.2 software interface showing the layout for the csv exported for a single parameter. The table displays numerical data for various parameters across multiple rows.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	m2	-40.00C	Li-35.00C	Li-30.00C	Li-25.00C	Li-20.00C	Li-15.00C	Li-10.00C	Li-5.00C	LiF 0.00C	LiF 5.00C	LiF 10.00C	LiF 15.00C	LiF 20.00C	LiF 25.00C	LiF 30.00C	LiF 35.00C	LiF 40.00C	LiF 45.00C	LiF 50.00C	LiF 55.00C
2	0.025	0.173	0.202	0.232	0.265	0.3	0.337	0.376	0.417	0.46	0.505	0.551	0.599	0.647	0.698	0.749	0.801	0.854	0.908	0.962	1.02
3	0.05	0.338	0.393	0.453	0.518	0.586	0.659	0.737	0.817	0.902	0.99	1.08	1.17	1.27	1.37	1.47	1.57	1.68	1.78	1.89	2
4	0.1	0.64	0.746	0.861	0.985	1.12	1.26	1.41	1.56	1.73	1.9	2.07	2.25	2.44	2.63	2.82	3.02	3.23	3.43	3.64	3.85
5	0.15	0.909	1.06	1.23	1.41	1.6	1.8	2.02	2.25	2.48	2.73	2.98	3.25	3.51	3.79	4.07	4.36	4.66	4.96	5.27	5.58
6	0.2	1.15	1.35	1.56	1.79	2.04	2.3	2.58	2.87	3.17	3.49	3.82	4.16	4.5	4.86	5.23	5.6	5.99	6.38	6.77	7.17
7	0.25	1.37	1.6	1.86	2.14	2.43	2.75	3.08	3.44	3.81	4.19	4.59	5.01	5.41	5.84	6.29	6.75	7.21	7.69	8.17	8.65
8	0.3	1.56	1.83	2.12	2.45	2.79	3.16	3.55	3.96	4.38	4.83	5.3	5.78	6.24	6.75	7.27	7.8	8.34	8.9	9.46	10
9	0.35	1.72	2.03	2.36	2.72	3.11	3.52	3.96	4.43	4.91	5.42	5.94	6.49	7	7.58	8.17	8.77	9.39	10	10.7	11.3
10	0.4	1.87	2.2	2.57	2.97	3.4	3.86	4.34	4.85	5.39	5.95	6.54	7.14	7.7	8.34	8.99	9.66	10.4	11	11.8	12.5
11	0.45	1.99	2.36	2.76	3.19	3.65	4.15	4.68	5.24	5.82	6.44	7.07	7.74	8.34	9.03	9.75	10.5	11.2	12	12.8	13.6
12	0.5	2.1	2.49	2.92	3.38	3.88	4.41	4.98	5.58	6.21	6.87	7.56	8.28	8.91	9.67	10.4	11.2	12	12.9	13.7	14.6
13	0.55	2.18	2.6	3.05	3.54	4.07	4.64	5.25	5.89	6.56	7.27	8	8.76	9.43	10.2	11.1	11.9	12.8	13.7	14.6	15.5
14	0.6	2.25	2.68	3.16	3.68	4.24	4.84	5.48	6.16	6.87	7.61	8.39	9.2	9.9	10.8	11.6	12.5	13.5	14.4	15.4	16.3
15	0.65	2.29	2.75	3.25	3.79	4.38	5.01	5.68	6.39	7.14	7.92	8.74	9.59	10.3	11.2	12.1	13.1	14.1	15.1	16.1	17.1
16	0.7	2.32	2.8	3.32	3.88	4.49	5.15	5.84	6.59	7.37	8.19	9.04	9.94	10.7	11.6	12.6	13.6	14.6	15.7	16.7	17.8
17	0.75	2.34	2.82	3.36	3.94	4.57	5.25	5.98	6.75	7.56	8.41	9.31	10.2	11	12	13	14	15.1	16.2	17.3	18.4
18	0.8	2.33	2.83	3.38	3.98	4.63	5.33	6.08	6.88	7.72	8.6	9.53	10.5	11.3	12.3	13.3	14.4	15.5	16.6	17.8	19
19	0.85	2.31	2.82	3.38	4	4.67	5.39	6.16	6.97	7.84	8.75	9.71	10.7	11.5	12.5	13.6	14.7	15.9	17.1	18.3	19.5
20	0.9	2.27	2.79	3.37	4	4.68	5.41	6.2	7.04	7.93	8.87	9.85	10.9	11.6	12.7	13.8	15	16.2	17.4	18.6	19.9
21	0.95	2.22	2.75	3.33	3.97	4.67	5.42	6.22	7.08	7.99	8.95	9.95	11	11.8	12.9	14	15.2	16.4	17.7	19	20.3
22	1	2.16	2.69	3.28	3.93	4.63	5.39	6.21	7.09	8.01	8.99	10	11.1	11.9	13	14.2	15.4	16.6	17.9	19.3	20.6
23	1.05	2.08	2.63	3.21	3.86	4.58	5.35	6.19	7.07	8.01	8.91	10.1	11.2	11.9	13.1	14.2	15.4	16.6	17.9	19.3	20.6

Figure 68. Layout for the csv exported for a single parameter

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- EXPORT ALL Feature:** Users can export the entire report containing all the parameters for all temperatures in to a csv file. The blue icon “EXPORT ALL” highlighted in **red rectangle** as shown in Figure 69, pops up a window that enables the user to save the file at a user specified location. The user can either choose the default name or can specify their own and hit on the “Save” button. A snapshot for the layout of the csv extracted for this feature is shown in Figure 70. The first column represents the solvent composition, the second column represents the salt composition, the third column represents the temperatures and the rest of the columns are for all the properties for a certain report. This capability can be reproduced for all other reports as well. This capability helps users for external analysis.

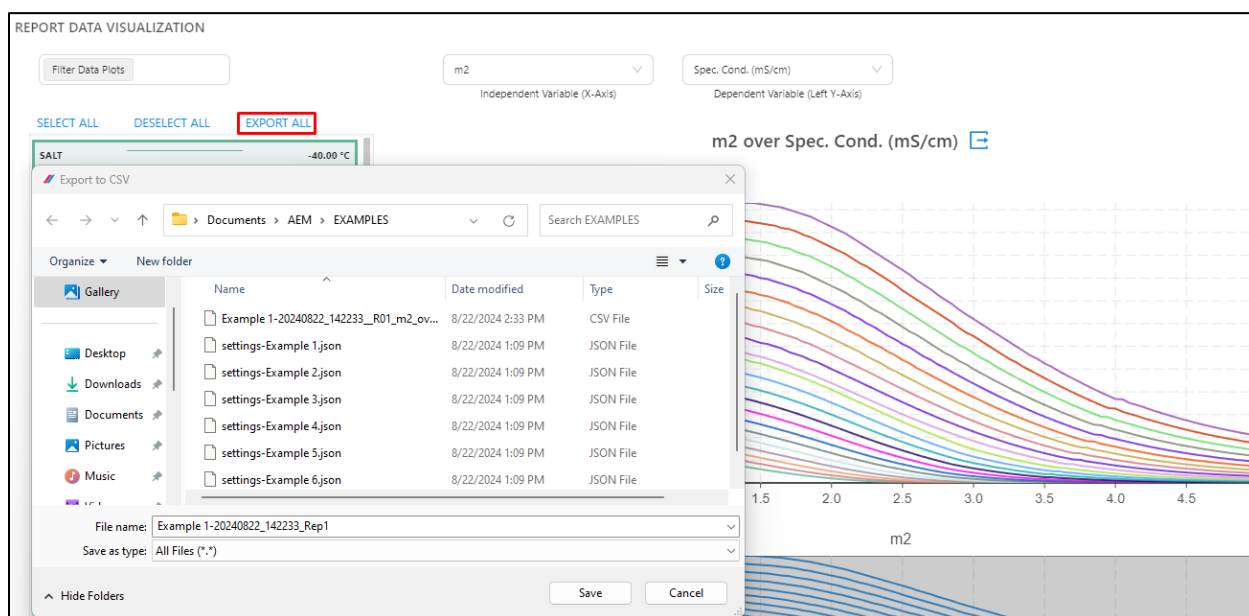


Figure 69. EXPORT ALL Feature

Tell me what you want to do																											
<																											

Figure 70. Layout for the csv exported for EXPORT ALL feature

3.4.8 Generate Special Report Button

The **Generate Special Report** button is located at the top left-hand corner of the **AEM Runs** section. This feature generates a structured .csv file that summarizes key electrolyte properties across regularly spaced molar salt concentration values (c2).

The screenshot displays the AEM software interface. On the left, the 'AEM RUNS' section is active, showing a list of runs with a dropdown menu set to 'Example 1-20260402_115601'. A red dashed box highlights the 'Generate Special Report' button. Below this, various input parameters are listed, including solvent composition (Ethyl Propionate, Dimethyl Carbonate, Ethylene Carbonate), salt (LiPF6), molal concentration (5 molal), temperature (30°C to 60°C), and ion stability method (Option 1). On the right, a preview of the generated CSV file is shown, with columns for solvent composition, salt composition, temperature, c2, density, viscosity, specific conductance, weight fraction of salt, EC mass fraction, DMC mass fraction, and EP mass fraction. The data is organized into rows for each concentration point, with columns grouped by property type.

Figure 71. AEM Special Report.

Unlike standard AEM output reports, which are primarily designed for detailed electrochemical and transport analysis, the Special Report is intended for formulation development, data analysis, and direct comparison with experimental measurements. It extracts and organizes model results into a user-friendly format suitable for spreadsheet-based workflows and rapid screening studies.

The generated report includes the following parameters at each concentration point:

- Solvent Composition
- Salt Composition
- Temperature
- Density
- Viscosity
- Specific Conductance of the Electrolyte
- Weight Fraction of Salt
- Mass Fractions of Solvents

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This output enables users to:

- Evaluate trends in conductivity, viscosity, and density as a function of salt concentration
- Compare AEM predictions directly with laboratory measurements
- Identify optimal formulation regions for further experimental validation
- Extract property values at specific concentration ranges without manually parsing standard reports

Clicking the button will automatically generate the .csv file and save it in the corresponding **AEM Run** folder under the name "Special Report.csv". The file can also be accessed through the **AEM Reports** section within the interface.

Note: The Special Report is generated from single-run simulations and is not available for Matrix runs.

3.5 AEM Compare

The third tab in the AEM GUI is the **AEM Compare** tab. This tab allows the users to compare the results of different AEM simulations that have already been run. This is a great feature as it allows to directly compare useful outputs against different cases and make qualitative and quantitative analysis. This feature can also be used to find the optimum electrolyte among the selected candidates for which the AEM simulations have been performed. First, the users need to select the runs they want to compare from the Select Run drop down (blue box) as shown in Figure 72.

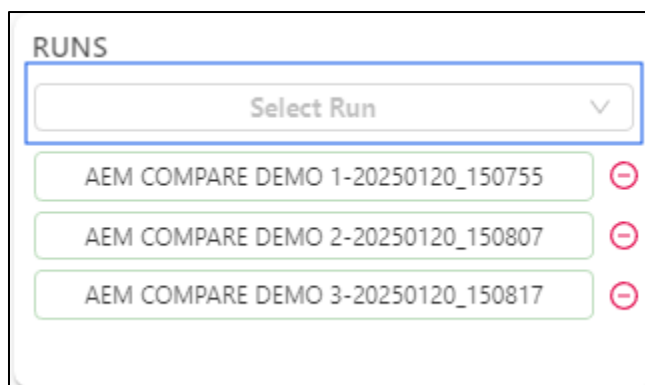


Figure 72. AEM Compare- Select Run

Once, the runs are selected, the users need to select the report from which they want to perform the comparisons. As, different reports contain different set of parameters, this selection is important and is show in Figure 73.

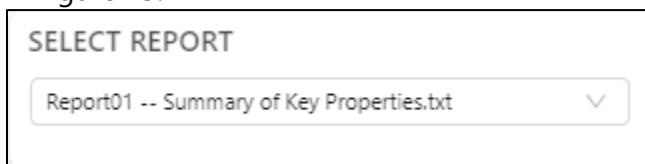


Figure 73. AEM Compare- Select Report

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Once the report has been selected, the users then need to select the X-Axis and Y-Axis parameters of interest. This is followed by selection of the curves to plot from the filter list. Figure 74 shows the visual of the AEM Compare tab with some pre run results for reference.

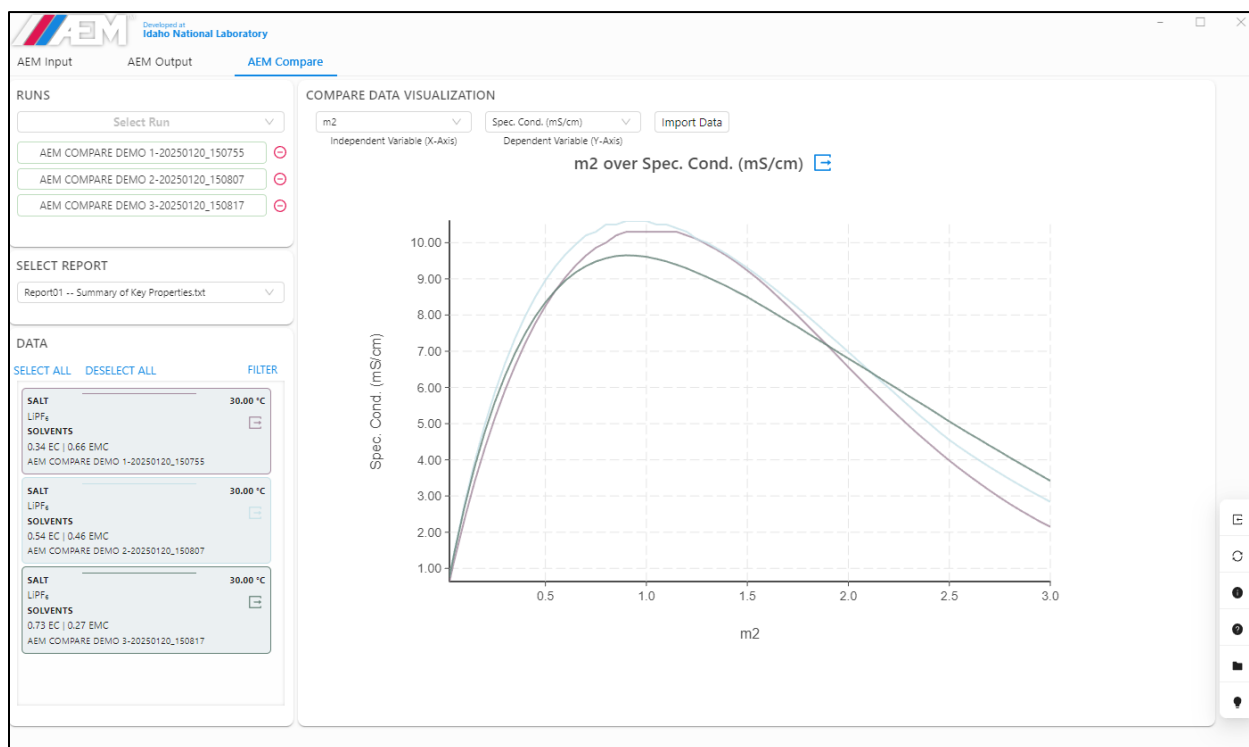


Figure 74. AEM Compare Tab

3.6 Global Features & Icons

There are seven (7) primary operating buttons that can be considered as global features. The program also supports a global input that is used to save a list of all defined simulation conditions in a unique .json file, that can be stored for future use. A brief overview of each global feature is outlined below:

1. **Import Button:** This button allows a user to import a full set of simulation conditions from a .json file that is stored in the **C:\Users\user-name\Documents\AEM\EXAMPLES** subdirectory. When selected, this button will open a Windows File Explorer for the user to search for a .json file from a particular simulation for the inputs defined by the user. This feature is very useful when trying to rerun previous AEM simulations as well as existing examples covered in the User Guide. After a .json file is selected for import, the program will prepopulate all simulation conditions to be specified in each tab.



Figure 75. Icon for GUI Import Button.

2. **Reset Inputs Button:** This button can be pressed to start a new simulation from scratch. It will clear all the initial inputs supplied by the user and reset the AEM GUI to start over.



Figure 76. Icon for GUI Reset Inputs Button.

3. **Information Button:** This button will generate a pop-up window and tell the user what version of the GUI they are working with.

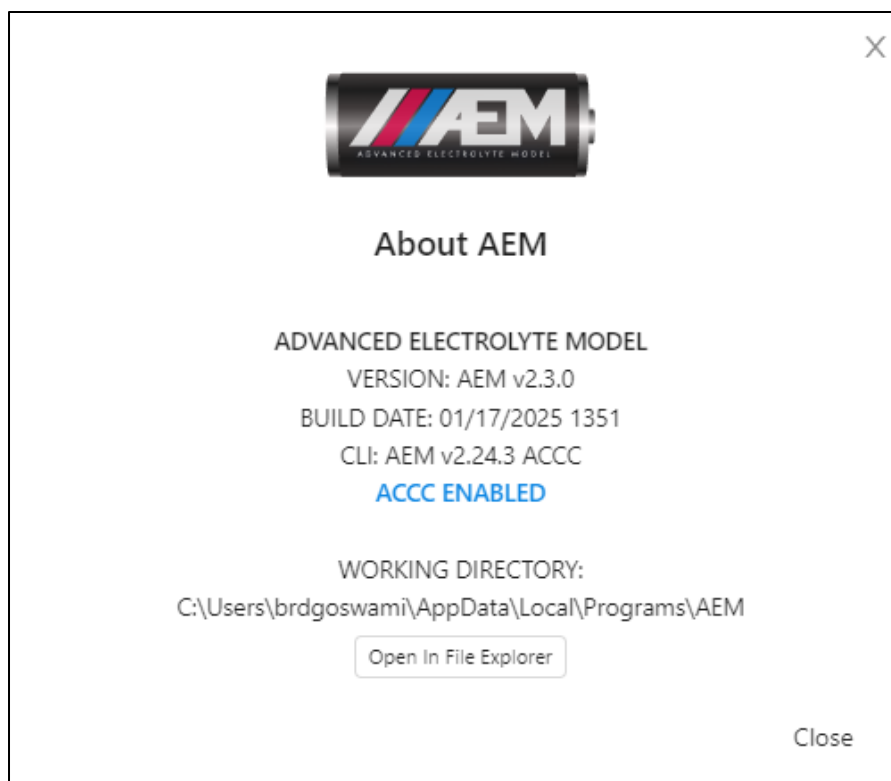


Figure 77. Icon and pop-up message for GUI Information Button.

4. **Help Button:** This button will open a pop-up window with hyperlinks to create a Support Ticket and open the User Guide.



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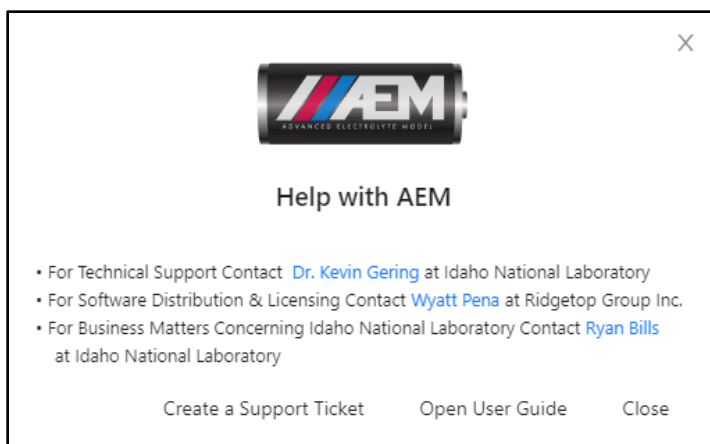


Figure 78. Icon and pop-up message for GUI Help Button.

5. **Expanded / Compact Format Button:** This button will change the format of the AEM GUI to an expanded or compact view when selecting Solvents and Salts.



Advanced Electrolyte Model (AEM)

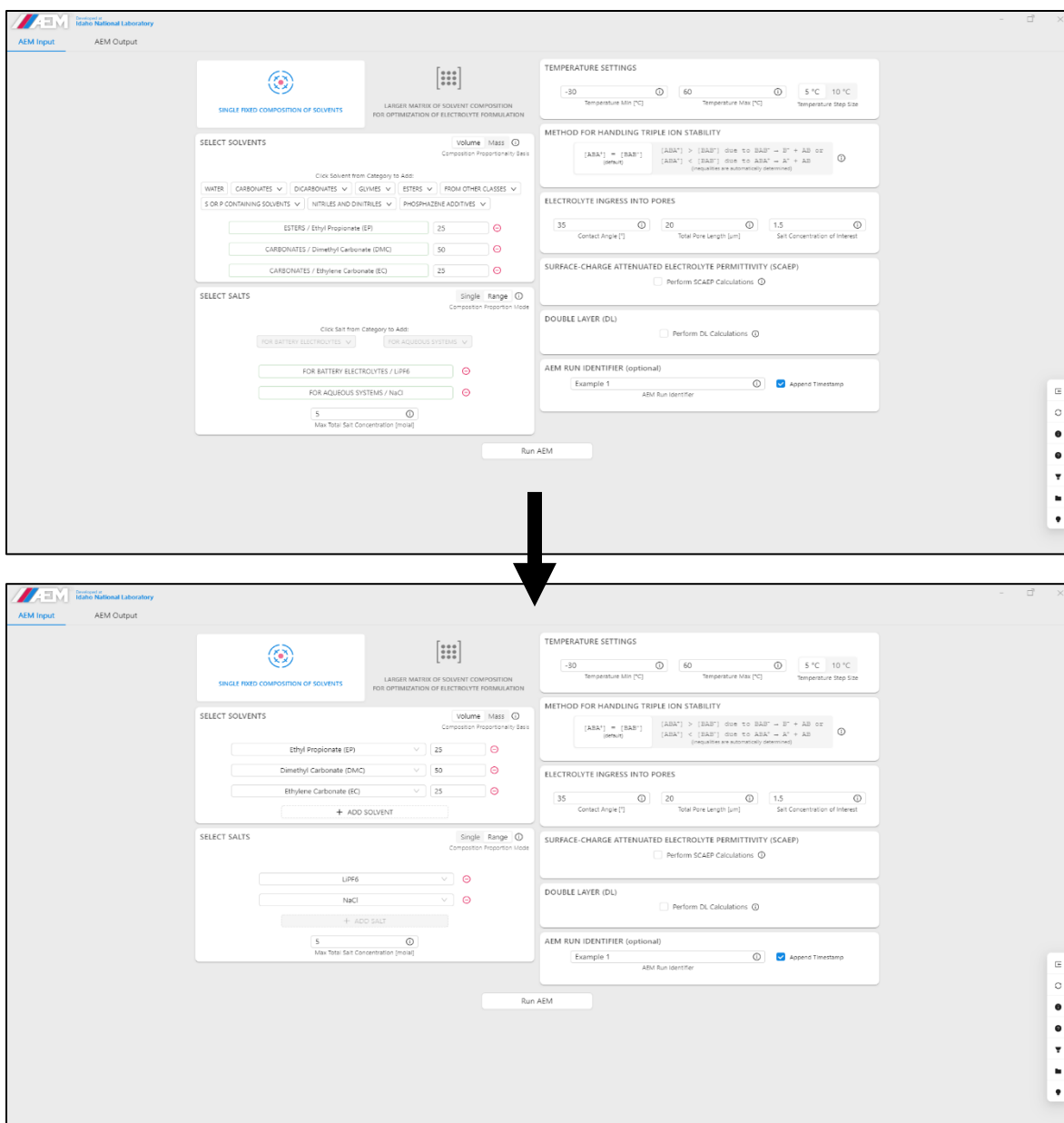


Figure 79. Icon and visual aid for Expanded Format (Top) and Compact Format (Bottom).

- Select New User Directory Button:** This button will open a pop-up window where the user can specify the location for the default folder where the AEM directory needs to be placed.



Advanced Electrolyte Model (AEM)

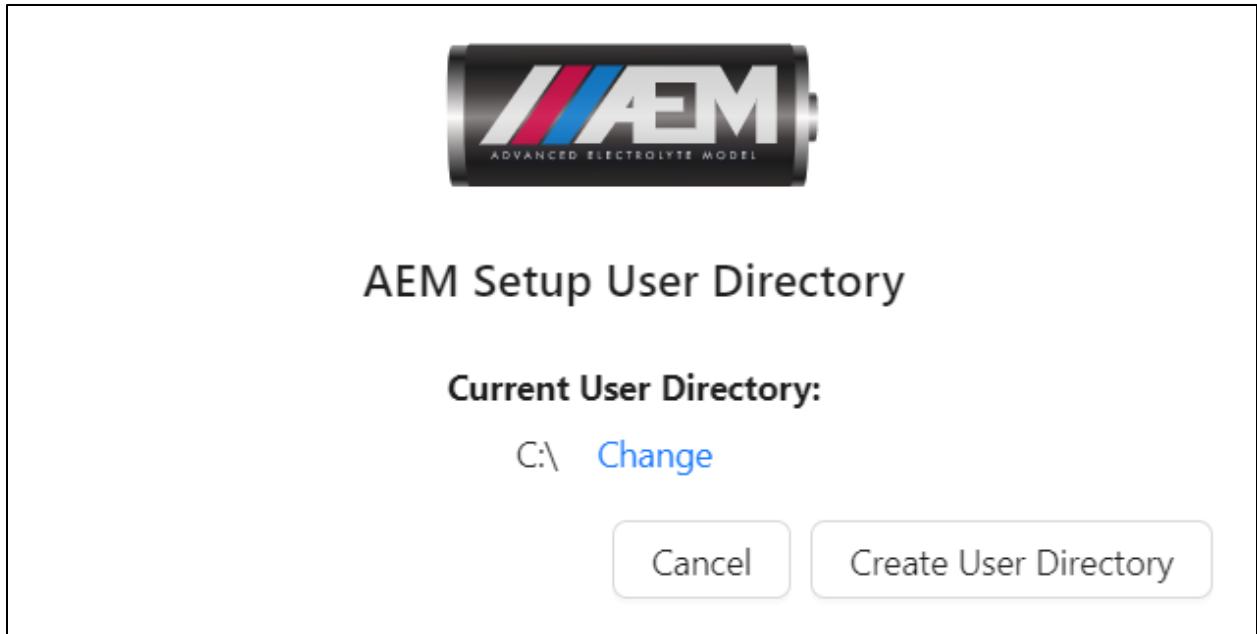


Figure 80. Icon and pop-up message when changing the default program directory.

Important Note: It is recommended that the AEM folder be inside the **My Documents** or **C:** directories.

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- Dark Mode / Light Mode Toggle Button:** This button can be pressed to switch the AEM GUI to a dark mode. This is ideal for using the AEM GUI at night or in locations where too much stray light is not desired.

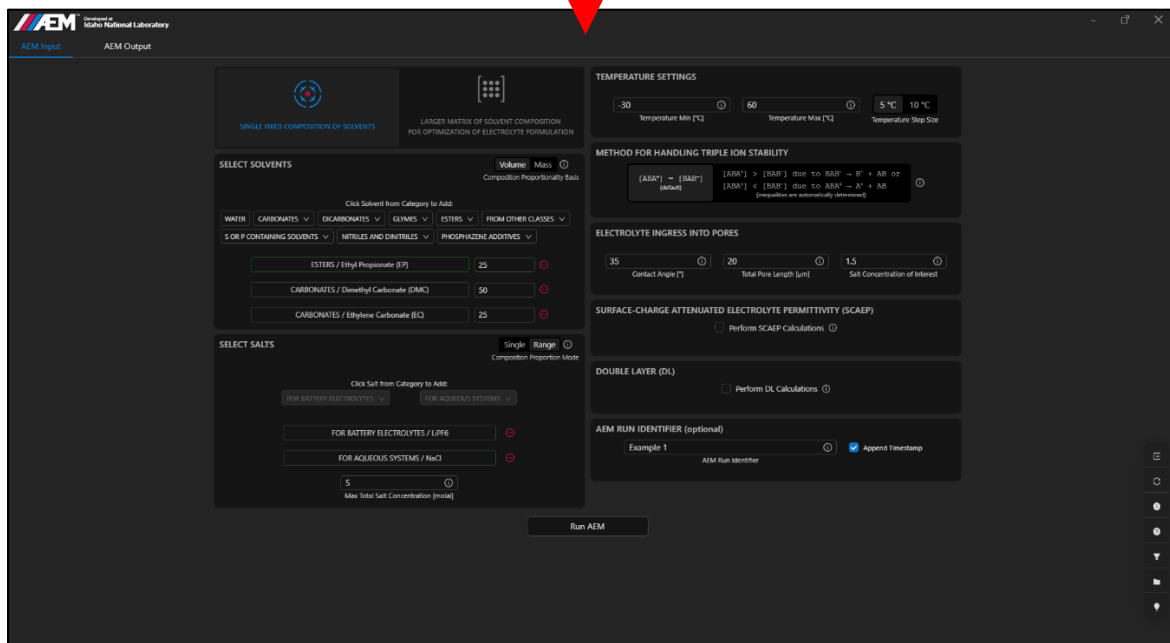
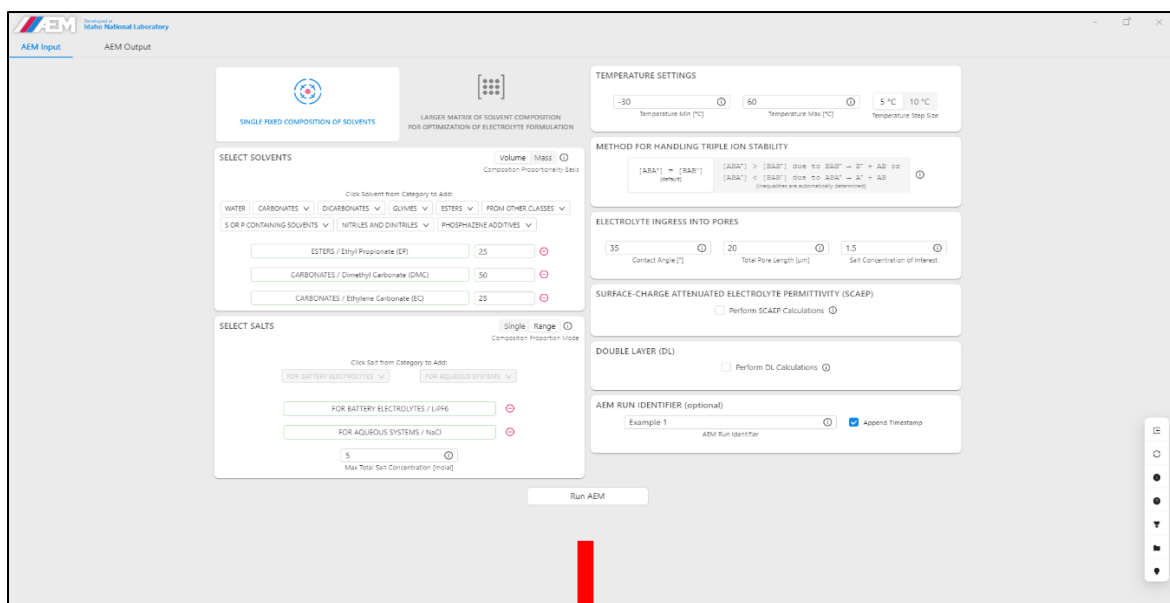


Figure 81. Icon and visual aid for GUI in Light Mode (Top) and Dark Mode (Bottom).

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8. **Exit Button:** This button will prompt the user if they would like to close and exit the program as shown below. The simulation conditions that have been selected will not be remembered, so a user should only exit the program when they wish to close it. If the user selects **OK** the GUI will close and if the user selects **Cancel**, the prompt will close itself.

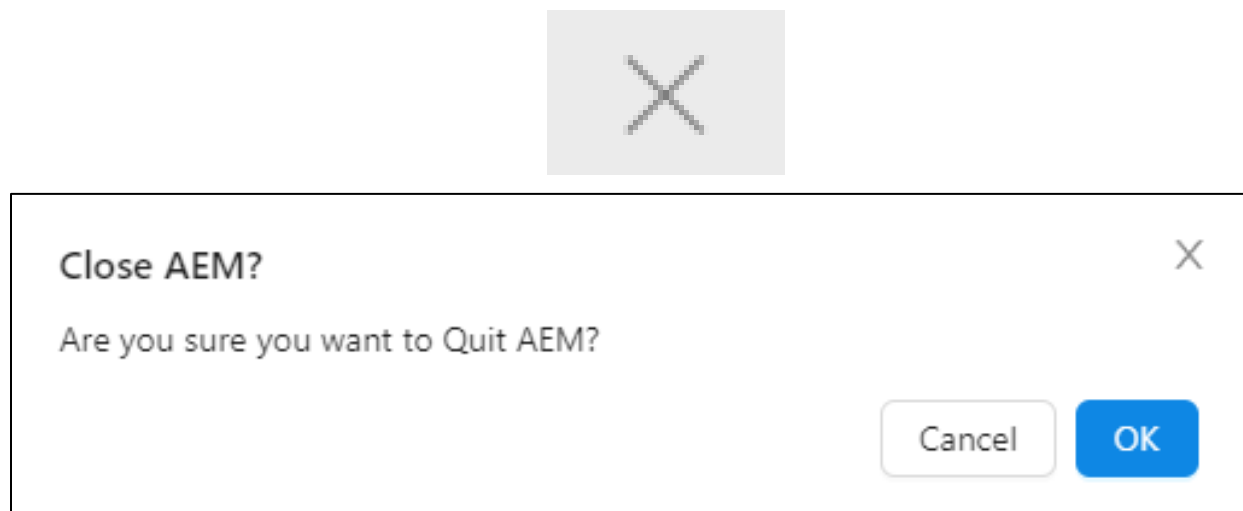


Figure 82. Icon and pop-up message for the Exit Button.

4 Examples

Five electrolyte examples are explored herein to provide useful simulation details to the user and to provide context for the selection process. Through this the user can anticipate simulation outcomes for diverse examples ranging from simple to complex, non-aqueous and aqueous.

While AEM is a robust code, please note that not every conceivable input condition or electrolyte combination will be viable within the chemical physics of AEM. Much effort has been devoted to allowing for wide ranges of conditions for temperature, salt concentration and solvent composition, to support an enormous amount of territory to be explored with AEM. Some simple guidance: cases where AEM doesn't run or terminates early are rare, but if the program doesn't run and terminate as expected, this could be due to excessively low temperatures or salt concentrations that are too high. If the code does not terminate as normal in such cases, the Report files often contain output up to the point of termination. To reduce the occurrence of premature termination, try increasing the low temperature limit and/or decreasing the maximum salt concentration.

4.1 Example 1: Fixed Volume Proportions for 3-Solvent, 1-Salt System

This example considers the three-solvent system EC, DMC, EP at a fixed volume fraction composition (25, 50, 25 vol %) with LiPF_6 . The salt concentration is allowed to go to 5 molal. Other factors behind the simulation include:

- $T = -30^\circ\text{C}$ to 60°C with 5°C steps,
- Used default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- No SCAEP; No DL.

The parameters for electrolyte ingress are relevant estimations for battery electrode materials in general and can be updated to reflect more finely tuned attributes of the porous media.

This relatively simple electrolyte represents a lower-viscosity, higher-conductivity system that would facilitate higher current densities in lithium-ion type cells. This electrolyte system has benefits toward conditions like moderately-fast charging and higher-power applications.

4.1.1 Example 1: Importing Example 1 Parameters

We start off Example 1 by first using the **Import** Button from Section 3.66 to prepopulate each of the AEM input conditions for Example 1. This can be achieved by following the list of below instructions and referencing Figure 83 and Figure 84.

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- Click on the **Import** (≡) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- Select the **settings-Example 1.json** file and click the **Open** button.

Important Note: An error message will pop up if the "settings-Example 1.json" file was not imported successfully.

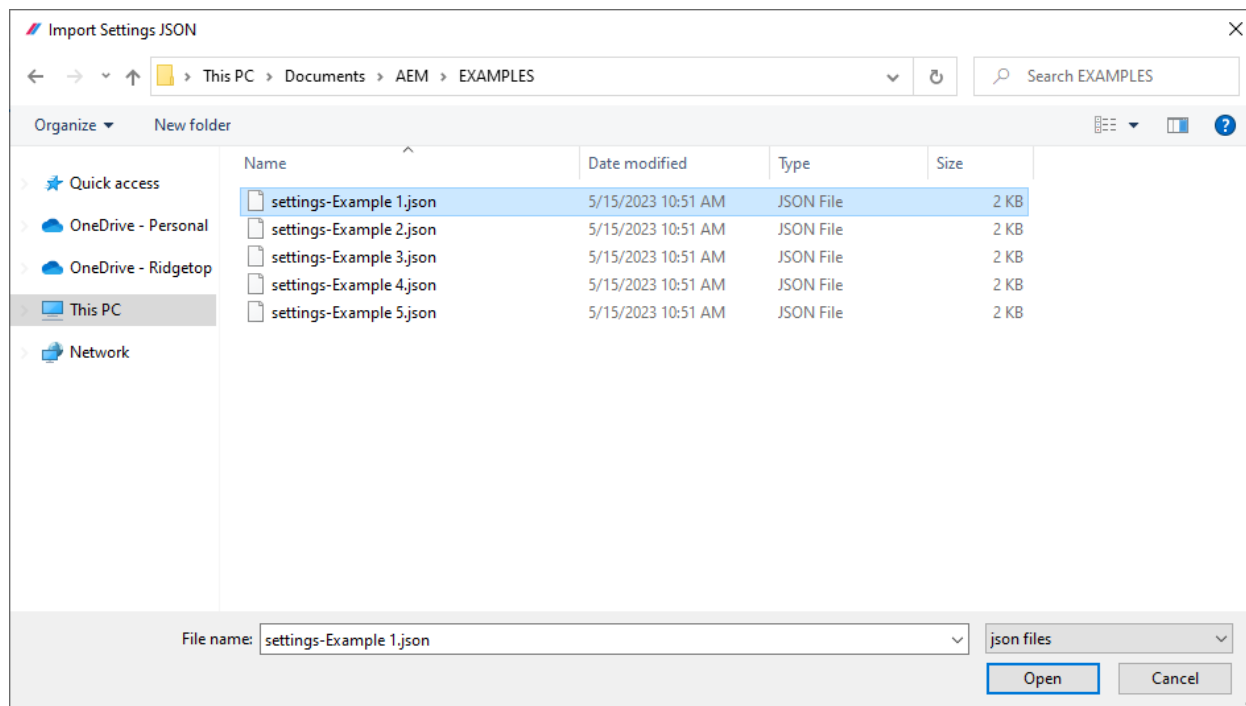


Figure 83. Importing Example 1 Input Condition using the settings-Example 1.json file.

4.1.2 Example 1: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, select the following input conditions:

- EP, DMC, EC at a fixed volume fraction composition (25, 50, 25 vol %)
- LiPF₆ with salt concentration maximum of 5 molal.
- T = -30°C to 60°C with 5°C steps,
- Choose default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = 20μm, salt conc. of interest = 1.5 molal,
- Choose No SCAEP; No DL.

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The screenshot displays the AEM Input tab with the following sections:

- TEMPERATURE SETTINGS:** Temperature Min: -30 °C, Temperature Max: 60 °C, Temperature Step Size: 5 °C.
- METHOD FOR HANDLING TRIPLE ION STABILITY:**
 - [ABA⁺] = [BAB⁺] (default)
 - [ABA⁺] > [BAB⁺] due to BAB⁺ → B⁻ + AB or [ABA⁺] < [BAB⁺] due to ABA⁺ → A⁺ + AB (inequalities are automatically determined)
- ELECTROLYTE INGRESS INTO PORES:** Contact Angle: 35 °, Total Pore Length: 20 μm, Salt Concentration of Interest: 1.5 molal.
- SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP):** ☐ Perform SCAEP Calculations
- DOUBLE LAYER (DL):** ☐ Perform DL Calculations
- AEM RUN IDENTIFIER (optional):** Example 1, ☒ Append Timestamp
- SELECT SOLVENTS:**
 - Click Solvent from Category to Add (Max: 10)
 - WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, FROM OTHER CLASSES, S OR P CONTAINING SOLVENTS, NITRILES AND DINITRILES, PHOSPHAZENE ADDITIVES, ACCC SOLVENTS, SEARCH ALL SOLVENTS
 - ESTERS / Ethyl Propionate (EP): 25
 - CARBONATES / Dimethyl Carbonate (DMC): 50
 - CARBONATES / Ethylene Carbonate (EC): 25
- SELECT SALTS:**
 - Click Salt from Category to Add (Max: 2)
 - FOR BATTERY ELECTROLYTES, FOR AQUEOUS SYSTEMS, ACCC SALTS, SEARCH ALL SALTS
 - FOR BATTERY ELECTROLYTES / LiPF₆: 5 molal
 - Max Total Salt Concentration
- Run AEM:** ☐ Append Multiple Runs

Figure 84. Example 1 – AEM Input tab.

4.1.3 Example 1: Data Output & Discussion

Simulation outputs are plotted in Figure 86 - Figure 91 for the transport suite of properties (salt diffusivity, conductivity, viscosity, cation transference number). General goals for battery electrolytes are to achieve high diffusivity and conductivity, low viscosity and high cation transference. Since most property metrics exhibit a strong reliance on salt concentration, an effort should be made to identify the range of salt concentration that maximizes the foremost properties of interest. Note that this range should also consider the possible concentration polarization (CP) that can occur over more prolonged cycling elements, since CP will cause a salt depletion at one electrode, while causing an enrichment at the opposite electrode. Given these factors in concert, a feasible overall operational salt concentration range could be defined as 0.5 to 2.0 molal. Note that the effective temperature range for this example is -40 to 70 °C, since AEM adds temperature steps to enable calculation of activation energies over the user-specified range. Also note that for the cation transference values (Report01), the following is given:

- (a) t_+ evaluated with single-ion cations vs all charged species,
- (b) t_+ evaluated with all cation-bearing charged species vs all charged species.

The trends seen over these two approaches are very similar, where their behavior over salt concentration reflect the relative change in the local solvating structure around the lithium cations in relation to that of the anions.

Advanced Electrolyte Model (AEM)

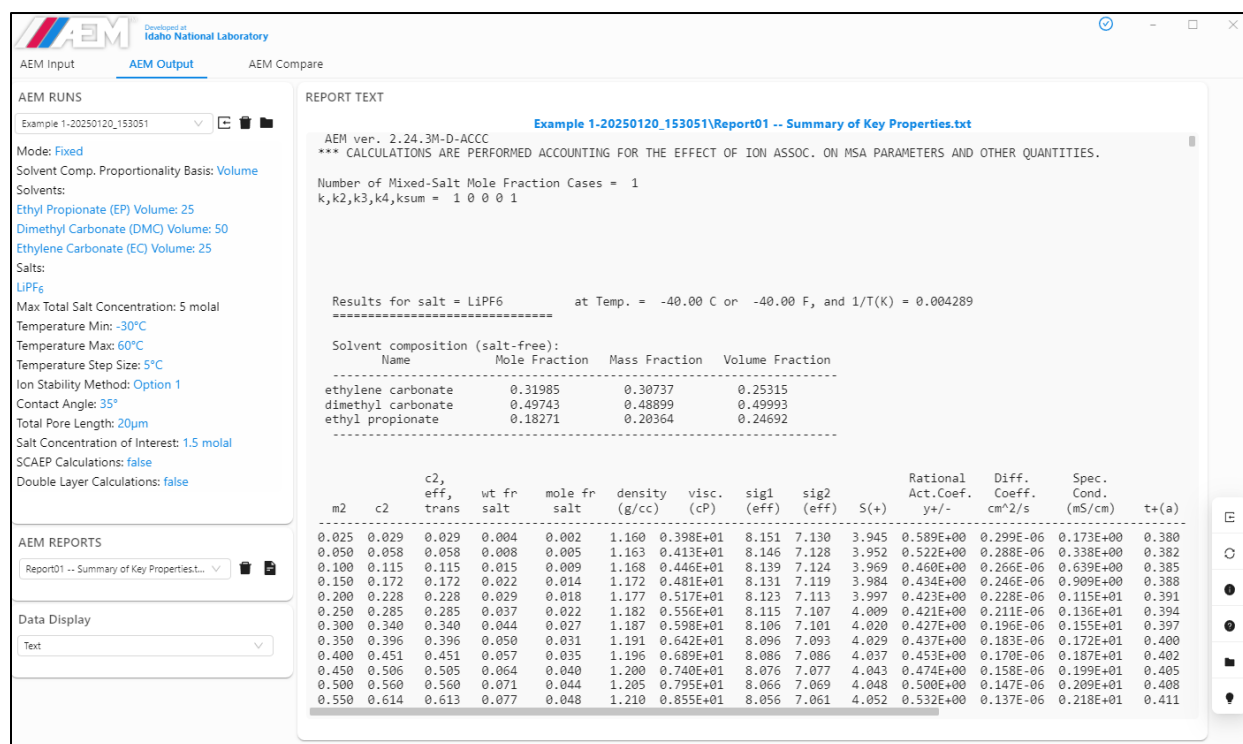


Figure 85. Example 1 – AEM Output tab – REPORT TEXT section to view Report01-Report20.

Advanced Electrolyte Model (AEM)

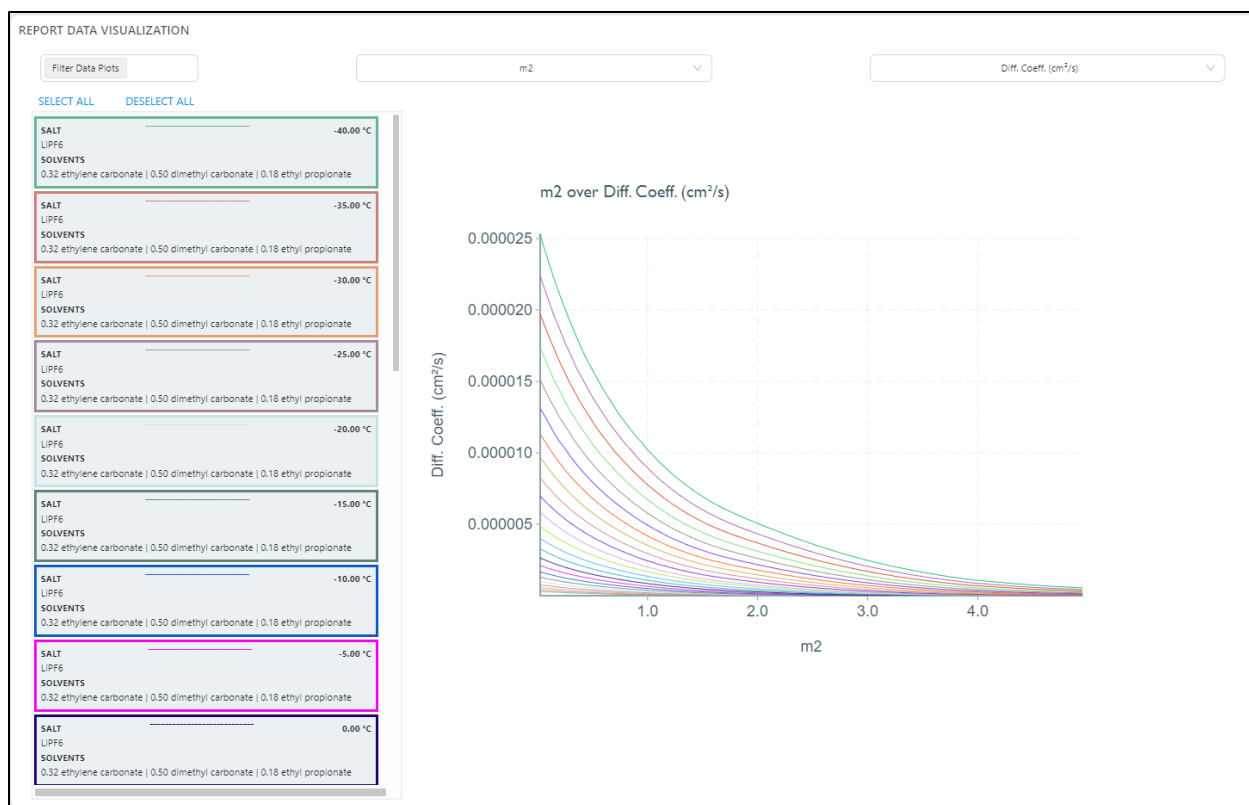


Figure 86. Example 1 – Diffusivity/Diffusion Coefficient (cm²/s) v/s Molal Salt Conc. (m₂).

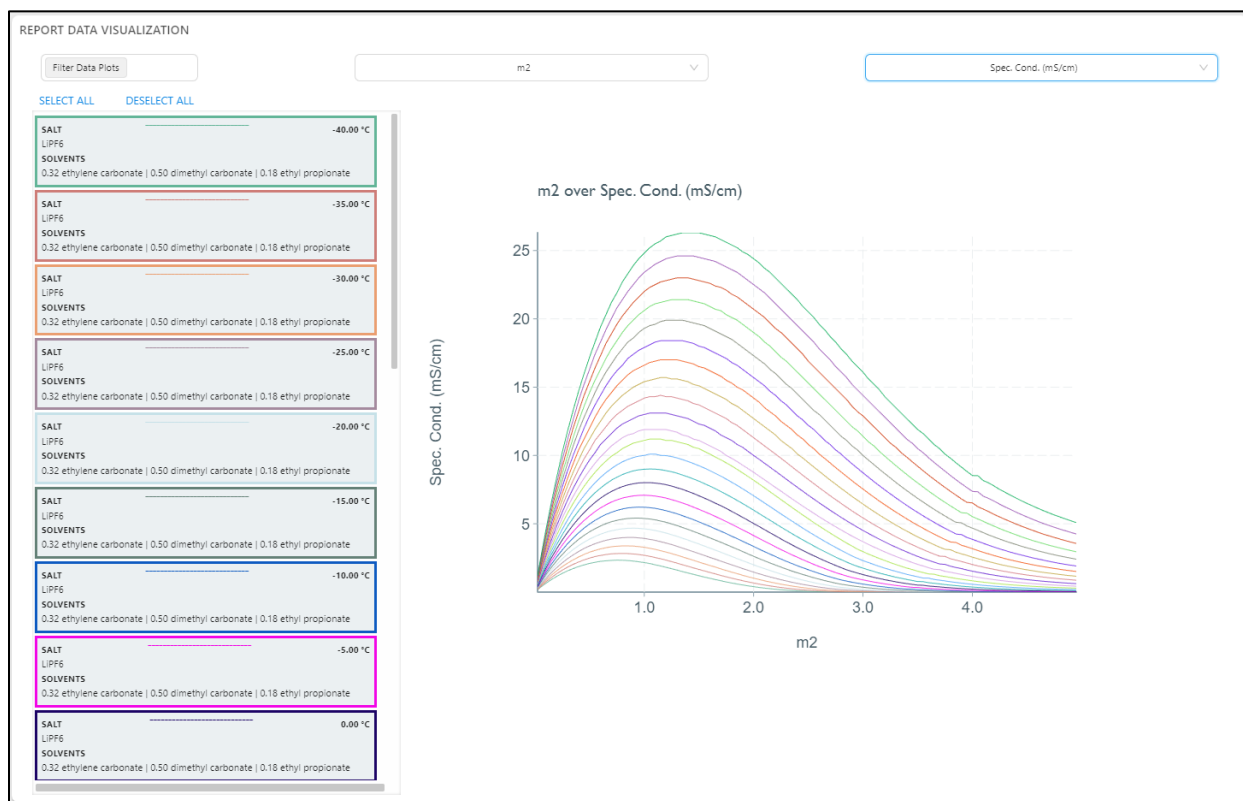


Figure 87. Example 1 – Specific Conductivity (mS/cm) v/s Molal Salt Conc. (m₂).

Advanced Electrolyte Model (AEM)

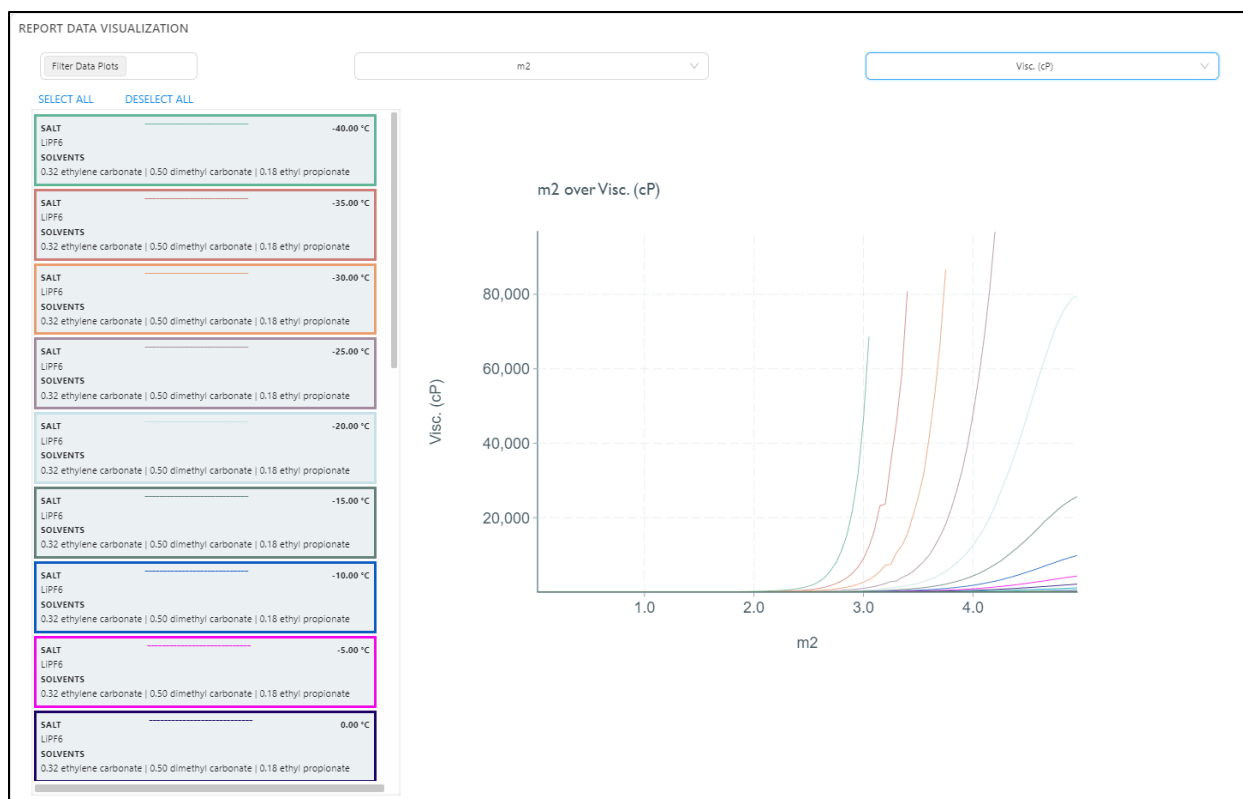


Figure 88. Example 1 – Viscosity (cP) v/s Molal Salt Conc. (m2).

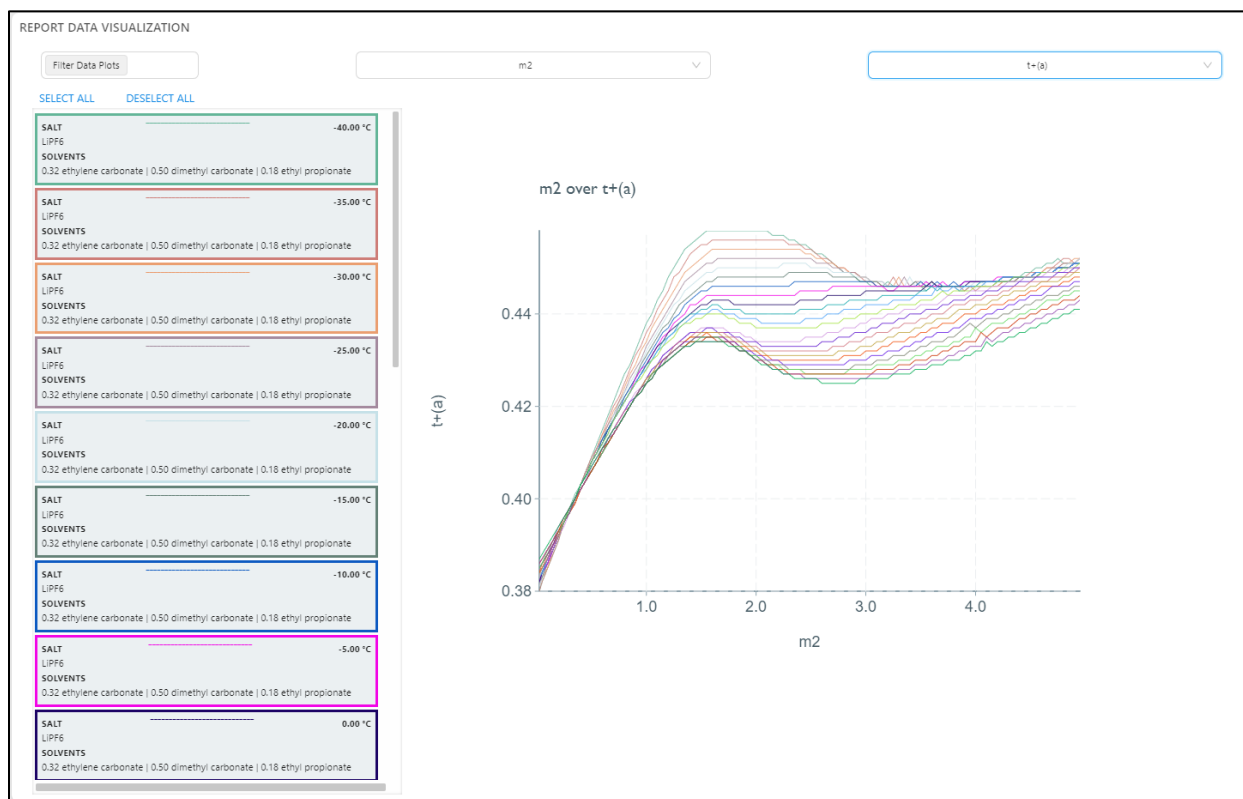


Figure 89. Example 1 – Cation Transference No. (t+(a)) v/s Molal Salt Conc. (m2).

Advanced Electrolyte Model (AEM)

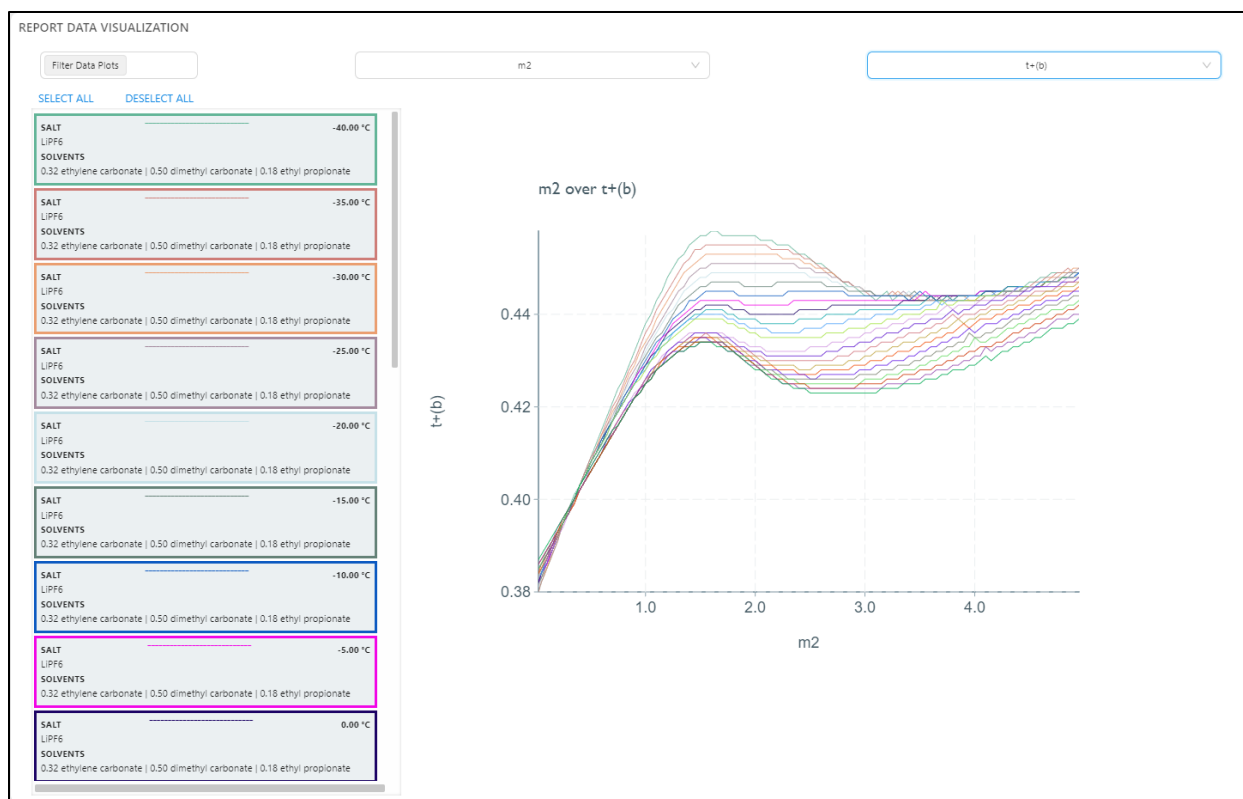


Figure 90. Example 1 – Cation Transference No. ($t_+(b)$) v/s Molal Salt Conc. (m_2).

Advanced Electrolyte Model (AEM)

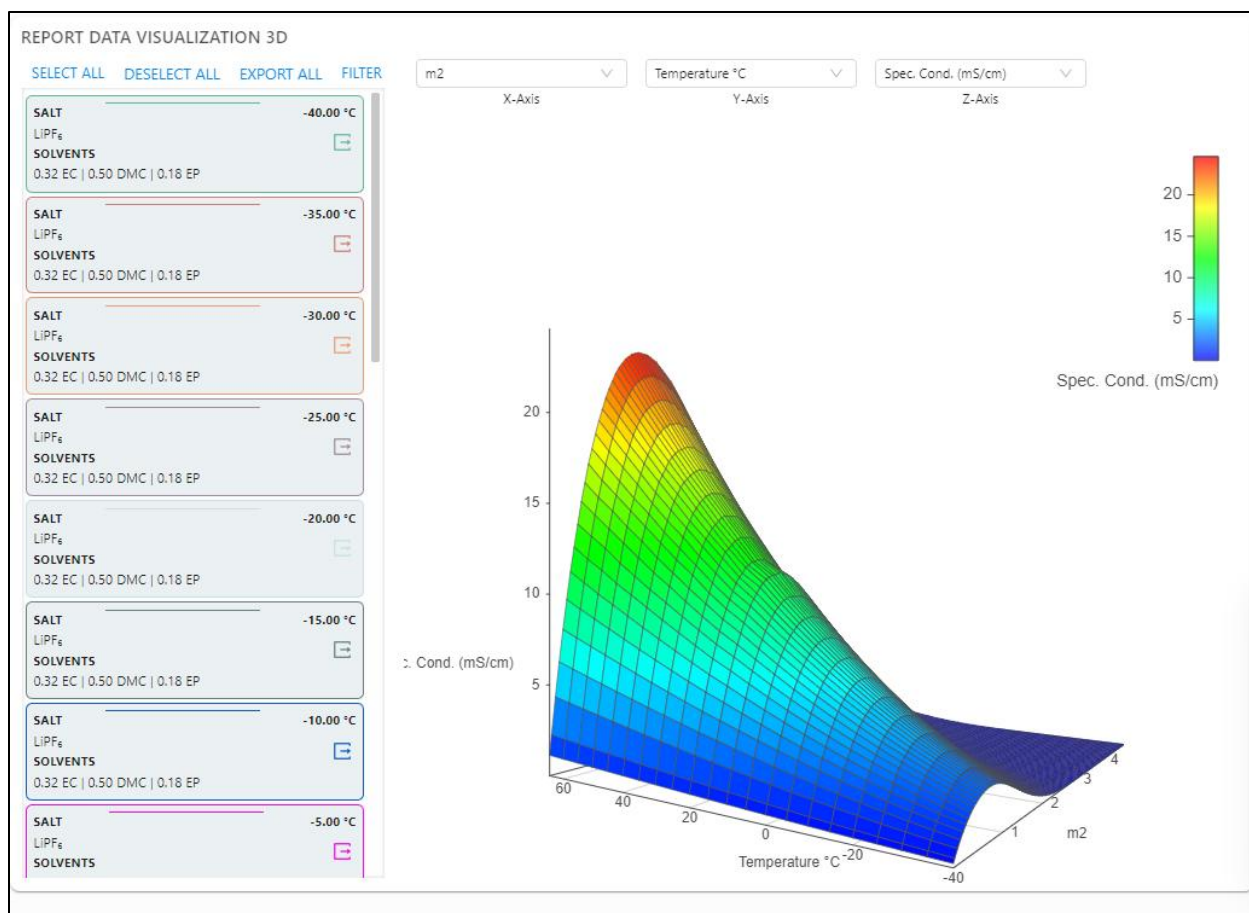


Figure 91. Example 1- Specific Conductivity (mS/cm) vs Molal Salt Conc. (m2) vs Temperature (deg C) 3D contour

4.2 Example 2: Fixed Mass Proportions for 8-Solvent, 2-Salt System

This example considers a complex eight-solvent system with EC, PC, EMC, MP, DEGDME, BN, TMP, VEC at fixed proportions (10, 10, 15, 20, 25, 12, 5, 3 mass %) with a dual salt LiPF_6 and LiFSI that covers the full dual salt matrix of eleven proportions. The total salt concentration is allowed to go to 5 molal. Other factors behind the simulation include:

- $T = 0^\circ\text{C}$ to 50°C with 10°C steps,
- Used default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- No SCAEP; No DL.

The parameters for electrolyte ingress are relevant estimations for battery electrode materials in general and can be updated to reflect more finely tuned attributes of the porous media.

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This complex electrolyte represents a lower-viscosity, higher-conductivity system that would facilitate higher current densities in lithium-ion type cells, while adding stability to passivation films (SEI). This electrolyte system has benefits toward conditions like moderately-fast charging and higher-power applications, where the formulation can be fine-tuned through the salt matrix.

4.2.1 Example 2: Importing Example 2 Parameters

We start off Example 2 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 2. This can be achieved by following the list of instructions below and referencing Figure 92 and Figure 93.

- Click on the **Import** (📄) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- Select the **settings-Example 2.json** file and click the **Open** button.

Important Note: An error message will pop up if the "settings-Example 2.json" file was not imported successfully.

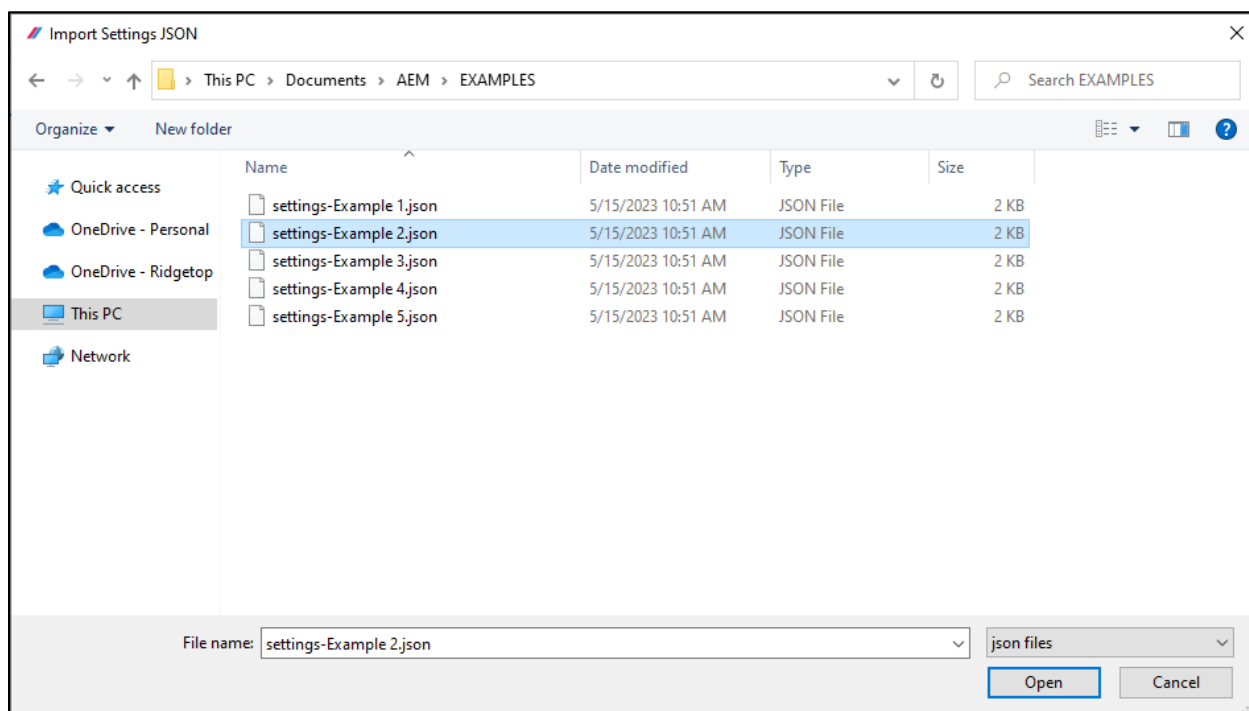


Figure 92. Importing Example 2 Input Condition using the settings-Example 2.json file.

4.2.2 Example 2: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, select the following

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- Select the solvents EC, PC, EMC, MP, DEGDME, BN, TMP, VEC at fixed proportions (10, 10, 15, 20, 25, 12, 5, 3 mass %),
- Select dual salt system LiPF_6 and LiFSI with a total maximum concentration of 5 molal. Choose full dual salt matrix of eleven proportions.
- $T = 0^\circ\text{C}$ to 50°C with 10°C steps,
- Use default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- No SCAEP; No DL.

The screenshot displays the AEM Input tab interface, organized into several sections:

- TEMPERATURE SETTINGS:** Includes input fields for Temperature Min (0 °C), Temperature Max (50 °C), and Temperature Step Size (5 °C).
- METHOD FOR HANDLING TRIPLE ION STABILITY:** Provides a dropdown menu to select between different stability methods, with a default option and a note about automatic determination of inequalities.
- ELECTROLYTE INGRESS INTO PORES:** Features input fields for Contact Angle (35 °), Total Pore Length (20 μm), and Salt Concentration of Interest (1.5 molal).
- SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP):** Includes a checkbox to 'Perform SCAEP Calculations'.
- DOUBLE LAYER (DL):** Includes a checkbox to 'Perform DL Calculations'.
- AEM RUN IDENTIFIER (optional):** Includes a text field for the identifier (Example 2) and a checkbox to 'Append Timestamp'.
- SELECT SOLVENTS:** A section with a 'Click Solvent from Category to Add (Max: 10)' instruction. It lists various solvent categories (WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, FROM OTHER CLASSES, S OR P CONTAINING SOLVENTS, NITRILES AND DINITRILES, PHOSPHAZENE ADDITIVES, ACCC SOLVENTS) and a search bar. Below this, a table shows selected solvents and their proportions:

Solvent	Proportion
CARBONATES / Ethylene Carbonate (EC)	10
CARBONATES / Propylene Carbonate (PC)	10
CARBONATES / Ethylmethyl Carbonate (EMC)	15
ESTERS / Methyl Propionate (MP)	20
GLYMES / DEGDME (diglyme)	25
NITRILES & DINITRILES / Butyronitrile (BN)	12
S OR P CONTAINING / Trimethyl Phosphate (TMP)	5
CARBONATES / Vinyl Ethylene Carbonate (VEC)	3
- SELECT SALTS:** A section with a 'Click Salt from Category to Add (Max: 2)' instruction. It lists categories (FOR BATTERY ELECTROLYTES, FOR AQUEOUS SYSTEMS, ACCC SALTS) and a search bar. Below this, a table shows selected salts and their proportions:

Salt	Proportion
FOR BATTERY ELECTROLYTES / LiPF_6	5
FOR BATTERY ELECTROLYTES / LiFSI	5

 A 'Max Total Salt Concentration' field is set to 5 molal.
- Run AEM:** A button to execute the simulation, with an option to 'Append Multiple Runs'.

Figure 93. Example 2 – AEM Input tab.

4.2.3 Example 2: Data Output & Discussion

Simulation outputs are plotted in Figure 95 - Figure 100 for the transport suite of properties (salt diffusivity, conductivity, viscosity, cation transference number). Results are plotted over seven (7) temperatures (0, 10, 20, 30, 40, 50, 60 °C), where at each set of temperature the matrix of eleven (11) dual-salt proportions are included. General goals for battery electrolytes are to achieve high

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diffusivity and conductivity, low viscosity and high cation transference. While there are similarities between Examples 1 and 2 in the transport properties, Example 2 has the advantage of the additional matrix of dual-salt proportions. Note that favorable salt proportions can change over total salt concentration and temperature. Since most property metrics exhibit a strong reliance on salt concentration, an effort should be made to identify the range of salt concentration that maximizes the foremost properties of interest. Note that this range should also consider the possible concentration polarization (CP) that can occur over more prolonged cycling elements, since CP will cause a salt depletion at one electrode, while causing an enrichment at the opposite electrode. Given these factors in concert, a feasible overall operational salt concentration range could be defined as 0.5 to 2.1 molal. Note that the effective output temperature range for this example is 0 to 60 °C. Also note that for the cation transference values (Report01), the following is given:

- (a) t_+ evaluated with single-ion cations vs all charged species,
- (b) t_+ evaluated with all cation-bearing charged species vs all charged species.

The trends seen over these two approaches are very similar, where their behavior over salt concentration reflect the relative change in the local solvating structure around the lithium cations in relation to that of the anions. In Example 2, more variability is seen in t_+ over salt concentration compared to Example 1 owing to (1) the more diverse field of solvents that can act as solvators, and (2) the impact of the dual-salt formulations and how the presence of two types of anions affects cation solvation and the corresponding solvated diameter over salt concentration.

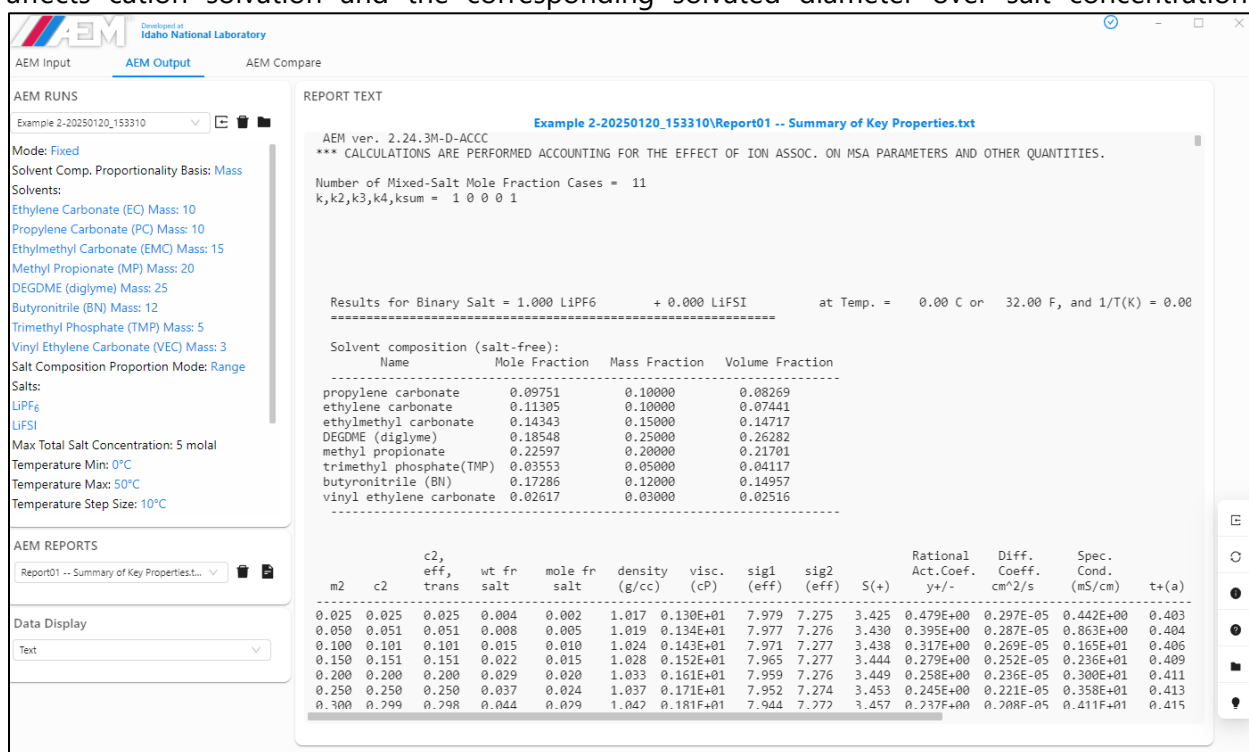


Figure 94. Example 2 – AEM Output tab – REPORT TEXT section to view Report01-Report20.

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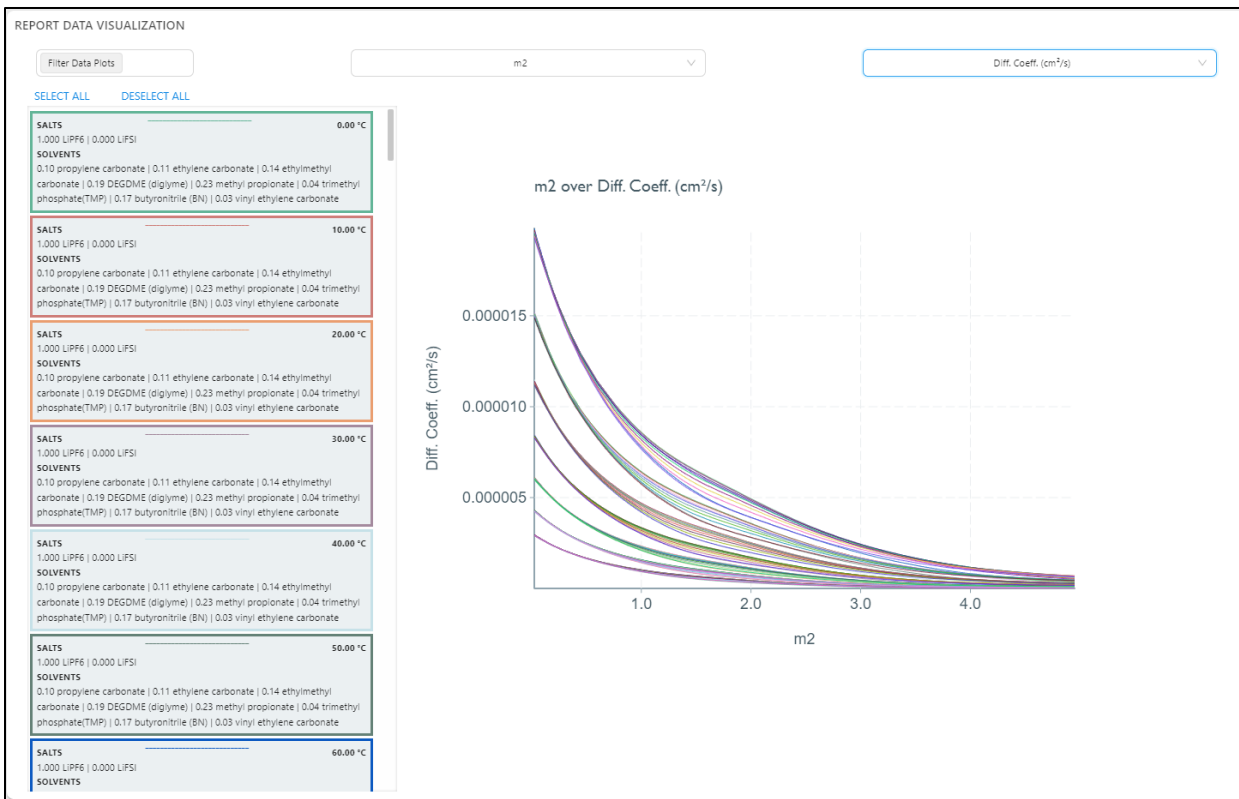


Figure 95. Example 2 – Diffusivity/Diffusion Coefficient (cm²/s) v/s Molal Salt Conc. (m2).

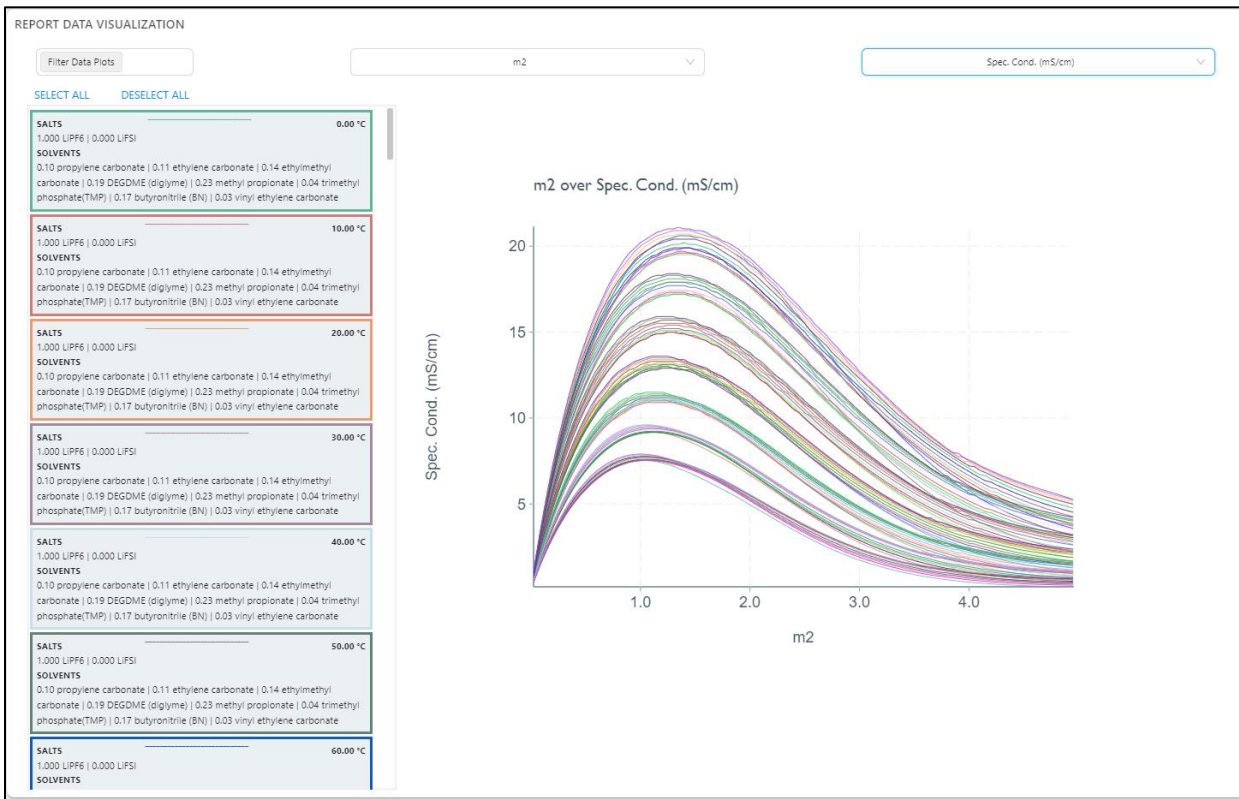


Figure 96. Example 2 – Specific Conductivity (mS/cm) v/s Molal Salt Conc. (m2).

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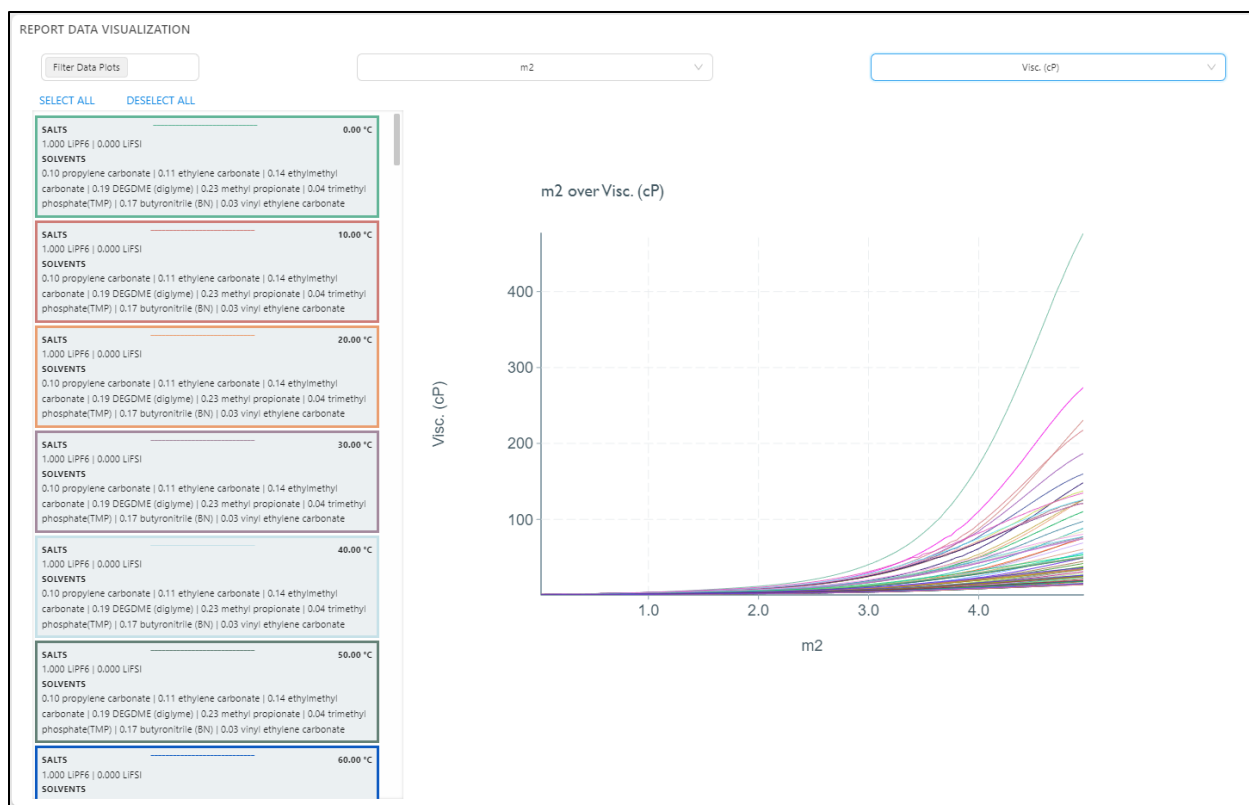


Figure 97. Example 2 – Viscosity (cP) v/s Molal Salt Conc. (m2).

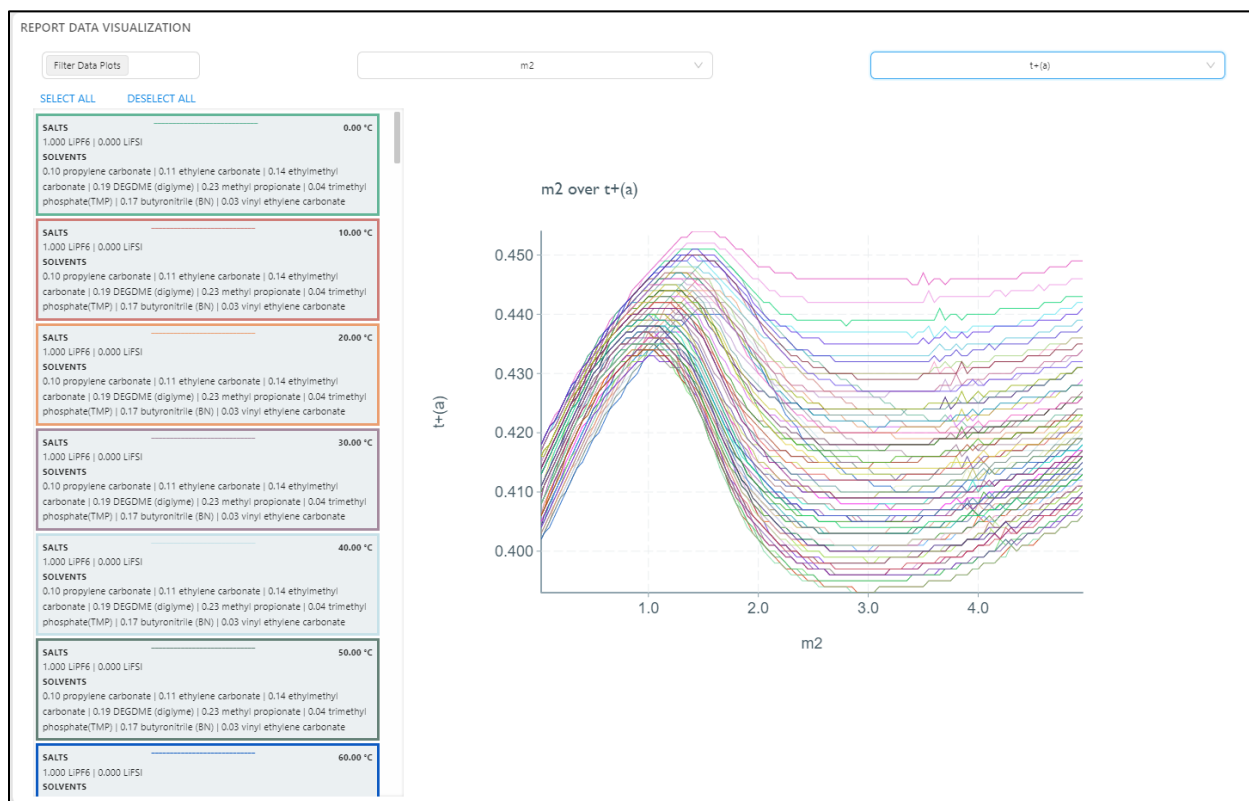


Figure 98. Example 2 – Cation Transference No. (t+(a)) v/s Molal Salt Conc. (m2).

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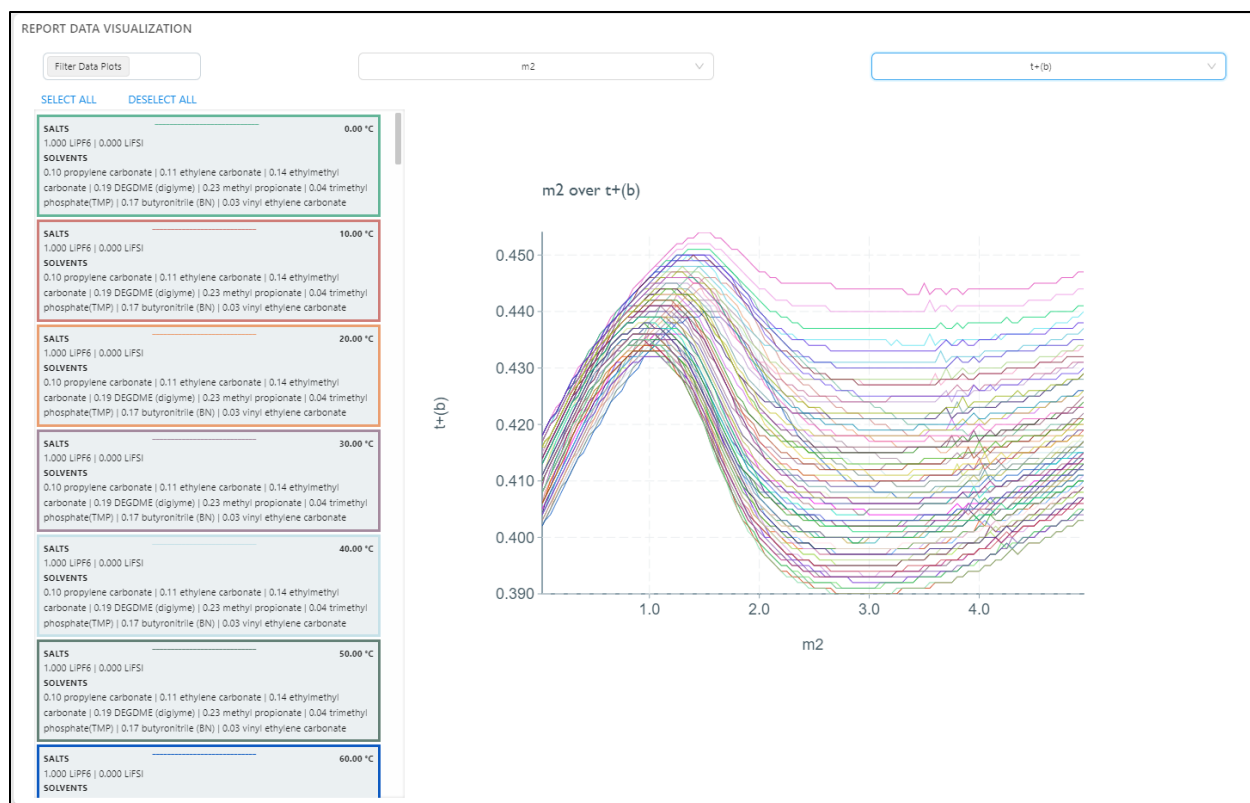


Figure 99. Example 2 – Cation Transference No. ($t_+(b)$) v/s Molal Salt Conc. (m_2).

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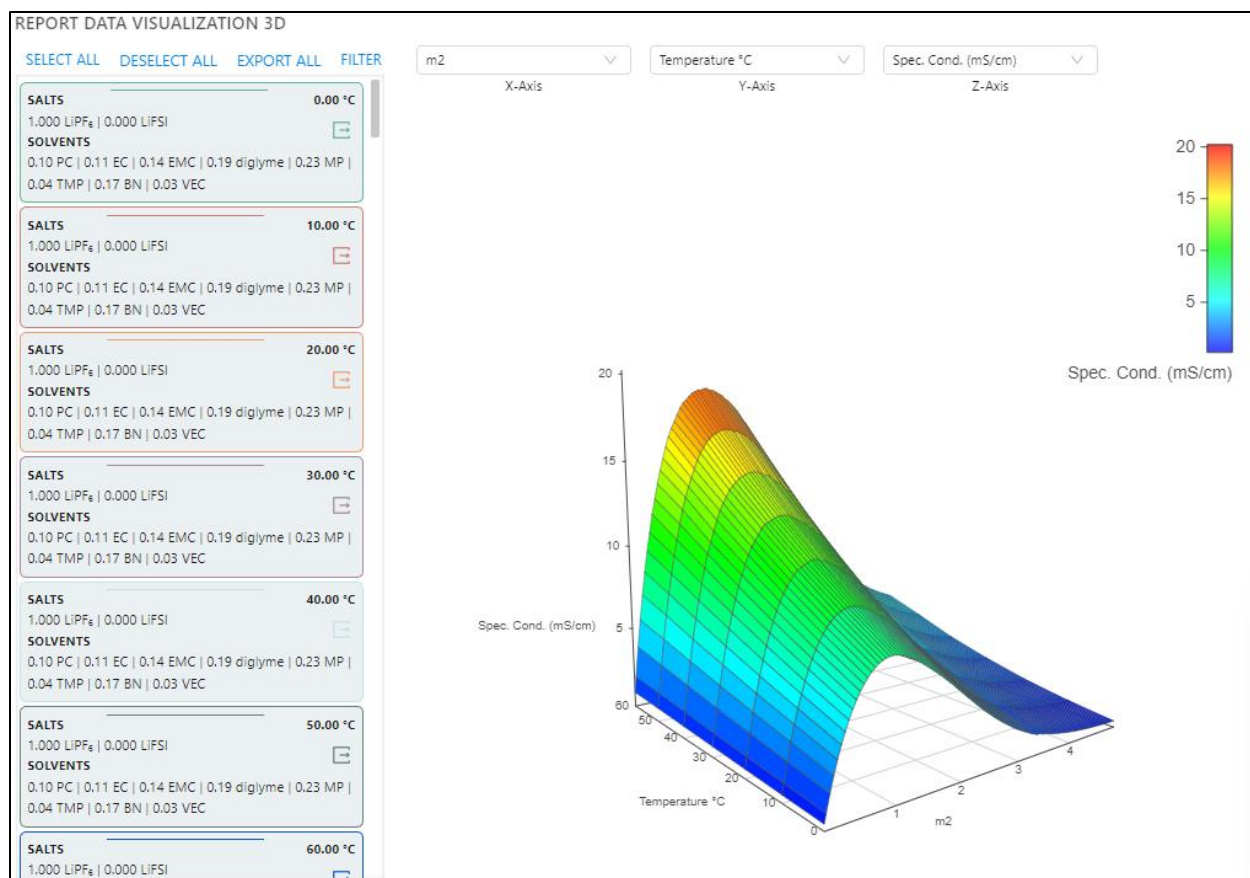


Figure 100. Example 2- Specific Conductivity (mS/cm) vs Molal Salt Conc. (m2) vs Temperature (deg C) 3D contour

4.3 Example 3: 3-Solvent with Fixed Volume Proportions, 2-Salt System

This example is similar to Example 1, yet with a fixed dual salt and higher maximum salt concentration. Considered is the three-solvent system EC, DMC, EP at a fixed volume fraction composition (25, 50, 25 vol %) with LiPF_6 and LiBF_4 . The total salt concentration is allowed to go to 7 molal. Other factors behind the simulation include:

- $T = -20^\circ\text{C}$ to 50°C with 5°C steps,
- Used default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- Yes SCAEP; No DL.

This relatively simple electrolyte represents a lower-viscosity, higher-conductivity system that would facilitate higher current densities in lithium-ion type cells, with possible benefits past Example 1 due to the mixed salts. This electrolyte system has benefits toward conditions like moderately-fast charging and higher-power applications. A key difference in this example is that at the higher salt concentrations (generally past 5 molal) the AEM software detects more cases of “possible onset of solid phase formation”, which impacts calculations thereafter that may discover interesting trends in terms like viscosity and conductivity.

4.3.1 Example 3: Importing Example 3 Parameters

We start off Example 3 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 3. This can be achieved by following the below list of instructions and referencing Figure 101 and Figure 102.

- Click on the **Import** (📁) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- Select the **settings-Example 3.json** file and click the **Open** button.

Important Note: An error message will pop up if the “settings-Example 3.json” file was not imported successfully.

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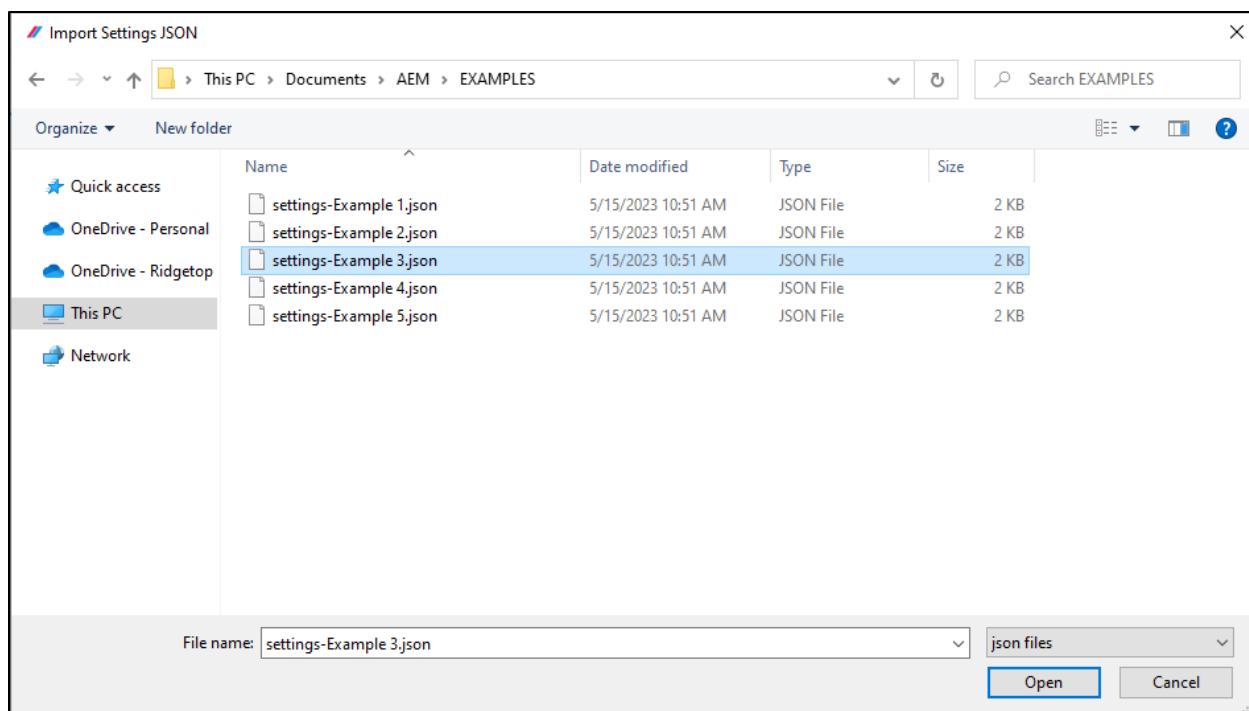


Figure 101. Importing Example 3 Input Condition using the settings-Example 3.json file.

4.3.2 Example 3: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, select the following

- Select the three-solvent system EC, DMC, EP at a fixed volume fraction composition (25, 50, 25 vol %)
- Select the dual salt LiPF_6 and LiBF_4 (70:30 molar proportions), where the maximum salt concentration is 7 molal,
- Choose $T = -20^\circ\text{C}$ to 50°C with 5°C steps,
- Used default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- Yes SCAEP; No DL. This will generate Report10, which contains information on how the magnitude of the electrode surface charge density will attenuate the relative permittivity of the neighboring electrolyte. The following SCAEP inputs are used:
 - a. Mode 1 (discharge)
 - b. Voltage = 4.2V
 - c. Conc. = 2.0
 - d. SEI thickness = 250.
 - e. SEI relative permittivity = 12.
 - f. SEI porosity = 0.3

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The screenshot displays the AEM input interface, organized into several functional sections:

- Top Navigation:** Includes a logo, a 'SINGLE FIXED COMPOSITION OF SOLVENTS' button, and a 'LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION' button.
- SELECT SOLVENTS:** Features a 'Click Solvent from Category to Add (Max: 10)' instruction. It includes dropdown menus for 'WATER', 'CARBONATES', 'DICARBONATES', 'GLYMES', 'ESTERS', and 'FROM OTHER CLASSES'. Below these are buttons for 'S OR P CONTAINING SOLVENTS', 'NITRILES AND DINITRILES', and 'PHOSPHAZENE ADDITIVES'. A search bar 'SEARCH ALL SOLVENTS' is also present. Three solvents are currently selected: 'CARBONATES / Ethylene Carbonate (EC)' at 25, 'CARBONATES / Dimethyl Carbonate (DMC)' at 50, and 'ESTERS / Ethyl Propionate (EP)' at 25.
- SELECT SALTS:** Includes a 'Click Salt from Category to Add (Max: 2)' instruction. It has dropdowns for 'FOR BATTERY ELECTROLYTES', 'FOR AQUEOUS SYSTEMS', and 'ACCC SALTS', along with a 'SEARCH ALL SALTS' button. Two salts are selected: 'FOR BATTERY ELECTROLYTES / LiPF₆' at 70 and 'FOR BATTERY ELECTROLYTES / LiBF₄' at 30. A 'Max Total Salt Concentration' of 7 molal is specified.
- TEMPERATURE SETTINGS:** Contains input fields for 'Temperature Min' (-20 °C), 'Temperature Max' (50 °C), and 'Temperature Step Size' (5 °C).
- METHOD FOR HANDLING TRIPLE ION STABILITY:** Provides a dropdown to select a method, with options for '[ABA*] = [BAB*]' (default) and '[ABA*] > [BAB*]' due to $BAB^- \rightarrow B^- + AB$ or $[ABA*] < [BAB*]$ due to $ABA^+ \rightarrow A^+ + AB$.
- ELECTROLYTE INGRESS INTO PORES:** Includes fields for 'Contact Angle' (35 °), 'Total Pore Length' (20 μm), and 'Salt Concentration of Interest' (1.5 molal).
- SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP):** Features a checkbox for 'Perform SCAEP Calculations' (checked). It includes fields for 'Type of Pulse Condition' (Discharge), 'Cell Voltage of Interest' (4.2 V), 'Bulk Salt Concentration' (2.0 molal), 'Thickness of Cathode SEI' (250 Å), 'Rel. Permittivity of Cathode SEI' (12 %), and 'Ave. Porosity of Cathode SEI' (0.3 ε).
- DOUBLE LAYER (DL):** Includes a checkbox for 'Perform DL Calculations'.
- AEM RUN IDENTIFIER (optional):** Contains a text field for 'AEM Run Identifier' (Example 3) and a checkbox for 'Append Timestamp' (checked).
- Bottom Controls:** Includes a 'Run AEM' button and a checkbox for 'Append Multiple Runs'.

Figure 102. Example 3 – AEM Input tab.

4.3.3 Example 3: Data Output & Discussion

Discussion of this example is similar to Example 1, with the following distinction. The presence of LiBF₄ (30% molar in total salt) causes a higher incidence of ion association due to the smaller anion size compared to PF₆⁻, which reduces the amount of free ion charge carriers in solution. As a result, ionic conductivity values are lower than for Example 1, yet the viscosity is generally lower, and the salt diffusivity is higher. Thus, this example showcases the impact of introducing a salt like LiBF₄ that has an anion with a smaller size. Aside from transport properties, potential benefits may arise from mixed salt electrolytes in terms of reduced cation desolvation energies. Figure 103 - Figure 109 cover this example case.

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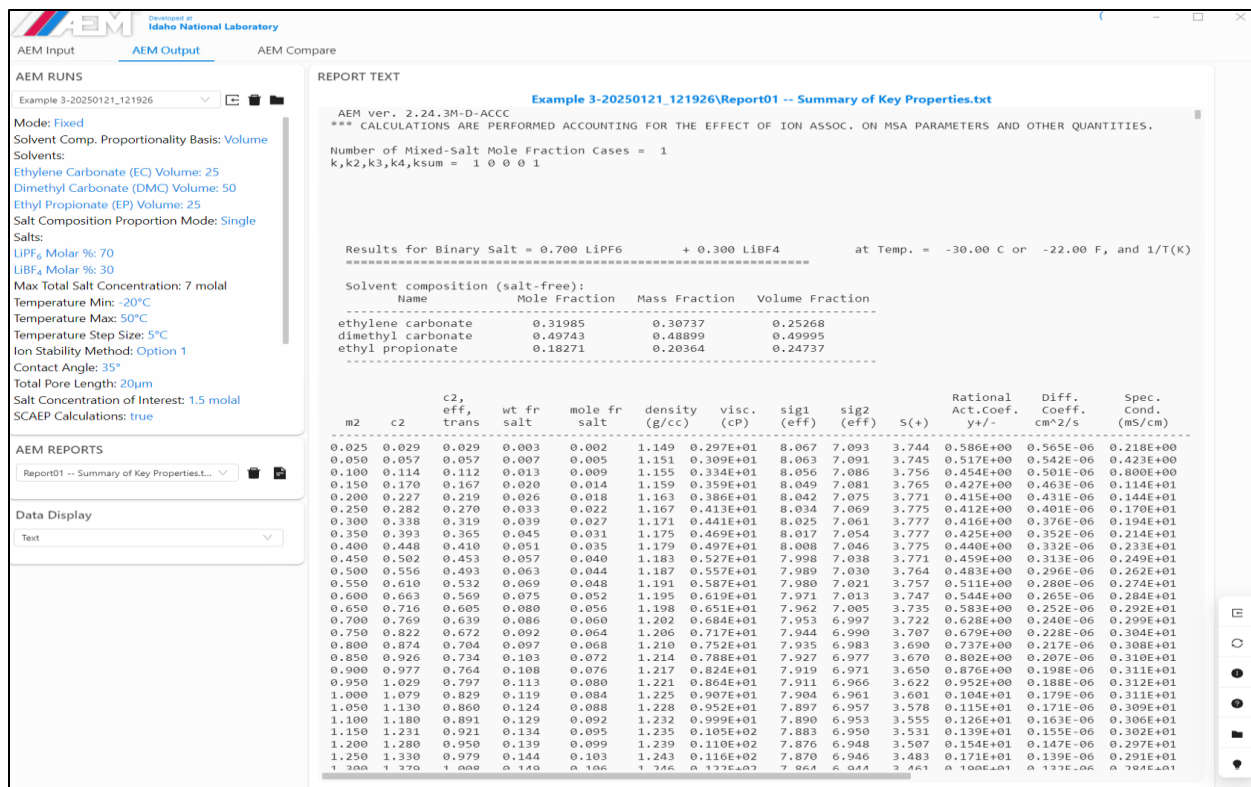


Figure 103. Example 3 – AEM Output tab – REPORT TEXT section to view Report01-Report20.

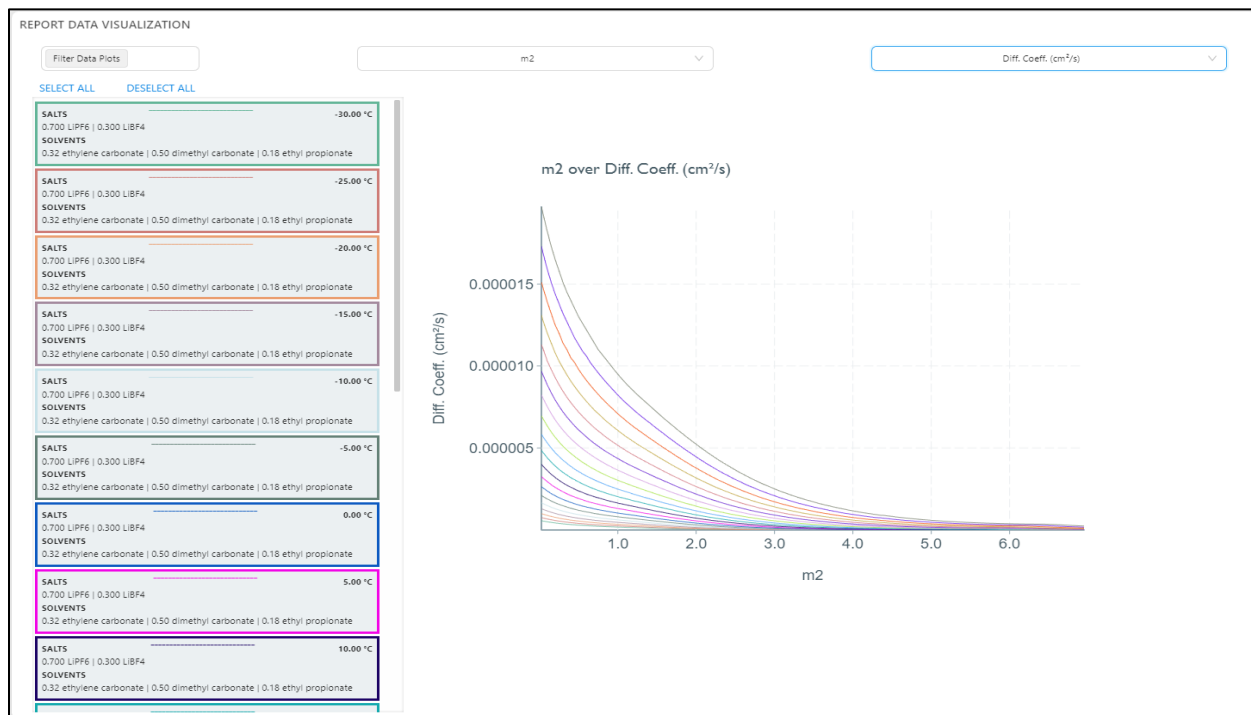


Figure 104. Example 3 – Diffusivity/Diffusion Coefficient (cm²/s) v/s Molal Salt Conc. (m²).

Advanced Electrolyte Model (AEM)

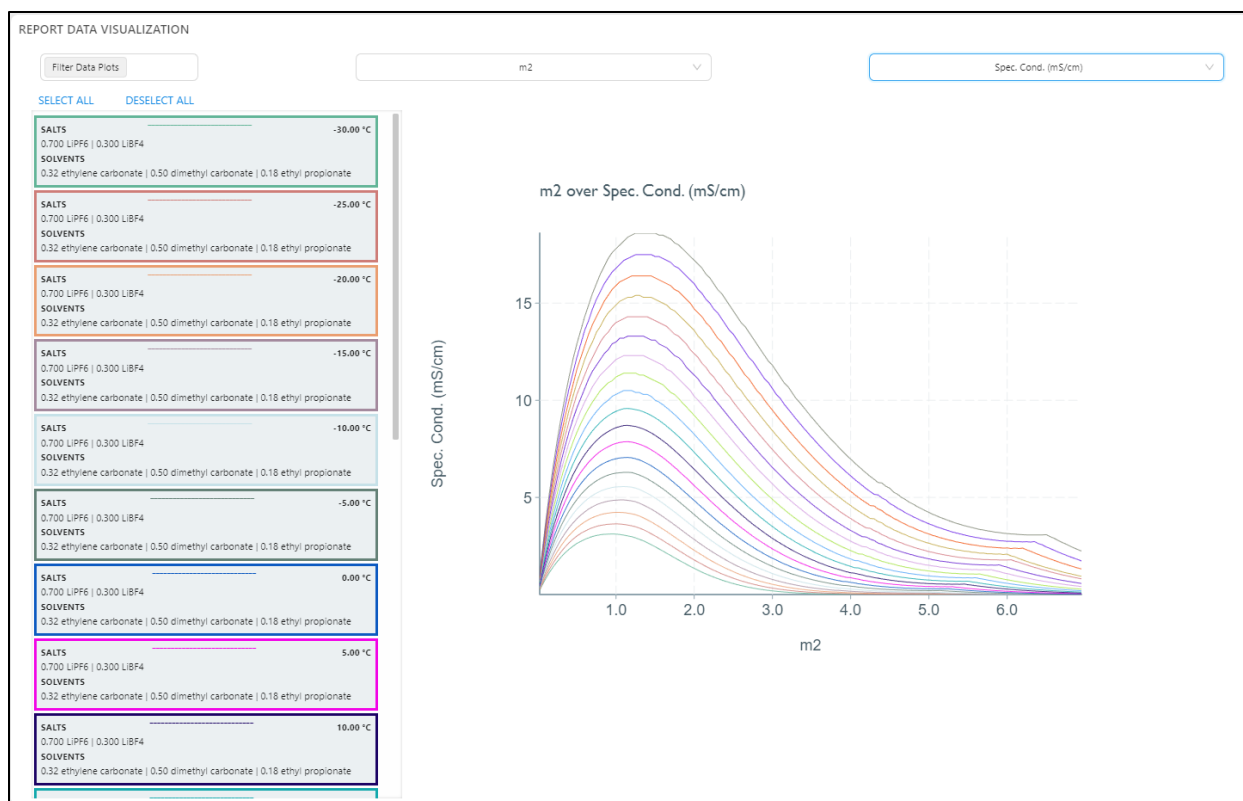


Figure 105. Example 3 – Specific Conductivity (mS/cm) v/s Molal Salt Conc. (m2).

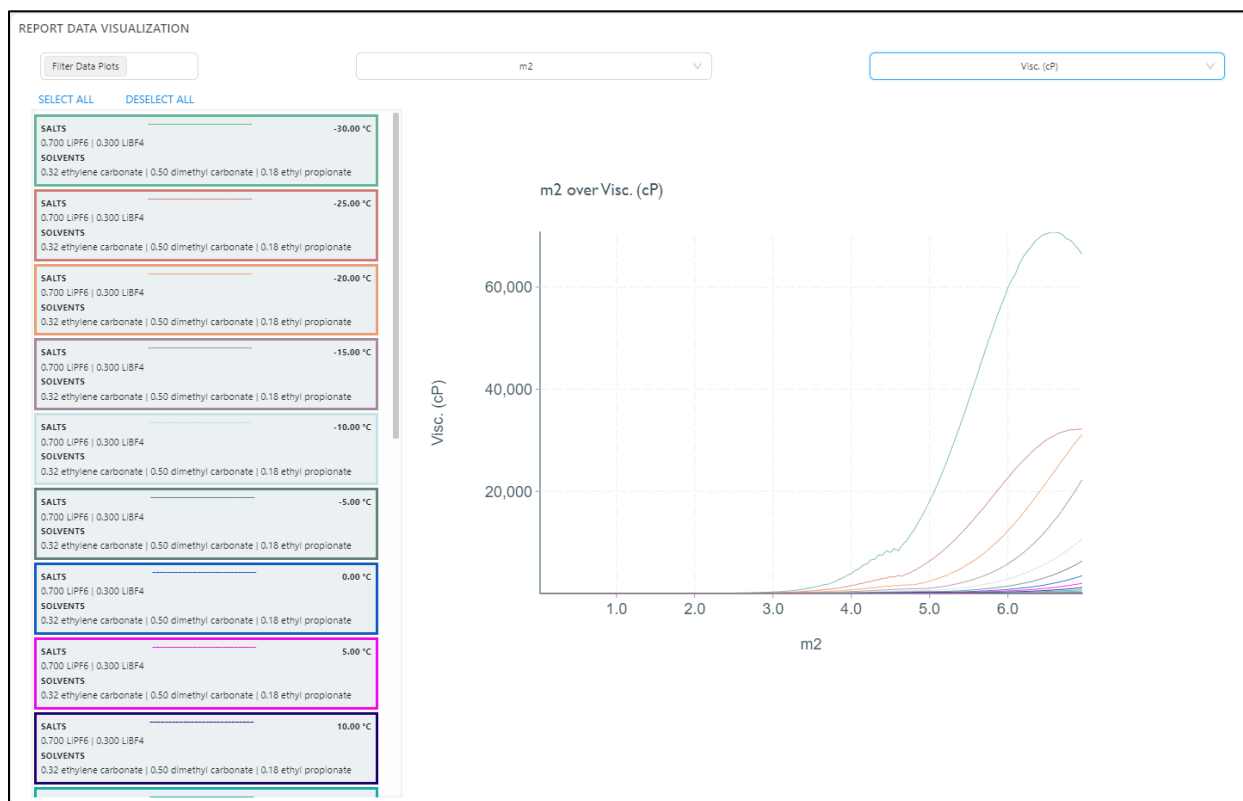


Figure 106. Example 3 – Viscosity (cP) v/s Molal Salt Conc. (m2).

Advanced Electrolyte Model (AEM)

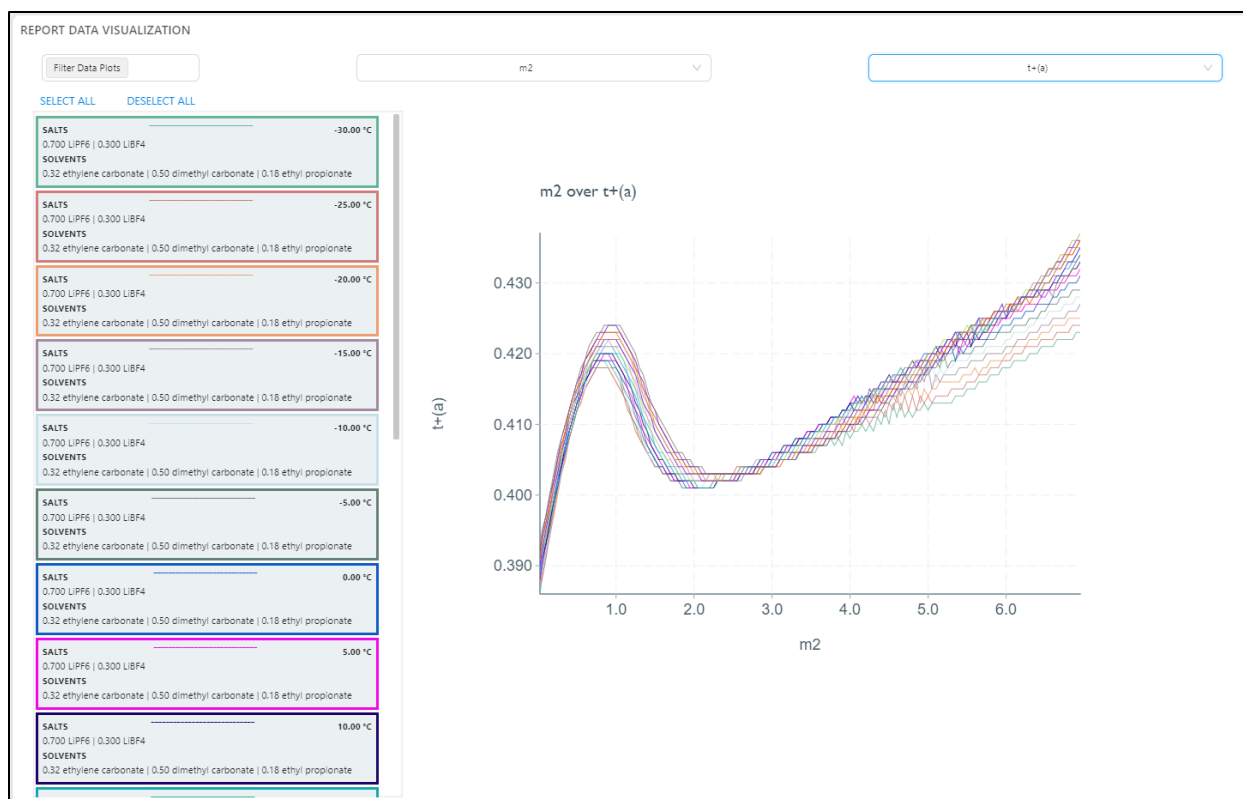


Figure 107. Example 3 – Cation Transference No. ($t_+(a)$) v/s Molal Salt Conc. (m_2).

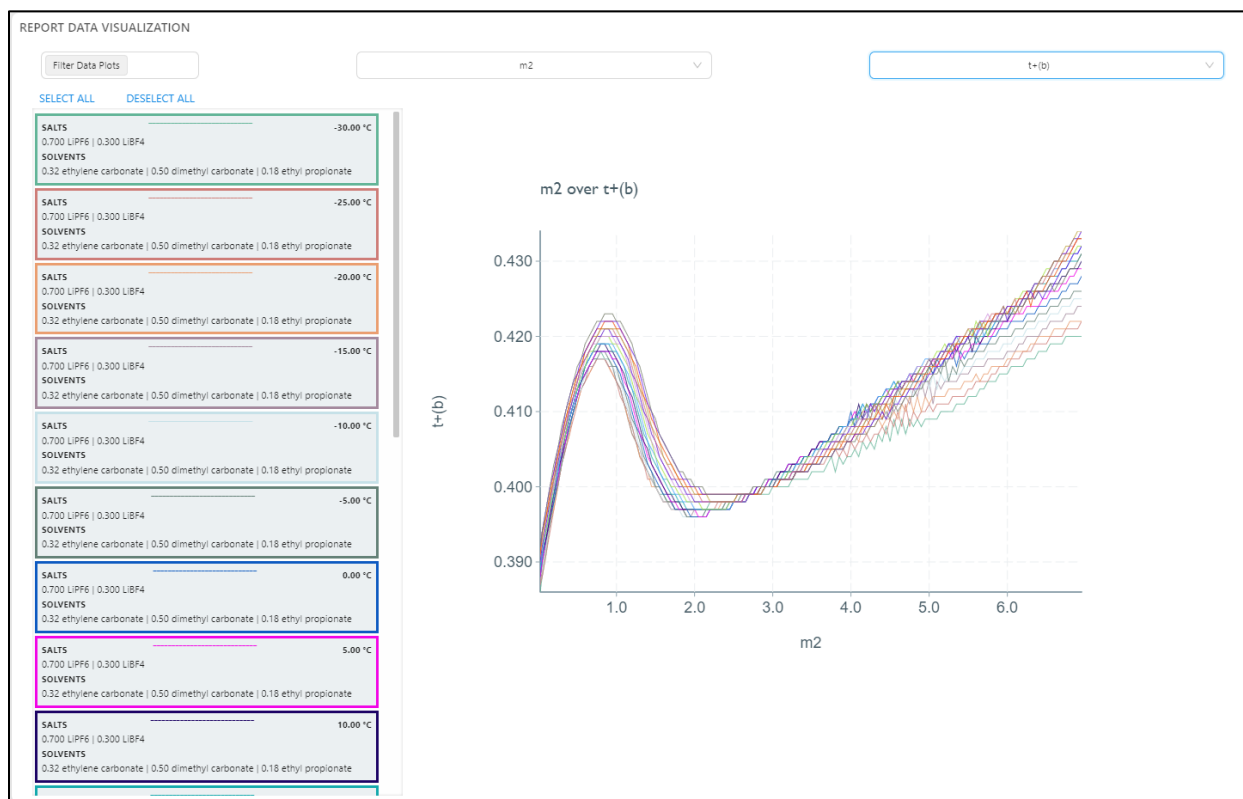


Figure 108. Example 3 – Cation Transference No. ($t_+(b)$) v/s Molal Salt Conc. (m_2).

Advanced Electrolyte Model (AEM)

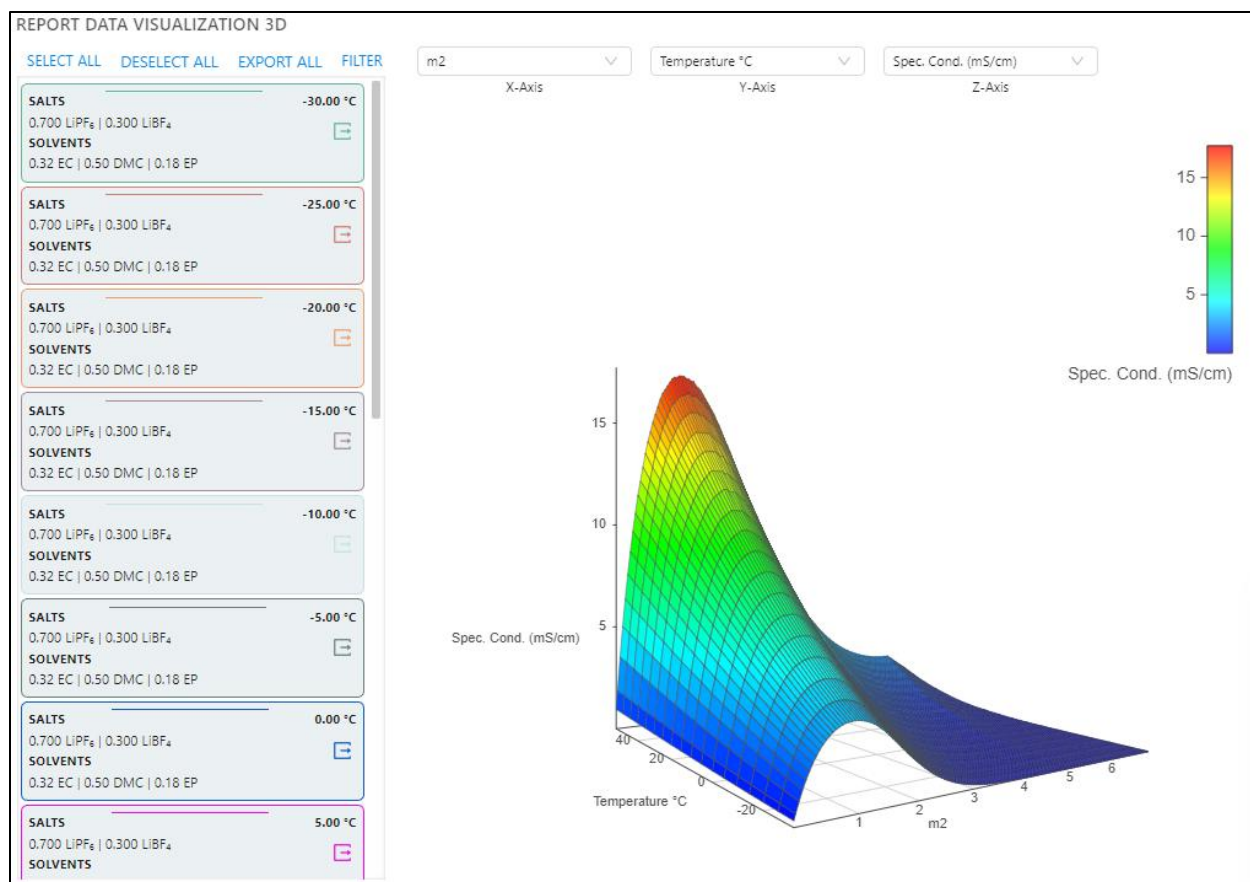


Figure 109. Example 3- Specific Conductivity (mS/cm) vs Molal Salt Conc. (m2) vs Temperature (deg C) 3D contour

4.4 Example 4: Water-based 2-Salt System with Fixed Molar Proportion

Aqueous chemistry research can see strong benefits from AEM. This example considers a dual salt system $\text{LiNO}_3 + \text{Na}_2\text{SO}_4$ of equimolar amounts in water. The effective total salt concentration approaches 12 molal. Other input factors for simulation of this system:

- $T = 0^\circ\text{C}$ to 70°C with 10°C steps;
- Use default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal
- No SCAEP; No DL.

An example of this type demonstrates AEM capabilities at higher salt concentrations than are ordinarily found for non-aqueous battery electrolytes.

4.4.1 Example 4: Importing Example 4 Parameters

We start off Example 4 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 4. This can be achieved by following the list of below instructions and referencing Figure 110 and Figure 111.

- Click on the **Import** (≡) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- Select the **settings-Example 4.json** file and click the **Open** button.

Important Note: An error message will pop up if the “settings-Example 4.json” file was not imported successfully.

Advanced Electrolyte Model (AEM)

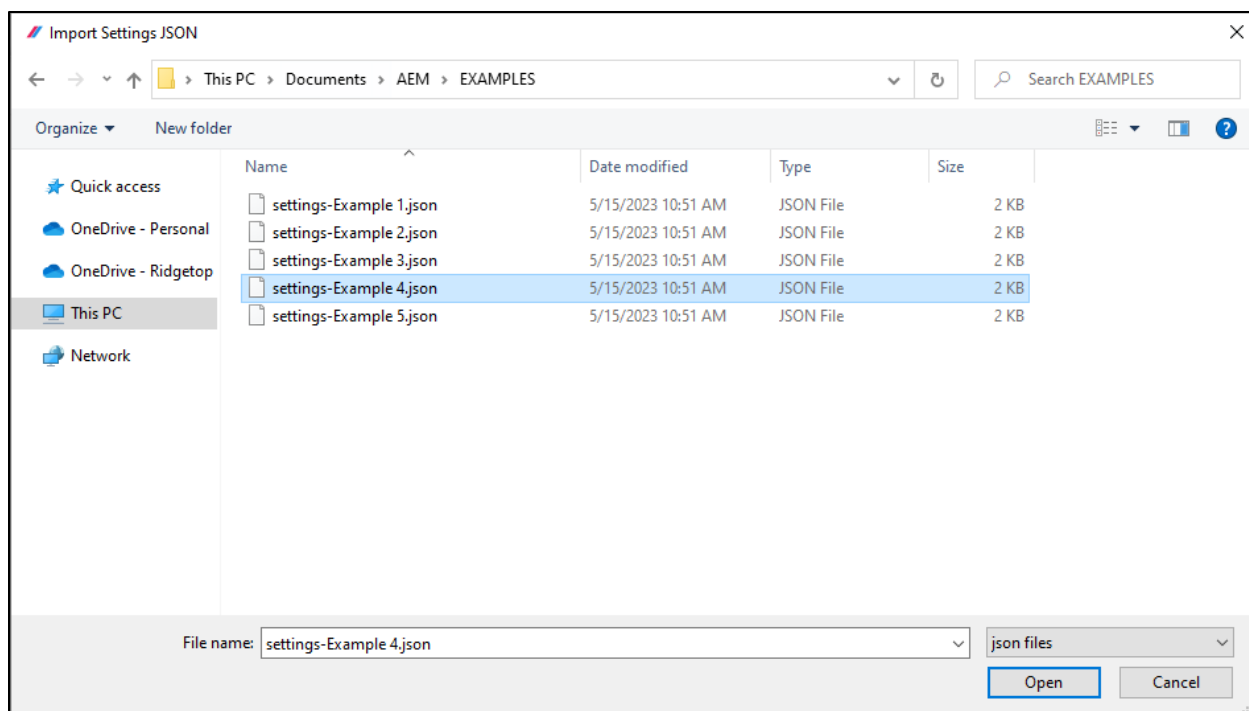


Figure 110. Importing Example 4 Input Condition using the settings-Example 5.json file.

4.4.2 Example 4: Overview of Simulation Input Conditions

- Choose water as the solvent,
- Choose the dual salt system $\text{LiNO}_3 + \text{Na}_2\text{SO}_4$ of equimolar amounts (50:50); specify 15 molal total salt (*this is different than the output range of approximately 12 molal, since AEM determines what is feasible given a combination of the 1:1 and 1:2 type salts*),
- Specify $T = 0^\circ\text{C}$ to 70°C with 10°C steps
- Use default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal
- Choose No SCAEP; No DL.

Advanced Electrolyte Model (AEM)

The screenshot displays the AEM input interface, organized into several functional sections:

- Top Bar:** Includes a logo on the left and a grid icon on the right.
- Left Column:**
 - SINGLE FIXED COMPOSITION OF SOLVENTS:** A section for selecting solvents.
 - SELECT SOLVENTS (+):** Features a "Composition Proportionality Basis" toggle (Volume/Mass) and a list of solvent categories (WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, FROM OTHER CLASSES, S OR P CONTAINING SOLVENTS, NITRILES AND DINITRILES, PHOSPHAZENE ADDITIVES, ACCC SOLVENTS, SEARCH ALL SOLVENTS). A search bar contains "Water".
 - SELECT SALTS (+):** Features a "Composition Proportion Mode" toggle (Single/Range) and a list of salt categories (FOR BATTERY ELECTROLYTES, FOR AQUEOUS SYSTEMS, ACCC SALTS, SEARCH ALL SALTS). Two entries are shown: "FOR AQUEOUS SYSTEMS / LiNO₃" and "FOR AQUEOUS SYSTEMS / Na₂SO₄", both with a value of 50. A "Max Total Salt Concentration" field is set to 15 molal.
- Right Column:**
 - TEMPERATURE SETTINGS:** Includes fields for Temperature Min (0 °C), Temperature Max (70 °C), and Temperature Step Size (5 °C, 10 °C).
 - METHOD FOR HANDLING TRIPLE ION STABILITY:** Contains a dropdown menu with options for [ABA*] = [BAB*] (default), [ABA*] > [BAB*], and [ABA*] < [BAB*].
 - ELECTROLYTE INGRESS INTO PORES:** Includes fields for Contact Angle (35 °), Total Pore Length (20 μm), and Salt Concentration of Interest (1.5 molal).
 - SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP):** Includes a checkbox for "Perform SCAEP Calculations".
 - DOUBLE LAYER (DL):** Includes a checkbox for "Perform DL Calculations".
 - AEM RUN IDENTIFIER (optional):** Includes a text field for "Example 4" and a checked checkbox for "Append Timestamp".

At the bottom center, there is a "Run AEM" button and a checkbox for "Append Multiple Runs".

Figure 111. Example 4 – AEM Input tab.

4.4.3 Example 4: Data Output & Discussion

Figure 113 - Figure 118 summarize the transport plots for this example. Whether for aqueous battery systems or water treatment applications, the collection of output properties generated within the twenty (20) reports gives genomic data for aqueous systems. This example demonstrates that AEM can operate at elevated salt content while considering different salt stoichiometries (here, 1:1 and 1:2 in the same system).

Advanced Electrolyte Model (AEM)

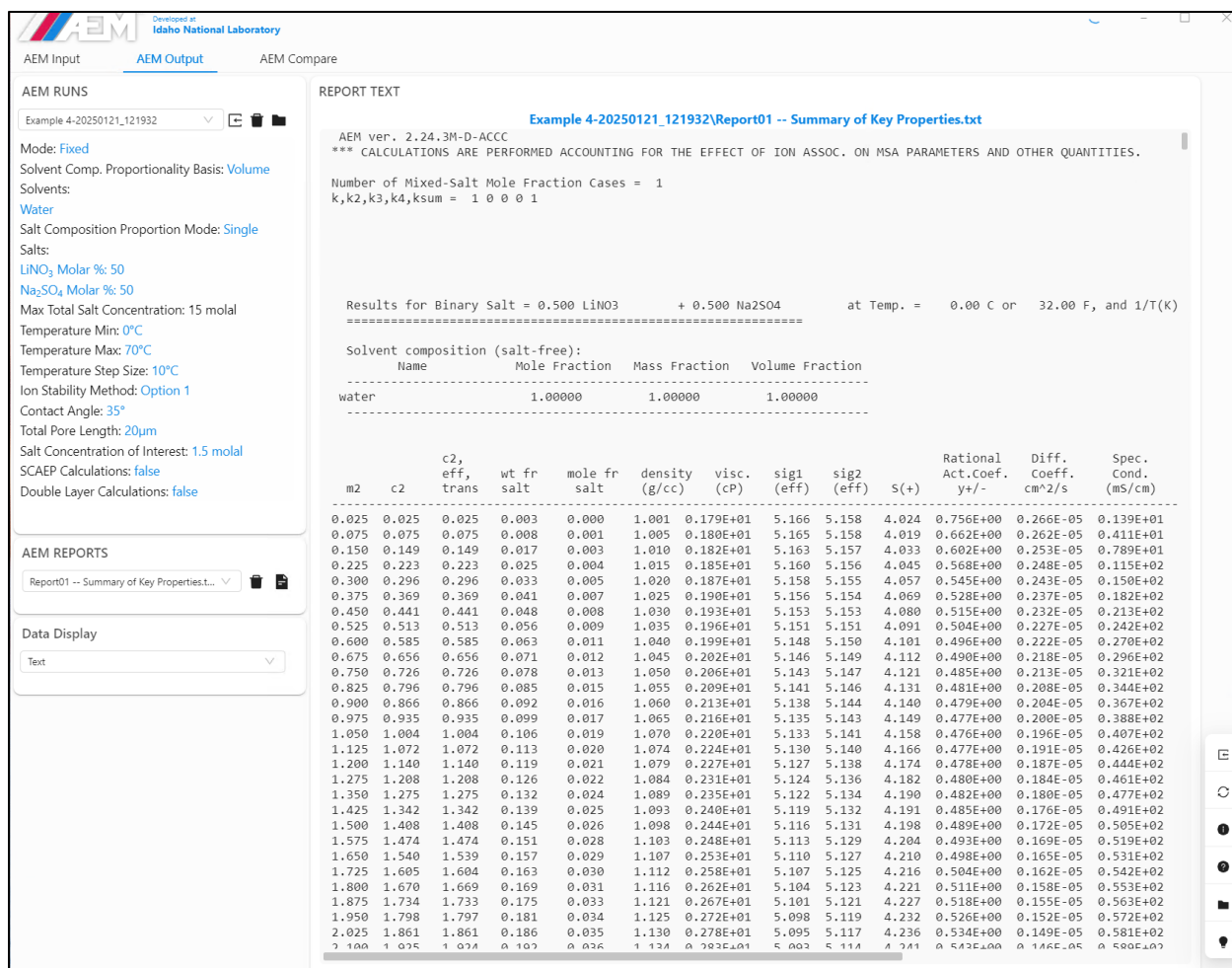


Figure 112. Example 4 – AEM Output tab – REPORT TEXT section to view Report01-Report20.

Advanced Electrolyte Model (AEM)

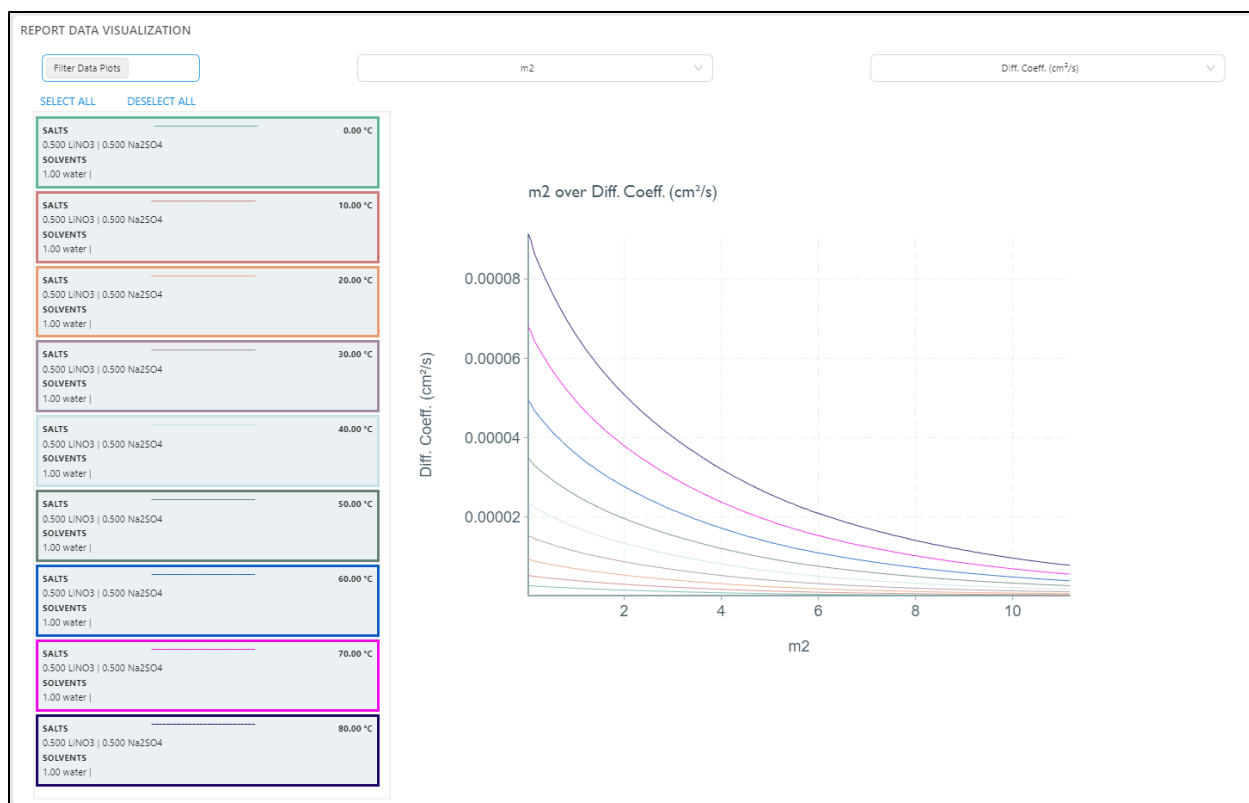


Figure 113. Example 4 – Diffusivity/Diffusion Coefficient (cm²/s) v/s Molal Salt Conc. (m2).

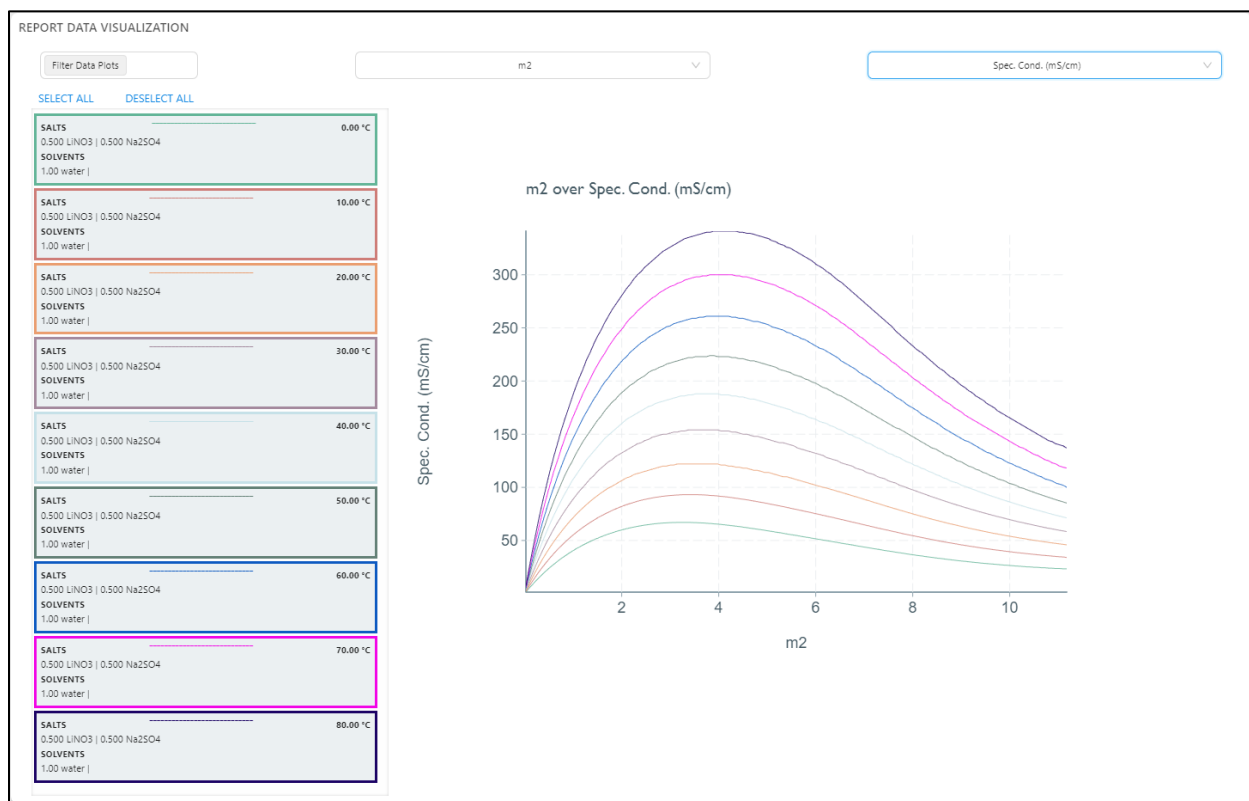


Figure 114. Example 4 – Specific Conductivity (mS/cm) v/s Molal Salt Conc. (m2).

Advanced Electrolyte Model (AEM)

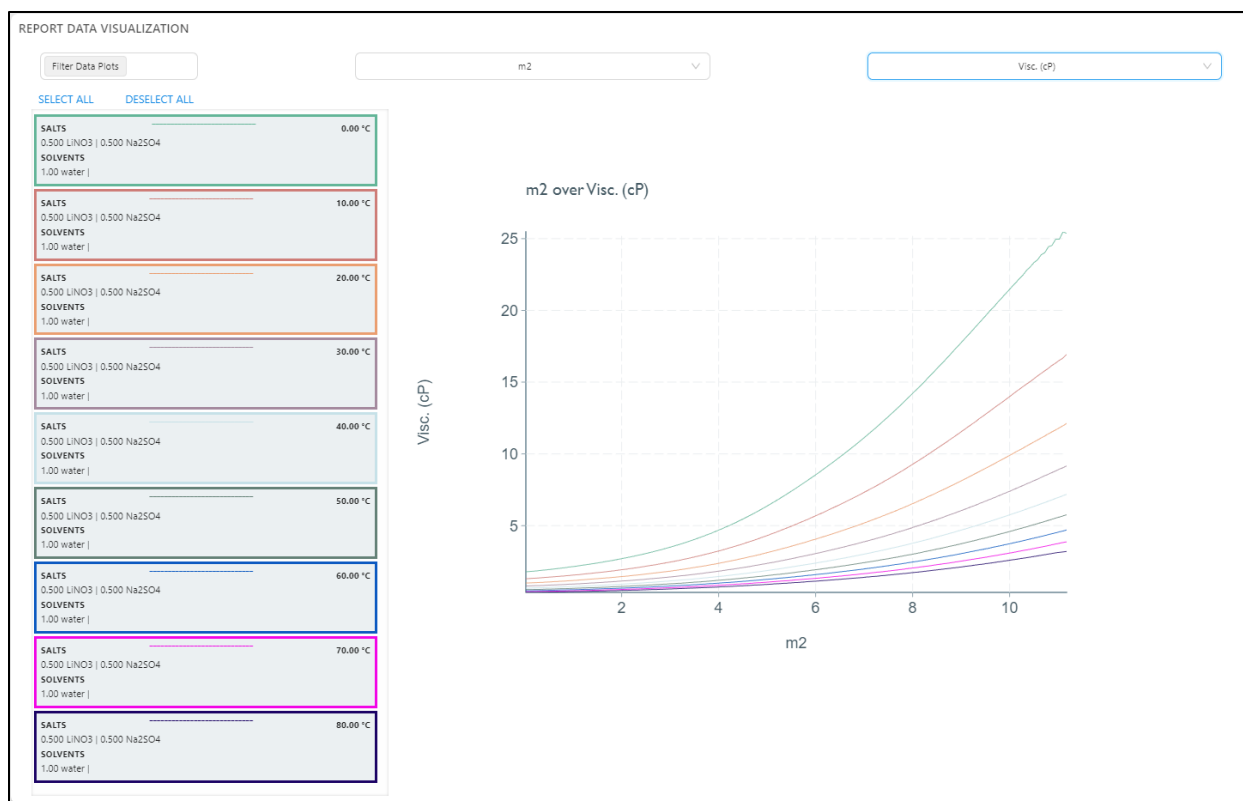


Figure 115. Example 4 – Viscosity (cP) v/s Molal Salt Conc. (m2).

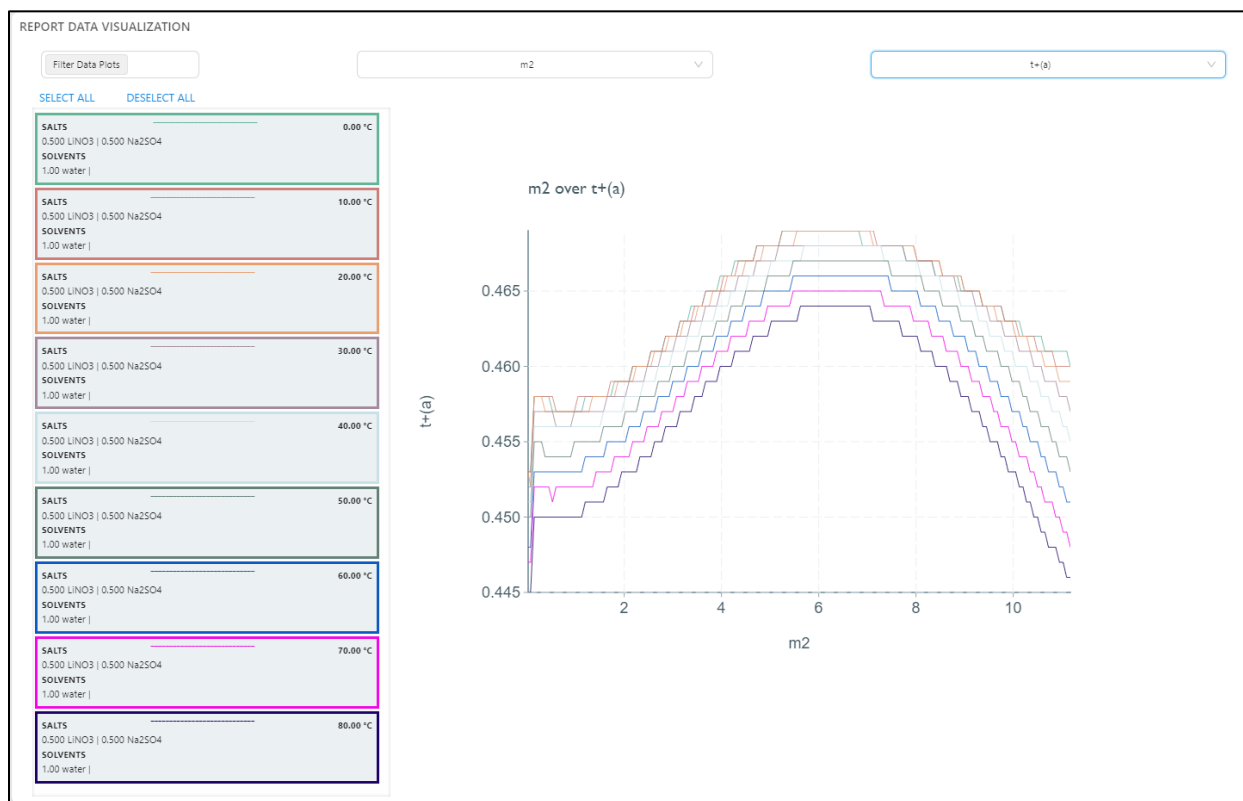


Figure 116. Example 4 – Cation Transference No. (t+(a)) v/s Molal Salt Conc. (m2).

Advanced Electrolyte Model (AEM)

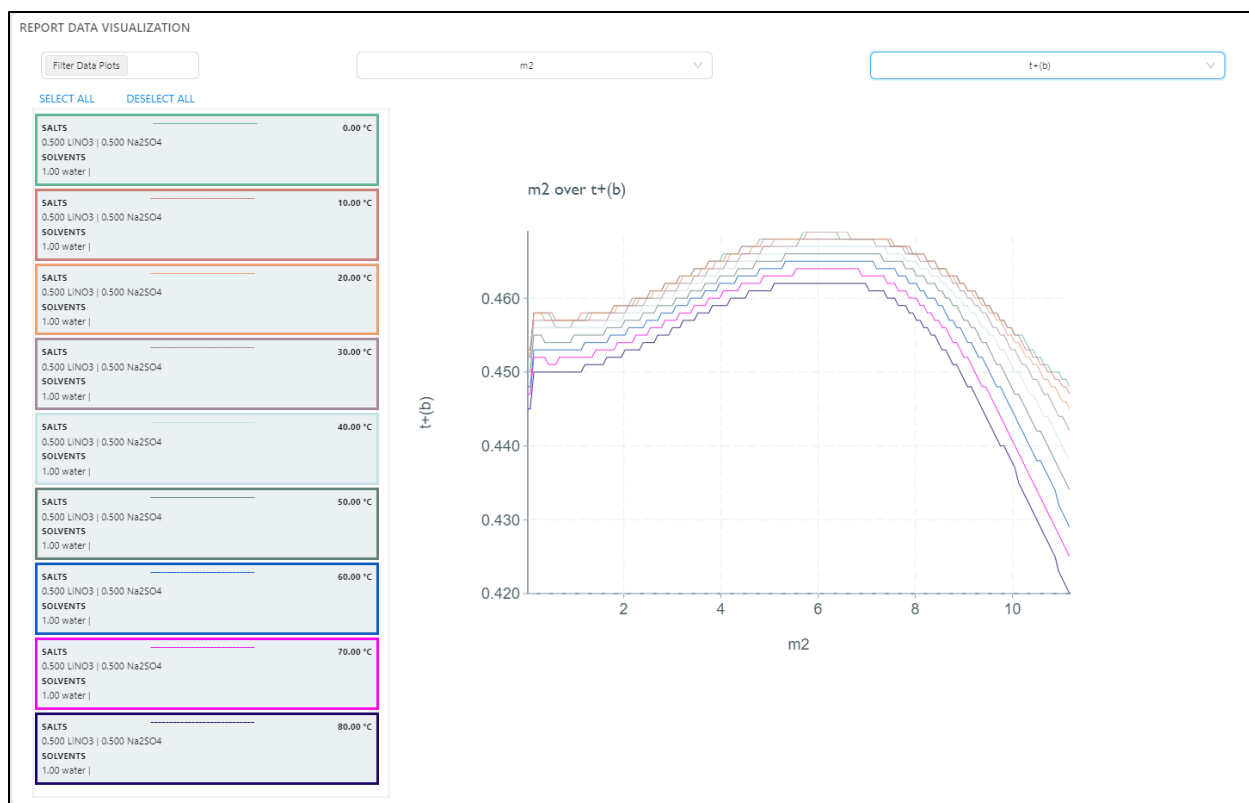


Figure 117. Example 4 – Cation Transference No. ($t_+(b)$) v/s Molal Salt Conc. (m_2).

Advanced Electrolyte Model (AEM)

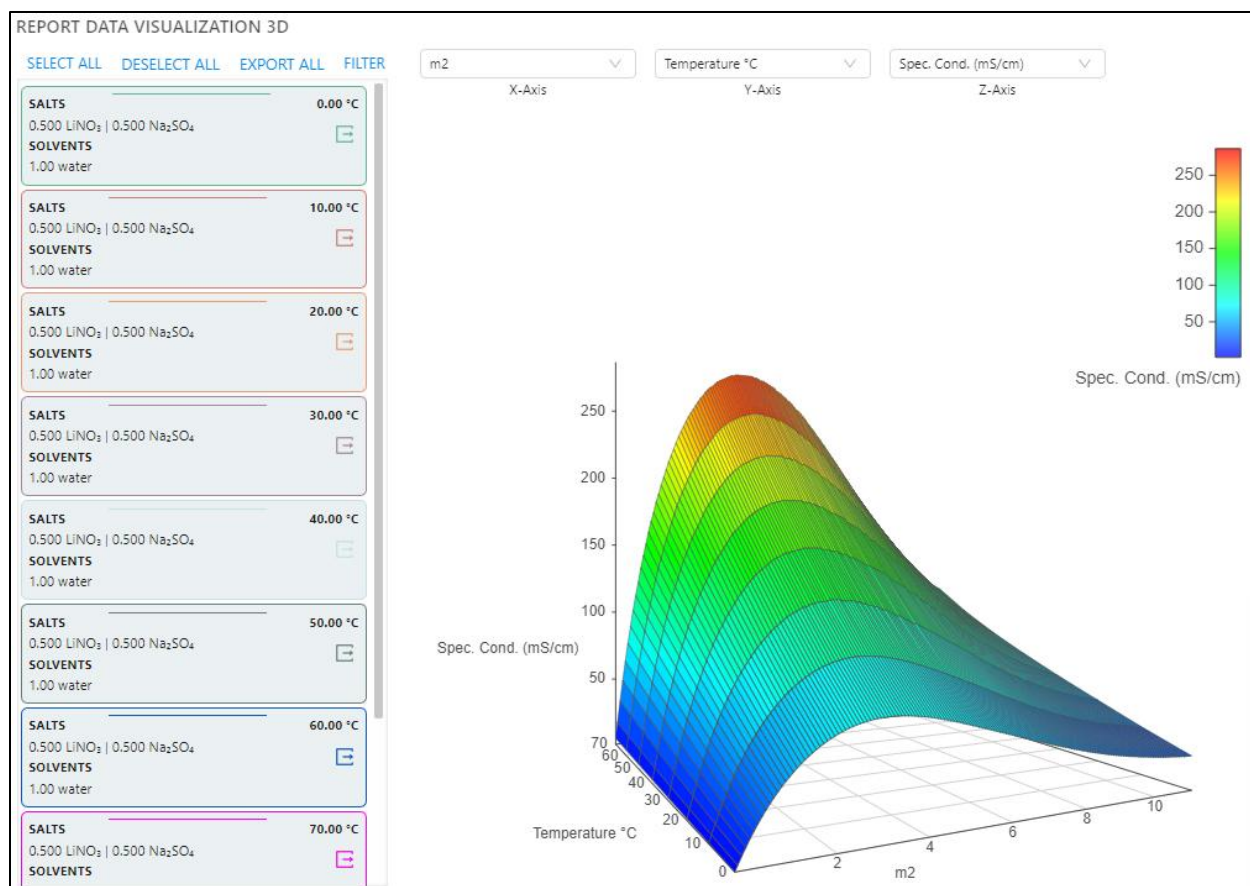


Figure 118. Example 4- Specific Conductivity (mS/cm) vs Molal Salt Conc. (m2) vs Temperature (deg C) 3D contour

4.5 Example 5: Matrix Optimization of Electrolyte Formulation with 4 Solvent, 2 Salt System

This example demonstrates a use case that performs optimization of a multi-solvent and dual salt electrolyte in terms of preferred solvent formulations and salt proportions. Optimization is done over (C_{salt} , T), allowing the user to have high-fidelity information that supports fine-tuning of the electrolyte according to anticipated conditions of the application. AEM large-scale optimization offers a means to screen millions of formulations and conditions to survey different property landscapes for the most promising combinations. This gives the AEM user a powerful research accelerator. Simulation inputs include the following:

- Four solvents (choose optimization) with PC, MP, BN, TMP. TMP will have a set mass fraction of 0.25
- Two-salt system with $LiPF_6$ and $LiTFSI$, using full matrix of salt proportions, up to 2.0 molal,
- $T = 0^\circ C$ to $50^\circ C$ with $10^\circ C$ steps,
- Use default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu m$, salt conc. of interest = 1.5 molal
- No SCAEP; No DL.

It should be noted that depending on input conditions, AEM optimization runs can take several minutes to hours depending on your computer setup. AEM optimization also produce large output files, some exceeding 1 GB in size. Upon run completion, Report07 contains an optimization summary.

4.5.1 Example 5: Importing Example 5 Parameters

We start off Example 5 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 5. This can be achieved by following the list of instructions below and referencing Figure 119 and Figure 120.

- Click on the **Import** (≡) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- Select the **settings-Example 5.json** file and click the **Open** button.

Important Note: An error message will pop up if the "settings-Example 5.json" file was not imported successfully.

Advanced Electrolyte Model (AEM)

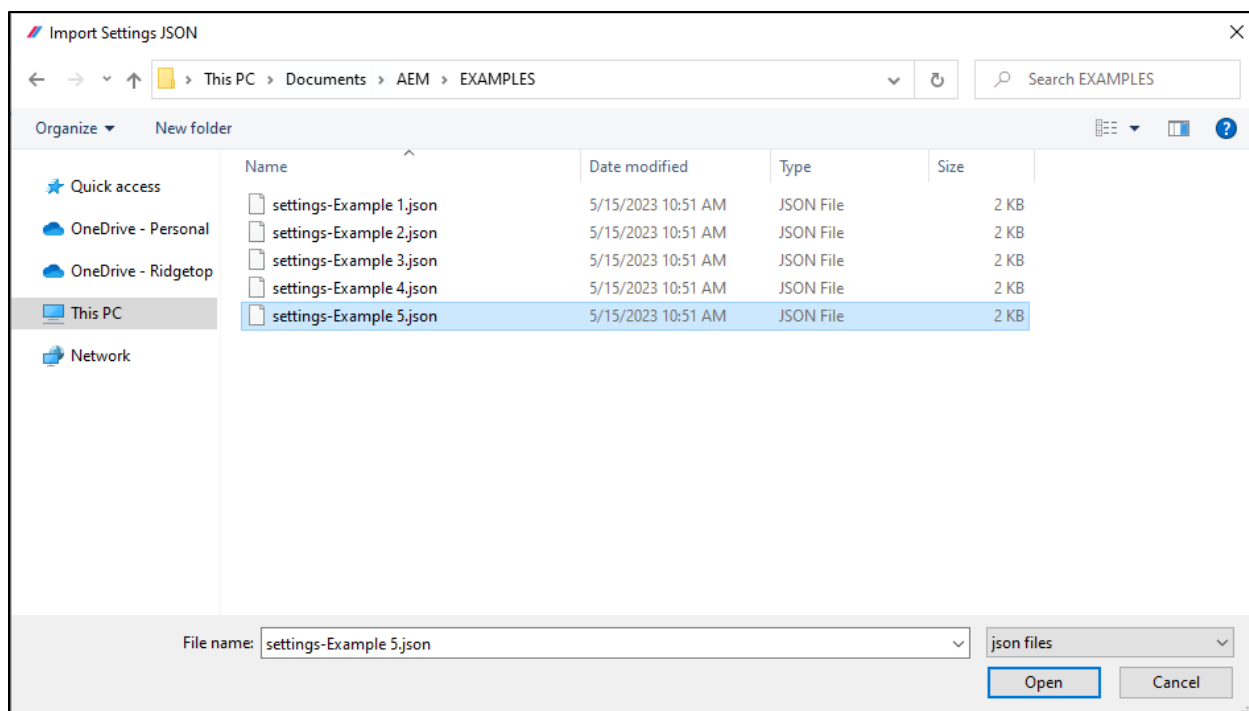


Figure 119. Importing Example 5 Input Condition using the settings-Example 5.json file.

4.5.2 Example 5: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, select the following

- Select the four solvents PC, MP, BN, TMP. Choose optimization, where TMP has a set mass fraction of 0.25. As such, the software will determine optimal proportions of PC, MP and BN in relation to the set TMP fraction.
- Specify a two-salt system with LiPF_6 and LiTFSI , using full matrix of salt proportions. Set maximum salt concentration to 2.0 molal. Optimization will also include the influence of the mixed salt proportions,
- Select $T = 0^\circ\text{C}$ to 50°C with 10°C steps,
- Use default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal
- Choose No SCAEP; No DL.

Important Note: This example can take several minutes to hours to complete depending on your computational hardware. Please wait for the simulation to complete and then compare your data outputs in the discussion section.

Advanced Electrolyte Model (AEM)

The screenshot displays the AEM input interface, organized into several functional sections:

- Top Bar:** Includes a logo on the left and a button labeled "LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION" on the right.
- TEMPERATURE SETTINGS:** Features three input fields: "Temperature Min" (0 °C), "Temperature Max" (50 °C), and "Temperature Step Size" (5 °C).
- METHOD FOR HANDLING TRIPLE ION STABILITY:** Contains a dropdown menu with options like "[ABA⁺] = [BAB⁺]" and "[ABA⁺] > [BAB⁺]" with explanatory text about ion stability.
- ELECTROLYTE INGRESS INTO PORES:** Includes inputs for "Contact Angle" (35 °), "Total Pore Length" (20 μm), and "Salt Concentration of Interest" (1.5 molal).
- AEM RUN IDENTIFIER (optional):** Has a text field for the run identifier (set to "Example 5") and a checkbox for "Append Timestamp".
- SELECT SOLVENTS:** A large section with multiple categories (WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, FROM OTHER CLASSES, S OR P CONTAINING SOLVENTS, NITRILES AND DINITRILES, PHOSPHAZENE ADDITIVES, ACCC SOLVENTS, SEARCH ALL SOLVENTS). It lists selected solvents: CARBONATES / Propylene Carbonate (PC), ESTERS / Methyl Propionate (MP), NITRILES & DINITRILES / Butyronitrile (BN), and S OR P CONTAINING / Trimethyl Phosphate (TMP). Each entry has a "Set Constant Mass Fraction" checkbox and a value field (0.25 for TMP).
- SELECT SALTS:** Includes categories (FOR BATTERY ELECTROLYTES, FOR AQUEOUS SYSTEMS, ACCC SALTS, SEARCH ALL SALTS) and selected salts: FOR BATTERY ELECTROLYTES / LiPF₆ and FOR BATTERY ELECTROLYTES / LiTFSI. It also has a "Max Total Salt Concentration" input (2 molal).
- Bottom Controls:** A "Run AEM" button and an "Append Multiple Runs" checkbox.

Figure 120. Example 5 – AEM Input tab.

4.5.3 Example 5: Data Output & Discussion

It is estimated that nearly 15,000 unique formulations were surveyed in this optimization run, each run over other matrices involving 7 temperatures and about 50 salt concentrations. That totals to about 5.25M total conditions. Any number of duplicate conditions are in addition to this number. Because these types of data output reports are so large and can be several GB in size, the AEM GUI has a built-in file size limit of 500MB. Any reports larger than that will not be automatically parsed to visualize the output graphs as shown in Examples 1-4.

For this Example, Report07 contains an optimization results summary toward the end, where an assortment of optimization parameters is considered. The optimization outcomes can vary according to the target. For example, optimization in terms of conductivity over particular (Csalt, T) may yield a different general result than that for diffusivity and may differ as well from that obtained for the molecular-based process of cation desolvation (i.e., Li-STEP terms).

Results from Report07 indicate that low-to-no PC is preferred for transport properties such as conductivity and diffusivity, while higher amounts of BN are noted (about 56% by mass). This is due to how the presence of PC increases the solution viscosity. However, for considerations of cation desolvation (see optimization metrics tied to Li-STEP, Lithium Solvation and Transport Electrolyte Parameter), a low amount of BN is preferred, with a clear preference for optimizing around MP and TMP. This is generally due to how nitriles are aggressive solvators of cation

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species, and the relatively high cation solvation number that is obtained with PC. In some cases, the amount of desirable PC increases at higher salt levels, possibly due to the higher relative permittivity of PC that allows greater dissociation of salt.

Regarding the preferred amount of each salt, greater proportions of LiTFSI are noted for conductivity at relatively low salt concentrations, while the preference switches toward having more LiPF₆ for the diffusion coefficient, as well as for cation desolvation terms.

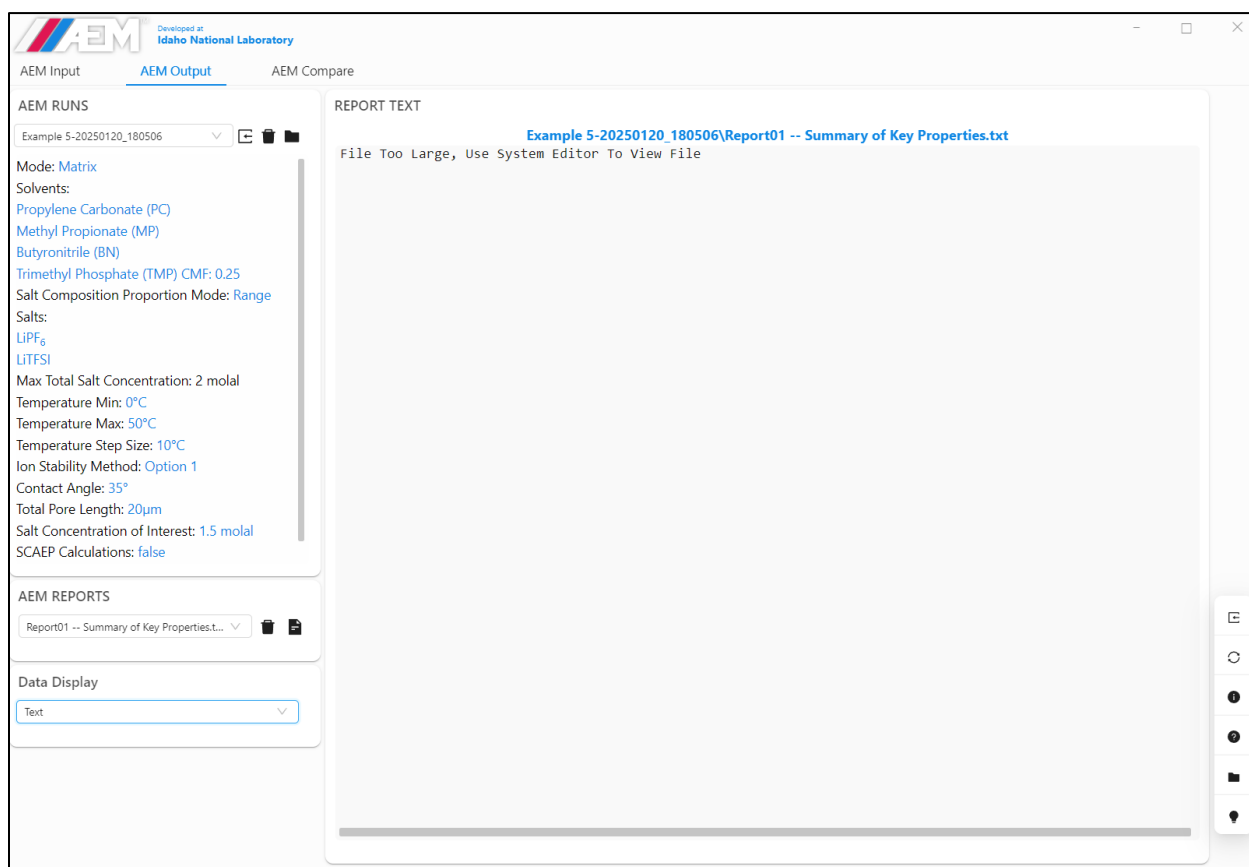


Figure 121. Example 5 – AEM Output tab – REPORT TEXT section to view Report01-Report20.

4.6 Example 6: ACCC Simulation with one ACCC Salt Definition

This example considers the two-solvent system from the ACCC Solvents EC EMC at a fixed volume fraction composition (30:70 vol %) with ACCC salt class as LiPF₆_w_EC EMC. The salt concentration is allowed to go to 5 molal. Other factors behind the simulation include:

- T = -40°C to 60°C with 5°C steps,
- Used default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35°, Pore length = 20µm, salt conc. of interest = 1.5 molal,
- No SCAEP; No DL.

The parameters for electrolyte ingress are relevant estimations for battery electrode materials in general and can be updated to reflect more finely tuned attributes of the porous media.

This relatively simple electrolyte represents a lower-viscosity, higher-conductivity system that would facilitate higher current densities in lithium-ion type cells. This electrolyte system has benefits toward conditions like moderately-fast charging and higher-power applications.

4.6.1 Example 6: Importing Example 6 Parameters

We start off Example 6 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 6. This can be achieved by following the list of instructions below and referencing Figure 122 and Figure 123.

- a) Click on the **Import** (≡) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- b) Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- c) Select the **settings-Example 6.json** file and click the **Open** button.

Important Note: An error message will pop up if the "settings-Example 6.json" file was not imported successfully.

Advanced Electrolyte Model (AEM)

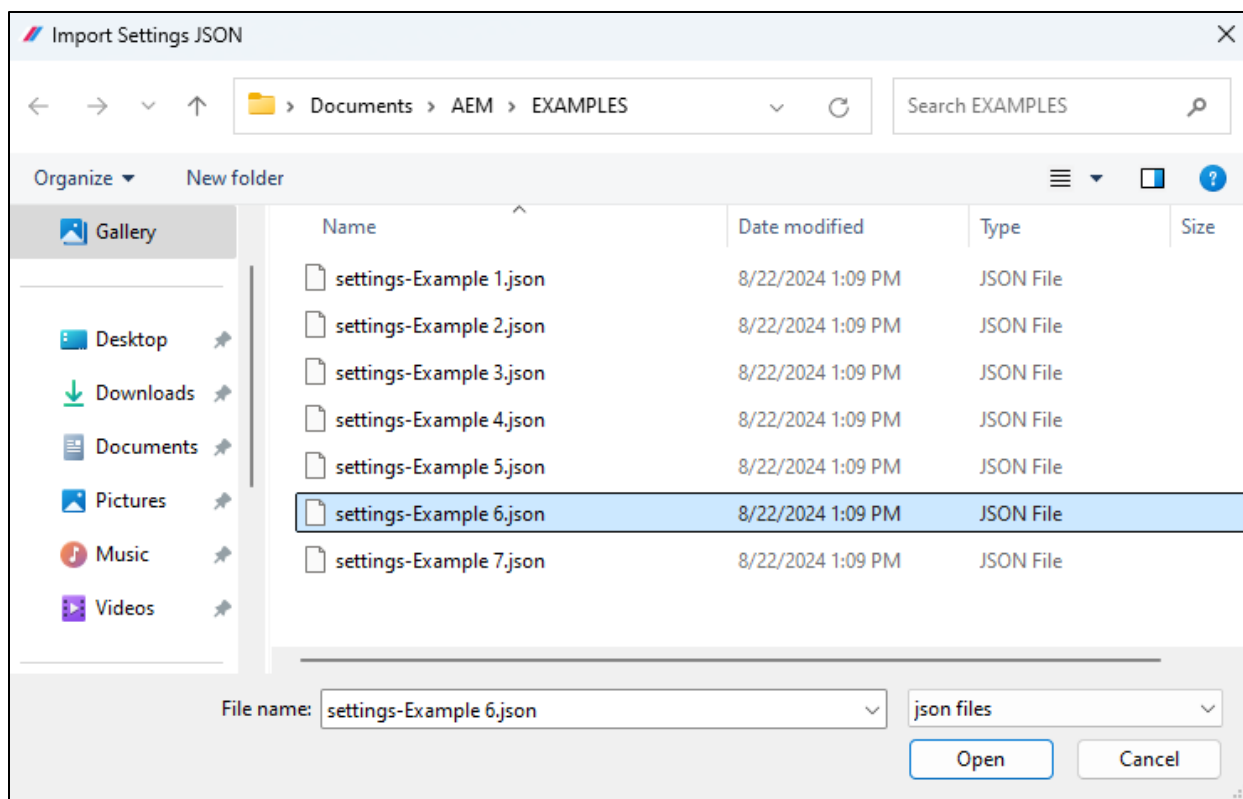


Figure 122. Importing Example 6 Input Condition using the settings-Example 6.json file.

4.6.2 Example 6: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, select the following

- Select the EC EMC from the ACCC SOVENTS module. Provide 30:70 volume %
- Specify salt from the ACCC SALTS module as $\text{LiPF}_6\text{-w-EC-EMC}$. Set maximum salt concentration to 5 molal.
- $T = -40^\circ\text{C}$ to 60°C with 5°C steps,
- Choose default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- Choose No SCAEP; No DL.

Advanced Electrolyte Model (AEM)

The screenshot displays the AEM input interface, organized into several panels:

- TEMPERATURE SETTINGS:** Includes input fields for Temperature Min (-30 °C), Temperature Max (60 °C), and Temperature Step Size (5 °C).
- METHOD FOR HANDLING TRIPLE ION STABILITY:** Provides a dropdown for the method, with a default of $[ABA^+] + [BAB^-]$ and an option for a more complex stability criterion.
- ELECTROLYTE INGRESS INTO PORES:** Features input fields for Contact Angle (35 °), Total Pore Length (20 μm), and Salt Concentration of Interest (1.5 molal).
- SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP):** Includes a checkbox to "Perform SCAEP Calculations".
- DOUBLE LAYER (DL):** Includes a checkbox to "Perform DL Calculations".
- AEM RUN IDENTIFIER (optional):** Includes a text field for the identifier (set to "Example 6") and a checked checkbox for "Append Timestamp".
- SELECT SOLVENTS:** A large section with multiple dropdown menus for selecting solvents from categories like WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, and FROM OTHER CLASSES. It also includes a search bar and a list of selected solvents with their respective concentrations.
- SELECT SALTS:** A section for selecting salts, with dropdowns for categories like FOR BATTERY ELECTROLYTES, FOR AQUEOUS SYSTEMS, and ACCC SALTS. It includes a search bar and a list of selected salts with their concentrations.
- Run AEM:** A large button at the bottom right to execute the simulation.

Figure 123. Example 6 – AEM Input tab.

4.6.3 Example 6: Data Output & Discussion

Simulation outputs are plotted in Figure 125 - Figure 130 for the transport suite of properties (salt diffusivity, conductivity, viscosity, cation transference number). General goals for battery electrolytes are to achieve high diffusivity and conductivity, low viscosity and high cation transference. Since most property metrics exhibit a strong reliance on salt concentration, an effort should be made to identify the range of salt concentration that maximizes the foremost properties of interest. Note that this range should also consider the possible concentration polarization (CP) that can occur over more prolonged cycling elements, since CP will cause a salt depletion at one electrode, while causing an enrichment at the opposite electrode. Given these factors in concert, a feasible overall operational salt concentration range could be defined as 0.5 to 2.0 molal. Note that the effective temperature range for this example is -40 to 70 °C, since AEM adds temperature steps to enable calculation of activation energies over the user-specified range. Also note that for the cation transference values (Report01), the following is given:

- (a) t_+ evaluated with single-ion cations vs all charged species,
- (b) t_+ evaluated with all cation-bearing charged species vs all charged species.

The trends seen over these two approaches are very similar, where their behavior over salt concentration reflect the relative change in the local solvating structure around the lithium cations in relation to that of the anions.

Advanced Electrolyte Model (AEM)

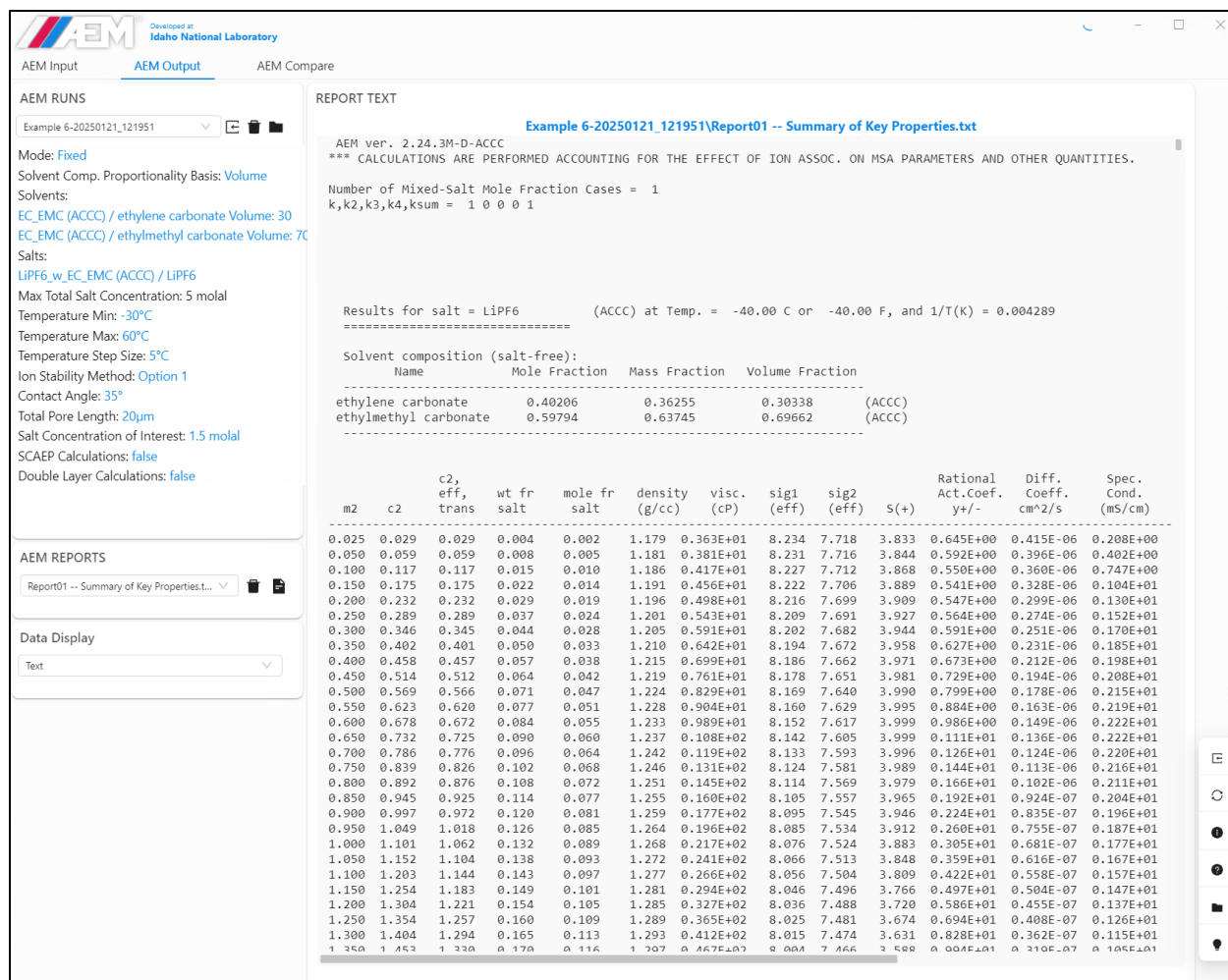


Figure 124. Example 6 – AEM Output tab – REPORT TEXT section to view Report01-Report20.

Advanced Electrolyte Model (AEM)

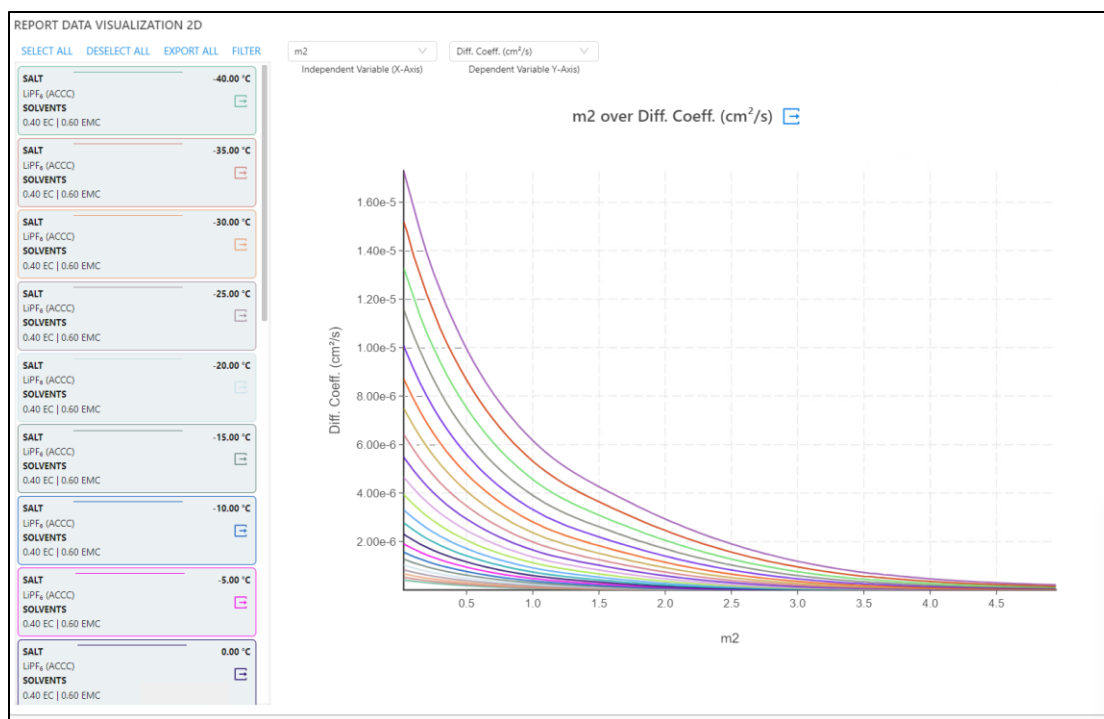


Figure 125. Example 6 – Diffusivity/Diffusion Coefficient (cm²/s) v/s Molal Salt Conc. (m2).

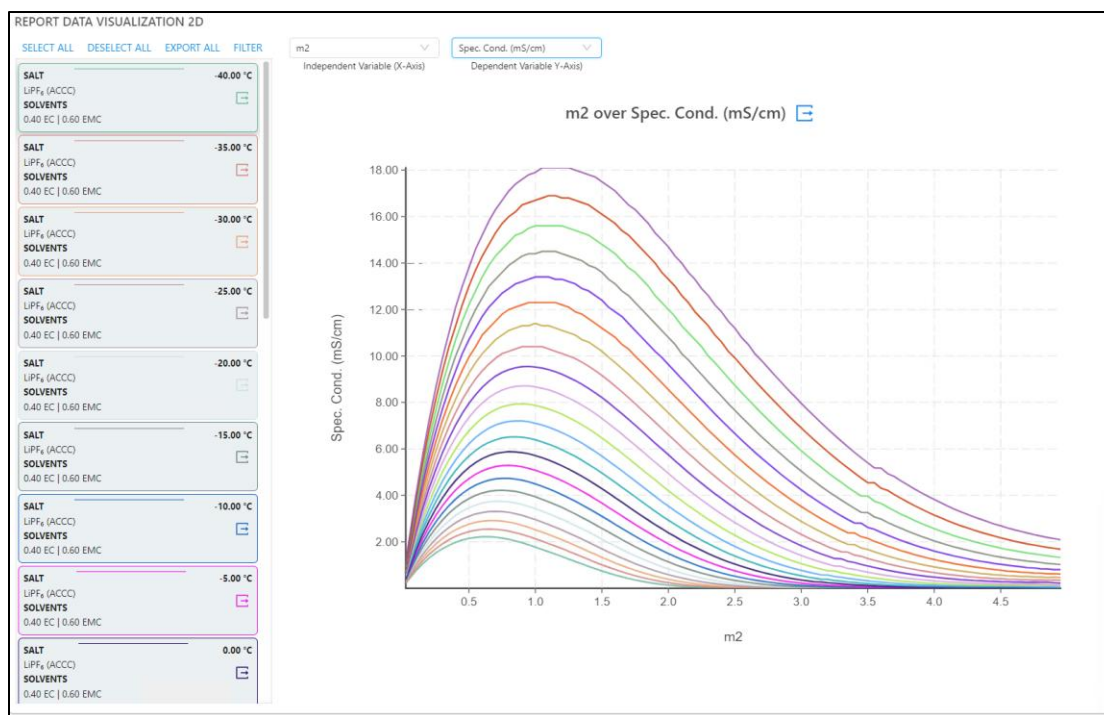


Figure 126. Example 6 – Specific Conductivity (mS/cm) v/s Molal Salt Conc. (m2).

Advanced Electrolyte Model (AEM)

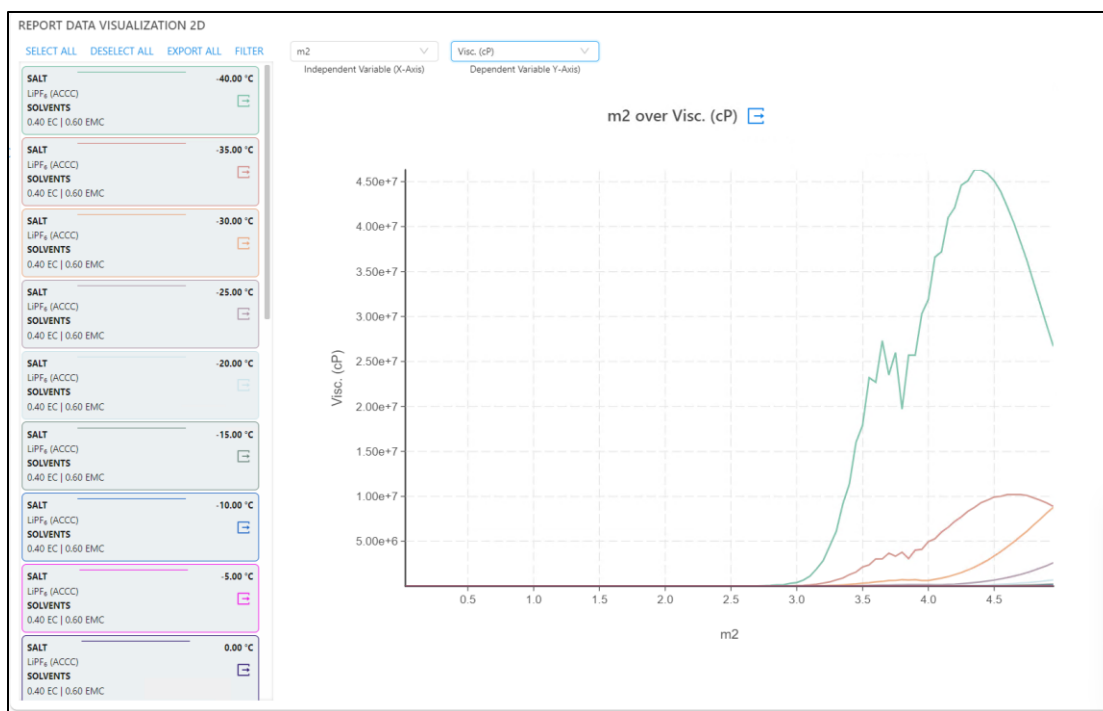


Figure 127. Example 6 – Viscosity (cP) v/s Molal Salt Conc. (m2).

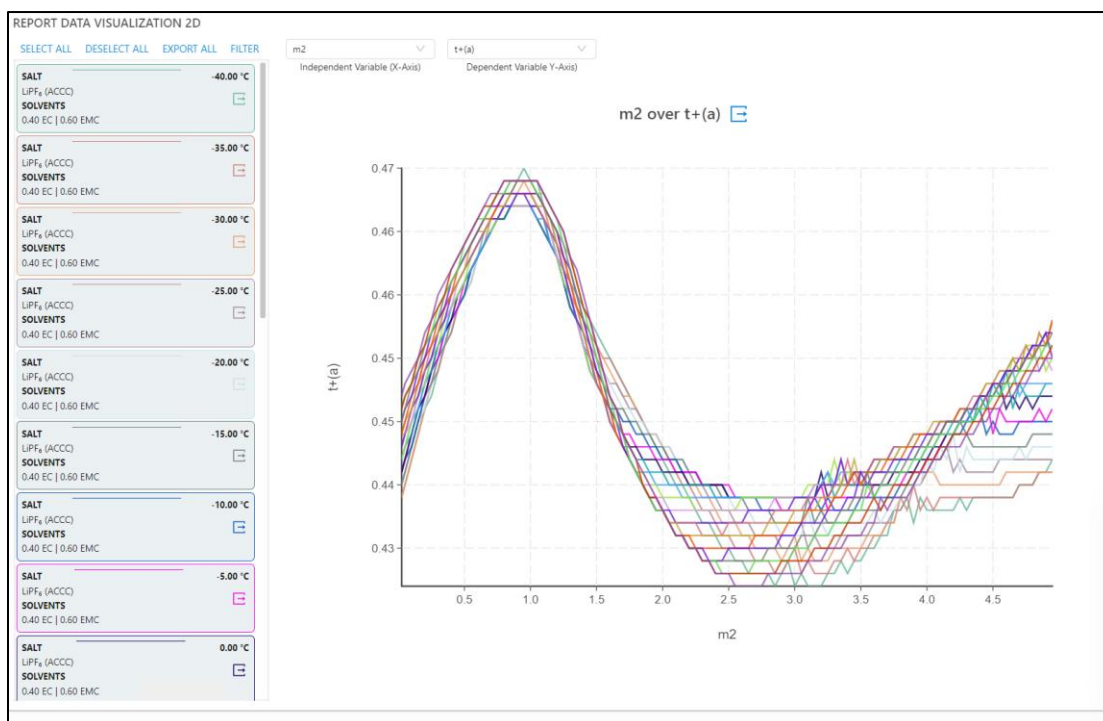


Figure 128. Example 6 – Cation Transference No. (t+(a)) v/s Molal Salt Conc. (m2).

Advanced Electrolyte Model (AEM)

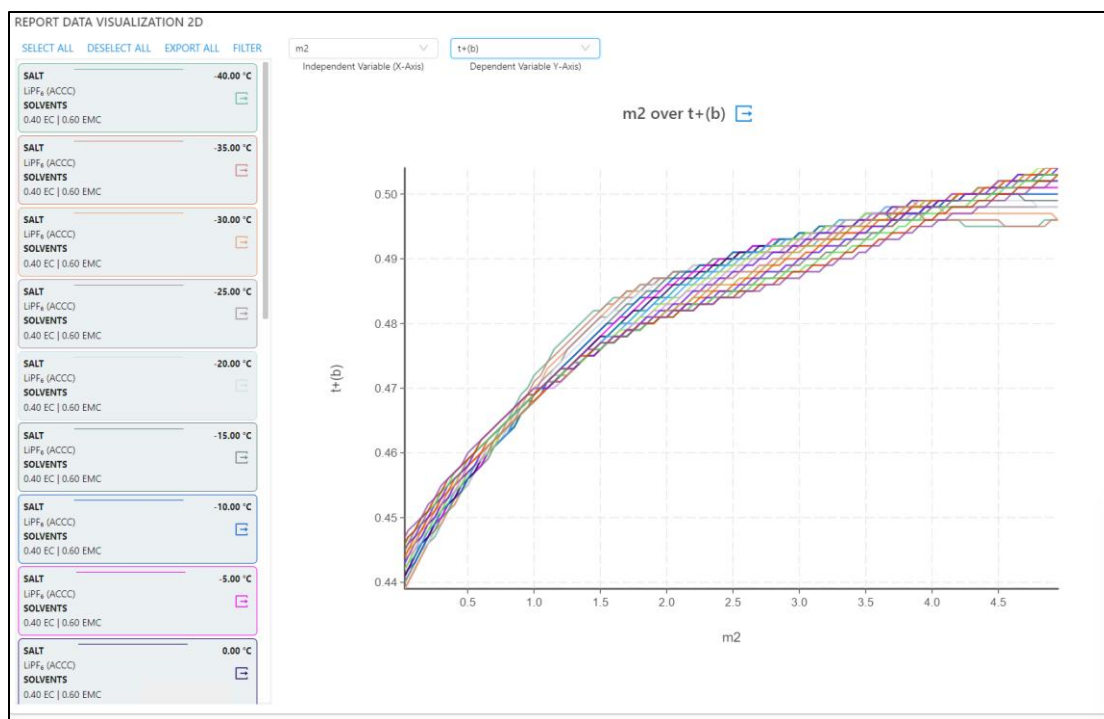


Figure 129. Example 6 – Cation Transference No. ($t+(b)$) v/s Molal Salt Conc. (m_2).

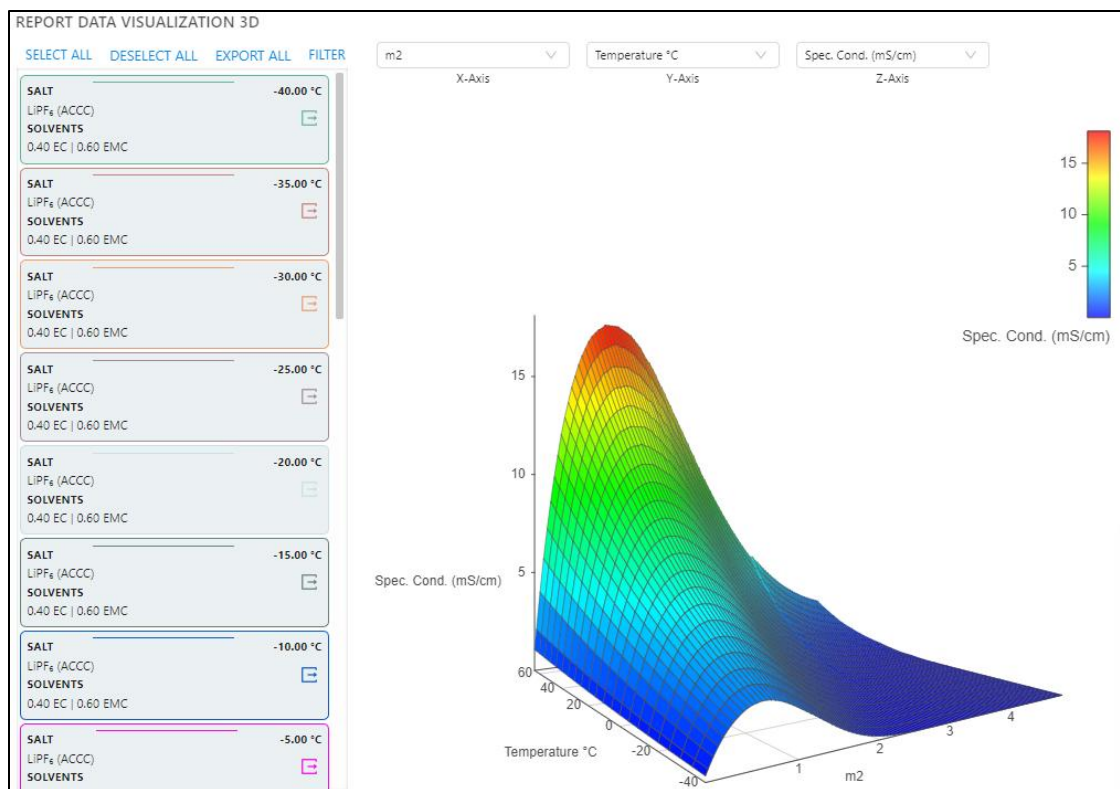


Figure 130. Example 6- Specific Conductivity (mS/cm) vs Molal Salt Conc. (m_2) vs Temperature (deg C) 3D contour

4.7 Example 7: Two Salt Simulation with one Default Library Salt and one ACCC Salt

This example considers the two-salt system. One salt from the ACCC Salt class LiBOB_w_PC_EA and the second one from the default library LiPF6 at a fixed molar proportion (50:50). The salt concentration is allowed to go to 3 molal. Other factors behind the simulation include:

- $T = -20^{\circ}\text{C}$ to 50°C with 5°C steps,
- Used default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- No SCAEP; No DL.

The parameters for electrolyte ingress are relevant estimations for battery electrode materials in general and can be updated to reflect more finely tuned attributes of the porous media.

This relatively simple electrolyte represents a lower-viscosity, higher-conductivity system that would facilitate higher current densities in lithium-ion type cells. This electrolyte system has benefits toward conditions like moderately-fast charging and higher-power applications.

4.7.1 Example 7: Importing Example 7 Parameters

We start off Example 7 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 7. This can be achieved by following the list of instructions below and referencing Figure 131 and Figure 132.

- Click on the **Import** () Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- Select the **settings-Example 7.json** file and click the **Open** button.

Important Note: An error message will pop up if the "settings-Example 7.json" file was not imported successfully.

Advanced Electrolyte Model (AEM)

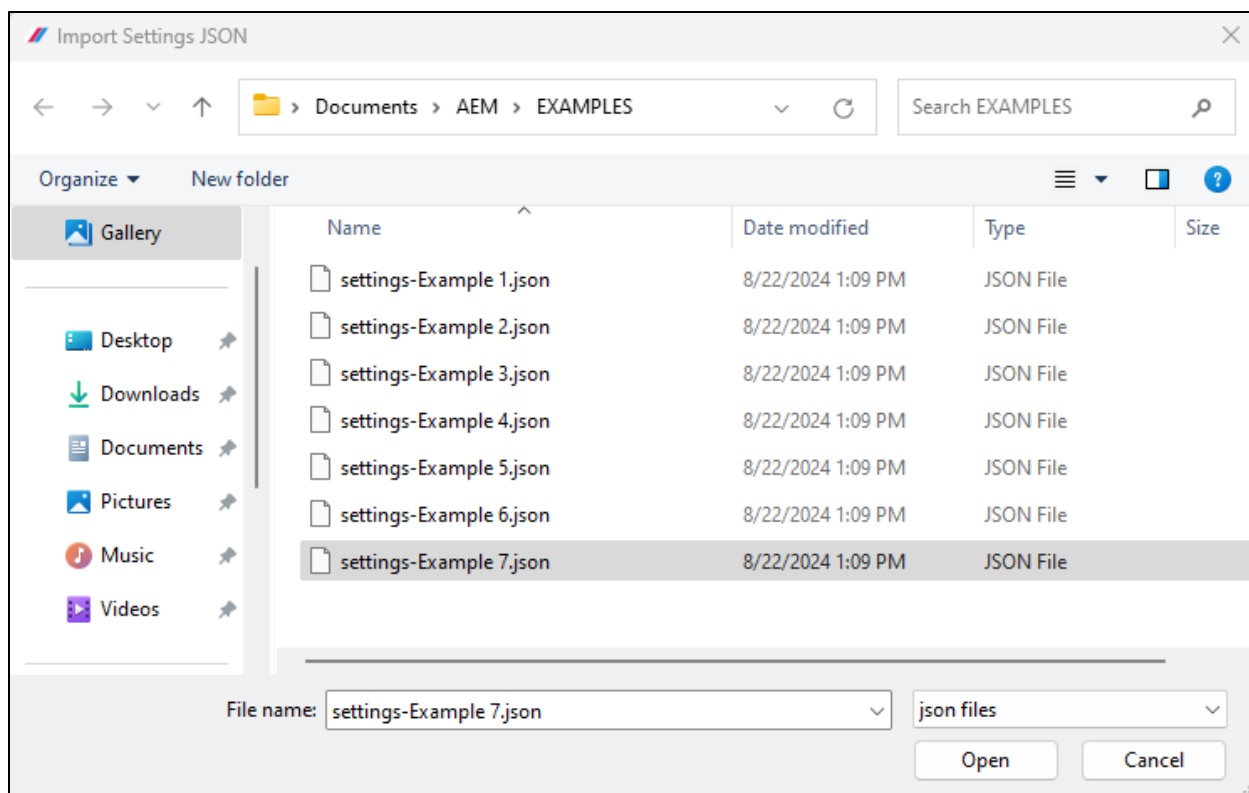


Figure 131. Importing Example 7 Input Condition using the settings-Example 7.json file.

4.7.2 Example 7: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, select the following

- Select a single solvent PC from the default solvent library.
- Select the two salts, LiBOB_w_PC_EA from the ACCC salt class and LiPF6 from the default salt class with 50:50 molar proportion. Set maximum salt concentration to 3 molal,
- Select $T = -20^{\circ}\text{C}$ to 50°C with 10°C steps,
- Use default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal
- Choose No SCAEP; No DL.

Advanced Electrolyte Model (AEM)

The screenshot displays the AEM input interface, organized into several functional sections:

- Top Bar:** Includes a logo, a 'SINGLE FIXED COMPOSITION OF SOLVENTS' button, and a 'LARGER MATRIX OF SOLVENT COMPOSITION FOR OPTIMIZATION OF ELECTROLYTE FORMULATION' button.
- TEMPERATURE SETTINGS:** Features three input fields: 'Temperature Min' (set to -20 °C), 'Temperature Max' (set to 50 °C), and 'Temperature Step Size' (with options for 5 °C and 10 °C).
- METHOD FOR HANDLING TRIPLE ION STABILITY:** Contains a dropdown menu with '[ABA+] = [BAB-]' (default) and a checkbox for '[ABA+] > [BAB-] due to BAB- + B- + AB or [ABA+] < [BAB-] due to ABA+ + A+ + AB'.
- ELECTROLYTE INGRESS INTO PORES:** Includes inputs for 'Contact Angle' (35 °), 'Total Pore Length' (20 μm), and 'Salt Concentration of Interest' (1.5 molal).
- SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP):** Features a checkbox for 'Perform SCAEP Calculations'.
- DOUBLE LAYER (DL):** Includes a checkbox for 'Perform DL Calculations'.
- AEM RUN IDENTIFIER (optional):** Contains a text field for 'AEM Run Identifier' (set to 'Example 7') and a checked checkbox for 'Append Timestamp'.
- SELECT SOLVENTS:** A section with multiple dropdown menus for categories like WATER, CARBONATES, DICARBONATES, GLYMES, ESTERS, and FROM OTHER CLASSES. It also includes a search bar and a 'Run AEM' button.
- SELECT SALTS:** A section with dropdown menus for categories like FOR BATTERY ELECTROLYTES, FOR AQUEOUS SYSTEMS, and ACCC SALTS. It includes a search bar and a 'Run AEM' button.

Figure 132. Example 7 – AEM Input tab.

4.7.3 Example 7: Data Output & Discussion

Simulation outputs are plotted in Figure 134 - Figure 139 for the transport suite of properties (salt diffusivity, conductivity, viscosity, cation transference number). General goals for battery electrolytes are to achieve high diffusivity and conductivity, low viscosity and high cation transference. Since most property metrics exhibit a strong reliance on salt concentration, an effort should be made to identify the range of salt concentration that maximizes the foremost properties of interest. Note that this range should also consider the possible concentration polarization (CP) that can occur over more prolonged cycling elements, since CP will cause a salt depletion at one electrode, while causing an enrichment at the opposite electrode. Note that the effective temperature range for this example is -30 to 55 °C, since AEM adds temperature steps to enable calculation of activation energies over the user-specified range. Also note that for the cation transference values (Report01), the following is given:

- (a) t_+ evaluated with single-ion cations vs all charged species,
- (b) t_+ evaluated with all cation-bearing charged species vs all charged species.

The trends seen over these two approaches are very similar, where their behavior over salt concentration reflect the relative change in the local solvating structure around the lithium cations in relation to that of the anions.

Advanced Electrolyte Model (AEM)

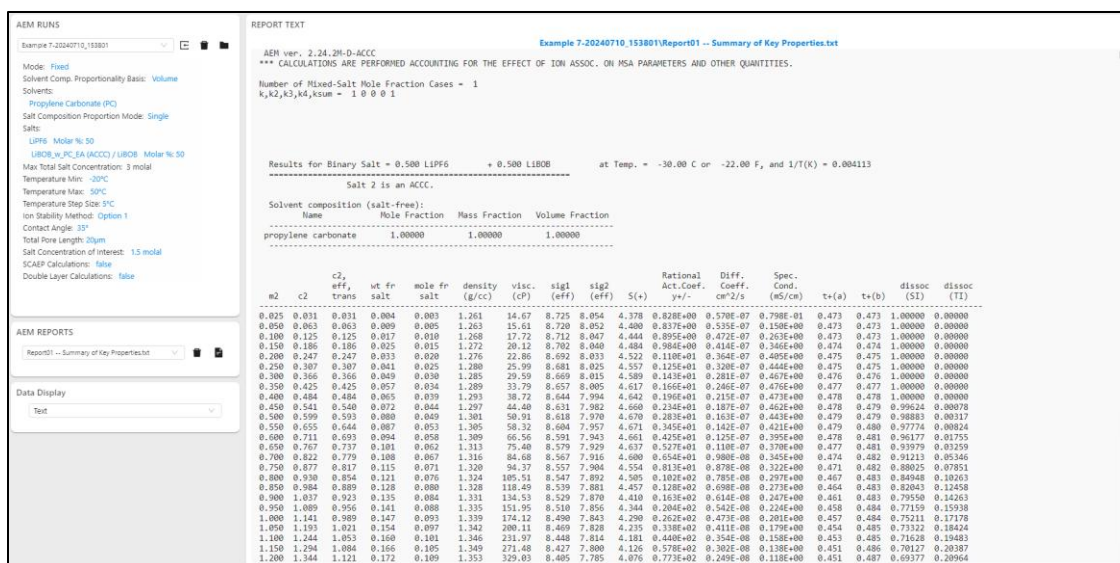


Figure 133. Example 7 - AEM Output tab – REPORT TEXT section to view Report01-Report20.

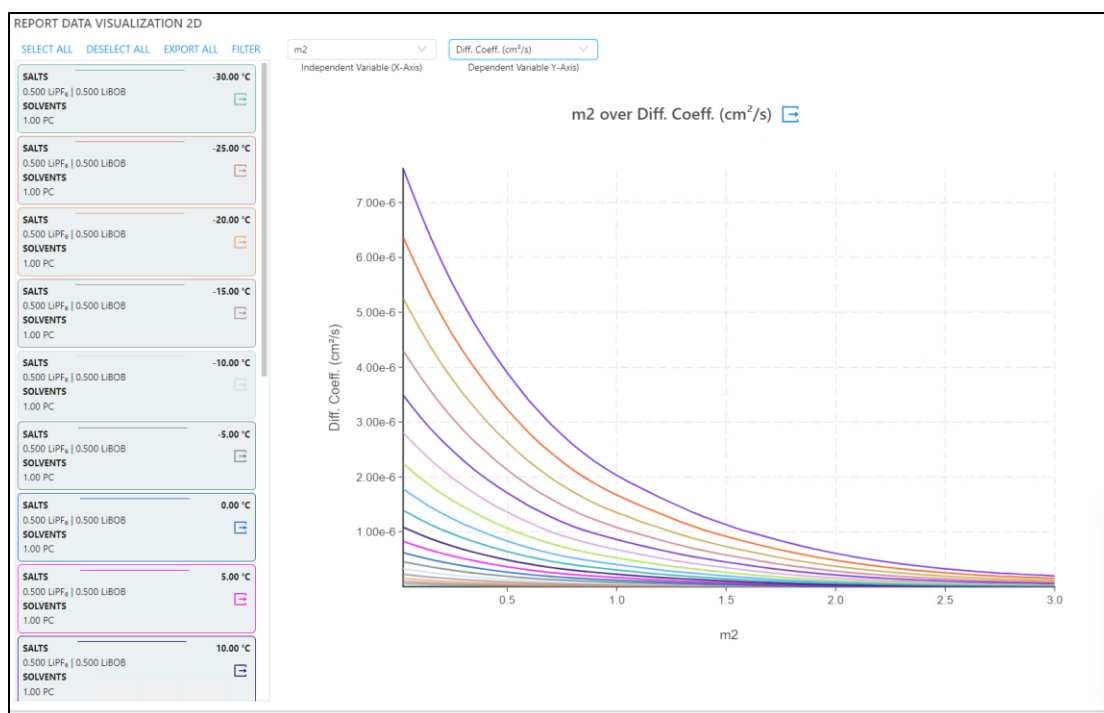


Figure 134. Example 7 – Diffusivity/Diffusion Coefficient (cm²/s) v/s Molal Salt Conc. (m²).

Advanced Electrolyte Model (AEM)

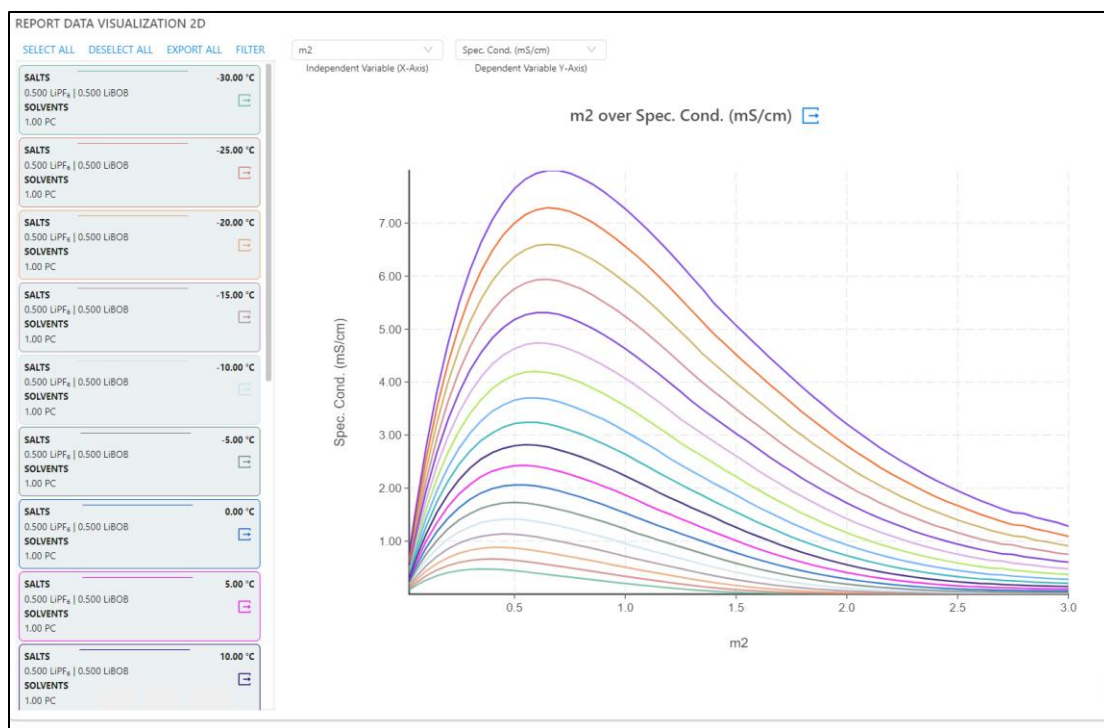


Figure 135. Example 7 – Specific Conductivity (mS/cm) v/s Molal Salt Conc. (m2).

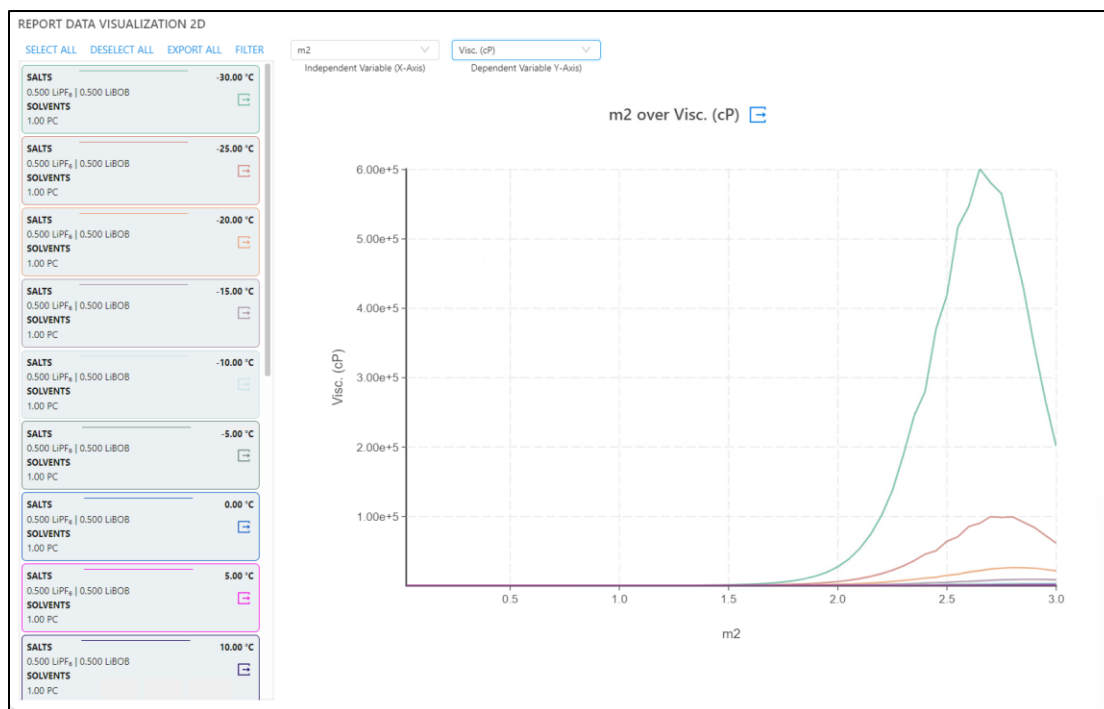


Figure 136. Example 7 – Viscosity (cP) v/s Molal Salt Conc. (m2).

Advanced Electrolyte Model (AEM)

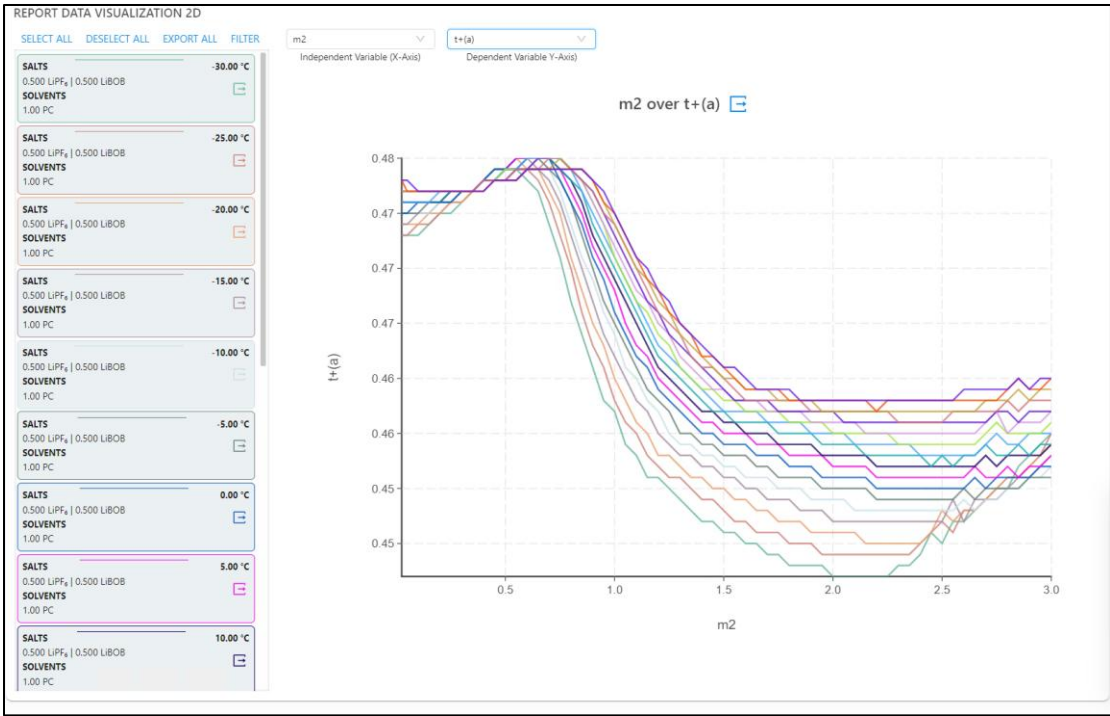


Figure 137. Example 7 – Cation Transference No. (t_+) v/s Molal Salt Conc. (m_2).

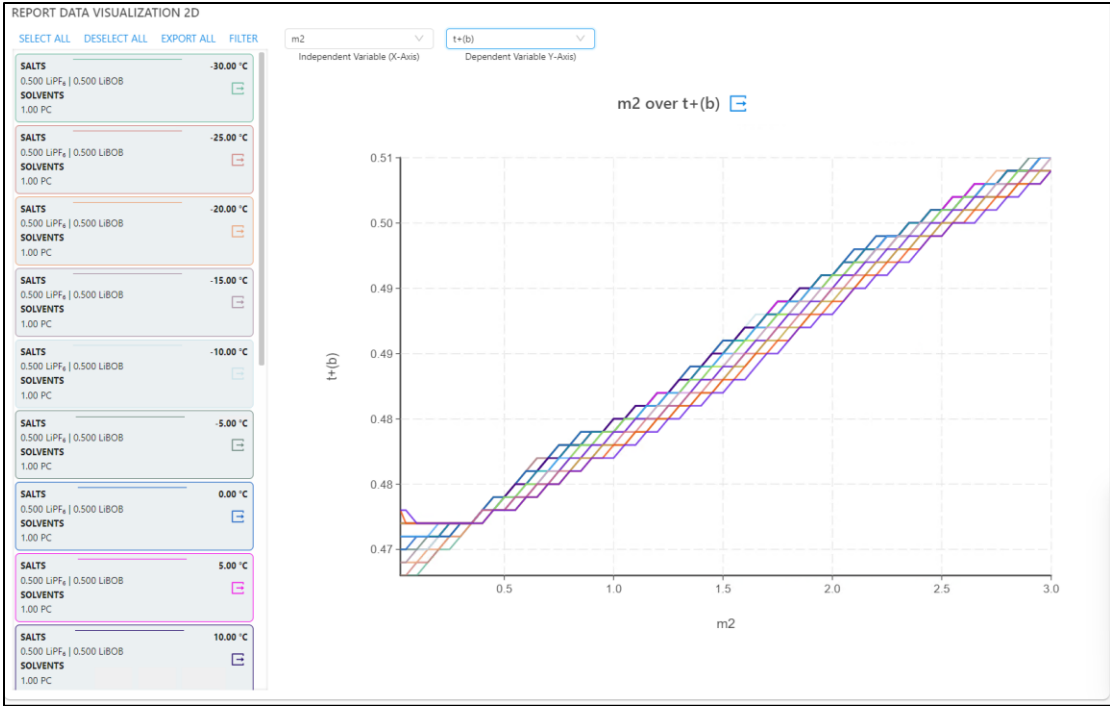


Figure 138. Example 7 – Cation Transference No. (t_+ (b)) v/s Molal Salt Conc. (m_2).

Advanced Electrolyte Model (AEM)

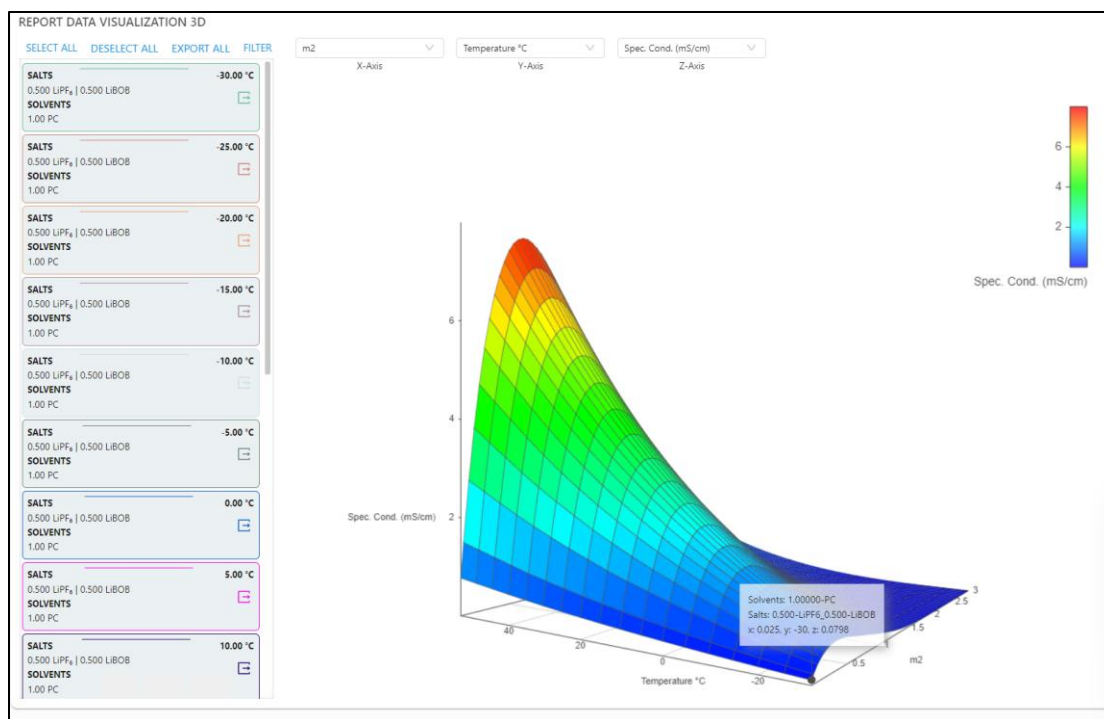


Figure 139. Example 7- Specific Conductivity (mS/cm) vs Molal Salt Conc. (m2) vs Temperature (deg C) 3D contour

4.8 Example 8: AEM Compare: Multi-Simulation Analysis

This example showcases the AEM Compare feature that has been introduced to provide quantitative and qualitative insights by comparing various simulations in the same plot. This considers a two-solvent system from the default library solvents EC and EMC with default salt class as LiPF₆. The salt concentration is allowed to go to 5 molal. Since, this is a multi-simulation analysis, let's consider three simulations with varying volume fractions for the solvents. The various volume fractions combinations considered are:

- EC:EMC (30:70 vol %) (Simulation 1)
- EC:EMC (50:50 vol %) (Simulation 2)
- EC:EMC (70:30 vol %) (Simulation 3)

Other factors behind the simulation include:

- T = 0°C to 60°C with 10°C steps,
- Used default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35°, Pore length = 20µm, salt conc. of interest = 1.5 molal,
- No SCAEP; No DL.

The parameters for electrolyte ingress are relevant estimations for battery electrode materials in general and can be updated to reflect more finely tuned attributes of the porous media.

This relatively simple electrolyte represents a lower-viscosity, higher-conductivity system that would facilitate higher current densities in lithium-ion type cells. This electrolyte system has benefits toward conditions like moderately-fast charging and higher-power applications.

4.8.1 Example 8: Importing Example 8 Parameters

We start off Example 8 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 8. Since there are three simulations, hence, we need to import three files individually, run them and then we can compare. This can be achieved by following the list of instructions below and referencing Figure 140 Figure 122 and Figure 141.

- Click on the **Import** (≡) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- Select the **settings-Example 8a.json** file and click the **Open** button.

Note : Repeat the above steps while importing the other two files named **settings-Example 8b.json** and **settings-Example 8c.json** after running each of them.

Important Note: An error message will pop up if the "settings-Example 8a.json" file was not imported successfully.

Advanced Electrolyte Model (AEM)

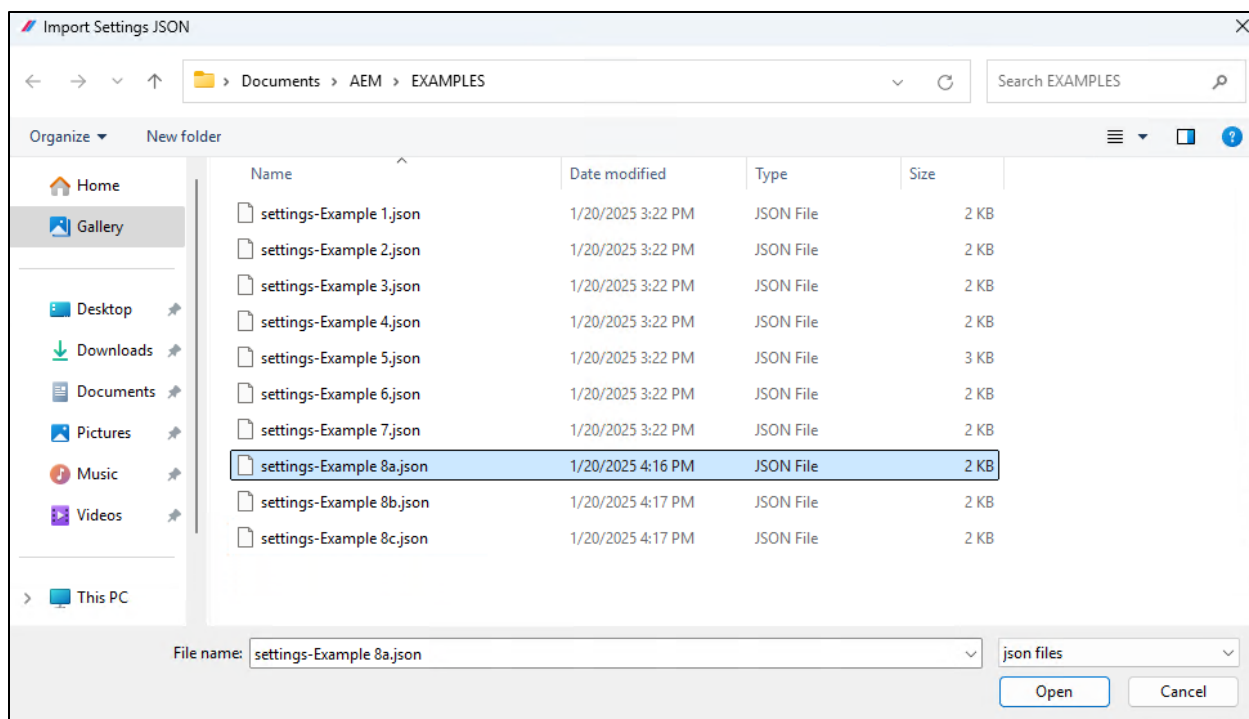


Figure 140. Importing Example 8 simulation 1 (EC:EMC 30:70 %v) Input Condition using the settings-Example 8a.json file.

4.8.2 Example 8: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, select the following

- Select the EC and EMC from the DEFAULT SOVENTS module. Provide 30:70 volume % for the first simulation, followed by 50:50 and 70:30 for the subsequent ones.
- Specify salt from the DEFAULT SALTS module as LiPF_6 . Set maximum salt concentration to 5 molal.
- $T = 0^\circ\text{C}$ to 60°C with 10°C steps,
- Choose default method for handling triple ion stability,
- Inputs for electrolyte ingress into pores: contact angle = 35° , Pore length = $20\mu\text{m}$, salt conc. of interest = 1.5 molal,
- Choose No SCAEP; No DL.

Advanced Electrolyte Model (AEM)

Advanced Electrolyte Model (AEM)

SELECT SOLVENTS (+)

Click Solvent from Category to Add (Max: 10)

WATER CARBONATES DICARBONATES GLYMES ESTERS FROM OTHER CLASSES

S OR P CONTAINING SOLVENTS NITRILES AND DINITRILES PHOSPHAZENE ADDITIVES

ACCC SOLVENTS SEARCH ALL SOLVENTS

CARBONATES / Ethylene Carbonate (EC) 70

CARBONATES / Ethylmethyl Carbonate (EMC) 30

SELECT SALTS (+)

Click Salt from Category to Add (Max: 2)

FOR BATTERY ELECTROLYTES FOR AQUEOUS SYSTEMS ACCC SALTS SEARCH ALL SALTS

FOR BATTERY ELECTROLYTES / LiPF₆ 5 molal

Max Total Salt Concentration

TEMPERATURE SETTINGS

0 °C 60 °C 5 °C 10 °C

Temperature Min Temperature Max Temperature Step Size

METHOD FOR HANDLING TRIPLE ION STABILITY

[ABA⁺] = [BAB⁺] (default)

[ABA⁺] > [BAB⁺] due to BAB⁺ + B⁻ + AB or [ABA⁺] < [BAB⁺] due to ABA⁺ + A⁺ + AB (inequalities are automatically determined)

ELECTROLYTE INGRESS INTO PORES

35 ° 20 μm 1.5 molal

Contact Angle Total Pore Length Salt Concentration of Interest

SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP)

☐ Perform SCAEP Calculations

DOUBLE LAYER (DL)

☐ Perform DL Calculations

AEM RUN IDENTIFIER (optional)

Example 8c

AEM Run Identifier

☐ Append Timestamp

☐ Append Multiple Runs

Figure 141. Example 8- AEM Input Tab for simulation 1 (EC:EMC 30:70 %v)

4.8.3 Example 8: Data Output & Discussion

Simulation comparisons are plotted in Figure 142 - Figure 144 for the transport suite of properties (salt diffusivity, conductivity and viscosity). These comparisons are obtained using the AEM Compare tab where the desired runs to be compared are selected as per instructions given in Section 3.5. General goals for battery electrolytes are to achieve high diffusivity and conductivity and low viscosity. For the comparison purpose we choose a fixed temperature of 30° C from the **FILTER** button within the GUI. Figure 142 shows the visual output for AEM Compare tab showcasing the comparison for Diffusivity/Diffusion Coefficient (cm²/s) for various volume fractions with the same solvents (EC, EMC). The comparison shows that the first simulation (8a) with EC:EMC 30:70 vol % shows higher diffusivity across a wider range of molal concentrations. Hence, this particular formulation is mostly favored for a better battery electrolyte among the three. However, this analysis need further validation from the other parameters shown in Figure 143 and Figure 144.

Advanced Electrolyte Model (AEM)

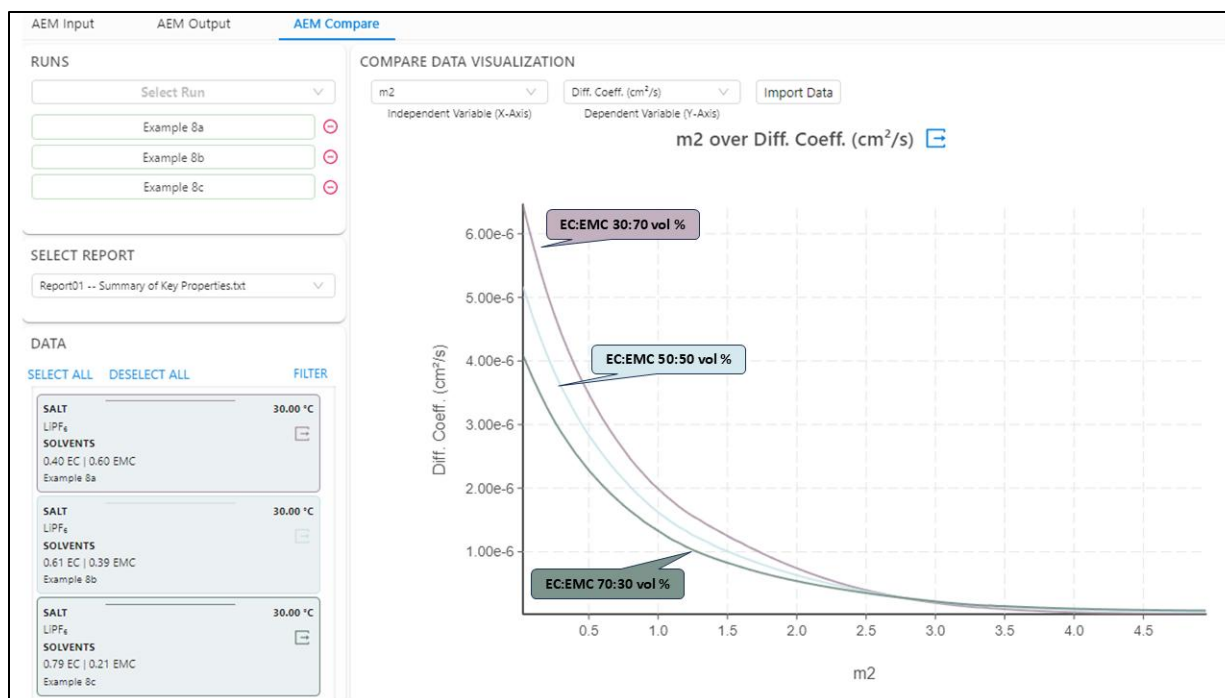


Figure 142. Example 8 – Comparing Diffusivity/Diffusion Coefficient (cm²/s) for various volume fractions with the same solvents at 30°C

Figure 143 shows the visual output for AEM Compare tab showcasing the comparison for Specific Conductivity (mS/cm) for various volume fractions with the same solvents (EC, EMC). The comparison shows that the first simulation (8a) with EC:EMC 30:70 vol % shows higher diffusivity up to 1.8 molal concentration. However, beyond that it drops and the third simulation with EC:EMC 70:30 vol% takes over and achieves the greater value among the three.

Advanced Electrolyte Model (AEM)

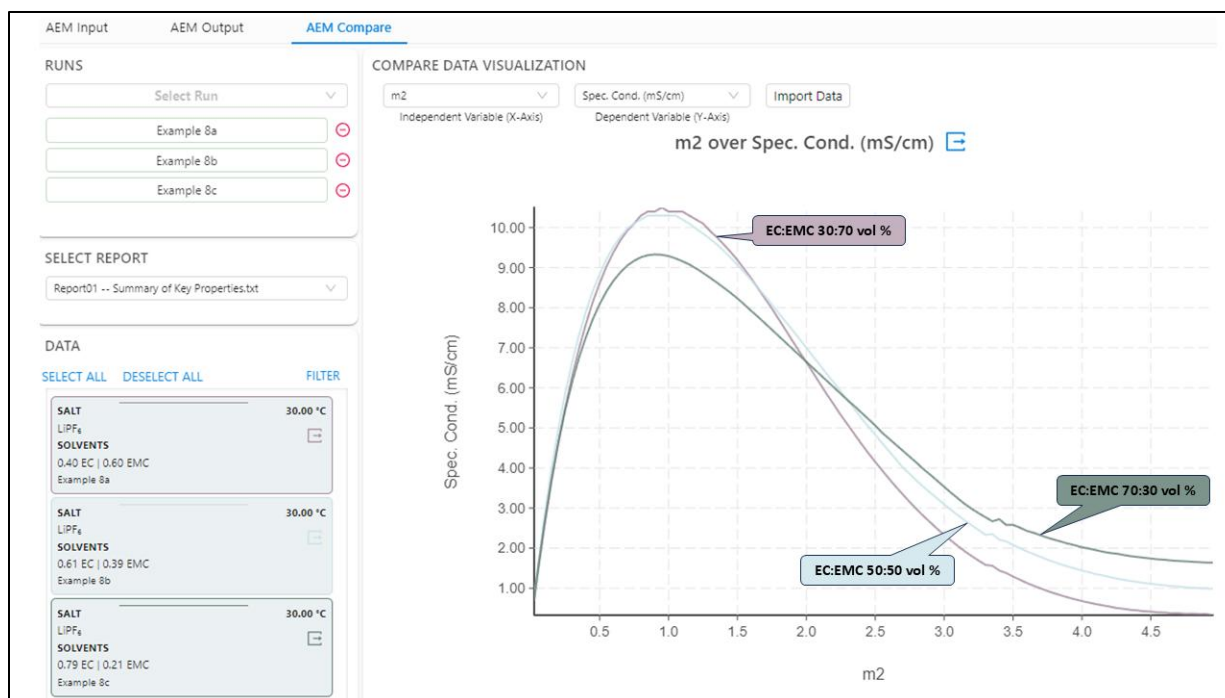


Figure 143. Example 8 – Comparing Specific Conductivity (mS/cm) for various volume fractions with the same solvents at 30°C

Figure 144 shows the visual output for AEM Compare tab showcasing the comparison for Viscosity (cP) for various volume fractions with the same solvents (EC, EMC). The comparison shows that the first simulation with EC:EMC 30:70 vol % has lower viscosity among the three up to 2.7 molal. However, beyond that range, EC:EMC 70:30 (Simulation 3) has the lowest viscosity, indicating better flow characteristics at higher concentrations.

Advanced Electrolyte Model (AEM)

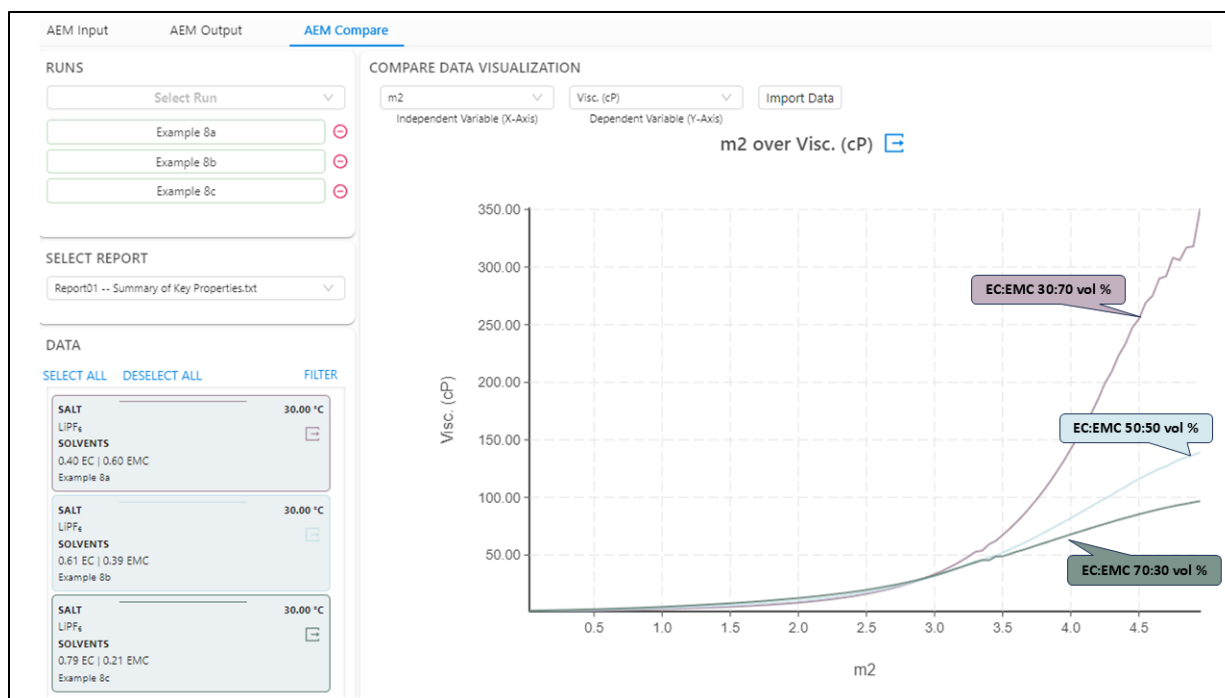


Figure 144. Example 8 – Comparing Viscosity (cP) for various volume fractions with the same solvents at 30°C

The choice of electrolyte depends on the specific application requirements and the concentration range at which the electrolyte will be used. If the application prioritizes broad diffusivity performance across a wide range of molal concentrations, **Simulation 1 (EC:EMC 30:70)** is the most suitable option due to its consistently high diffusivity. For applications operating within moderate concentration ranges, **Simulation 1** is again recommended as it exhibits the highest conductivity and lowest viscosity up to 1.8 molal and 2.7 molal, respectively. However, for high-concentration scenarios, **Simulation 3 (EC:EMC 70:30)** emerges as the better choice, offering superior conductivity beyond 1.8 molal and maintaining the lowest viscosity beyond 2.7 molal. These attributes make it particularly advantageous for high-performance applications at elevated molal concentrations.

Simulation 1 (EC:EMC 30:70) stands out as a versatile option, providing a balanced performance across diffusivity, conductivity, and viscosity, especially in lower-to-moderate concentration ranges. This makes it well-suited for a wide array of applications where consistent ionic transport and low resistance to flow are critical. On the other hand, **Simulation 3 (EC:EMC 70:30)** is optimized for higher concentration regimes, making it a strong candidate for use cases where enhanced ionic mobility and reduced viscosity at elevated concentrations are essential. When selecting an electrolyte, it is also important to consider factors such as thermal stability and long-term performance under the intended operating conditions, particularly for applications that involve battery safety and extended cycling. These considerations will help ensure the electrolyte aligns with both performance and safety requirements.

4.9 Example 9: Custom Recipe Input Mode

This example demonstrates AEM's new **Custom Recipe Input Mode**, enabling mass-% inputs for solvents and molality (mol/kg solvent) for salts—directly aligned with typical laboratory practices. The recipe below showcases a high-performance Li-ion electrolyte designed for fast charging and high-voltage operation.

- **Primary solvents (wt%):**
 - Ethylene Carbonate (EC): 35%
 - Dimethyl Carbonate (DMC): 30%
 - Fluoroethylene Carbonate (FEC): 15%
- **Co-solvents (wt%):**
 - 1,2-Dimethoxyethane (DME): 15%
 - Tetrahydrofuran (THF): 5%
- **Salts (molal):**
 - LiPF₆: 0.8 m
 - LiFSI: 0.4 m

Target use case: Temperature performance from –30 °C to 60 °C, relevant for high-power Li-ion cells. AEM handles full mass and molal conversions internally, preserving experimental realism while using mass- and mole-based inputs for simulation.

4.9.1 Example 9: Importing Example 9 Parameters

We start off Example 9 by first using the **Import** Button from Section 3.6 to prepopulate each of the AEM input conditions for Example 9. This can be achieved by following the list of instructions below and referencing Figure 145 and Figure 146 .

- d) Click on the **Import** (≡) Button in the **Global Features & Icons** section in the lower left corner of the GUI.
- e) Verify that the GUI prompts to the **C:\Users\user-name\Documents\AEM\EXAMPLES** directory in Windows File Explorer.
- f) Select the **settings-Example 9.json** file and click the **Open** button.

Important Note: An error message will pop up if the "settings-Example 9.json" file was not imported successfully.

Advanced Electrolyte Model (AEM)

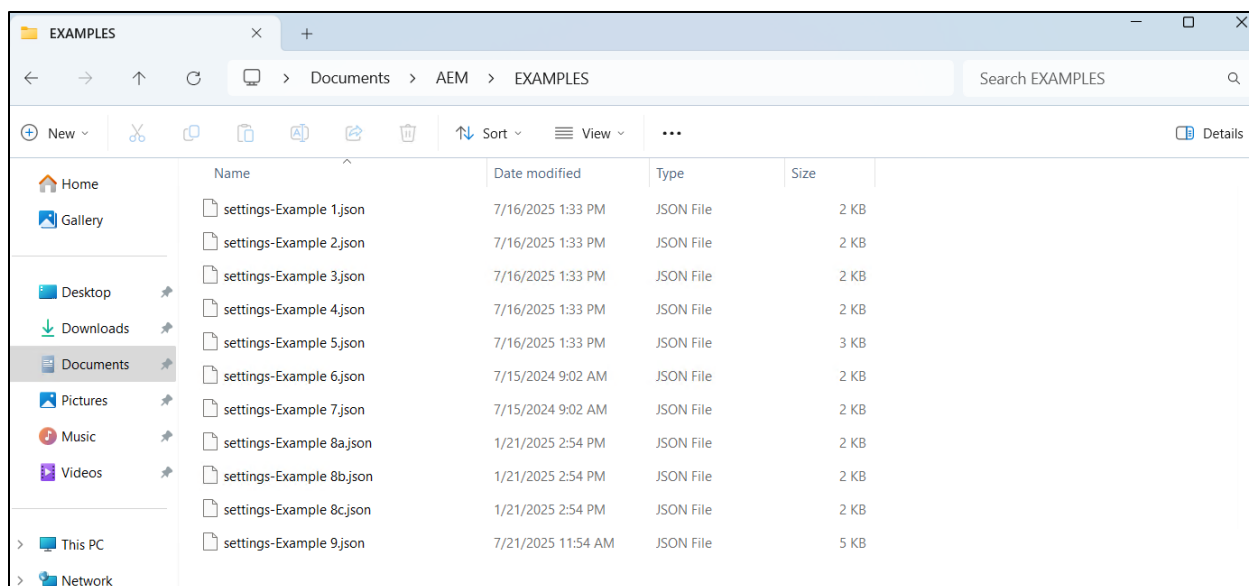


Figure 145. Importing Example 9 for a simulation using the Custom Recipe Input mode as defined in the settings-Example 9.json file.

4.9.2 Example 9: Overview of Simulation Input Conditions

Based on the conditions mentioned at the beginning of this example, verify the following input conditions have been entered:

- Custom Recipe Mode in AEM has been selected.
- Primary Solvent have been defined as:
 - EC:DMC:FEC (35%: 30%: 15% in wt%)
- Co-solvents have been define as:
 - DME, THF (15%: 5% in wt%)
- Salts have been specified by molality:
 - LiPF_6 (0.8 m), LiFSI (0.4 m)
- In AEM Conversion Table:
 - Verify primary solvent mass is defaulted to 1 kg and preview conversion results for mass fractions, molar fractions, total molality, densities, molar masses, etc. as shown in Figure 146.
- Proceed to simulation setup: max total salt molality (5 molal), temperature range (–30 to 60 °C), etc.

Advanced Electrolyte Model (AEM)

SINGLE FIXED COMPOSITION OF SOLVENTS

LARGER MATRIX OF SOLVENT COMPOSITION
FOR OPTIMIZATION OF ELECTROLYTE FORMULATION

Fixed Composition Solvent/Salt Input Mode

Default Input Mode Recipe Input Mode

SELECT PRIMARY SOLVENTS +

Click Solvent from Category to Add

WATER CARBONATES DICARBONATES GLYMES ESTERS FROM OTHER CLASSES
 S OR P CONTAINING SOLVENTS NITRILES AND DIINITRILES PHOSPHAZENE ADDITIVES
 ACC SOLVENTS SEARCH ALL SOLVENTS

Ethylene Carbonate (EC)
CARBONATES

35 %

⊖

Ethylmethyl Carbonate (EMC)
CARBONATES

30 %

⊖

Diethyl Carbonate (DEC)
CARBONATES

15 %

⊖

SELECT CO-SOLVENTS +

Click Solvent from Category to Add

WATER CARBONATES DICARBONATES GLYMES ESTERS FROM OTHER CLASSES
 S OR P CONTAINING SOLVENTS NITRILES AND DIINITRILES PHOSPHAZENE ADDITIVES
 ACC SOLVENTS SEARCH ALL SOLVENTS

1,2 Dimethoxyethane (DME)
GLYMES

15 %

⊖

Tetrahydrofuran (THF)
FROM OTHER CLASSES

5 %

⊖

SELECT SALTS +

Click Salt from Category to Add (Max 2)

FOR BATTERY ELECTROLYTES FOR AQUEOUS SYSTEMS FOR AQUEOUS FLOW BATTERIES
 ACC SOLTS SEARCH ALL SALTS

LiPF₆
FOR BATTERY ELECTROLYTES

0.8 m

⊖

LiFSI
FOR BATTERY ELECTROLYTES

0.4 m

⊖

5 molal

Max Total Salt Concentration

RECIPE BREAKDOWN

Composition Summary Solvents/Salts Properties

1000 grams

Primary Solvent Mass

Category	Component	Input Solvent Mass %	Input Salt Molality	Mass (g)	Moles	Total Mass %	Total Moles %	AEM Solvent Mass %	AEM Salt Molar %
Solvent	EC	20%	---	200.00	2.27	13.93%	17.23%	16.67%	---
Solvent	EMC	30%	---	300.00	2.88	20.90%	21.86%	25.00%	---
Solvent	DEC	50%	---	500.00	4.23	34.83%	32.11%	41.67%	---
Co-Solvent	DME	15%	---	150.00	1.66	10.45%	12.63%	12.50%	---
Co-Solvent	THF	5%	---	50.00	0.69	3.48%	5.26%	4.17%	---
Salt	LiPF ₆	---	0.8m	145.82	0.96	10.16%	7.28%	---	66.67%
Salt	LiFSI	---	0.4m	89.79	0.48	6.25%	3.64%	---	33.33%
Total		120%	1.2000000	1435.62	13.18	100.00%	100.00%	100.00%	100.00%

TEMPERATURE SETTINGS

-30 °C 60 °C 5 °C 10 °C

Temperature Min Temperature Max Temperature Step Size

METHOD FOR HANDLING TRIPLE ION STABILITY

[ABA*] = [BAB*] (default)
 [ABA*] > [BAB*] due to BAB* → B* + AB or
 [ABA*] < [BAB*] due to ABA* → A* + AB
 (inequalities are automatically determined)

ELECTROLYTE INGRESS INTO PORES

35 ° 20 μm 1.5 molal

Contact Angle Total Pore Length Salt Concentration of Interest

SURFACE-CHARGE ATTENUATED ELECTROLYTE PERMITTIVITY (SCAEP)

☐ Perform SCAEP Calculations

DOUBLE LAYER (DL)

☐ Perform DL Calculations

AEM RUN IDENTIFIER (optional)

Example 9 ☒ Append Timestamp

AEM Run Identifier

Figure 146. Example 9- AEM Input Tab for Custom Recipe Simulation with primary solvents EC:EMC:DEC (35%: 30%: 15%), co-solvents of DME:THF (15%:5%), and salts LiPF₆:LiFSi (0.8m: 0.4m).

4.9.3 Example 9: Data Output & Discussion

As seen in previous AEM simulations, the simulation outputs include temperature-dependent electrolyte properties (viscosity, conductivity, ion diffusivity, transference numbers) across a user defined temperature range. A preview of Conductivity is shown in Figure 145.

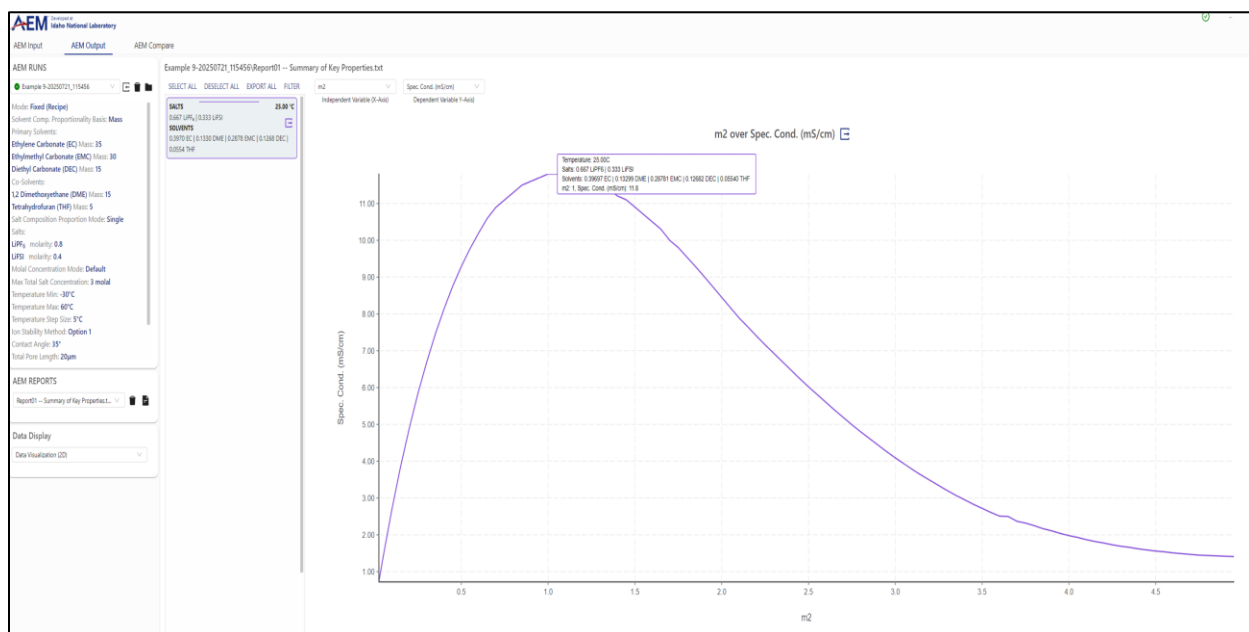


Figure 147. Example 9 – AEM output preview for conductivity (filtered at 25 °C).

Upon running the simulation using the Custom Recipe Input Mode, AEM generates a complete set of temperature-dependent electrolyte properties over the range of -30°C to 60°C in 5°C increments. These include key transport and thermodynamic metrics such as ionic conductivity, salt diffusivity, viscosity, cation transference number, and related figures of merit essential for evaluating electrolyte performance across a broad operating envelope.

AEM's automated conversion engine first processes the mass- and molality-based user inputs into internally required formats, then outputs a comprehensive conversion table (available in .txt or .json) summarizing:

- Mass and molar fractions of all solvents and salts
- Total molality and ionic strength
- Reference compound metadata (molecular weight, density, CAS#, SMILES)
- Computed values for derived properties such as free liquid fraction, Debye length, and activation energies (if applicable)

This simulation underscores the value of the Custom Recipe Mode for modeling advanced multi-component electrolyte systems. By accurately capturing the interactions between

carbonate and ether-based solvents and multiple salts, AEM provides high-resolution insight into formulation-performance relationships. In particular:

1. The combination of LiPF_6 and LiFSI leverages the conductivity of PF_6^- systems with the electrochemical and thermal stability of FSI^- anions.
2. The inclusion of co-solvents like DME and THF enhances low-temperature fluidity and helps offset viscosity increases from EC/FEC, extending performance into sub-zero conditions.
3. The model's melting point and free liquid calculations are critical in capturing physical transitions that impact ion transport near and below 0 °C.

Together, these outputs support electrolyte engineers in identifying optimized formulations for fast-charging, high-voltage, and low-temperature Li-ion applications.

5 AEM Command Line Interface (CLI)

5.1 AEM CLI Overview

The AEM Graphical User Interface (GUI) covered in Section 3 and Section 4, utilizes the original AEM Command Line Interface (CLI) executable program. Within both AEM programs, the software is organized based on the composition of the substituents that go in the user's desired electrolyte as well as other key simulation conditions.

For the AEM CLI program there are six sections of settings: A. Input Electrolyte Component List and Proportions, B. Input Other Simulation Conditions, C. Choose whether Surface-Charge Attenuated Electrolyte Permittivity (SCAEP) Calculations are Performed, D. Choose whether Double Layer (DL) Calculations are Performed, and E. Output Files.

Important Note: The traditional AEM CLI program is not available in the AEM Trial Version of the software. Contact Ridgetop Group or INL directly if you wish to upgrade to the full product version of AEM.

5.2 Running Example 1 with AEM CLI

Important Note: This section only describes running Example 1 with the AEM CLI. The user can simulate other examples by following the process outlined for Example 1. If further technical support or guidance is needed, contact a Ridgetop or INL representative directly.

5.2.1 Setup to Run Example 1 with AEM CLI

We start off Example 1 by first locating the AEM CLI executable in the AEM program directory. The executable file is named **aem-2242m-d-acc** and is in **...\\AEM\\CLI**. A visual of this file is shown in Figure 148, but do note that **"2242"** corresponds to the AEM CLI version number and your version may be different. To execute AEM on the CLI, right click on the file and select **Open**

Advanced Electrolyte Model (AEM)

or **Run as Administrator**, or alternatively **double-click it**. Next, verify that the command prompt opens with the following welcome message as shown in Figure 149.

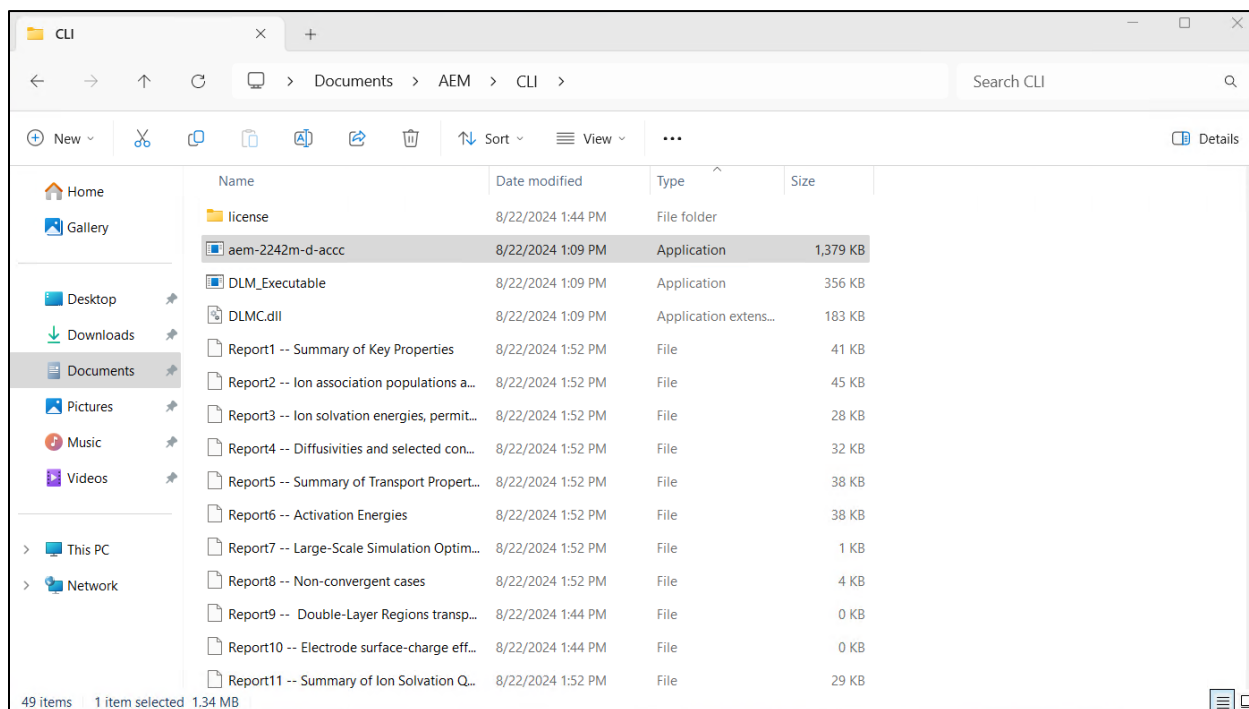


Figure 148. aem-2242m-d-acc CLI Executable in ...\\AEM\\CLI\\.

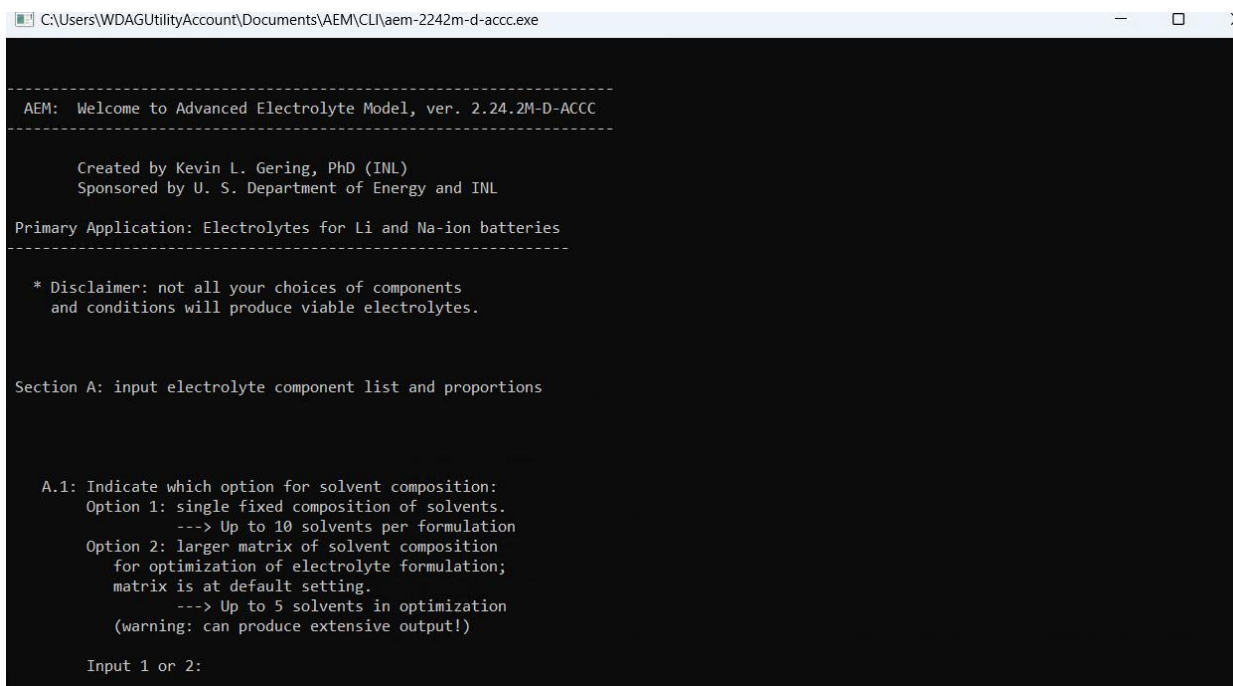


Figure 149. Welcome Message prompt when AEM is first run.

5.2.2 Section A: Input Electrolyte Component List and Proportions:

This section defines the user's desired electrolyte's input components and their proportion. This section contains 6 subsections based on the user's desired qualitative and quantitative inputs.

A.1: Indicate which option for solvent composition:

This is where the user chooses from the two available options for solvent composition. As shown in Figure 150, for Example 1, we use **Option 1: single fixed composition of solvents**, which will run a finite number of solvents to formulate the composition of the electrolyte.

```
A.1: Indicate which option for solvent composition:
      Option 1: single fixed composition of solvents.
                ---> Up to 10 solvents per formulation
      Option 2: larger matrix of solvent composition
                for optimization of electrolyte formulation;
                matrix is at default setting.
                ---> Up to 5 solvents in optimization
                (warning: can produce extensive output!)

      Input 1 or 2:
                    1
```

Figure 150. Section A.1 from AEM Fortran Version 2.24.2

When **Option 1** is chosen, the program prompts the user to choose the composition proportionality basis as either **Volume** or **Mass**. As shown in Figure 151, for Example 1, we use **Volume** which will fix the volume proportions of our solvents to formulate the composition of the electrolyte.

```
For Option 1, choose the composition proportionality basis:
  1 = Volume
  2 = Mass

      Input 1 or 2:
                    1
```

Figure 151. Prompt to Input 1 for Section A.1 from AEM Fortran Version 2.24.2.

A.2: Indicate number of solvents, integer up to 10:

This is where the user defines the number of solvents that are combined to formulate the desired electrolyte. The maximum number of solvents allowed as inputs to the program using **Option 1: single fixed composition of solvents**, is **ten (10)**. As shown in Figure 152, for Example 1, we use **three (3)** solvents.

```
A.2: Indicate number of solvents, integer up to 10
      3
```

Figure 152. Section A.2F from AEM Fortran Version 2.24.2.

A.3: Choose solvent(s) from the following library list:

This is where the user defines the type of solvent (based on existing chemistries in the AEM library) for each solvent specified in A.3. If applicable, this is also where the user can define how many ACCC Solvents are included in the simulation. As shown in Figure 153 and Figure 154, for Example 1, we do not include any ACCC solvents and use **3: ethylene carbonate (EC)**, **8: dimethyl carbonate (DMC)**, and **12: ethyl propionate (EP)**.

```
A.3: Choose solvent(s) from the following library list:

      1: water

      CARBONATES:
        2: propylene carbonate (PC)
        3: ethylene carbonate (EC)
        6: ethylmethyl carbonate (EMC)
        7: diethyl carbonate (DEC)
        8: dimethyl carbonate (DMC)
        15: fluoroethylene carbonate (FEC)
        39: vinylene carbonate (VC)
        45: vinyl ethylene carbonate (VEC)

      DICARBONATES:
        49: Dimethyl 2,5, Dioxahexanedioate (DMOHC)
        50: Diethyl 2,5, Dioxahexanedioate (DEOHC)

      GLYMES
        5: 1,2-dimethoxyethane (DME)
        59: 1,2-dimethoxypropane (DMP)
        20: DEGDME (diglyme)
        21: 3-TEGDME (triglyme)
        22: 4-TEGDME (tetraglyme)

      S or P CONTAINING SOLVENTS:
        16: ethylmethoxyethyl sulfone (EMES)
        28: ethylmethyl sulfone (EMS)
        34: sulfolane (TMS)
        35: trimethyl phosphate (TMP)
        55: triethyl phosphate (TEP)
        46: (proprietary, unavailable)
        60: dimethyl sulfite (DMS)

      FROM OTHER CLASSES:
        4: gamma-butyrolactone (GBL)
        10: 1,3 dioxolane
        14: ethylene glycol (EG)
        17: Dowanol-PMA (DOW-PMA)
        18: 2-methoxyethyl acetate (MEA)
        19: 1NM2-siloxane (1NM2)
        47: Tris(2,2,2-trifluoroethyl)
           -orthoformate (TFEO)
        51: 7-oxabicyclo[2.2.1]heptane
           -2-carbonitrile (OCN)
        52: tetrahydrofuran (THF)
        53: (proprietary, unavailable)
        54: (proprietary, unavailable)
        56: Bis(2,2,2-trifluoroethyl) ether (BTFE)
        57: 1,1,2,2-tetrafluoroethyl-
           2,2,3,3-tetrafluoropropyl ether (TTE)
        63: Fluorobenzene (FBZ)

      ESTERS:
        9: ethyl acetate (EA)
        11: methyl butyrate (MB)
        12: ethyl propionate (EP)
        13: n-propyl acetate (nPA)
        29: methyl acetate (MA)
        30: methyl propionate (MP)
        31: methyl formate (MF)
        32: ethyl formate (EF)
        37: methyl monofluoroacetate (M1FA)
        38: methyl difluoroacetate (M2FA)
        58: ethyl difluoroacetate (E2FA)
        40: n-propyl propionate (nPP)
        42: n-butyl acetate (nBA)
        43: n-butyl propionate (nBP)
        44: n-butyl butyrate (nBB)

      NITRILES AND DINITRILES:
        26: glutaronitrile (GLN)
        27: adiponitrile (ADN)
        33: propionitrile (PN)
        36: acetonitrile (AN)
        41: butyronitrile (BN)
        48: succinonitrile (SN)
        61: isovaleronitrile (IVN)
        62: isobutyronitrile (IBN)

      PHOSPHAZENE ADDITIVES:
        23: INL Additive FM2
        24: INL Additive SM4
        25: INL Additive PA5
```

Figure 153. Section A.3 (1) from AEM Fortran Version 2.24.2.

Advanced Electrolyte Model (AEM)

```
--> You have an option to designate some solvents as Arbitrarily-Chosen
    Chemical Compounds (ACCC), which may or may not be listed in the solvent
    library above. ACCC parameters will be housed in and read from the file
        "ACCC Inputs (Solvent Class)--suffix"
    that you provide (input structure per K. Gering).

    How many ACCC solvents are included in the overall solvent number?

        Input Number of ACCC Solvents: 0 to 3
0
    Choose non-ACCC solvents from library above:

        Input number for solvent 1 :
3
        Input number for solvent 2 :
8
        Input number for solvent 3 :
12
```

Figure 154. Selecting solvents in CLI Example 1.

In this section, the CLI also prompts the user to select the volume percentages of the selected solvents. As shown in Figure 155, for Example 1, we use **25% EC**, **50% DMC**, and **25% EP**.

```
Input Volume Percent of solvent 1 :
    25
Input Volume Percent of solvent 2 :
    50
Input Volume Percent of solvent 3 :
    25
Percentage sums not equal to 100 will be derived by relative input proportions.
```

Figure 155. Section A.3 (2) from AEM Fortran Version 2.24.2.

A.4: Indicate number of salts (1 or 2):

This is where the user defines the number of salts used to formulate the desired electrolyte. The maximum number of salts allowed as inputs to the program is **two (2)**. As shown in Figure 156. For Example 1, we use **one (1)** salt.

```
A.4: Indicate number of salts (1 or 2)

    Input 1 or 2:
        1
```

Figure 156. Section A.4 from AEM Fortran Version 2.24.2.

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A.5: Choose salt(s) from the following partial sample list:

This is where the user defines the type of salt (based on existing chemistries in the AEM library) for each solvent specified in A.4. As shown in Figure 157, for Example 1, we do not include any ACCC salts and use **7: LiPF₆**.

```
A.5: Choose salt(s) from the following partial sample list:

For Battery Electrolytes:
  7: LiPF6
  8: LiBF4
  9: LiBOB
 10: LiTFSI
 11: Li2(B12F12)
 12: LiDFOB
 38: LiDFBOP
 13: LiFSI
 14: Li-Triflate
 15: LiAsF6
 16: LiPO2F2 (additive only)
 17: NaClO4
 18: NaPF6
 19: NaBF4
 20: NaTFSI
 37: NaFSI
 21: PA5-PF6 (RTIL)
 22: PYR13-FSI (RTIL)
 23: EMI-FSI or EMIM-FSI (RTIL)
 24: CsPF6
 25: Mg(TFSI)2

For Aqueous Systems:
 17: NaClO4
 26: NaCl
 27: NaNO3
 28: Na2SO4
 29: LiNO3
 30: Li2SO4
 31: MgSO4
 32: Al(NO3)3
 33: LiBr
 34: NaOH
 35: KOH
 36: LiOH
 37: NaFSI

Note1: aqueous systems outside salts 25-31 have not been validated.
Note2: if an RTIL is chosen in a salt mixture, input the RTIL FIRST.

--> You have the option to designate a salt as an Arbitrarily-Chosen
      Chemical Compound (ACCC), which may or may not be listed in the salt
      library above. You may designate ONE ACCC salt per run, which can
      be part of a two-salt mixture with a library salt.

      ACCC salt parameters will be housed in and read from the file
          "ACCC Inputs (Salt Class)--suffix"
      that you provide (input structure per K. Gering).

      Do you wish to include one ACCC salt?

          0 for No,      1 for Yes

Input:
0

Choose non-ACCC salt(s) from library above:

Input number for salt 1 :
7
```

Figure 157. Section A.5 from AEM Fortran Version 2.24.2.

A.8: Indicate maximum total salt concentration of interest:

This is where the user defines the maximum concentration of salts. For battery electrolytes, this value falls between **6m to 8m**, where **m is molal units**. This value can reach upwards of **15m** when dealing with **aqueous systems**. As shown in Figure 158 for Example 1, we use **5m**.

```
A.8 Indicate maximum total salt concentration of interest.
(molal units, typically reaching 6 to 8m for battery electrolytes).
NOTE1: use care for conditions exceeding 5 molal total salt (non-aqueous).
NOTE2: can input up to 15 molal for some 1:1 salts (aqueous).

Value:
      5
```

Figure 158. Section A.8 from AEM Fortran Version 2.24.2.

5.2.3 Section B: Input Other Simulation Conditions:

This section defines some other key parameters like the temperature profile, the method to handle Triple Ion Stability (ABA+ vs BAB-), and the inputs for electrolyte ingress into pores which are essential in simulating the user's desired electrolyte. This section contains 4 subsections described below.

B.1: Temperature Range.

This is where the user defines the temperature range in which the electrolyte will be used. There are two inputs for the temperature range, as follows.

- Minimum Temperature (**Input Tmin (C):**)
- Maximum Temperature (**Input Tmax (C):**)

The temperature range allowed by AEM will cover all conditions for electrolyte systems under conventional use conditions. A typical range for non-aqueous electrolytes is **-30°C to 60°C** with lower and higher temperatures possible. It is recommended not to exceed 100 °C. As shown in Figure 159, for Example 1, we use default inputs - **-30°C** for **Input Tmin (C)** and **60°C** for **Input Tmax (C)**.

```
B.1: Temperature Range. Calculation of properties over temperature
is necessary to derive thermodynamic quantities.
Specify minimum and maximum temperature of interest (degrees C),
spanning at least 20 degrees.
==> Recommended typical range: -30 to 60 C. Do not exceed 100 C.

Note: Inputs will be adjusted to the nearest decade.

Input Tmin (C):
      -30
Input Tmax (C):
      60
```

Figure 159. Section B.1 from AEM Fortran Version 2.24.2.

B.2: Input temperature stepsize for matrix and for outputs:

This is where the user defines the step size for the temperature range that was specified in B.1. As shown in Figure 160, for Example 1, we use the **5°C** temperature step size.

```
B.2: Input temperature stepsize in matrix and for outputs:
      5 degrees C (finer grid; slightly better fidelity in Arrhenius calcs)
      10 degrees C (reduces matrix by nearly half; coarser Arrhenius calcs)

Value:
      5
```

Figure 160. Section B.2 from AEM Fortran Version 2.24.2.

B.3: Input method for handling Triple Ion Stability (ABA+ vs BAB-):

This is where the user chooses the preferred approach for triple ion stability, whether the default is used ($[ABA+] = [BAB-]$) or whether there is subsequent conversion of either $[ABA+]$ or $[BAB-]$ into a single ion and an ion pair. As shown in Figure 161, for Example 1, we use **Option 1: $[ABA+] = [BAB-]$ (default)**, which will run

```
B.3: Input method for handling Triple Ion Stability (ABA+ vs BAB-):
      Option 1: [ABA+] = [BAB-] (default)
      Option 2: [ABA+] > [BAB-] due to BAB- --> B- + AB, or
                [ABA+] < [BAB-] due to ABA+ --> A+ + AB
                (these inequalities are automatically determined)

Input Option (1 or 2):
      1
```

Figure 161. Section B.3 from AEM Fortran Version 2.24.2.

B.4: Inputs for Electrolyte Ingress into Pores:

This is where the user defines three inputs correlating to the permeation of the electrolyte in porous regions, as follows. As shown in Figure 162, for Example 1 following are the inputs used.

- Contact Angle (**1. Contact Angle (0 to 90°)**) = **35°**
- Total Length of Pore (**2. Total Pore Length (0.1 to 50 micron)**) = **20μm**
- Concentration of Salt (**3. Salt Concentration of Interest (molal), must be less or equal to the maximum conc.)** = **1.5m**

```

1
B.4: Inputs for Electrolyte Ingress into Pores:
  1. Contact Angle (0 to 90 degrees)
  2. Total Pore Length (0.1 to 50 micron)
  3. Salt Concentration of Interest (molal),
      must be less or equal to the maximum conc.

Input Contact Angle (value of 0 to 90):
      35

Input Total Pore Length (value of 0.1 to 50):
      20

Input Salt Concentration of Interest (value of 0.1 to 5.00000000 ):
      1.5

```

Figure 162. Section B.4 from AEM Fortran Version 2.24.2.

5.2.4 Section C: Choose whether Surface-Charge Attenuated Electrolyte Permittivity (SCAEP) Calculations are Performed:

This section defines user inputs for calculation of surface-charge effects on permittivity for electrolyte neighboring an electrode surface. Local electrolyte permittivity will be reduced by the presence of an electrode surface charge as would be present during actual battery operation. As shown in Figure 163, for Example 1, we use **Input Choice: 0 for No** which will not run SCAEP calculations.

```

Section C: Choose whether Surface-Charge Attenuated Electrolyte Permittivity
(SCAEP) calculations are performed. These calculations will be
done at the solvent composition specified above and at conditions
chosen by the user. Other conditions are pre-set.
WARNING: SCAEP calculations take much CPU time and produce
large outputs

Input Choice:    0 for No,    1 for Yes

Input:
      0

```

Figure 163. Section C from AEM Fortran Version 2.24.2.

5.2.5 Section D: Choose whether Double Layer (DL) Calculations are Performed:

This section defines user inputs to facilitate double-layer (DL) calculations. These calculations are done over a fine grid of spatial (1D, 5 nm steps) and time dimensions, allowing determination of electrolyte gradients and related properties in the anode and cathode double layers. These calculations are done over relatively short time periods to reveal the emergent dynamic of the electrolyte gradients. As shown in Figure 164, for Example 1, we use Input Choice: 0 for No which will not run DL calculations.

```

Section D: Choose whether Double Layer (DL) calculations are performed.
          DL calculations will be done at a the starting composition of
          solvent proportions specified above and a chosen salt concentration.
          Basis: constant-current conditions over pulse time (pre-set).
          Default cell voltage of 4.0V is assigned.
          WARNING: DL calulations take much CPU time, produce large outputs,
                  and not every choice of conditions will converge.

          Input Choice:    0 for No,      1 for Yes

Input:
                        0
    
```

Figure 164. Section D from AEM Fortran Version 2.24.2.

5.2.6 Section E: Output Files

When all the simulations settings are supplied, AEM starts the simulation run. AEM then updates the data output report files with the simulation results. As shown in **Figure 165**, the AEM CLI will prompt the user to run another set of conditions (**1 = Yes**) which will append the data output files or exit the program (**0 = No**).

Note that if the software is restarted, then previously generated data output files will be overwritten. These files should be copied to another folder for future use if desired or use the GUI which organizes them in a timestamped directory.

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```
Depending on input choices, AEM is generating Reports covering Key Topics:

Report1 -- Highlighted Quantities
Report2 -- Ionic Speciation, Thermodynamic and Equilibrium Quantities
Report3 -- GFE and Ion Solvation Energies; Permittivity Terms
Report4 -- Diffusivities and selected conductivity terms
Report5 -- Transport Properties (viscosity, conductivity, etc.)
Report6 -- Arrhenius Analyses for Key Quantities
Report7 -- Optimization Results (multi-solvent matrix)
Report8 -- Non-convergent cases
Report9 -- Double Layer (DL) Analyses (if chosen; extensive CPU time)
Report10 -- Surface Charge Effects on Electrolyte Permittivity
Report11 -- Comprehensive Ion Solvation Quantities
Report12 -- Preferential Ion Solvation Quantities
Report13 -- Factors that Impact Ionic Conductivity (cation and anion)
Report14 -- Li-STEP terms
Report15 -- Cation and solvent velocities and transit times per dV/dx
Report16 -- Surface Tension and pore filling time over salt conc.
Report17 -- Percent pore length filled over time
Report18 -- Ligand-wise cation desolvation energy and time
Report19 -- Ligand-wise cation desolvation energy and time (accounting for CS)
Report20 -- Terms relating to structure and Communal Ion Solvation (CS)

All inputs received.  Calculations have started.....

.....Calculations have finished.

Do you wish to perform another set of conditions?
0 = No
1 = Yes

(additional results will be added at the end of current output files)

Input 0 or 1
0
```

Figure 165. Section E from AEM Fortran Version 2.24.2.

The final output reports can be found in ...**\AEM\CLI** as shown in Figure 166.

Congratulations! You have successfully run Example 1 using the AEM CLI if you make it to this stage without any errors.

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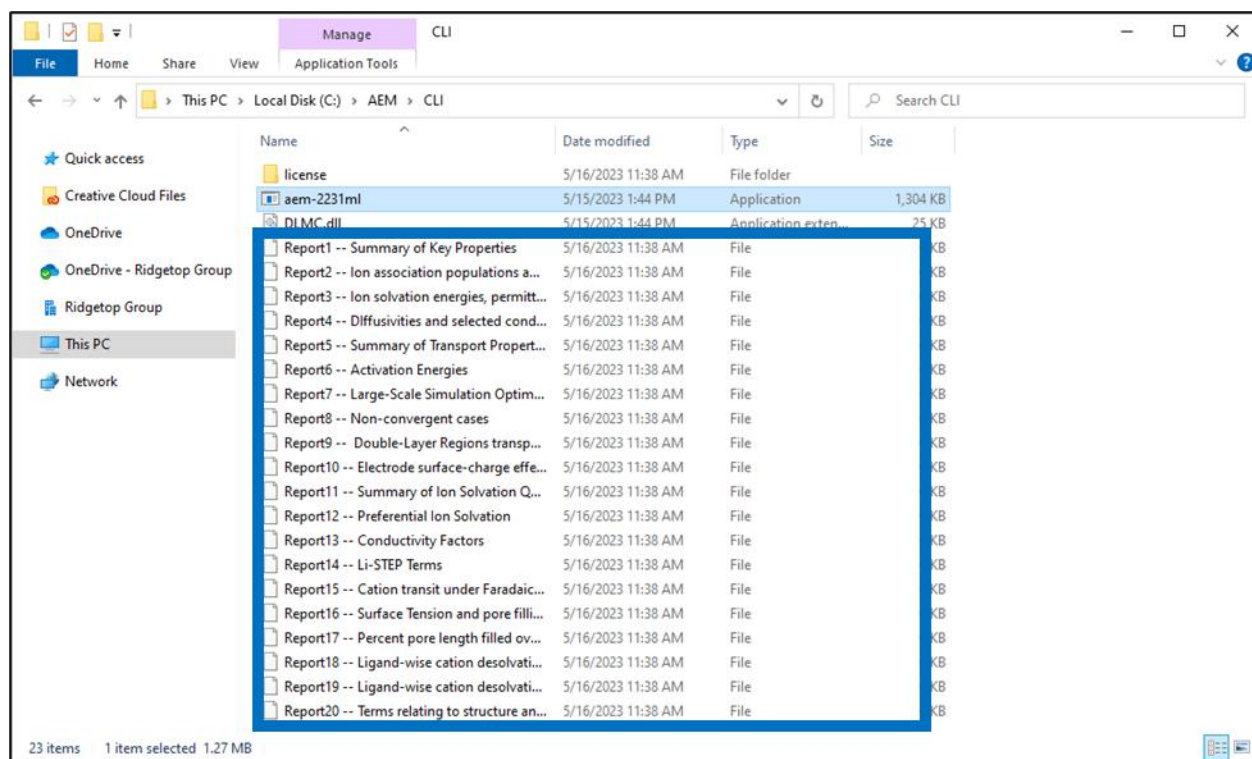


Figure 166. The different output files generated by AEM are highlighted inside the blue box.

6.1 API Architecture



On the **user level**, there are three major inputs - **Electrolyte Composition**, **ACCC Electrolyte Composition**, and **Input Parameters**. In the Electrolyte Composition, the user can define up to 10 solvents (fixed mode) or 5 solvents (matrix mode) and up to two salts. In the ACCC Electrolyte Composition, the user can define up to 10 ACCC solvents (fixed mode) or 5 ACCC solvents (matrix mode) and only one ACCC salt. The ACCC salt chosen must contain all the identifiers for the chosen ACCC solvent. The user can choose between a pure non-ACCC composition, a pure ACCC composition, or a cross composition between non-ACCC and ACCC components. Next, Input Parameters lists the various inputs required for running AEM CLI executable. These inputs are

essential for configuring AEM simulation runs and determining the specific conditions under which the model operates.

Within the **API itself (AEM_API.py)** there are three classes - the **ElectrolyteComposition Class**, the **ACCCElectrolyteComposition Class**, and the **AEM_API Class**. The ElectrolyteComposition Class includes functions for managing the non-ACCC composition, whereas the ACCCElectrolyteComposition Class includes functions for managing the ACCC composition. The AEM_API Class contains functions for reading inputs, generating compositions, running simulations using the AEM CLI program, saving logs, exporting files and results, processing data, and saving processed data.

The **Outputs** from the API are organized into the **AEM_API_Output** directory in run directories with the name **AEMRun-RunID-RunDate-RunTime**. Each run directory includes the **Reports** directory, the **Plots** directory, and a **run log .json file (AEMRun-RunID-RunDate-RunTime-Log.json)**. The Reports folder contains different types of reports generated in .txt, .csv, and .json formats. The .csv files are saved in the **Reports\csv** folder, whereas the .json files are saved in the **Reports\json** folder. These reports cover a wide range of analyses, from key properties summaries and thermodynamic terms to density, viscosity, conductivity, and cation desolvation. Other reports focus on dielectric analysis, transport properties, molar volume, ion-pair formation, self-diffusion coefficients, and ligand-wise analyses, among others. Each report provides detailed insights into specific aspects of the electrolyte compositions and their behaviors under various conditions. The Plots folder contains the **data plot .png file (AEMOutput-ReportNo-y_vs_x.png)**. The user has a choice to plot any desired quantities within the 20 reports.

6.2 Requirements and Dependencies

This section outlines the necessary requirements and dependencies required to run the Advanced Electrolyte Model (AEM) API. Ensuring these dependencies are installed correctly is essential for running the AEM API as defined in this User Guide.

- **Visual Studio Code:** Visual Studio Code Version 1.92 or higher ([Download Visual Studio Code - Mac, Linux, Windows](#))
- **Python:** Python 3.7 or higher ([Download Python | Python.org](#))
- **Git:** 2.46.0 or higher ([Git - Downloading Package \(git-scm.com\)](#))

Below is a list of the Python libraries and dependencies required to run the AEM API. Each library serves a specific purpose in the workflow.

1. **subprocess:** This module allows you to spawn new processes, connect to their input/output/error pipes, and obtain their return codes. It is used for running external commands and programs from within the script. is part of the standard Python library, so no separate installation is required.

2. **pandas:** A powerful data manipulation and analysis library. It is used for handling and processing data structures like DataFrames. Can be installed by running - `pip install pandas`
3. **matplotlib:** A plotting library used for creating static, animated, and interactive visualizations in Python. Can be installed by running - `pip install matplotlib`
4. **re:** The re module provides regular expression matching operations. It is part of the standard Python library, so no separate installation is required.
5. **collections:** The collections module implements specialized container datatypes like OrderedDict. It is part of the standard Python library.
6. **OrderedDict:** This class is a dictionary that remembers the order in which its contents are added.
7. **datetime:** This module supplies classes for manipulating dates and times. It is part of the standard Python library.
8. **json:** This module provides an easy way to encode and decode data in JSON format. It is part of the standard Python library.
9. **uuid:** This module provides immutable UUID objects (universally unique identifiers) and the functions to generate them. It is part of the standard Python library.
10. **os:** This module provides a way of using operating system-dependent functionality like reading or writing to the file system. It is part of the standard Python library.
11. **sys:** This module provides access to some variables used or maintained by the Python interpreter and to functions that interact strongly with the interpreter. It is part of the standard Python library.
12. **shutil:** This module offers a number of high-level operations on files and collections of files. It is part of the standard Python library.

6.3 Installation and Execution

Step 1: Preparation

These are some initial steps to be followed before cloning the AEM API repository from GitHub to your local machine.

1. Download and install each main dependency defined in Section 6.2 for Visual Studio Code, Python, and Git. Note that the "Add python.exe to PATH" checkbox should be selected as shown in Figure 168.

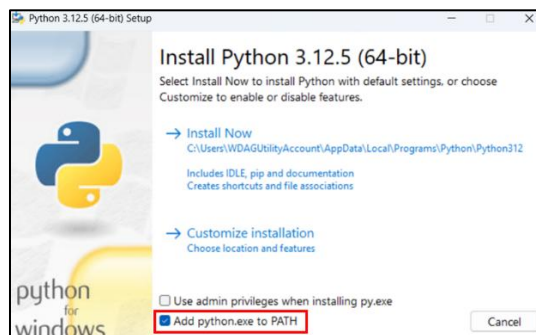


Figure 168. Selecting the option python.exe checkbox during the Python installation.

Advanced Electrolyte Model (AEM)

2. Install AEM as described in Section 2.
3. Make sure default AEM directory is set to **C:\Users\user-name\Documents\AEM** and the directories shown below exist within it as shown in Figure 169.

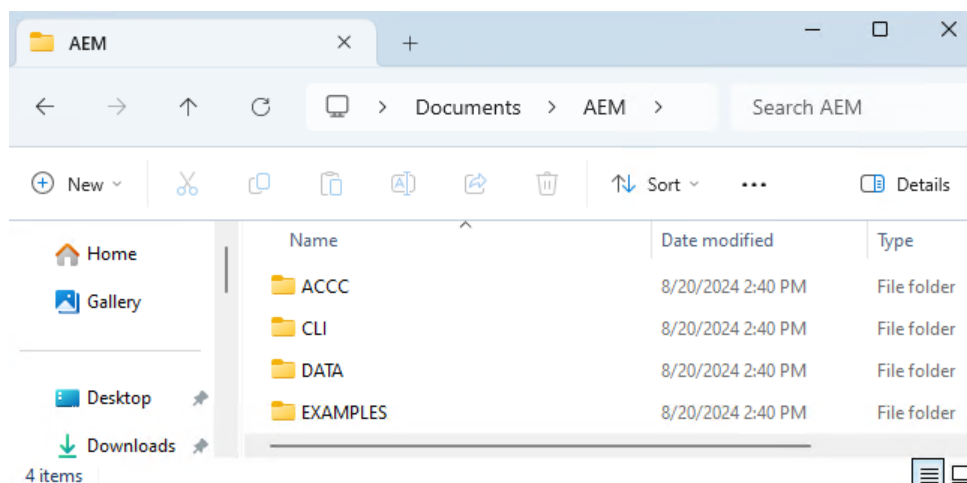


Figure 169. The folder structure for the default AEM directory.

4. Copy all/required ACCC compositions as shown in Figure 170, within the **ACCC** directory, to the **CLI** directory.

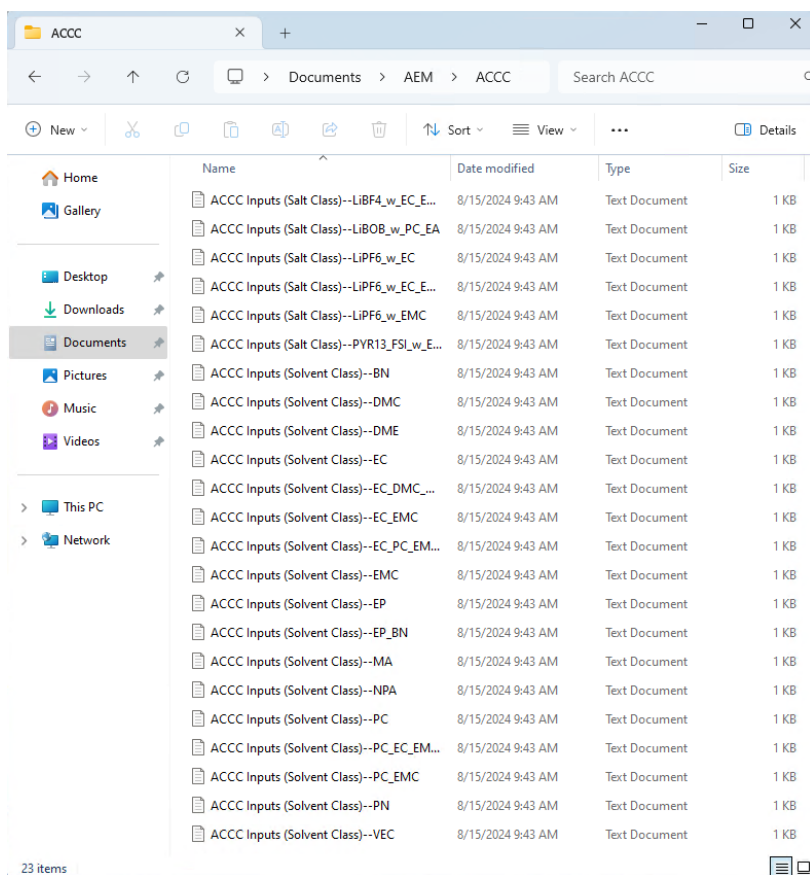


Figure 170. ACCC compositions within the ACCC\ directory.

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5. Note that default AEM directory is organized with the following subdirectories:

C:\Users\user-name\Documents\AEM\	Executable subdirectory
1. ACCC\	AEM ACCC Composition Files
2. CLI\	AEM Command Line Interface Directory
3. DATA\	AEM Data Output Files
4. EXAMPLES\	AEM Example Files

Step 2: Open Visual Studio Code and Clone the Repository from GitHub

1. Next, open Visual Studio Code, select File>Open Folder, and open the default AEM directory as shown in Figure 171. Note, that the Windows PowerShell / Terminal can be added to the bottom of the VS Code window by selecting View>Terminal.

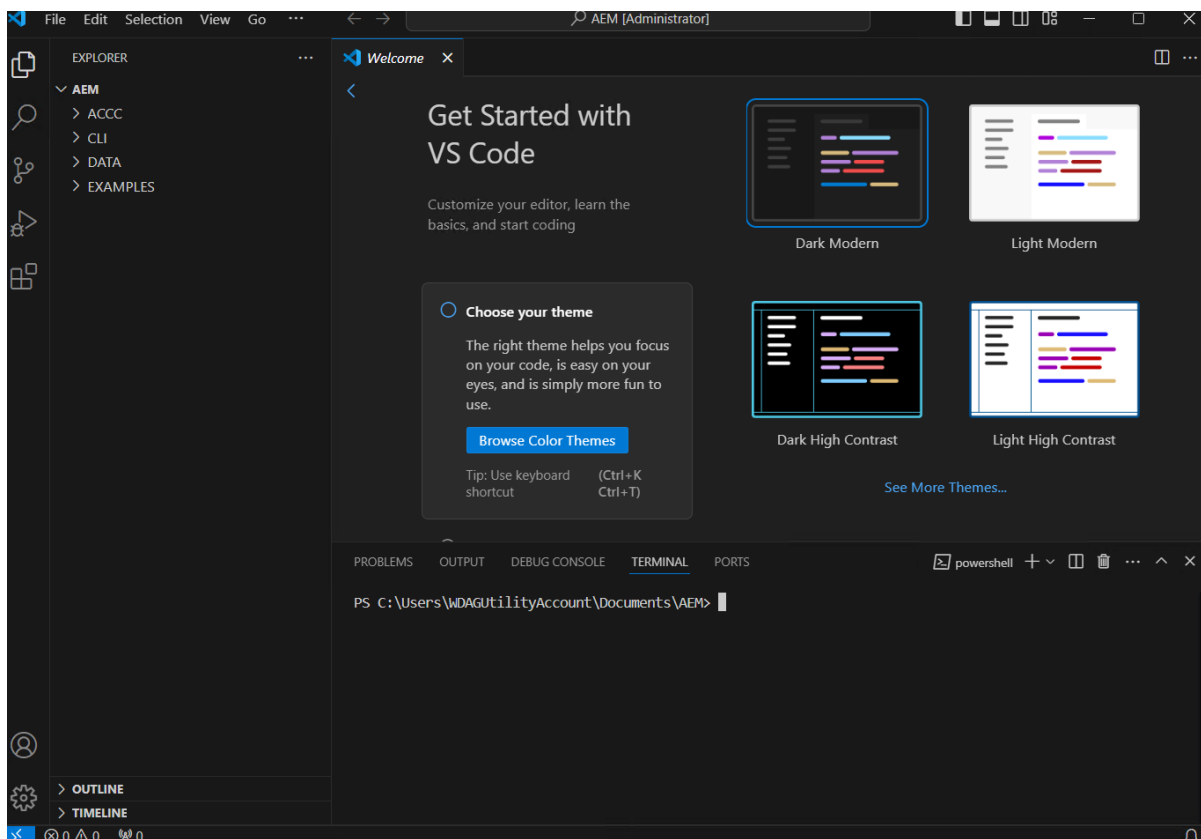


Figure 171. Opening default AEM user directory in VS Code.

2. Run the following command to clone from GitHub:

```
git clone https://github.com/RidgetopGroupInc/AEM-API.git
```

3. Verify this command created a directory named **AEM-API** in the **AEM** directory with all the files from the repository as shown in Figure 172.

Advanced Electrolyte Model (AEM)

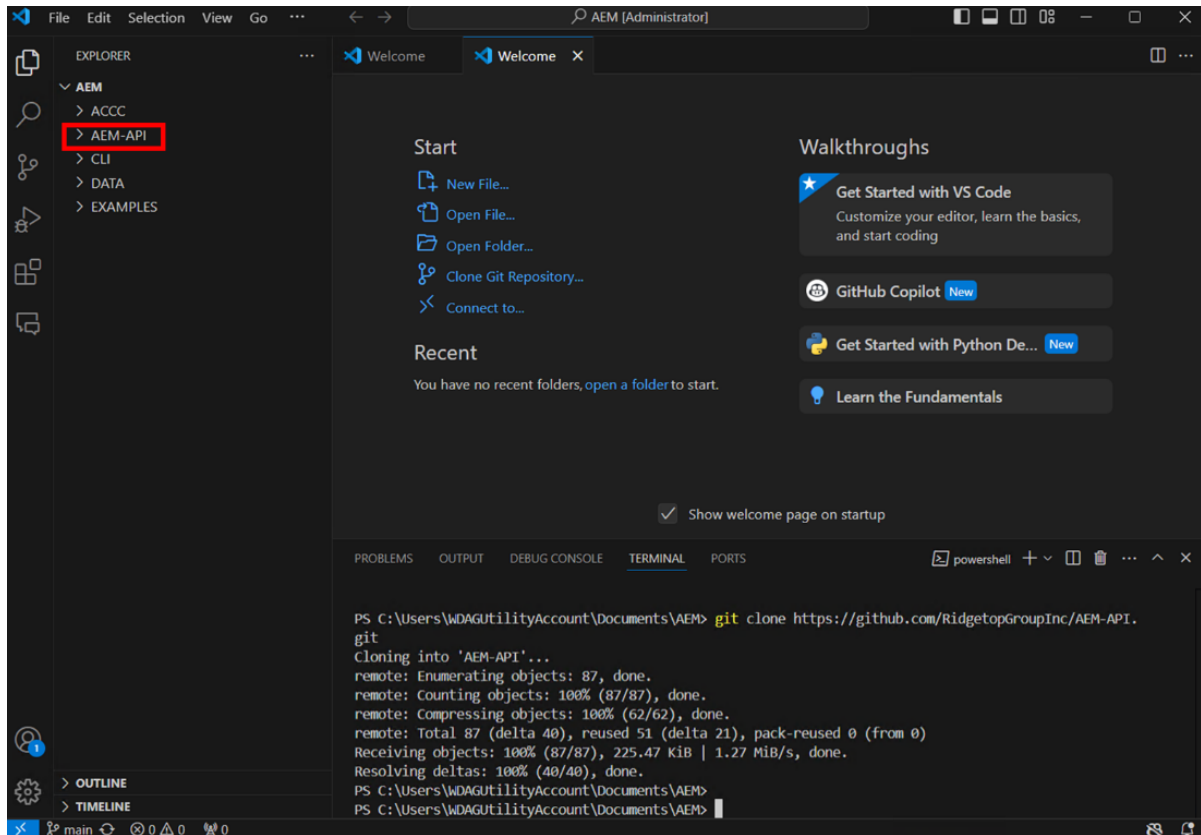


Figure 172. View of VS Code Window after cloning AEM-API from GitHub.

- Using the Terminal in VS Code, navigate to the **AEM-API** repository directory and install the required dependencies using the following commands.

```
cd AEM-API
pip install -r requirements.txt
```

Important Note: If you encounter any errors, verify that the "Add Python to environment variables" checkbox was selected during the Python installation and confirm that the **requirements.txt** file is located in the root directory of the cloned repository, listing all required dependencies.

- Verify that the Terminal in VS Code shows a successful installation for all dependencies in the **requirements.txt** file as shown in Figure 173.

```
PS C:\Users\WDAGUtilityAccount\Documents\AEM\AEM-API> pip install -r requirements.txt
Requirement already satisfied: numpy in c:\users\wdagutilityaccount\appdata\local\programs\python\python312\lib\site-packages (from -r requirements.txt (line 1)) (2.1.0)
Collecting pandas (from -r requirements.txt (line 2))
  Downloading pandas-2.2.2-cp312-cp312-win_amd64.whl.metadata (19 kB)
Collecting matplotlib (from -r requirements.txt (line 3))
  Downloading matplotlib-3.9.2-cp312-cp312-win_amd64.whl.metadata (11 kB)
Collecting python-dateutil>=2.8.2 (from pandas->-r requirements.txt (line 2))
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas->-r requirements.txt (line 2))
  Downloading pytz-2024.1-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas->-r requirements.txt (line 2))
  Downloading tzdata-2024.1-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib->-r requirements.txt (line 3))
  Downloading contourpy-1.2.1-cp312-cp312-win_amd64.whl.metadata (5.8 kB)
Collecting cyclor>=0.10 (from matplotlib->-r requirements.txt (line 3))
  Downloading cyclor-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib->-r requirements.txt (line 3))
  Downloading fonttools-4.53.1-cp312-cp312-win_amd64.whl.metadata (165 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib->-r requirements.txt (line 3))
  Downloading kiwisolver-1.4.5-cp312-cp312-win_amd64.whl.metadata (6.5 kB)
Collecting packaging>=20.0 (from matplotlib->-r requirements.txt (line 3))
  Downloading packaging-24.1-py3-none-any.whl.metadata (3.2 kB)
Collecting pillow>=8 (from matplotlib->-r requirements.txt (line 3))
  Downloading pillow-10.4.0-cp312-cp312-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=2.3.1 (from matplotlib->-r requirements.txt (line 3))
  Downloading pyparsing-3.1.2-py3-none-any.whl.metadata (5.1 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas->-r requirements.txt (line 2))
  Downloading six-1.16.0-py2.py3-none-any.whl.metadata (1.8 kB)
Downloading pandas-2.2.2-cp312-cp312-win_amd64.whl (11.5 MB)
 11.5/11.5 MB 32.7 MB/s eta 0:00:00
Downloading matplotlib-3.9.2-cp312-cp312-win_amd64.whl (7.8 MB)
 7.8/7.8 MB 48.5 MB/s eta 0:00:00
Downloading contourpy-1.2.1-cp312-cp312-win_amd64.whl (189 kB)
Downloading cyclor-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.53.1-cp312-cp312-win_amd64.whl (2.2 MB)
 2.2/2.2 MB 20.5 MB/s eta 0:00:00
Downloading kiwisolver-1.4.5-cp312-cp312-win_amd64.whl (56 kB)
Downloading packaging-24.1-py3-none-any.whl (53 kB)
Downloading pillow-10.4.0-cp312-cp312-win_amd64.whl (2.6 MB)
 2.6/2.6 MB 49.8 MB/s eta 0:00:00
Downloading pyparsing-3.1.2-py3-none-any.whl (103 kB)
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading pytz-2024.1-py2.py3-none-any.whl (505 kB)
Downloading tzdata-2024.1-py2.py3-none-any.whl (345 kB)
Downloading six-1.16.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: pytz, tzdata, six, pyparsing, pillow, packaging, kiwisolver, fonttools, cyclor, contourpy, python-dateutil, pandas, matplotlib
Successfully installed contourpy-1.2.1 cyclor-0.12.1 fonttools-4.53.1 kiwisolver-1.4.5 matplotlib-3.9.2 packaging-24.1 pandas-2.2.2 pillow-10.4.0 pyparsing-3.1.2 python-dateutil-2.9.0.post0 pytz-2024.1 six-1.16.0 tzdata-2024.1
```

Figure 173. A visual of the command line while installing dependencies.

Step 3: Running the AEM API Scripts

The following Python scripts are provided to simulate different AEM scenarios.

1. **AEM_API_woACCC.py** – No ACCC Functionality, with pure non-ACCC composition
2. **AEM_API_wACCC_nonACCCcomp.py** – ACCC, with pure non-ACCC composition
3. **AEM_API_wACCC_allACCCcomp.py** – ACCC, with pure ACCC composition
4. **AEM_API_wACCC_allACCCcomp_multiple.py** – ACCC, with pure ACCC composition (multiple solvents)
5. **AEM_API_wACCC_ACCCSolvents+nonACCCSalt.py** – ACCC, 2 ACCC Solvents + 1 Non-ACCC Salt
6. **AEM_API_wACCC_nonACCCSolvents+ACCCSalt.py** – ACCC, 2 Non-ACCC Solvents + 1 ACCC Salt

They can be run by passing the following command from within the **AEM-API** directory.

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```
python AEM_API_ScriptName.py
```

An example is shown in Figure 174, where the AEM_API_wACCC_allACCCcomp.py is run from the Terminal in VS Code.

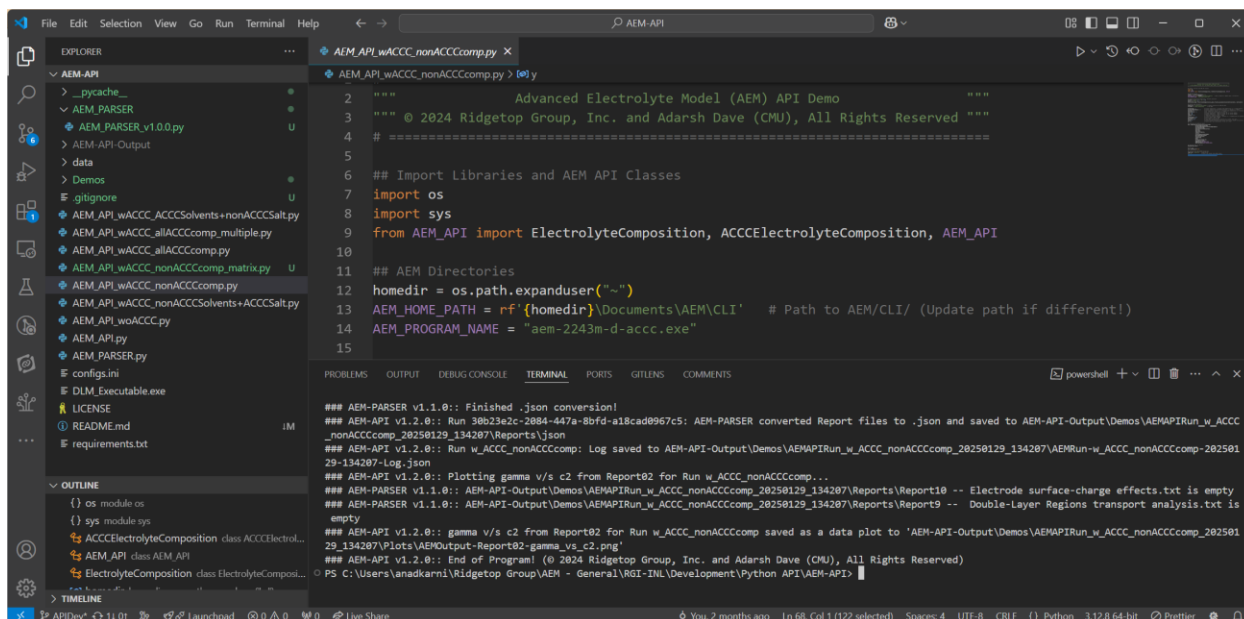


Figure 174. A visual from VS Code when the when AEM_API_wACCC_allACCCcomp.py script is run from the Terminal.

The output results can be viewed by opening a File Explorer and navigating to the following directory:

- C:\Users\user-name\Documents\AEM\AEM-API\AEM-API-Output\Demos\

Note that the AEM-API saves the output reports and automatically processed data plots in a time stamped directory as shown in Figure 175:

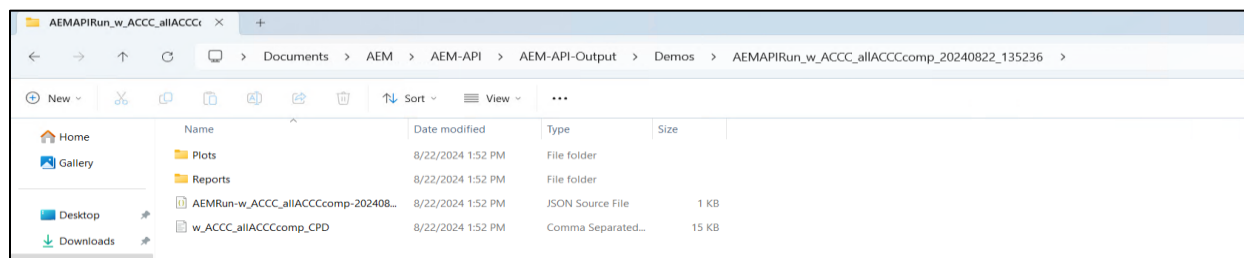


Figure 175. View of data output directory from AEM simulation using Python API.

6.4 AEM API Usage

6.4.1 AEM Home Path and Program Name Setup

To begin with, ensure that the AEM home path and program name are correctly set up in your Python script.

Example configuration in your Python script:

```
## AEM Directories
homedir = os.path.expanduser("~")
AEM_HOME_PATH = rf'{homedir}\Documents\AEM\CLI'
AEM_PROGRAM_NAME = "aem-2243m-d-accs.exe"
```

Replace **AEM_HOME_PATH** to the correct path where the AEM executable is located if it is not set to **default (C:\Users\<username>\Documents\AEM\CLI)**.

Important Note: The current AEM Program Name is **aem-2243m-d-accs.exe** which corresponds to v2.24.3 ACCC. The API is backward compatible for any version of AEM higher than v2.1.0. Get in touch with Ridgetop should you need a backward compatible version of the API.

6.4.2 Define Electrolyte Compositions

The first step is to define the composition of the solvents and salts you will use in your electrolyte model. The user can define either a non-ACCC or ACCC compositions, or both.

Non-ACCC Electrolyte Composition

In this example we define the solvents EMC and EC with mass fractions of 0.7 and 0.3, respectively, and the salt LiPF6 with a molality of 1.

```
## Define Electrolyte Composition - Solvents and Salts
solvents = {'EMC': 0.7, 'EC': 0.3}
salts = {'LiPF6': 1}
electrolyte_comp = ElectrolyteComposition.by_mass_fraction_and_molality(solvents=solvents, salts=salts)
```

ACCC Electrolyte Composition

Important Note: Only 1 ACCC Salt can be used with any given number of ACCC solvents.

In this example we define the ACCC solvent EC with mass percentage of 100% and the ACCC salt LiPF6 with a mass percentage of 100% as well.

```
## Define ACCC Electrolyte Composition - Solvents and Salts
solvents = {'EC': (100)}
salts = {'LiPF6_w_EC': 100}
accs_electrolyte = ACCCElectrolyteComposition(solvents=solvents, salts=salts)
```

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In the example below, a case with multiple ACCC solvents is shown where we define the ACCC solvent EC EMC, consisting of EC and EMC, with mass percentages of 70% and 30%, respectively, and the ACCC salt LiPF6_w_EC EMC with a mass percentage of 100%.

Important Note: If multiple ACCC solvents are being used, the ACCC solvent names ($S_1S_2...S_n$) should match the ACCC salt name (Salt_w_ $S_1S_2...S_n$).

```
## Define ACCC Electrolyte Composition - Solvents and Salts
solvents = {'EC EMC': (70, 30)}
salts = {'LiPF6_w_EC EMC': 100}
accc_electrolyte = ACCCElectrolyteComposition(solvents=solvents, salts=salts)
```

6.4.3 Define Input Parameters

Next, set the necessary parameters for the AEM API. These parameters control the conditions and settings for your model and are essential for defining the conditions under which the electrolyte properties will be evaluated.

```
## Define Input Parameters
output_dir = 'AEM-API-Output\\Demos' # Output Directory for AEM API Runs
run_name = 'w_ACCC_allACCCcomp' # Run Name
```

```
solventcomp = 1 # Solvent Composition: 1 (Single Fixed Composition) or 2 (Larger Matrix)
solventcompbasis = 2 # Solvent Composition Proportionality Basis: 1 (Volume) or 2 (Mass)
saltcomp = 1 # Salt Composition [Salts > 1]: 1 (Single Fixed Composition) or 2 (Several Proportions)
totalsaltconc = 1 # Max. Total Salt Concentration of Interest
tmin = 10 # Minimum Temperature: -30deg to 60 deg (Do not exceed 100deg)
tmax = 60 # Maximum Temperature: -30deg to 60 deg (Do not exceed 100deg)
stepsize = 10 # Temperature Stepsize: 5deg or 10deg
tis = 1 # Triple-Ion Stability Method: 1 (default) or 2 (inequalities are automatically determined)
contactangle = 90 # Contact Angle: 0deg to 90deg
porelength = 50 # Pore Length: 0.1um to 50um
saltconc = 0.1 # Input Salt Concentration of Interest: 0.1 to Max. Salt Conc.
scaep = 0 # Surface-Charge Attenuated Electrolyte Permittivity (SCAEP) Calculations: 0 (No) or 1 (Yes)
dl = 0 # Double Layer (DL) Calculations: 0 (No) or 1 (Yes)
```

6.4.4 Initialize and Run the AEM API

Initialize the AEM API object with the defined electrolyte composition and input parameters, and then run the model. Here, we initialize the **AEM_API** object with the electrolyte composition and input parameters, then the **generate_cues()** method prepares the necessary cues for running the AEM CLI, and the **runAEM()** method executes the AEM model. The **quiet=True** parameter suppresses verbose output.

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```
## Initialize the AEM-API Object
aem = AEM_API(accc_electrolyte=accc_electrolyte,
              solventcomp=solventcomp,
              solventcomppropbasis=solventcomppropbasis,
              saltcomp=saltcomp,
              totalsaltconc=totalsaltconc,
              tmin=tmin,
              tmax=tmax,
              stepsize=stepsize,
              tis=tis,
              contactangle=contactangle,
              porelength=porelength,
              saltconc=saltconc,
              scaep=scaep,
              dl=dl,
              output_dir=output_dir,
              run_name=run_name,
              AEMHomePath=AEM_HOME_PATH,
              AEMProgramName=AEM_PROGRAM_NAME)

## Generate Cues for AEM CLI
aem.generate_cues()

## Run AEM
aem.runAEM(quiet=False)
```

Visualize the Output Data

After running the AEM model, the next steps are to visualize the output data. Here, the **aem.plot_parsed_data()** method takes the user inputs of Report Number and x and y variables from the reports (column names), parses required data, displays the plot, and saves the plot to the **Plots** folder.

```
## Visualize the AEM_PARSER Parsed Data
report_no = "Report02"    # Report01-20
x = "c2"                  # x-axis: Any column from chosen report
y = "gamma"               # y-axis: Any column from chosen report
aem.plot_parsed_data(x=x, y=y, report_number=report_no)
```

Following is an example of a plot generated by the program.

Advanced Electrolyte Model (AEM)

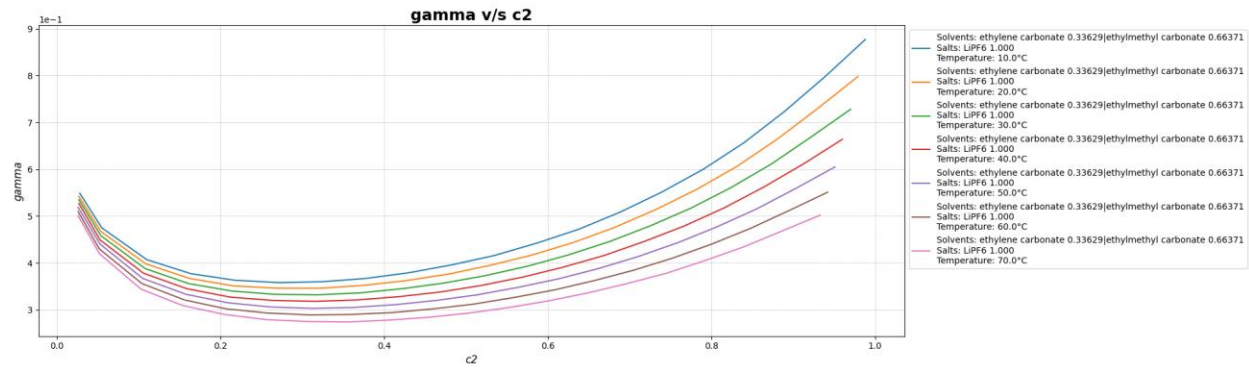


Figure 176. Example of data plot outputted by the AEM API.

7 Appendix

7.1 AEM Data Output Summary

7.1.1 Summary Table of AEM Outputs

Quantity	Definition/Description	Importance and Application Areas
Solvated Ion Sizes (cation, anion) - Hard sphere basis - Effective transport basis	Chemical Physics property directly relevant to extent of ion solvation over conditions of state (temperature, salt conc., solvent composition)	Ionic transport and conductivity in electrochemical systems; metal ion transport in metabolic processes. Also influences thermodynamic metrics such as activity coefficients
Solvent-to-ion Binding Energies (cation, anion)	Thermodynamic property directly relevant to extent of ion solvation over conditions of state (temperature, salt conc., solvent composition)	Ion desolvation energy requirements in charge transfer process in electrochemical systems, and in metabolic processes
Solvation Numbers (cation, anion, ion-associated species)	Thermodynamic property that represents the number of solvent species that reside in close proximity to ions.	Impacts kinetics and energetics of ion desolvation processes in electrochemical systems, as well as solvent co-transport in a number of electrolyte-centric processes.
Ion Desolvation Energy	Net energy involved in solvent-wise ion desolvation based on the solvation number at a given condition (cation).	Quantifies the energy required to overcome lithium solvation prior to lithium (or other) ion charge transfer and insertion into host. Analogs exist in terms of metabolic processes.
Ion Desolvation Kinetics	Net time required for solvent-wise ion desolvation based on the solvation number at a given condition (cation).	Quantifies the time required to overcome lithium solvation prior to lithium (or other) ion charge transfer and insertion into host. Analogs exist in terms of metabolic processes.

Advanced Electrolyte Model (AEM)

Quantity	Definition/Description	Importance and Application Areas
Comprehensive evaluation of species populations: cation, anion, ion pairs, triple ions, solid solvates	Correct equilibrium speciation of all components by thermodynamic Mass Action Law analyses	Impacts macro-scale properties that are dependent on the extent of free ions (conductivity, diffusivity) and net solvation behavior (viscosity). This is a crucial metric within electrochemical double layers.
Solution Permittivity Solvent Permittivity	Represents true dielectric of solvent and solution according to “dielectric depression” that occurs due to the presence of ionic fields. Calculated over conditions of state (temperature, salt conc., solvent composition)	Dielectric depression will influence the extent of ion association, which in turn impacts properties like conductivity and diffusivity.
Relative molal enthalpy	Thermophysical property tied to heat capacity, calculated over conditions of state (temperature, salt conc., solvent composition)	Important to thermal issues involving electrolyte, such a heat transfer.
Density	Mass electrolyte per unit volume	Related to partial molar volume of electrolyte.
Mean activity coefficient of salt - Rational basis - Practical basis	Describes the extent of non-ideality introduced by presence of solvated charged species	Crucial thermodynamic metric for issues pertaining to electrolyte equilibrium per state conditions
Osmotic coefficient (and Osmotic Pressure)	Characterizes how the presence of a dissolved solute causes deviation of a solvent from ideal behavior, using a reference state such as pure solvent	Crucial thermodynamic metric for issues pertaining to the equilibrium state per state conditions; related to solvent activity

Advanced Electrolyte Model (AEM)

Quantity	Definition/Description	Importance and Application Areas
Viscosity	Physical property that characterizes a fluid's resistance to flow	Key transport property inversely related to conductivity and diffusivity. Viscosity is most problematic at low temperatures and high salt concentrations.
Conductivity (ion-wise, net)	Electronic property that characterizes the ability to pass current, here through ionic mobility; inverse to resistivity.	The premier transport property in terms of battery electrolytes. Used extensively in transport analysis. One of the most-of measured and published properties of electrolytes of electrochemical systems.
Diffusivity (ion-wise; solvent)	Physical property tied to the diffusive propensity (Fickian) of labile species.	Also used in chemical and electrochemical transport calculations. Important where concentration gradients emerge (double layers) due to battery cycling conditions.
Transference Numbers	Electrochemical property that represents the fraction of the total electric current that cations and anions carry in passing through an electrolytic solution. The cationic term is usually of interest.	High cationic transference numbers indicate that a battery electrolyte is more suited for higher-power applications.
Ion Hopping - cation - anion	The statistical fraction of path length wherein an ion is minimally solvated under Faradaic transport.	Ion hopping is one of the most efficient modes of transport in electrolyte systems. The goal is to find electrolytes and conditions therein that promote ion hopping.

Advanced Electrolyte Model (AEM)

Quantity	Definition/Description	Importance and Application Areas
Communal Solvation (CS)	The fraction of solvent in an electrolyte that is involved in multi-ion solvation or simultaneous solvation per solvent molecule.	CS contributes to higher electrolyte viscosity because it introduces structure on the molecular scale, and it can complicate the ion desolvation process. In some applications (batteries) low values of CS are sought, while in other applications (solvent-based extraction) high values are desirable.
Li-STEP	Lithium ion Solvation and Transport Electrolyte Parameter is a term that incorporates salient macro-scale transport terms with molecular scale desolvation energy and kinetics.	Serves as a convenient overall screening metric for evaluating electrolyte systems for batteries.
Electrolyte properties in double layer (DL) under conditions of non-electroneutrality	While performing transport calculations AEM calculates the dynamic change of electrolyte conc. and properties within DL regions as concentration polarization causes speciation over pulse time.	Non-electroneutrality can occur in a double layer, and the electrolyte properties of such a resultant region are nearly impossible to predict until now. AEM reveals complex behavior in DL.
Large-scale Optimization (LSO) of multi-component electrolytes	When chosen as an option, LSO performs calculations over matrices of electrolyte formulations to determine optimized values to achieve best performance under the optimization metrics (e.g., conductivity).	Greatly reduces time to identify optimized formulations and their performance and reduces laboratory measurements.

Advanced Electrolyte Model (AEM)

Quantity	Definition/Description	Importance and Application Areas
Walden Parameters	Walden terms shed light on the true interplay between viscosity and ionic conductivity in electrolytes.	The x and y axes for Walden plots are given in one of the output reports, covering each condition of temperature, solvent composition and salt concentration.
Preferential Ion Solvation	This metric considers the probability of finding each particular solvent in near proximity to an ion, given a solvent mixture. Covers both cation and anion species.	Preferential solvation is important to describing the solvation shell occupancy of ions. This matters, for example in effective solvated ion sizes, during early-life SEI formation processes and also as cells become polarized during operation.
Cation Velocity over a range of potential gradients, Solvent diffusion velocity Both are f(conc., T)	The range of velocities of cation species in a battery determines how well power delivery can be sustained. It is a function of the electrolyte transport properties and the imposed potential gradient driving force. Solvent diffusion velocity determines how readily solvent can back-diffuse after it is Faradaically carried with the cation or anion.	These terms allow consideration of some concentration polarization (CP) effects. CP causes detrimental changes in electrolyte properties in double-layer regions. Electrolytes with higher velocities over a larger salt concentration range will be less prone toward CP development and related CP impedance increases.

Advanced Electrolyte Model (AEM)

Quantity	Definition/Description	Importance and Application Areas
Surface Tension	Surface tension reflects the cohesive energy an electrolyte has within itself. In AEM surface tension is calculated as a function of solvent mixture, salt concentration, and temperature.	Surface tension is a key parameter for looking at liquid permeation into porous media, where there exists both cohesive and adhesive (liquid-to-solid) considerations under competition.
Electrolyte Migration into Porous Media	This set of metrics considers the behavior of electrolyte migration into pores of a given dimension (diameter and length) and roughness. The user is queried to input the anticipated pore-related contact angle and other key terms.	Together with electrolyte properties (e.g., surface tension, viscosity, density, etc.) real porous media attributes will influence the rate of liquid infiltration. This is a key consideration for the wetting of porous solids and the effective liquid migration rates into porous structures having a wide range of dimensions.
Solvent Activity	Thermodynamic indicator of solvent availability, or conversely, indicator of solvent unavailability.	Ion solvation will diminish solvent activity, more so at higher salt conc. Important metric for SEI and HCE.
Ligand-wise cation desolvation energy and time	These terms represent the energy and kinetic requirements of solvent-wise removal from solvated cations.	Cation desolvation is a foremost consideration for applications that much achieve a bare cation in one or more processing steps.

IMPORTANT NOTE: There are additional output quantities that are found in the report files.

7.1.2 Data Output Reports and Definition of Terms

♦ Report1 -- Summary of Key Properties

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)
c2, eff, trans	effective molar salt concentration available for cationic transport
wt fr salt (2)	weight fraction salt in electrolyte
mole fr salt	mole fraction salt in electrolyte
density (g/cc)	electrolyte density
visc. (cP)	electrolyte viscosity
sig1 (eff)	solvated cation diameter, effective Stokes transport basis, Angstroms
sig2 (eff)	solvated anion diameter, effective Stokes transport basis, Angstroms
<u>S</u> (+)	cation solvation number
Rational Act. Coef. $\gamma_{\pm}$	electrolyte salt activity coefficient, rational (molar) basis
Diff. Coeff. (cm ² /s)	diffusion coefficient or diffusivity for salt in electrolyte
Spec. Cond. (mS/cm)	specific conductivity of electrolyte
t+ (a)	cation transference number based on SI cations vs all charged species
t+ (b)	cation transference number based on all cation-bearing charged species vs all charged species
dissoc (SI)	fraction of salt dissociated into free single ions
dissoc (IP)	fraction of salt existing as ion pairs
dissoc (TI)	fraction of salt existing as triple ions

Advanced Electrolyte Model (AEM)

◆ Report2 -- Ion association populations and other thermodynamic terms

m2	molal salt conc.
c2	molar salt conc.
cmeff	effective molar concentration of all charged species
alphanet	total molar fraction of salt utilized in producing charged species
cation	mole fraction salt existing as cations
anion	mole fraction salt existing as anions
IP	mole fraction salt existing as ion pairs
TI	mole fraction salt existing as triple ions
fcip	fraction of ion pairs that are contact-type ion pairs
gamma+/-	molal-scale salt activity coefficient
y+/-	molar-scale salt activity coefficient
y+/--bar	molar-scale salt activity coefficient, using ion-averaged properties Osmotic
Osmotic Coeff. (m)	Osmotic coefficient, molal basis
Osmotic Coeff. (M)	Osmotic coefficient, molar basis
Solvent Activity	Fraction of solvent "available" for associative and reactive processes
Kip	Ion Pair association equilibrium constant
Kti	Triple Ion association equilibrium constant
Ksol	Solid solvate association equilibrium constant
Adj. Solvent Activity	solvent activity adjusted for effects of ion solvation dynamics and communal solvation
Adj. Phi, m	molal osmotic coefficient adjusted for effects of ion solvation dynamics and communal solvation
Adj. Gamma, m	molal mean activity coefficient of salt adjusted for effects of ion solvation dynamics and communal solvation
fvpd	fraction of vapor pressure depression of vapor phase in equilibrium with liquid electrolyte

◆ Report3 -- Ion solvation energies, permittivity and cation desolvation

m/c	molal/molar salt conc.
GFE cation	Gibbs free energy of cation solvation based on integral form
GFE anion	Gibbs free energy of anion solvation based on integral form
Solvation E, cation	cation solvation (binding) energy based on cation solvation number at salt conc.
Solvation E, anion	anion solvation (binding) energy based on anion solvation number at salt conc.
Rel. Permittivity, solution	relative permittivity for entire electrolyte solution
Rel. Permittivity, solvent	relative permittivity for electrolyte solvent mixture only
alpha1	fraction of salt dissociated into single (non-associated) ions
alpha3	fraction of salt dissociated into triple ions
Cation Desolv. Energy Sum	Ligand-wise summation of cation desolvation energies
Cation Desolv. Energy Ave	Ligand-wise average of cation desolvation energies
Cation Desolv. T	Ligand-wise summation of cation desolvation times
Polarizability, Solution Basis ($\text{\AA}^3$)	polarizability of electrolyte solution (Clausius–Mossotti)
Polarizability, Solvent Basis ($\text{\AA}^3$)	polarizability of solvent mixture only (Clausius–Mossotti)

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◆ Report4 -- Diffusivities and selected conductivity terms

salt molality	moles salt per kg pure solvent
1/visc	inverse viscosity
Cond.	specific conductivity
Diffusivity	diffusivity of salt in bulk electrolyte (Nernst basis as in Report4)
Li-step (full)	Lithium Solvation and Transport Electrolyte Parameter, full form*
Li-step (solv.)	Lithium Solvation and Transport Electrolyte Parameter, solvation terms-only form*
Net BE, solvent-cation	net solvent-cation binding energy
Net Desolvation Energy, cation	net cation desolvation energy
Rel. Perm., solution	relative permittivity of electrolyte solution
Rel. Perm., solvent	relative permittivity of solvent mixture only
Thermal Cond.	thermal conductivity
Thermal Diff.	thermal diffusivity
Specific Heat Cap.	specific heat capacity
Surface Tension	surface tension of electrolyte
Cation Hopping	cation hopping rate parameter
Anion Hopping	anion hopping rate parameter
Kip	Ion Pair association equilibrium constant
Kti	Triple Ion association equilibrium constant
Ksolv	Solid solvate association equilibrium constant
Solvent Activity	Fraction of solvent "available" for associative and reactive processes
gamma+/-	molal-scale salt activity coefficient

* See supplemental Li-STEP information from K. Gering

◆ Report5 -- Summary of Transport Properties and Walden analysis

m2	molal salt conc.
c2	molar salt conc.
c2,pseudo	pseudo-molar salt concentration (moles per liter pure solvent)
density (g/cc)	electrolyte density
visc. (cP)	electrolyte viscosity
Rational Act. Coef. $\gamma_{\pm}$	electrolyte salt activity coefficient, rational (molar) basis Diff.
Coeff. (cm ² /s)	diffusion coefficient or diffusivity for salt in electrolyte Spec.
Cond. (mS/cm)	specific conductivity of electrolyte
t+ (a)	cation transference number based on SI cations vs all charged species
t+ (b)	cation transference number based on all cation-bearing charged species vs all charged species fhop (+)
	quantity that represents the relative prevalence of bare cation hopping as a cationic conduction mechanism, as defined over the conductive path length
fhop (-)	quantity that represents the relative prevalence of bare anion hopping as a anionic conduction mechanism, as defined over the conductive path length
Pos	Osmotic pressure in atmosphere units
Walden x	log(1/visc) is used as the x axis for Walden-type plots
Walden y	log(cond.) is used as the y axis for Walden-type plots; note that "cond." is conductance
Walden Product	product of electrolyte conductance times viscosity
Thermal Diffusivity	Describes the rate at which heat spreads through the electrolyte. It reflects how quickly temperature gradients equilibrate and is influenced by thermal conductivity, density, and specific heat.
Specific Heat Capacity	Measures the amount of heat required to raise the temperature of a unit mass of electrolyte by one degree Kelvin. It is essential for understanding thermal management and energy storage in electrochemical systems.

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◆ Report6 -- Activation Energies

salt molality	moles salt per kg pure solvent
1/visc	inverse viscosity
k*t+	product of specific conductivity times cation transference number
Diffusivity	diffusivity of salt in electrolyte (Nernst basis as in Report4)
Li-step (full)	Lithium Solvation and Transport Electrolyte Parameter, full form*
Li-step (solv.)	Lithium Solvation and Transport Electrolyte Parameter, solvation terms-only form*
Kip	Ion Pair association equilibrium constant
Kti	Triple Ion association equilibrium constant
Ksolv	Solid solvate association equilibrium constant
Solvent Activity	Fraction of solvent "available" for associative and reactive processes
gamma+/-	molal-scale salt activity coefficient

◆ Report7 -- Large-Scale Simulation Optimization

Output quantities for Report7 are self-explanatory. Note that both mole and mass fractions are used in reporting within Report7.

◆ Report8 -- Non-convergent cases

Non-convergent cases will be listed between the temperature and compositional legends. Non-convergent notations are generally of two types:

- '***' If MSA-type chemical physics calculations or equilibrium determinations do not converge,
- Permittivity notification if solution or solvent permittivity calculations do not converge.

Upon non-convergence, the best information available at that point is utilized for remaining AEM calculations. If all cases converge between two temperatures, there will be no such notations.

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◆ Report9 -- Double-Layer Regions transport analysis

The main tables generated in this report are defined with three regions of interest:

- Transport Region for Releasing Electrode (anode if discharge conditions, cathode if charge conditions)
- Separator Region
- Transport Region for Receiving Electrode (anode if charge conditions, cathode if discharge conditions)

Each table is printed at a particular elapsed time from the beginning of the charge or discharge event.

x	distance from the Releasing Electrode surface, Angstroms
C+	cation molar concentration at each x position
C-	anion molar concentration at each x position
Cs	solvent molar concentration at each x position
I	current at each x position, A
R	resistance at each x position, Ohm
ns+	cation solvation number at each x
ns-	anion solvation number at each x
v+	cation transport velocity at each x
v-	anion transport velocity at each x
dens.	electrolyte density at each x
visc.	electrolyte viscosity at each x
eff. cond.	effective electrolyte conductivity at the conditions of each x
t+	cation transference number at each x
'other'	generally refers to factors related to ionic hopping by cations and anions

◆ Report10 -- Electrode surface-charge effects

Calculations are done at each temperature, covering a matrix of Surface Charge Densities at the target (cation receiving) electrode surface. Each temperature is considered since involved properties such as electrolyte permittivity are $f(T)$.

r	distance from target electrode surface, Angstroms
Eff Surface Ch Density at r	effective surface charge density at r due to electron population
Solution Rel. Perm.	Solution relative permittivity at r as determined from electrolyte ionic and surface charge
Ave.(r) Solution Rel. Perm.	Solution relative permittivity averaged over r as determined from electrolyte ionic and surface charge
Electric Field per SCh	electric field at r as driven by the surface charge density of electrons
Repulsive Energy	repulsive energy over r between electrode surface charge density and averaged negative solvent dipole charge.
Cell Voltage	local voltage over r in response to surface charge effects, based on assumed starting cell voltage.

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◆ Report11 -- Summary of Ion Solvation Quantities

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)
cation eff dia.	solvated cation diameter, effective Stokes transport basis, Angstroms solvated anion diameter, effective Stokes transport basis, Angstroms cation solvation number, thermodynamic basis
S ₋ (th)	anion solvation number, thermodynamic basis
Solvent Avail. (thermo.)	free solvent availability based on subtraction of thermodynamically determined solvation numbers
Solvent Avail. (MSA HS)	free solvent availability based on subtraction of solvation numbers calculated from MSA hard-sphere solvated ion diameters
Solvent BE to cation	solvent binding energy to cation for energy minimized (equilibrium) cation solvation states. Solvent BE to anion
anion	solvent binding energy to anion for energy minimized (equilibrium) anion solvation states.
Communal Solvation Factor	(1 – CS factor) describes the relative likelihood of solvent being shared as a solvator between two or more charged species.
Debye Relaxation Time	approximate time for solvent to recover or relax its position (orientation) after being in proximity to a local ionic field, nanoseconds
Fraction of Free Liquid in Solvent	Solvents with higher melting points will have complex phase behavior in relation to electrolyte salt content. This term estimates the fraction of solvent that exists as liquid under (c,T) conditions. This impacts the effective electrolyte viscosity and related transport properties. The intended target is electrolytes with a single solvent having a melting point above 0 °C.
Debye Length (Ang)	Represents the electrostatic screening distance in the electrolyte, influencing ion mobility, transport properties, and conductivity—especially in systems near solid-liquid phase transitions.

◆ Report12 -- Preferential Ion Solvation

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)

Other terms are self-explanatory. Results are provided for both the cation and anion, considering up to ten solvents. Preferential ion solvation calculations were upgraded to include local ion density effects.

◆ Report13 -- Conductivity Factors

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)

Other terms are self-explanatory. Values shown are factors as applied to the modified Stokes expression for specific ionic conductivity. Results are provided for both the cation and anion, considering the seven conductivity factors mentioned in: [K. L Gering, Prediction of electrolyte conductivity: results from a generalized molecular model based on ion solvation and a](#)

chemical physics framework, *Electrochim. Acta* **225** (2017) 175–189.
<http://dx.doi.org/10.1016/j.electacta.2016.12.083>.

◆ **Report14 -- Li-STEP Terms**

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)

Other terms are explained below. Higher values of Li-STEP are desirable.

$$\begin{aligned}
 Li - STEP &= \frac{\kappa t_+}{\Delta t_{Li^+ desolv.}} \left[\frac{\overline{BE}_{Li^+}}{RT} \right]^{-n_{s,Li^+}} \\
 &\simeq \kappa t_+ \left[\frac{\overline{BE}_{Li^+}}{RT} \bar{\tau}_{Li^+} \right]^{-n_{s,Li^+}}
 \end{aligned}$$

Lithium Solvation and Transport Electrolyte Parameter (Li-STEP)

Key transport and solvation terms are combined in a single parameter:

$$Li - STEP = \kappa t_+ \left[\frac{\overline{BE}_{Li^+}}{RT} \bar{\tau}_{Li^+} \right]^{-n_{s,Li^+}}$$

The inverse of this quantity is akin to an impedance term for Li⁺ transport.

Parameter

Conductivity (κ)
 Average Binding Energy ($\overline{BE}_{Li^+}$)
 Average solvent residence time ($\bar{\tau}_{Li^+}$)
 Cation solvation number (n_{s,Li^+})
 Cation transference number (t_+)

Solvation component

Justification

Key Transport Parameter
 Thermodynamics of Li⁺ solvation
 Kinetics of Li⁺ desolvation
 Extent of Li⁺ solvation
 Efficiency of Lithium transport

Note: $\overline{BE}_{Li^+}$ is a geometric average over all constituent ligands that comprise the cation solvation number. Thus, step-wise Li⁺ desolvation is described. n_{s,Li^+} is derived by a method adapted from Stokes and Robinson (1973).

This new approach enables us to judge the relative contribution of each parameter to the overall activation energy of lithium transport, and leads to intelligent selection of electrolyte systems for high-power applications.

Advanced Electrolyte Model (AEM)

◆ Report15 -- Cation transit under Faradaic conditions

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)
v1	cation transit velocity, cm/s
t1	cation transit time between electrodes (assuming no polarization), s
vsolv	solvent transit velocity (determined from solvent diffusion migration rates), cm/s
tsolv	solvent transit time between electrodes (assuming no polarization), s
dV/dx	potential gradient, V/cm

◆ Report16 -- Surface Tension and pore filling time over salt conc.

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)

Other terms are self-explanatory.

◆ Report17 -- Percent pore length filled over time

Report 17 provides predictions of electrolyte ingress into porous structures by reporting the percent pore length filled over time for a range of pore diameters. A value of 100% indicates full pore saturation, with filling rates dependent on pore size and electrolyte properties.

time (s)	time for electrolyte ingress into porous structure
0.02 micron	percent pore length filled for 0.02 μm pore radius
0.05 micron	percent pore length filled for 0.05 μm pore radius
0.1 micron	percent pore length filled for 0.1 μm pore radius
0.2 micron	percent pore length filled for 0.2 μm pore radius
0.5 micron	percent pore length filled for 0.5 μm pore radius
1 micron	percent pore length filled for 1 μm pore radius
2 micron	percent pore length filled for 2 μm pore radius
5 micron	percent pore length filled for 5 μm pore radius
10 micron	percent pore length filled for 10 μm pore radius
20 micron	percent pore length filled for 20 μm pore radius

◆ Report18 -- Ligand-wise cation desolvation energy and time

Herein, cation desolvation occurs in order of descending ligands:BE5 $\rightarrow$ BE4 $\rightarrow$ BE3 $\rightarrow$ BE2 $\rightarrow$ BE1 $\rightarrow$ bare cation, where the numbers represent quantity of solvators as determined through cation solvation numbers. Up to six ligands per cation are covered.

Advanced Electrolyte Model (AEM)

m2	molal salt conc. (moles salt per kg pure solvent)
c2	molar salt conc. (moles salt per liter electrolyte)
BE1....BE6	ligand-wise binding energy between solvent and cation, kJ/mole
BE, sum	summation of ligand-wise binding energies for cation, kJ/mole
dt1....dt6	ligand-wise desolvation time between solvent and cation, ns
dt, sum	summation of ligand-wise desolvation times for cation, ns
t-lambda	bulk screening term that reflects the many parameters that affect the cation desolvation process at an electrode interface and behavior of the local solvent population. The "Best" value is where t-lambda is minimized, indicating favorable conditions for cation desolvation and removal of shed solvent. t-lambda has units of ns.

◆ Report19 -- Ligand-wise cation desolvation energy and time (accounting for CS)

Report 19 provides a summary of cation desolvation predictions, including ligand-wise binding energies and desolvation times as a function of salt concentration. Results are based on first-order desolvation kinetics and include the effects of communal ion solvation (CS). The report also includes the t-lambda parameter, which serves as an indicator of overall desolvation favorability.

m2	molal salt concentration (moles salt per kg pure solvent)
c2	molar salt concentration (moles salt per liter electrolyte)
BE1 ... BE6	ligand-wise cation binding (desolvation) energies, kJ/mole
BE, sum	summation of ligand-wise cation binding energies, kJ/mole
dt1 ... dt6	ligand-wise cation desolvation times, ns
dt, sum	summation of ligand-wise cation desolvation times, ns
t-lambda (ns)	bulk screening parameter representing overall cation desolvation behavior; lower values indicate more favorable desolvation conditions

◆ Report20 -- Terms relating to structure and Communal Ion Solvation (CS)

Report 20 provides a summary of terms related to Communal Ion Solvation (CS), including the partitioning of salt and solvent between bulk, CS, and non-CS regions. The report also includes cation solvation numbers, solvent–cation binding energies, and bulk solvation metrics, which together describe how ion–solvent interactions evolve with concentration.

Advanced Electrolyte Model (AEM)

m2, bulk	total molal salt concentration in electrolyte
m2, non-CS	molal salt concentration outside communal solvation regions
m2, CS	molal salt concentration within communal solvation regions
y, free	fraction of free (unassociated) solvent
y, non-CS	fraction of solvent outside communal solvation regions
y, CS	fraction of solvent within communal solvation regions
CS factor	scaling factor representing extent of communal ion solvation
n,s+, bulk	cation solvation number in bulk electrolyte
n,s+, CS	cation solvation number within CS regions
n,s+, ave	average cation solvation number across all regions
BE,+, bulk	cation–solvent binding energy in bulk electrolyte, kJ/mole
BE,+, CS	cation–solvent binding energy within CS regions, kJ/mole
BE,+, ave	average cation–solvent binding energy, kJ/mole
n,s,CS,0	baseline solvent solvation number in CS regions
n,s,CS, ave	average solvent solvation number in CS regions
n,s,nonCS	solvent solvation number outside CS regions
Ratio n,solv,CS to m2,CS	ratio of solvent coordination in CS regions relative to CS salt concentration

◆ Report21 – Properties used in DFN Modeling

Report 21 provides electrolyte transport properties formatted for Doyle–Fuller–Newman (DFN) modeling, including concentration-dependent diffusivity, conductivity, transference number, thermodynamic factors, and activity coefficients. Both molal-based and molar-based formulations are included, with corrections for ion association and communal solvation effects.

m2	molal salt concentration (moles salt per kg solvent)
c2	molar salt concentration (moles salt per liter electrolyte)
c2,eff,trans	effective molar concentration for cation transport, accounting for ion association
D,salt	salt diffusivity in electrolyte, cm ² /s
Spec. Cond.	electrolyte ionic conductivity, mS/cm
t+	cation transference number
gamma+/- (molal)	molal-scale mean activity coefficient
TDF (molal)	thermodynamic factor based on molal concentration
D,app (molal)	apparent salt diffusivity (molal basis), cm ² /s
y+/- (molar)	molar-scale mean activity coefficient
TDF (molar)	thermodynamic factor based on molar concentration
D,app (molar)	apparent salt diffusivity (molar basis), cm ² /s

Additional Notes (Optional, for user guide)

- TDF (Thermodynamic Factor): reflects deviation from ideal solution behavior; does not include $(1 - t^+)$ in this report.
- c2,eff,trans: used directly in Newman-type transport models to account for ion pairing and clustering effects.
- Molal vs. Molar Basis: both are provided to support different modeling frameworks and numerical implementations.
- Communal Solvation Adjustment: activity coefficients and thermodynamic factors include corrections for multi-dentate ion solvation.

7.2 AEM ACCC Solvent Class Examples

Edit ACCC Solvent Class
✕

DMC

1

1. Solvent Class Name

2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

dimethyl carbonate

3. Solvent 1 Identifier

90.08 g/mole ⓘ	25. °C ⓘ	1.063 kg/L ⓘ	20. °C ⓘ
4. Molecular Weight	5. Reference Temperature For Density	6. Density at Reference Temperature	7. Reference Temperature for Relative Permittivity
3.3 ⓘ	-3.45e-03 ⓘ	25. °C ⓘ	0.59 cP ⓘ
8. Relative Permittivity at Reference Temperature	9. Parameter Related to the Temperature Dependence of Relative Permittivity	10. Reference Temperature for Viscosity	11. Viscosity at Reference Temperature
460. ⓘ	2. ⓘ	1.12 ns ⓘ	0.47 ns ⓘ
12. Parameter Related to the Temperature Dependence of Viscosity	13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.368 ⓘ	0.93 D ⓘ	0.5 ⓘ	0.232 ⓘ
16. Refractive Index (Na D line)	17. Dipole Moment, debye	18. π^* Solvatochromic parameter	19. Reichardt's Normalized Polarity Index $[E_T^N]$
2.5 ⓘ	2. ⓘ	17.5 ⓘ	8 ⓘ
20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	22. Donicity or Gutmann Donor Number	23. Index for Solvent Solvation Configuration Around Cation
8 ⓘ	1.4 ⓘ	1.57 ⓘ	12 ⓘ
24. Index for Solvent Solvation Configuration Around Anion	25. Parameter for Solvent Solvation Configuration Around Cation	26. Parameter for Solvent Solvation Configuration Around Anion	27. Number of Atoms in Solvent Molecule
1 ⓘ	3. ⓘ	2. ⓘ	2. ⓘ
28. Classification for Configurational Entropy Calculations as Related to Surface Tension	29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ	1 ⓘ		
32. Default Factor for Configurational Entropy, as Related to Surface Tension	33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

Close

Figure 177. ACCC Solvent Definition for DMC.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

DME

1

1. Solvent Class Name
2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

1,2 dimethoxyethane

3. Solvent 1 Identifier

90.12 g/mole ⓘ 4. Molecular Weight	25. °C ⓘ 5. Reference Temperature For Density	0.86125 kg/L ⓘ 6. Density at Reference Temperature	25. °C ⓘ 7. Reference Temperature for Relative Permittivity
7.154 ⓘ 8. Relative Permittivity at Reference Temperature	-3.85e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C ⓘ 10. Reference Temperature for Viscosity	0.41 cP ⓘ 11. Viscosity at Reference Temperature
380. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	2. ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	1. ns ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.3 ns ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.378 ⓘ 16. Refractive Index (Na D line)	1.61 D ⓘ 17. Dipole Moment, debye	0.53 ⓘ 18. π^* Solvatochromic parameter	0.231 ⓘ 19. Reichardt's Normalized Polarity Index [E_T^N]
2. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1.25 ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	18.4 ⓘ 22. Donicity or Gutmann Donor Number	7 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
2 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	2.4 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	1.5 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	16 ⓘ 27. Number of Atoms in Solvent Molecule
1 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	2. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	4. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	1. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	3 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

✕ Close

Figure 178. ACCC Solvent Definition for DME.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

EC

1

1. Solvent Class Name
2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

ethylene carbonate

3. Solvent 1 Identifier

88.06 g/mole ⓘ 4. Molecular Weight	39. °C ⓘ 5. Reference Temperature For Density	1.3214 kg/L ⓘ 6. Density at Reference Temperature	40. °C ⓘ 7. Reference Temperature for Relative Permittivity
90.36 ⓘ 8. Relative Permittivity at Reference Temperature	-5.175e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	40. °C ⓘ 10. Reference Temperature for Viscosity	1.93 cP ⓘ 11. Viscosity at Reference Temperature
800. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	1.5 ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.357 ns ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.286 ns ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.42 ⓘ 16. Refractive Index (Na D line)	4.51 D ⓘ 17. Dipole Moment, debye	0.84 ⓘ 18. π^* Solvatochromic parameter	0.552 ⓘ 19. Reichardt's Normalized Polarity Index [E_T^N]
4. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1. ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	16.4 ⓘ 22. Donicity or Gutmann Donor Number	10 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
2 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	1.28 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	1.22 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	10 ⓘ 27. Number of Atoms in Solvent Molecule
1 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	3. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	3. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	0. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

✕ Close

Figure 179. ACCC Solvent Definition for EC.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

ⓘ
 ⓘ

1. Solvent Class Name
 2. Number of Solvents

Sol1
Sol2
Sol3

ⓘ

3. Solvent 1 Identifier

88.06 ⓘ 4. Molecular Weight g/mole	39. ⓘ 5. Reference Temperature For Density °C	1.3214 ⓘ 6. Density at Reference Temperature kg/L	40. ⓘ 7. Reference Temperature for Relative Permittivity °C
90.36 ⓘ 8. Relative Permittivity at Reference Temperature	-5.175e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	40. ⓘ 10. Reference Temperature for Viscosity °C	1.93 ⓘ 11. Viscosity at Reference Temperature cP
800. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	1.5 ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.357 ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt ns	0.286 ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt ns
1.42 ⓘ 16. Refractive Index (Na D line)	4.51 ⓘ 17. Dipole Moment, debye D	0.84 ⓘ 18. π^* Solvatochromic parameter	0.552 ⓘ 19. Reichardt's Normalized Polarity Index [E_T^N]
4. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1. ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	16.4 ⓘ 22. Donicity or Gutmann Donor Number	10 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
2 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	1.28 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	1.22 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	10 ⓘ 27. Number of Atoms in Solvent Molecule
1 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	3. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	3. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	0. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Figure 180. ACCC Solvent Definition for EC-DMC-NPA.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

EC EMC

2

1. Solvent Class Name
2. Number of Solvents

Sol1
Sol2

IMPORT SOLVENT VALUES ▾

ethylene carbonate

3. Solvent 1 Identifier

88.06 g/mole 4. Molecular Weight	39. °C 5. Reference Temperature For Density	1.3214 kg/L 6. Density at Reference Temperature	40. °C 7. Reference Temperature for Relative Permittivity
90.36 8. Relative Permittivity at Reference Temperature	-5.175e-03 9. Parameter Related to the Temperature Dependence of Relative Permittivity	40. °C 10. Reference Temperature for Viscosity	1.93 cP 11. Viscosity at Reference Temperature
800. 12. Parameter Related to the Temperature Dependence of Viscosity	1.5 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.357 ns 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.286 ns 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.42 16. Refractive Index (Na D line)	4.51 D 17. Dipole Moment, debye	0.84 18. π^* Solvatochromic parameter	0.552 19. Reichardt's Normalized Polarity Index [E_T^N]
4. 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1. 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	16.4 22. Donicity or Gutmann Donor Number	10 23. Index for Solvent Solvation Configuration Around Cation
2 24. Index for Solvent Solvation Configuration Around Anion	1.28 25. Parameter for Solvent Solvation Configuration Around Cation	1.22 26. Parameter for Solvent Solvation Configuration Around Anion	10 27. Number of Atoms in Solvent Molecule
1 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	3. 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	3. 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	0. 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

Close

Figure 181. ACCC Solvent Definition for EC-EMC.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

EC_PC EMC_MA_BN

5

1. Solvent Class Name
2. Number of Solvents

Sol1
Sol2
Sol3
Sol4
Sol5

IMPORT SOLVENT VALUES ▾

ethylene carbonate

3. Solvent 1 Identifier

88.06 g/mole 4. Molecular Weight	39. °C 5. Reference Temperature For Density	1.3214 kg/L 6. Density at Reference Temperature	40. °C 7. Reference Temperature for Relative Permittivity
90.36 8. Relative Permittivity at Reference Temperature	-5.175e-03 9. Parameter Related to the Temperature Dependence of Relative Permittivity	40. °C 10. Reference Temperature for Viscosity	1.93 cP 11. Viscosity at Reference Temperature
800. 12. Parameter Related to the Temperature Dependence of Viscosity	1.5 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.357 ns 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.286 ns 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.42 16. Refractive Index (Na D line)	4.51 D 17. Dipole Moment, debye	0.84 18. π^* Solvatochromic parameter	0.552 19. Reichardt's Normalized Polarity Index [E_T^N]
4. 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1. 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	16.4 22. Donicity or Gutmann Donor Number	10 23. Index for Solvent Solvation Configuration Around Cation
2 24. Index for Solvent Solvation Configuration Around Anion	1.28 25. Parameter for Solvent Solvation Configuration Around Cation	1.22 26. Parameter for Solvent Solvation Configuration Around Anion	10 27. Number of Atoms in Solvent Molecule
1 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	3. 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	3. 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	0. 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

✕
Close

Figure 182. ACCC Solvent Definition for EC-PC-EMC-MA-BN.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

EMC

1

1. Solvent Class Name
2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

ethylmethyl carbonate

1

3. Solvent 1 Identifier

104.11 g/mole i 4. Molecular Weight	25. °C i 5. Reference Temperature For Density	1.007 kg/L i 6. Density at Reference Temperature	25. °C i 7. Reference Temperature for Relative Permittivity
2.958 i 8. Relative Permittivity at Reference Temperature	-3.65e-03 i 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C i 10. Reference Temperature for Viscosity	0.66 cP i 11. Viscosity at Reference Temperature
410. i 12. Parameter Related to the Temperature Dependence of Viscosity	2. i 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	1.01 ns i 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.45 ns i 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.376 i 16. Refractive Index (Na D line)	1.0 D i 17. Dipole Moment, debye	0.47 i 18. π^* Solvatochromic parameter	0.21 i 19. Reichardt's Normalized Polarity Index [E_T^N]
2. i 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	2. i 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	15.7 i 22. Donicity or Gutmann Donor Number	8 i 23. Index for Solvent Solvation Configuration Around Cation
2 i 24. Index for Solvent Solvation Configuration Around Anion	1.3 i 25. Parameter for Solvent Solvation Configuration Around Cation	1.6 i 26. Parameter for Solvent Solvation Configuration Around Anion	15 i 27. Number of Atoms in Solvent Molecule
1 i 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	3. i 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	3. i 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	3. i 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. i 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 i 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

Close

Figure 183. ACCC Solvent Definition for EMC.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

1. Solvent Class Name
2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

3. Solvent 1 Identifier

102.13 g/mole 4. Molecular Weight	16. °C 5. Reference Temperature For Density	0.895 kg/L 6. Density at Reference Temperature	25. °C 7. Reference Temperature for Relative Permittivity
5.65 8. Relative Permittivity at Reference Temperature	-3.65e-03 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C 10. Reference Temperature for Viscosity	0.51 cP 11. Viscosity at Reference Temperature
520. 12. Parameter Related to the Temperature Dependence of Viscosity	2. 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.9568 ns 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.3675 ns 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.381 16. Refractive Index (Na D line)	1.8 D 17. Dipole Moment, debye	0.5 18. π^* Solvatochromic parameter	0.25 19. Reichardt's Normalized Polarity Index [E_T^{NH}]
2. 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1. 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	16.7 22. Donicity or Gutmann Donor Number	7 23. Index for Solvent Solvation Configuration Around Cation
6 24. Index for Solvent Solvation Configuration Around Anion	1.1 25. Parameter for Solvent Solvation Configuration Around Cation	1.85 26. Parameter for Solvent Solvation Configuration Around Anion	17 27. Number of Atoms in Solvent Molecule
1 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	2. 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	5. 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	3. 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 33. Index for Solvent Viscosity Mixing Rule		

Figure 184. ACCC Solvent Definition for EP.

Advanced Electrolyte Model (AEM)

X

Edit ACCC Solvent Class

EP_BN

2

1. Solvent Class Name
2. Number of Solvents

Sol1
Sol2

IMPORT SOLVENT VALUES ▼

ethyl propionate

3. Solvent 1 Identifier

102.13 g/mole ⓘ 4. Molecular Weight	16. °C ⓘ 5. Reference Temperature For Density	0.895 kg/L ⓘ 6. Density at Reference Temperature	25. °C ⓘ 7. Reference Temperature for Relative Permittivity
5.65 ⓘ 8. Relative Permittivity at Reference Temperature	-3.65e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C ⓘ 10. Reference Temperature for Viscosity	0.51 cP ⓘ 11. Viscosity at Reference Temperature
540. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	2. ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.8 ns ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.3 ns ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.381 ⓘ 16. Refractive Index (Na D line)	1.8 D ⓘ 17. Dipole Moment, debye	0.5 ⓘ 18. n^* Solvatochromic parameter	0.25 ⓘ 19. Reichardt's Normalized Polarity Index [E_T^N]
2. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1. ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	16.7 ⓘ 22. Donicity or Gutmann Donor Number	7 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
6 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	1.35 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	1.45 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	17 ⓘ 27. Number of Atoms in Solvent Molecule
1 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	2. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	5. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	3. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

Close

Figure 185. ACCC Solvent Definition for EP-BN.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

1. Solvent Class Name

2. Number of Solvents

Sol1

3. Solvent 1 Identifier

74.08 g/mole i	25. °C i	0.928 kg/L i	20. °C i
4. Molecular Weight	5. Reference Temperature For Density	6. Density at Reference Temperature	7. Reference Temperature for Relative Permittivity
6.94 i	-4.1e-03 i	20. °C i	0.381 cP i
8. Relative Permittivity at Reference Temperature	9. Parameter Related to the Temperature Dependence of Relative Permittivity	10. Reference Temperature for Viscosity	11. Viscosity at Reference Temperature
400. i	2. i	0.95 ns i	0.67 ns i
12. Parameter Related to the Temperature Dependence of Viscosity	13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.361 i	1.69 D i	0.6 i	0.253 i
16. Refractive Index (Na D line)	17. Dipole Moment, debye	18. π^* Solvatochromic parameter	19. Reichardt's Normalized Polarity Index [E_T^{NH}]
2.5 i	1.5 i	17.7 i	7 i
20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	22. Donicity or Gutmann Donor Number	23. Index for Solvent Solvation Configuration Around Cation
4 i	1.56 i	1.2 i	11 i
24. Index for Solvent Solvation Configuration Around Anion	25. Parameter for Solvent Solvation Configuration Around Cation	26. Parameter for Solvent Solvation Configuration Around Anion	27. Number of Atoms in Solvent Molecule
1 i	2. i	3. i	1. i
28. Classification for Configurational Entropy Calculations as Related to Surface Tension	29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. i	1 i		
32. Default Factor for Configurational Entropy, as Related to Surface Tension	33. Index for Solvent Viscosity Mixing Rule		

Figure 186. ACCC Solvent Definition for MA.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

1. Solvent Class Name

2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

3. Solvent 1 Identifier

102.13 g/mole i	25. °C i	0.883 kg/L i	20. °C i
4. Molecular Weight	5. Reference Temperature For Density	6. Density at Reference Temperature	7. Reference Temperature for Relative Permittivity
4. i	-3.85e-03 i	25. °C i	0.55 cP i
8. Relative Permittivity at Reference Temperature	9. Parameter Related to the Temperature Dependence of Relative Permittivity	10. Reference Temperature for Viscosity	11. Viscosity at Reference Temperature
480. i	2. i	1.1 ns i	0.45 ns i
12. Parameter Related to the Temperature Dependence of Viscosity	13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.384 i	1.78 D i	0.45 i	0.21 i
16. Refractive Index (Na D line)	17. Dipole Moment, debye	18. π^* Solvatochromic parameter	19. Reichardt's Normalized Polarity Index [$E_T^{(N)}$]
2. i	1. i	15.7 i	7 i
20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	22. Donicity or Gutmann Donor Number	23. Index for Solvent Solvation Configuration Around Cation
6 i	1.1 i	1.6 i	17 i
24. Index for Solvent Solvation Configuration Around Anion	25. Parameter for Solvent Solvation Configuration Around Cation	26. Parameter for Solvent Solvation Configuration Around Anion	27. Number of Atoms in Solvent Molecule
1 i	2. i	5. i	2.5 i
28. Classification for Configurational Entropy Calculations as Related to Surface Tension	29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. i	1 i		
32. Default Factor for Configurational Entropy, as Related to Surface Tension	33. Index for Solvent Viscosity Mixing Rule		

Figure 187. ACCC Solvent Definition for NPA.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

PC

1

1. Solvent Class Name
2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

propylene carbonate

3. Solvent 1 Identifier

102.09 g/mole ⓘ 4. Molecular Weight	25. °C ⓘ 5. Reference Temperature For Density	1.199862 kg/L ⓘ 6. Density at Reference Temperature	25. °C ⓘ 7. Reference Temperature for Relative Permittivity
65.1848 ⓘ 8. Relative Permittivity at Reference Temperature	-3.65e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C ⓘ 10. Reference Temperature for Viscosity	2.54187 cP ⓘ 11. Viscosity at Reference Temperature
830. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	1.5 ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.47 ns ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.67 ns ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.4209 ⓘ 16. Refractive Index (Na D line)	4.98 D ⓘ 17. Dipole Moment, debye	0.83 ⓘ 18. π^* Solvatochromic parameter	0.472 ⓘ 19. Reichardt's Normalized Polarity Index $[E_T^{N(1)}$
4. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	2. ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	15.1 ⓘ 22. Donicity or Gutmann Donor Number	4 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
4 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	1.6 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	0.65 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	13 ⓘ 27. Number of Atoms in Solvent Molecule
2 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	3. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	4. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	1. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	2 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

✕ Close

Figure 188. ACCC Solvent Definition for PC

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

ⓘ
1. Solvent Class Name

ⓘ
2. Number of Solvents

Sol1
Sol2
Sol3
Sol4
Sol5

IMPORT SOLVENT VALUES ▾

ⓘ
3. Solvent 1 Identifier

102.09 g/mole ⓘ 4. Molecular Weight	25. °C ⓘ 5. Reference Temperature For Density	1.199862 kg/L ⓘ 6. Density at Reference Temperature	25. °C ⓘ 7. Reference Temperature for Relative Permittivity
65.1848 ⓘ 8. Relative Permittivity at Reference Temperature	-3.65e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C ⓘ 10. Reference Temperature for Viscosity	2.54187 cP ⓘ 11. Viscosity at Reference Temperature
830. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	1.5 ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.47 ns ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.67 ns ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.4209 ⓘ 16. Refractive Index (Na D line)	4.98 D ⓘ 17. Dipole Moment, debye	0.83 ⓘ 18. π^* Solvatochromic parameter	0.472 ⓘ 19. Reichardt's Normalized Polarity Index [E_T^N]
4. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	2. ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	15.1 ⓘ 22. Donicity or Gutmann Donor Number	4 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
4 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	1.6 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	0.65 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	13 ⓘ 27. Number of Atoms in Solvent Molecule
2 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	3. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	4. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	1. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	2 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Delete
Save
⌵ Close

Figure 189. ACCC Solvent Definition for PC-EC-EMC-MA-BN.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

1. Solvent Class Name

2. Number of Solvents

Sol1 Sol2

IMPORT SOLVENT VALUES ▾

3. Solvent 1 Identifier

102.09 i g/mole 4. Molecular Weight	25. i °C 5. Reference Temperature For Density	1.199862 i kg/L 6. Density at Reference Temperature	25. i °C 7. Reference Temperature for Relative Permittivity
65.1848 i	-3.65e-03 i	25. i °C 10. Reference Temperature for Viscosity	2.54187 i cP 11. Viscosity at Reference Temperature
830. i	1.5 i	0.47 i ns 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.67 i ns 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.4209 i	4.98 i D 17. Dipole Moment, debye	0.83 i	0.472 i 19. Reichardt's Normalized Polarity Index [E_T⁺]
4. i	2. i	15.1 i	4 i
4 i	1.6 i	0.65 i	13 i
2 i	3. i	4. i	1. i
1. i	2 i		

32. Default Factor for Configurational Entropy, as Related to Surface Tension

33. Index for Solvent Viscosity Mixing Rule

Figure 190. ACCC Solvent Definition for PC-EMC.

Advanced Electrolyte Model (AEM)

Edit ACCC Solvent Class

PN ⓘ

1 ⓘ

1. Solvent Class Name

2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

propionitrile (PrN) ⓘ

3. Solvent 1 Identifier

55.08 g/mole ⓘ 4. Molecular Weight	25. °C ⓘ 5. Reference Temperature For Density	0.777 kg/L ⓘ 6. Density at Reference Temperature	20. °C ⓘ 7. Reference Temperature for Relative Permittivity
28.86 ⓘ 8. Relative Permittivity at Reference Temperature	-4.3e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C ⓘ 10. Reference Temperature for Viscosity	0.411 cP ⓘ 11. Viscosity at Reference Temperature
580. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	2. ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.62 ns ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.27 ns ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.366 ⓘ 16. Refractive Index (Na D line)	4.04 D ⓘ 17. Dipole Moment, debye	0.71 ⓘ 18. π^* Solvatochromic parameter	0.398 ⓘ 19. Reichardt's Normalized Polarity Index [E_T^N]
3. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1.5 ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	16.1 ⓘ 22. Donicity or Gutmann Donor Number	4 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
14 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	1.4 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	1.45 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	9 ⓘ 27. Number of Atoms in Solvent Molecule
1 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	1. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	3. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	1. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	1 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Delete
Save
✕ Close

Figure 191. ACCC Solvent Definition for PN.

Advanced Electrolyte Model (AEM)

X

Edit ACCC Solvent Class

VEC ⓘ

1 ⓘ

1. Solvent Class Name
2. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

vinyl ethylene carbonate ⓘ

3. Solvent 1 Identifier

114.0993 g/mole ⓘ 4. Molecular Weight	20. °C ⓘ 5. Reference Temperature For Density	1.188 kg/L ⓘ 6. Density at Reference Temperature	25. °C ⓘ 7. Reference Temperature for Relative Permittivity
40. ⓘ 8. Relative Permittivity at Reference Temperature	-4.0e-03 ⓘ 9. Parameter Related to the Temperature Dependence of Relative Permittivity	25. °C ⓘ 10. Reference Temperature for Viscosity	2.2 cP ⓘ 11. Viscosity at Reference Temperature
820. ⓘ 12. Parameter Related to the Temperature Dependence of Viscosity	2. ⓘ 13. Reference Maximum Solvation Sites Within a Single Solvent Molecule that Promote Interaction With Cations	0.55 ns ⓘ 14. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C, Infinite Dilution Salt	0.6 ns ⓘ 15. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C, Infinite Dilution Salt
1.546 ⓘ 16. Refractive Index (Na D line)	3.5 D ⓘ 17. Dipole Moment, debye	0.83 ⓘ 18. π^* Solvatochromic parameter	0.51 ⓘ 19. Reichardt's Normalized Polarity Index [E_T^{N}]
4. ⓘ 20. Reference Solvation Number for Cation at Infinite Dilution (estimate)	2. ⓘ 21. Reference Solvation Number for Anion at Infinite Dilution (estimate)	15. ⓘ 22. Donicity or Gutmann Donor Number	10 ⓘ 23. Index for Solvent Solvation Configuration Around Cation
4 ⓘ 24. Index for Solvent Solvation Configuration Around Anion	1.3 ⓘ 25. Parameter for Solvent Solvation Configuration Around Cation	1.35 ⓘ 26. Parameter for Solvent Solvation Configuration Around Anion	14 ⓘ 27. Number of Atoms in Solvent Molecule
1 ⓘ 28. Classification for Configurational Entropy Calculations as Related to Surface Tension	4. ⓘ 29. Parameter for Oxygen (or S, P) Species in Configurational Entropy, as Related to Surface Tension Configuration Around Anion	4. ⓘ 30. Parameter for Carbon Species in Configurational Entropy, as Related to Surface Tension	2. ⓘ 31. Parameter for Molecular Pivot Points in Solvent for Configurational Entropy, as Related to Surface Tension
1. ⓘ 32. Default Factor for Configurational Entropy, as Related to Surface Tension	2 ⓘ 33. Index for Solvent Viscosity Mixing Rule		

Delete

Save

Close

Figure 192. ACCC Solvent Definition for VEC.

7.3 AEM ACCC Salt Class Examples

Edit ACCC Salt Class

×

LiBOB_w_PC_EA ⓘ

1. Salt Class Name

LiBOB ⓘ

2. Salt Identifier

Regular Salt

Ionic Liquid

3. Salt Type

1.36 ⓘ Å

4. Diameter of Cation from Crystalline Basis

6.396 ⓘ Å

5. Diameter of Anion from Crystalline Basis [Å]

3.358 ⓘ Å

6. Equivalent Diameter of Anion Accounting for Electrostatic Charge Distribution of Anion

2. ⓘ

7. Number of Charge Regions for Anion

1. ⓘ

8. Unit Charge of Cation

-1. ⓘ

9. Unit Charge of Anion

193.79 ⓘ g/mole

10. Molecular Weight of Salt

6.941 ⓘ g/mole

11. Molecular Weight of Cation

1.62 ⓘ g/cm³

12. Density of Pure Crystalline Salt

19.3 ⓘ

13. Relative Permittivity of Pure Crystalline Salt

0 ⓘ Å

14. Factor to Express Equivalent Diameter of Cation in RTIL Accounting for Electrostatic Charge Distribution of Cation

0 ⓘ

15. Number of Charge Regions for a Given Cation in RTIL

2 ⓘ

16. Number of Solvents

Sol1

Sol2

IMPORT SOLVENT VALUES ▾

0.47 ⓘ ns

17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C

0.5 ⓘ ns

18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C

4. ⓘ

19. Reference Solvation Number for Cation at Infinite Dilution (estimate)

2. ⓘ

20. Reference Solvation Number for Anion at Infinite Dilution (estimate)

4 ⓘ

21. Index for Solvent Solvation Configuration Around Cation

4 ⓘ

22. Index for Solvent Solvation Configuration Around Anion

1.6 ⓘ

23. Parameter for Solvent Solvation Configuration Around Cation

2.8 ⓘ

24. Parameter for Solvent Solvation Configuration Around Anion

Delete

Save

×

Close

Sol1

Sol2

IMPORT SOLVENT VALUES ▾

0.9 ⓘ ns

17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C

0.38 ⓘ ns

18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C

2. ⓘ

19. Reference Solvation Number for Cation at Infinite Dilution (estimate)

1.5 ⓘ

20. Reference Solvation Number for Anion at Infinite Dilution (estimate)

3 ⓘ

21. Index for Solvent Solvation Configuration Around Cation

4 ⓘ

22. Index for Solvent Solvation Configuration Around Anion

1.22 ⓘ

23. Parameter for Solvent Solvation Configuration Around Cation

4.2 ⓘ

24. Parameter for Solvent Solvation Configuration Around Anion

Delete

Save

×

Close

Figure 193. ACCC Salt Definition for LiBOB_w_PC_EA

Advanced Electrolyte Model (AEM)

×

Edit ACCC Salt Class

LiPF6_w_EC ⓘ

1. Salt Class Name

LiPF6 ⓘ

2. Salt Identifier

Regular Salt

Ionic Liquid

3. Salt Type

1.36 ⓘ Å

4. Diameter of Cation from Crystalline Basis

4.9 ⓘ Å

5. Diameter of Anion from Crystalline Basis [Å]

4.9 ⓘ Å

6. Equivalent Diameter of Anion Accounting for Electrostatic Charge Distribution of Anion

1 ⓘ

7. Number of Charge Regions for Anion

1 ⓘ

8. Unit Charge of Cation

-1 ⓘ

9. Unit Charge of Anion

151.90 ⓘ g/mole

10. Molecular Weight of Salt

6.941 ⓘ g/mole

11. Molecular Weight of Cation

2.6 ⓘ g/cm³

12. Density of Pure Crystalline Salt

19 ⓘ

13. Relative Permittivity of Pure Crystalline Salt

0 ⓘ Å

14. Factor to Express Equivalent Diameter of Cation in RTIL Accounting for Electrostatic Charge Distribution of Cation

0 ⓘ

15. Number of Charge Regions for a Given Cation in RTIL

1 ⓘ

16. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

0.357 ⓘ ns

17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C

0.286 ⓘ ns

18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C

4 ⓘ

19. Reference Solvation Number for Cation at Infinite Dilution (estimate)

1 ⓘ

20. Reference Solvation Number for Anion at Infinite Dilution (estimate)

10 ⓘ

21. Index for Solvent Solvation Configuration Around Cation

2 ⓘ

22. Index for Solvent Solvation Configuration Around Anion

1.28 ⓘ

23. Parameter for Solvent Solvation Configuration Around Cation

1.22 ⓘ

24. Parameter for Solvent Solvation Configuration Around Anion

Delete

Save

Close

Figure 194. ACCC Salt Definition for LiPF6_w_EC

Advanced Electrolyte Model (AEM)

Edit ACCC Salt Class

ⓘ
1. Salt Class Name

ⓘ
2. Salt Identifier

Regular Salt
Ionic Liquid

3. Salt Type

1.36 Å ⓘ 4. Diameter of Cation from Crystalline Basis	4.9 Å ⓘ 5. Diameter of Anion from Crystalline Basis [Å]	4.9 Å ⓘ 6. Equivalent Diameter of Anion Accounting for Electrostatic Charge Distribution of Anion	1. ⓘ 7. Number of Charge Regions for Anion
1. ⓘ 8. Unit Charge of Cation	-1. ⓘ 9. Unit Charge of Anion	151.90 g/mole ⓘ 10. Molecular Weight of Salt	6.941 g/mole ⓘ 11. Molecular Weight of Cation
2.6 g/cm³ ⓘ 12. Density of Pure Crystalline Salt	19. ⓘ 13. Relative Permittivity of Pure Crystalline Salt	0 Å ⓘ 14. Factor to Express Equivalent Diameter of Cation in RTIL Accounting for Electrostatic Charge Distribution of Cation	0 ⓘ 15. Number of Charge Regions for a Given Cation in RTIL

ⓘ
16. Number of Solvents
Sol1 Sol2

IMPORT SOLVENT VALUES ▼

0.357 ns ⓘ 17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C	0.286 ns ⓘ 18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C	4. ⓘ 19. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1. ⓘ 20. Reference Solvation Number for Anion at Infinite Dilution (estimate)
10 ⓘ 21. Index for Solvent Solvation Configuration Around Cation	2 ⓘ 22. Index for Solvent Solvation Configuration Around Anion	1.28 ⓘ 23. Parameter for Solvent Solvation Configuration Around Cation	1.22 ⓘ 24. Parameter for Solvent Solvation Configuration Around Anion

Delete
Save
✕ Close

Sol1 Sol2

IMPORT SOLVENT VALUES ▼

1.01 ns ⓘ 17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C	0.45 ns ⓘ 18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C	2. ⓘ 19. Reference Solvation Number for Cation at Infinite Dilution (estimate)	2. ⓘ 20. Reference Solvation Number for Anion at Infinite Dilution (estimate)
8 ⓘ 21. Index for Solvent Solvation Configuration Around Cation	2 ⓘ 22. Index for Solvent Solvation Configuration Around Anion	1.3 ⓘ 23. Parameter for Solvent Solvation Configuration Around Cation	1.6 ⓘ 24. Parameter for Solvent Solvation Configuration Around Anion

Delete
Save
✕ Close

Figure 195. ACCC Salt Definition for LiPF6_w_EC_EMC.

Advanced Electrolyte Model (AEM)

×

Edit ACCC Salt Class

LiPF6_w_EMC ⓘ

1. Salt Class Name

LiPF6 ⓘ

2. Salt Identifier

Regular Salt

Ionic Liquid

3. Salt Type

1.36 ⓘ Å ⓘ

4. Diameter of Cation from Crystalline Basis

4.9 ⓘ Å ⓘ

5. Diameter of Anion from Crystalline Basis [Å]

4.9 ⓘ Å ⓘ

6. Equivalent Diameter of Anion Accounting for Electrostatic Charge Distribution of Anion

1. ⓘ

7. Number of Charge Regions for Anion

1. ⓘ

8. Unit Charge of Cation

-1. ⓘ

9. Unit Charge of Anion

151.90 ⓘ g/mole ⓘ

10. Molecular Weight of Salt

6.941 ⓘ g/mole ⓘ

11. Molecular Weight of Cation

2.6 ⓘ g/cm³ ⓘ

12. Density of Pure Crystalline Salt

19. ⓘ

13. Relative Permittivity of Pure Crystalline Salt

0 ⓘ Å ⓘ

14. Factor to Express Equivalent Diameter of Cation in RTIL Accounting for Electrostatic Charge Distribution of Cation

0 ⓘ

15. Number of Charge Regions for a Given Cation in RTIL

1 ⓘ

16. Number of Solvents

Sol1

IMPORT SOLVENT VALUES ▾

1.01 ⓘ ns ⓘ

17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C

0.45 ⓘ ns ⓘ

18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C

2. ⓘ

19. Reference Solvation Number for Cation at Infinite Dilution (estimate)

2. ⓘ

20. Reference Solvation Number for Anion at Infinite Dilution (estimate)

8 ⓘ

21. Index for Solvent Solvation Configuration Around Cation

2 ⓘ

22. Index for Solvent Solvation Configuration Around Anion

1.3 ⓘ

23. Parameter for Solvent Solvation Configuration Around Cation

1.6 ⓘ

24. Parameter for Solvent Solvation Configuration Around Anion

🗑️ Delete

💾 Save

⌵ Close

Figure 196. ACCC Salt Definition for LiPF6_w_EC_EMC.

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Edit ACCC Salt Class
✕

ⓘ
1. Salt Class Name

ⓘ
2. Salt Identifier

Regular Salt
Ionic Liquid
3. Salt Type

1.776 Å ⓘ 4. Diameter of Cation from Crystalline Basis	5.66 Å ⓘ 5. Diameter of Anion from Crystalline Basis [Å]	4.811 Å ⓘ 6. Equivalent Diameter of Anion Accounting for Electrostatic Charge Distribution of Anion	1 ⓘ 7. Number of Charge Regions for Anion
1 ⓘ 8. Unit Charge of Cation	-1 ⓘ 9. Unit Charge of Anion	308.37 g/mole ⓘ 10. Molecular Weight of Salt	128.37 g/mole ⓘ 11. Molecular Weight of Cation
1.33 g/cm³ ⓘ 12. Density of Pure Crystalline Salt	22 ⓘ 13. Relative Permittivity of Pure Crystalline Salt	0.9 Å ⓘ 14. Factor to Express Equivalent Diameter of Cation in RTIL Accounting for Electrostatic Charge Distribution of Cation	1.2 ⓘ 15. Number of Charge Regions for a Given Cation in RTIL

ⓘ
16. Number of Solvents

Sol1 Sol2

IMPORT SOLVENT VALUES ▾
✕

1.15 ns ⓘ 17. Reference Value for Average Residence Time of Solvent Around Cation at 25 °C	0.32 ns ⓘ 18. Reference Value for Average Residence Time of Solvent Around Anion at 25 °C	5 ⓘ 19. Reference Solvation Number for Cation at Infinite Dilution (estimate)	1.5 ⓘ 20. Reference Solvation Number for Anion at Infinite Dilution (estimate)
10 ⓘ 21. Index for Solvent Solvation Configuration Around Cation	2 ⓘ 22. Index for Solvent Solvation Configuration Around Anion	1.28 ⓘ 23. Parameter for Solvent Solvation Configuration Around Cation	1.22 ⓘ 24. Parameter for Solvent Solvation Configuration Around Anion

Delete
Save
✕ Close

Figure 197. ACCC Salt Definition for PYR13-FSI_w_EC EMC.

7.4 ACCC Validation Examples

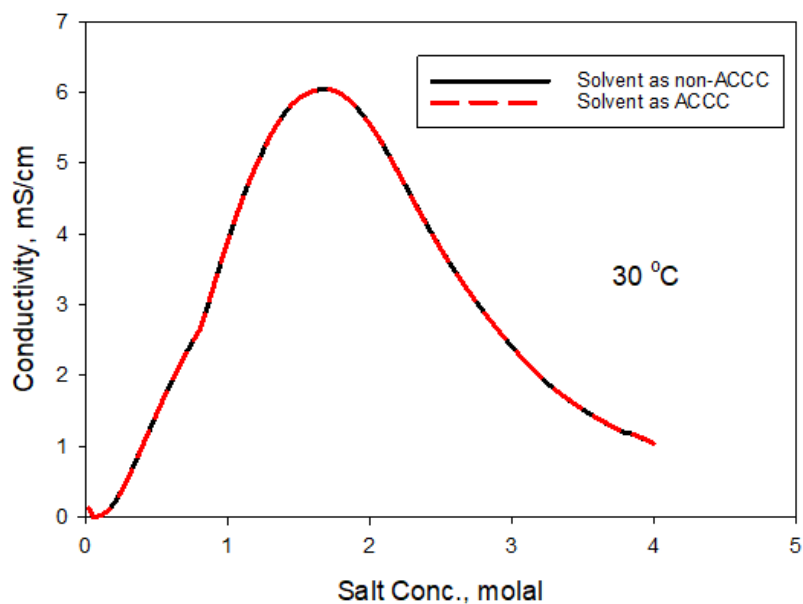


Figure 198. EMC+LiPF₆ at 30⁰

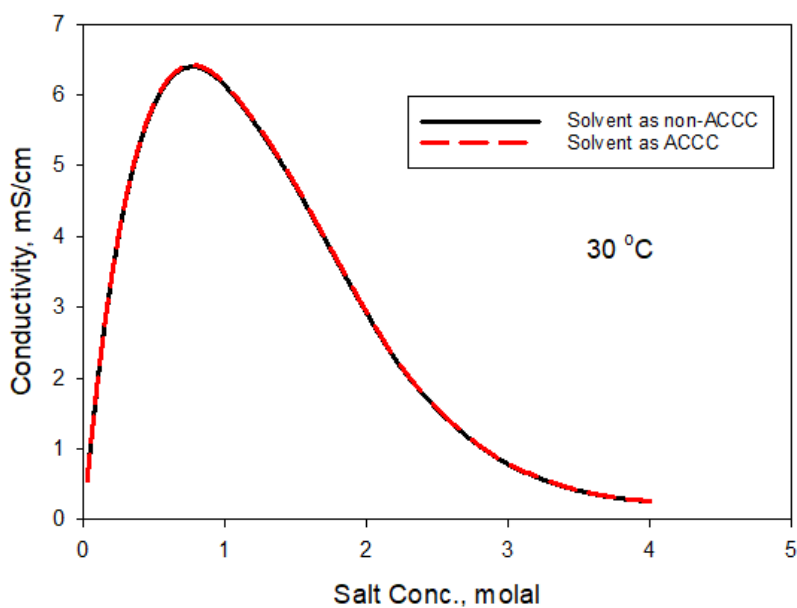


Figure 199. PC+LiPF₆ at 30⁰

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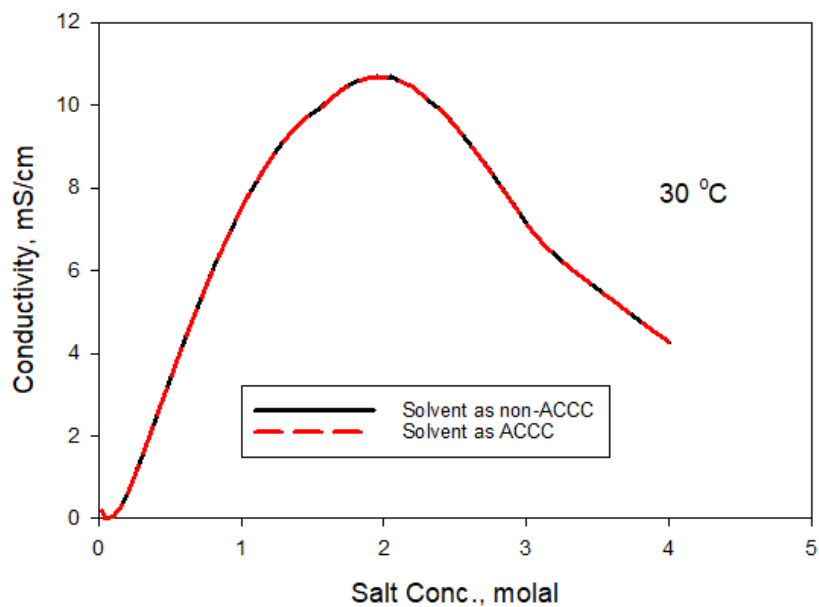


Figure 200. DMC+LiPF₆ at 30⁰

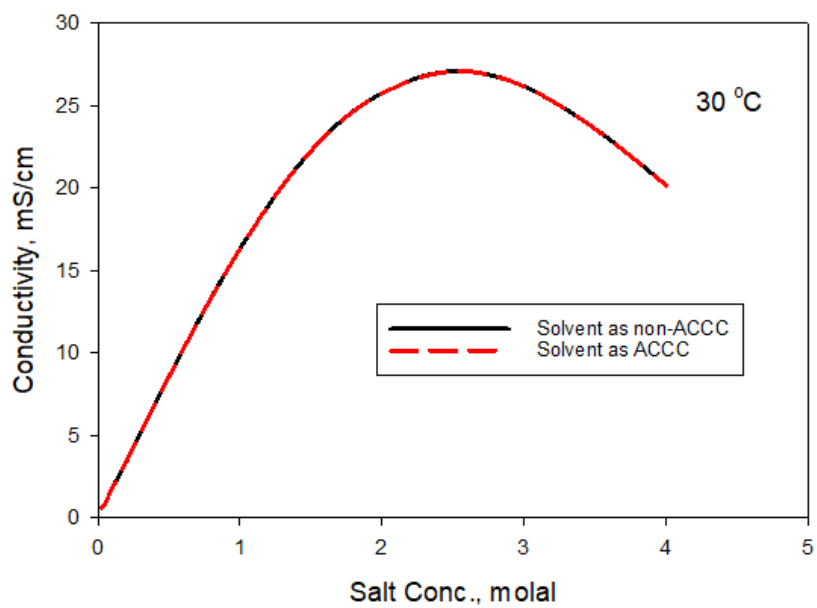


Figure 201. MA+LiPF₆ at 30⁰

Advanced Electrolyte Model (AEM)

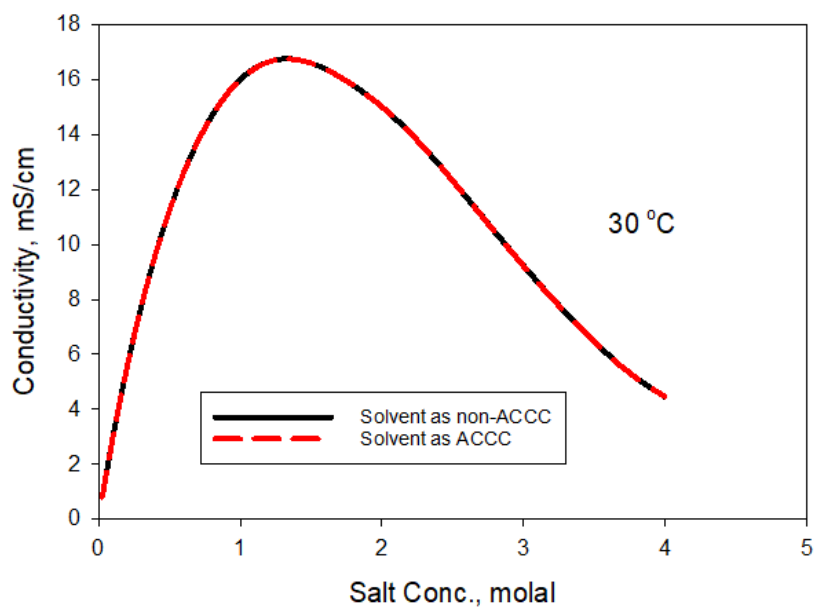


Figure 202. BN+LiPF₆ at 30⁰

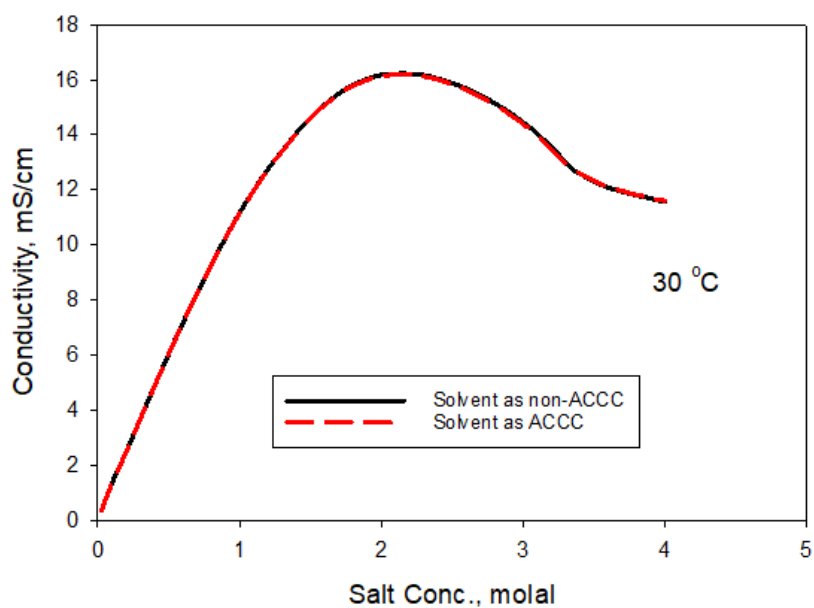


Figure 203. DME+LiPF₆ at 30⁰

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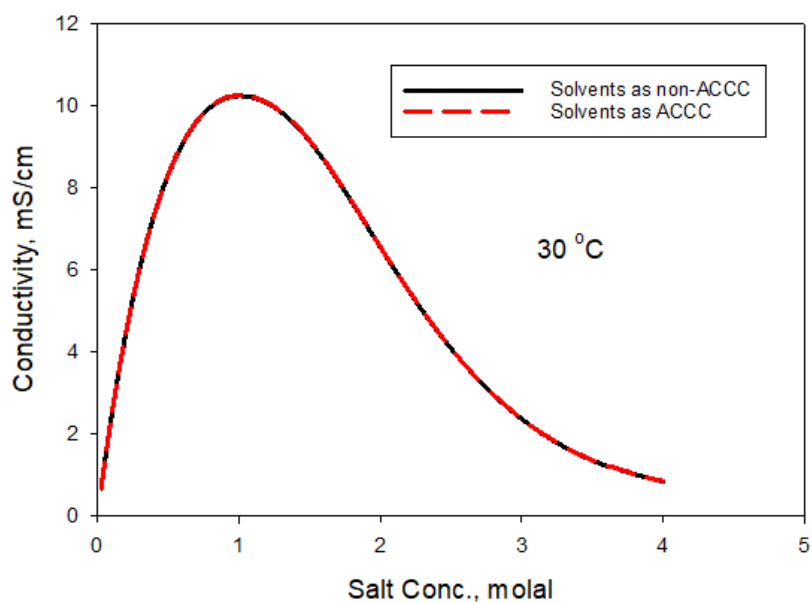


Figure 204. PC-EMC (4:6 by mass)+LiPF₆ at 30⁰

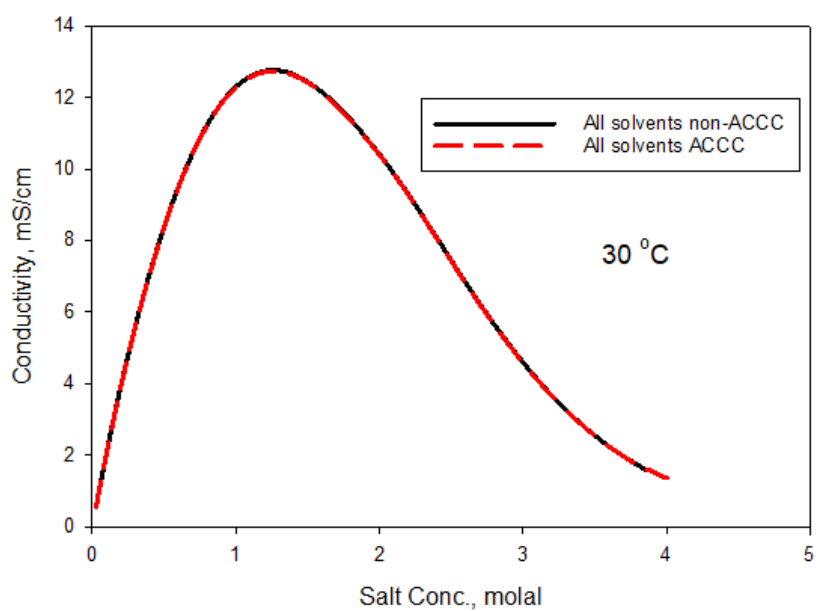


Figure 205. EC-DMC-nPA (2:5:3 by mass)+LiPF₆ at 30⁰

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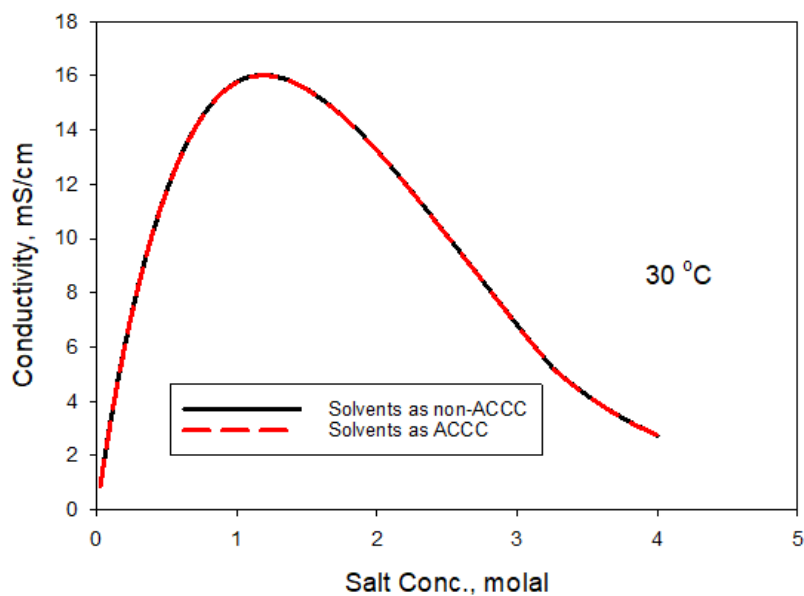


Figure 206. PC-EC-EMC-MA-BN (1:1:1:1 by mass) + LiPF₆ at 30°

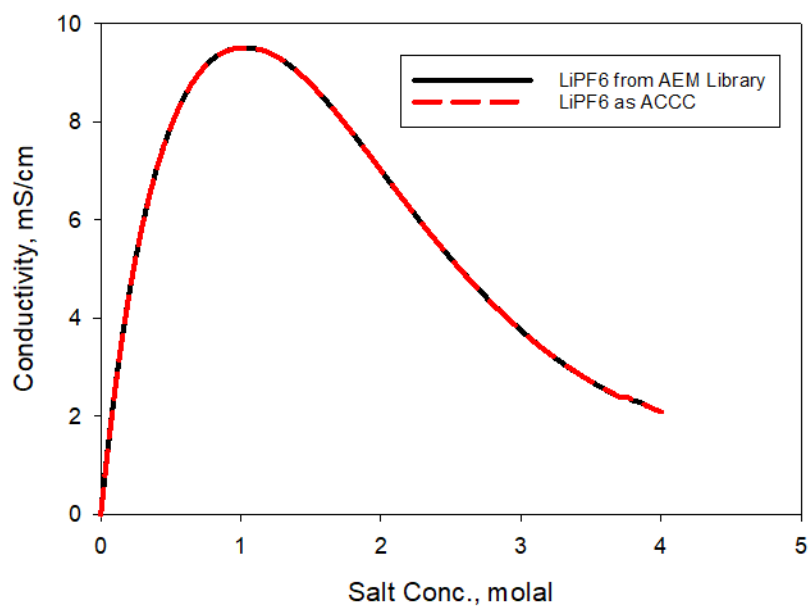


Figure 207. EC-EMC (3:7 by mass) + (0.5LiPF₆ + 0.5LiFSI)

Advanced Electrolyte Model (AEM)

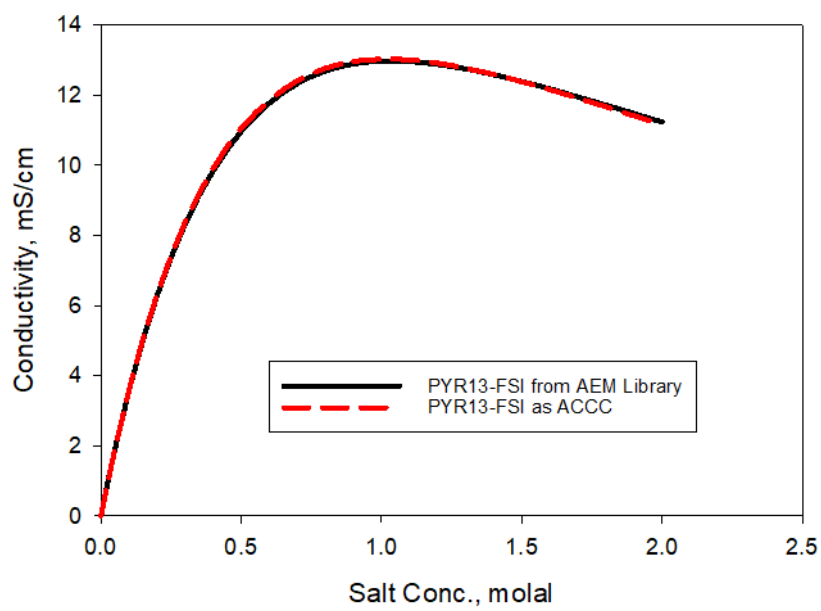


Figure 208. EC-EMC (3:7 by mass)+ PYR13-FSI

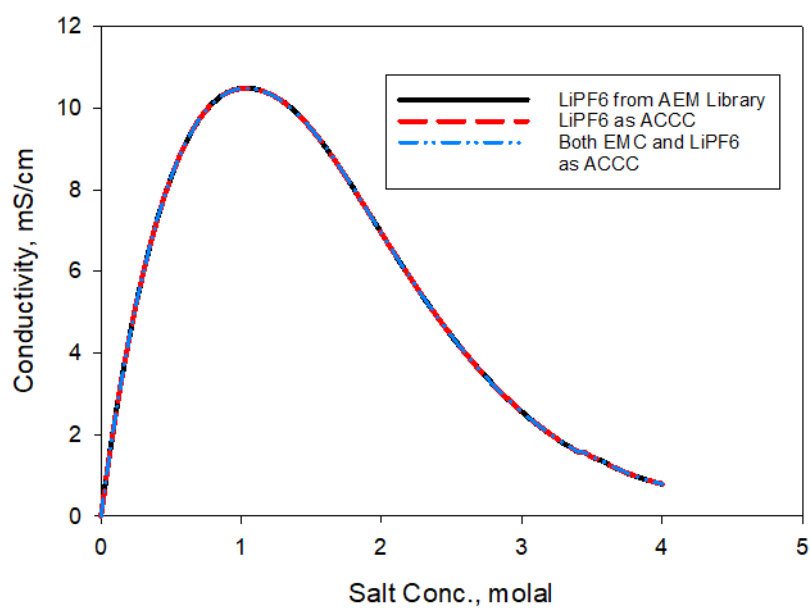


Figure 209. EC-EMC (3:7 by mass)+LiPF₆

Advanced Electrolyte Model (AEM)

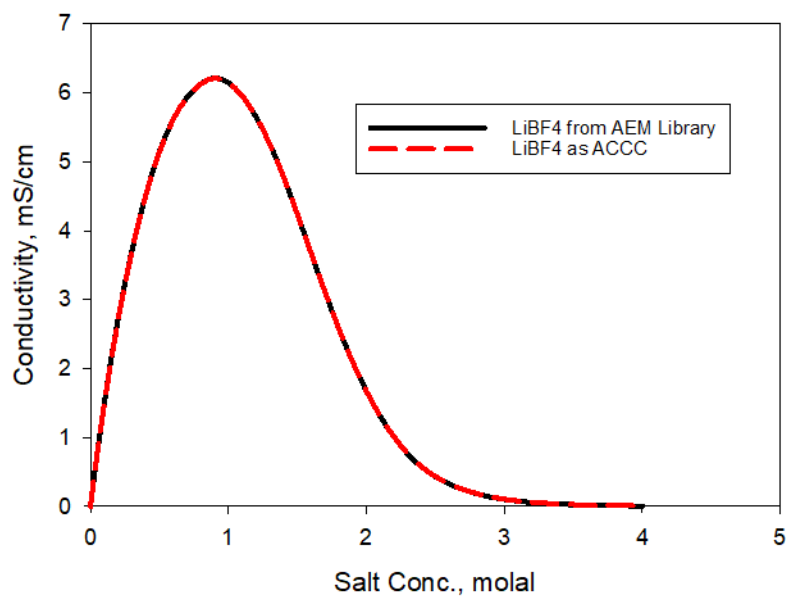


Figure 210. EC-EMC (2:8 by mass)+ LiBF₄

7.5 List of Relevant Publications

The following list provides a partial list of relevant AEM publications.

- A Critical Evaluation of the Advanced Electrolyte Model
- A Study of the Transport Properties of Ethylene Carbonate-Free Li Electrolytes
- Nonaqueous Liquid Electrolytes for Lithium-Based Rechargeable Batteries
- A Study of the Physical Properties of Li-ion Battery Electrolytes Containing Esters
- An automated system for performing continuous viscosity *versus* temperature measurements of fluids using an Ostwald viscometer
- Synergistic Effect of LiPF₆ and LiBF₄ as Electrolyte Salts in Lithium-Ion Cells
- Prediction of Electrolyte Conductivity: Results from a Generalized Molecular Model Based on Ion Solvation and a Chemical Physics Framework
- Macro-homogenous Modeling of Commercial, Primary Li/MnO₂ Coin Cells
- Prediction of electrolyte viscosity for aqueous and non-aqueous systems: Results from a molecular model based on ion solvation and a chemical physics framework
- Low-Temperature Performance Limitations of Lithium-ion Batteries
- Novel Method for Evaluation and Prediction of Capacity Loss Metrics in Li-Ion Electrochemical Cells
- An Experimental and Modeling Study of Sodium-ion Battery Electrolytes
- Methodologies for Design, Characterization and Testing of Electrolytes that Enable Extreme Fast Charging of Lithium-ion Cells

Additional publications may also be highlighted on the Ridgetop and INL product pages in accordance with the rules and regulations of publishing journals, conferences, and other related organizations.

- <https://www.ridgetopgroup.com/about-us/ridgetop-affiliates/advanced-electrolyte-model-aem-developed-by-idaho-national-laboratory/>
- <https://labpartnering.org/technologies/b361487d-0e61-4fe1-9ba8-4d004a8dc4a2>

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Software User Guide for the Advanced Electrolyte Model (AEM) Developed by Idaho National Laboratory

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